VU Medical Zone

- Admin: Amaan Khan

BIO 503 – Biological Physics

Lecture 76 To 150

Final term PPT

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Protein fractionation techniques

Protein fractionation techniques:

1. Homogenization
2. Centrifugation
3. Fractionation by precipitation
4. Ultra filtration
5. Chromatographic
6. Electrophoresis

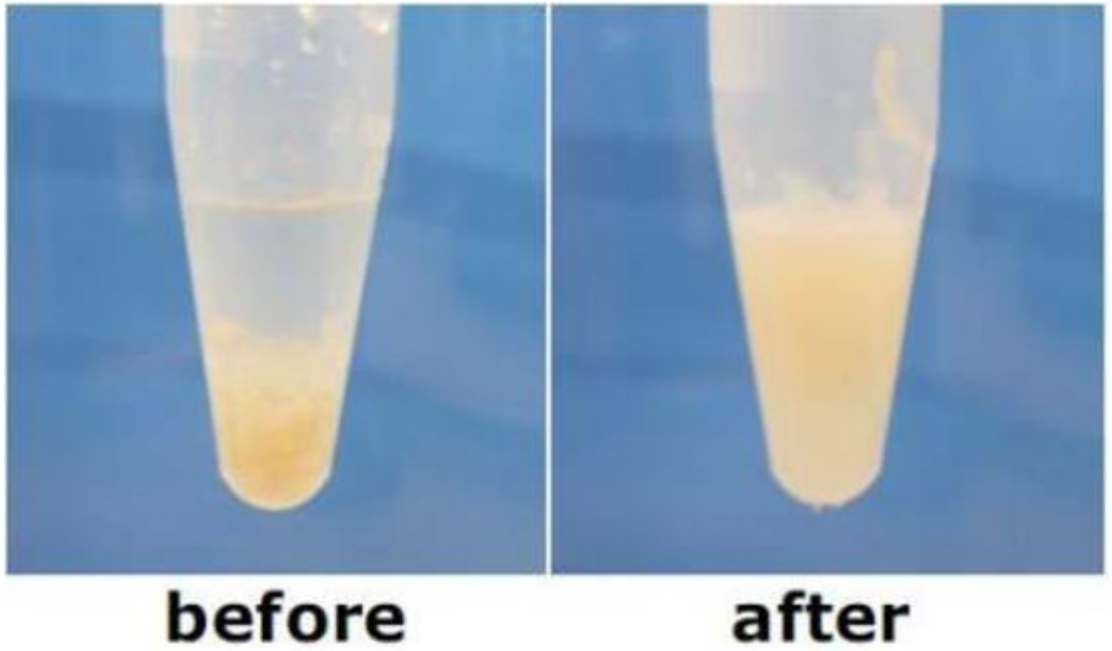
Homogenization

- Hydrophilic peptides/protein are generally extracted with homogenization in water or in solutions of organic acids whereas organic solvents are used to obtain highly hydrophobic peptides/protein.

- Homogenization in a mixture of organic solvents (chloroform/ methanol) can be used for peptide extraction as well as for the removal of sample interferences after producing a biphasic system

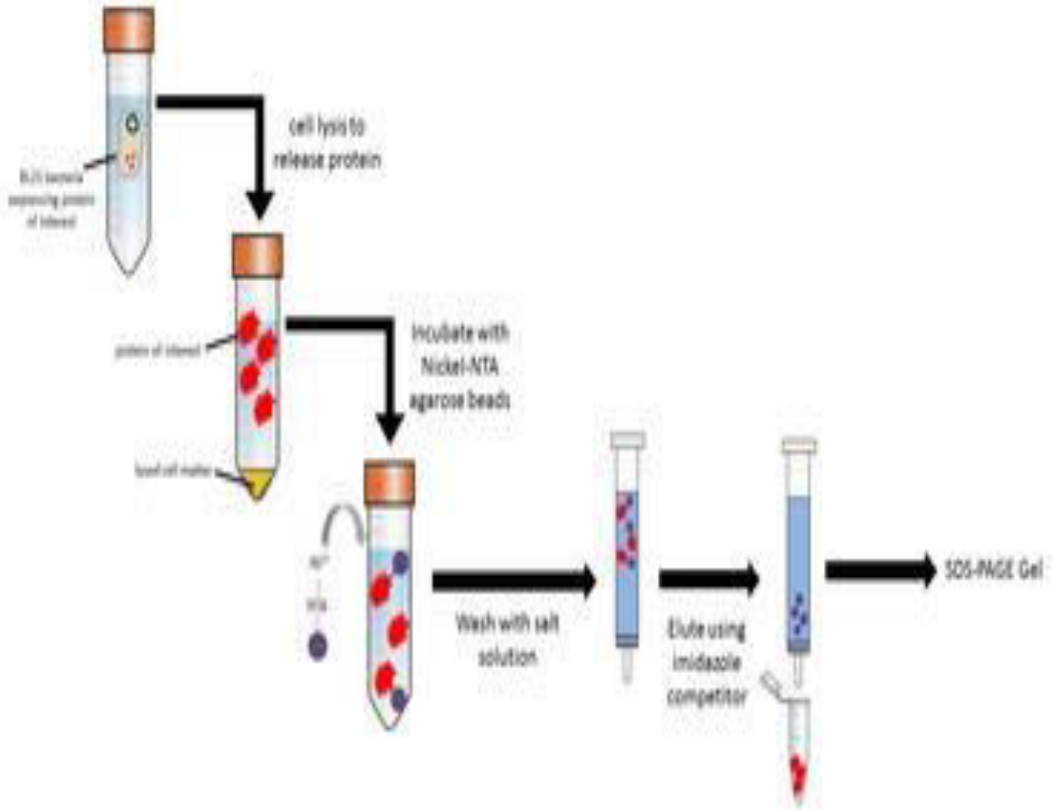
- Homogenization in a blender and in a Dounce are common and simple procedures form disruption soft tissues such as liver, heart, brain, and muscle or other sample.

Homogenization



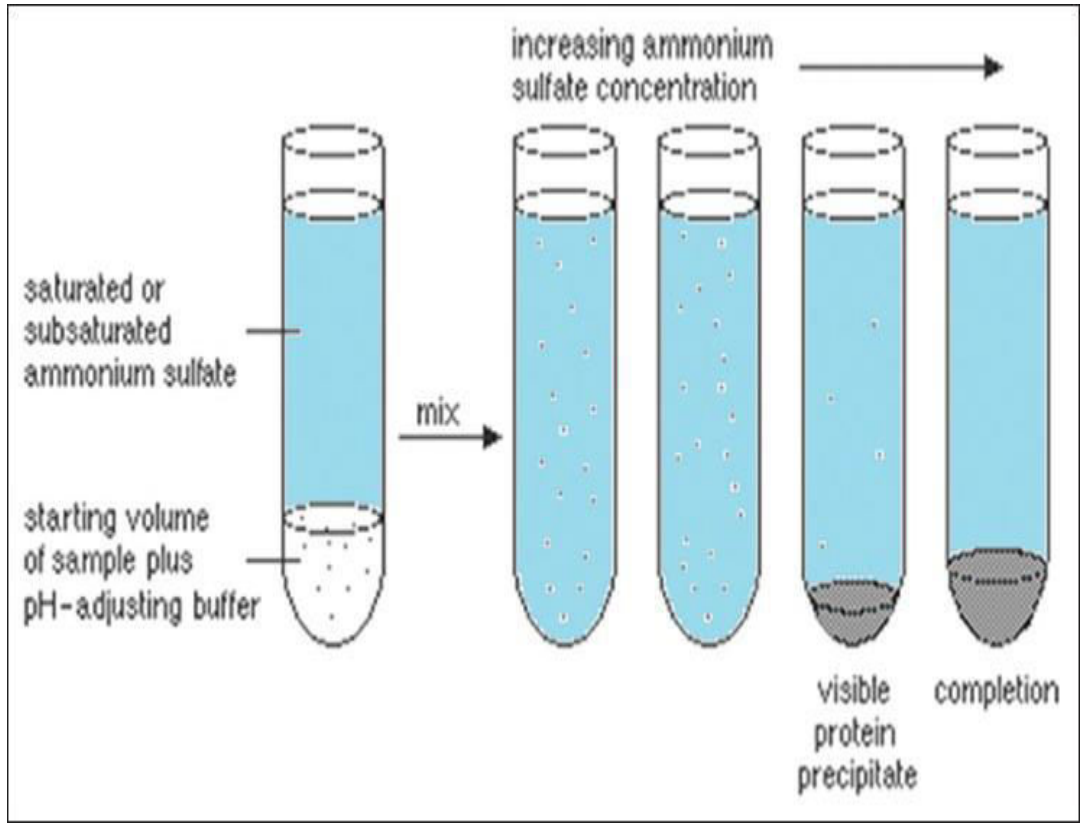
Centrifugation

- The use of centrifugation is one of the simplest methods used for isolation and fractionation of proteins.
- Centrifugation can be used for different purposes.
- It can be a first step to separate different cell substructures where our proteins of interest are locally concentrated, for instance, mitochondria, membrane, or nucleus



Precipitation

- It is recognized that among the different precipitants the most widespread is ammonium sulphate.
- The addition of high amounts of this salt or other such as sodium chloride into a protein solution provokes an increase of protein interactions followed by protein aggregation and finally precipitation.
- This is known as a salting-out process and, as the salt concentration needed for protein precipitation varies from one protein to another, it allows selective protein separation



Ultra Filtration

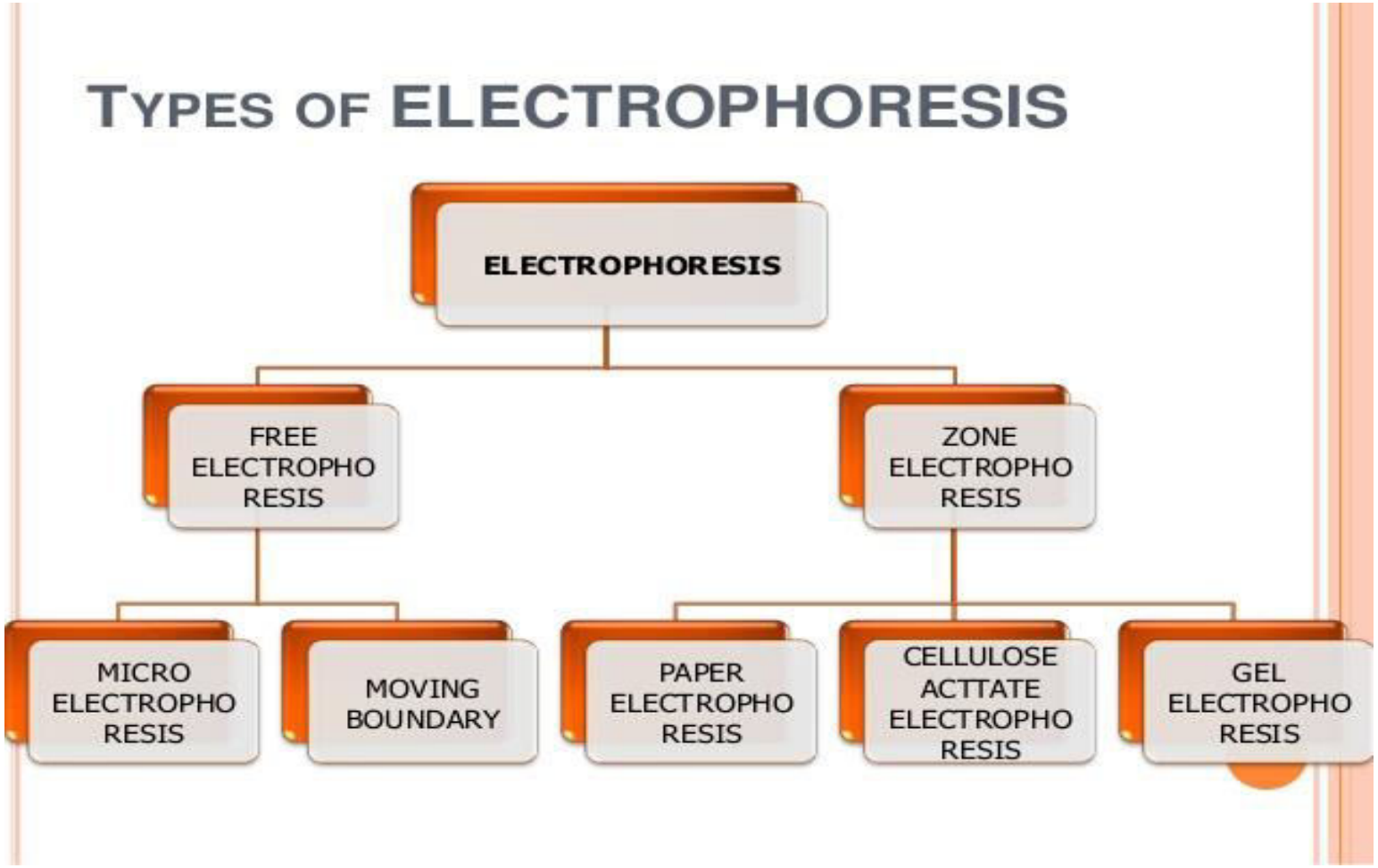
- Ultra filtration is mainly useful for fractionating peptides as well as the removal of proteins and other macromolecules based on their molecular size.
- Dedicated membranes are mostly made of polysulfone or cellulose derivatives.
- UF is used extensively in the dairy industry; particularly in the processing of cheese whey to obtain whey protein concentrate (WPC) and lactose-rich permeate.
- In a single stage, a UF process is able to concentrate the whey 10-30 times the feed.

Chromatography

- Analytical methods used to separate molecules.
- It involves a mobile and a stationary phase.
- Mobile phase is what the material to be separated is dissolved in.
- Stationary phase is a porous solid matrix which the mobile phase surrounds.
- Separation occurs because of the differing chemistries each molecule has with both the mobile and stationary phase.

Electrophoresis

Electrophoresis separates mixtures of proteins based on charge, charge/mass ratio or size.



Thank you

PROTEIN ISOLATION AND PURIFICATION

3. CRUDE SAMPLE PREPARATION:

After the protein-containing fraction has been obtained, several relatively crude methods may be used to enhance purification.

Processes like,

1. Salting out
2. Dialysis
3. Gel extrusion chromatographywere used for this purpose.

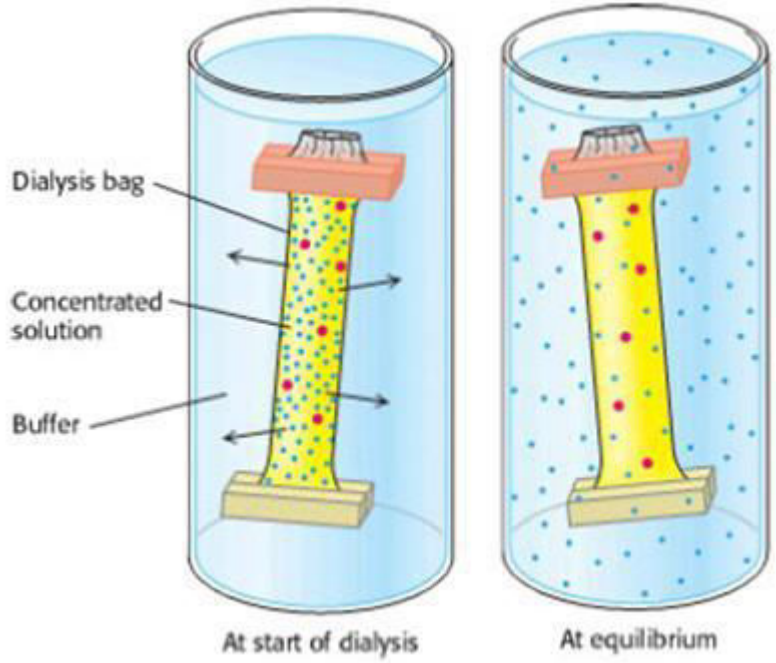
a. Salting out:

- It is a technique in which high concentrations of salts such as ammonium sulfate [(NH4)2SO4] are used to precipitate proteins. Because each protein has a characteristic salting-out point, this technique removes many impurities.
- Unwanted proteins that remain in solution are then discarded when the liquid is decanted. When proteins are tightly bound to membrane, organic solvents or detergents often aid in their extraction.
- For example, 0.8 M ammonium sulfate precipitates fibrinogen, a blood-clotting protein, whereas a concentration of 2.4 M is needed to precipitate serum albumin.
- Salting out is also useful for concentrating dilute solutions of proteins, including active fractions obtained from other purification steps. Dialysis can be used to remove the salt if necessary.

b. Dialysis:

- Proteins can be separated from small molecules by dialysis through a semipermeable membrane, such as a cellulose membrane with pores.
- Molecules having dimensions significantly greater than the pore diameter are retained inside the dialysis bag, whereas smaller molecules and ions traverse the pores of such a membrane and emerge in the dialysate outside the bag.
- This technique is useful for removing a salt or other small molecule, but it will not distinguish between proteins effectively. The speed of dialysis might be increased by using hot water and electrical field.
- Ultrafiltration is an improvement on the dialysis principle. Filters having pore sizes over the range of bimolecular dimensions are used to filter solutions to select for molecules in a particular size range.
- Because the pore sizes in these filters are microscopic, high pressures are often required to force the solution through the filter. This technique is useful for concentrating dilute solutions of macromolecules. The concentrated protein can then be diluted into the solution of choice.

b. Dialysis:


Dialysis bag
Concentrated solution
Buffer
At start of dialysis
At equilibrium

c. Gel Filtration Chromatography:

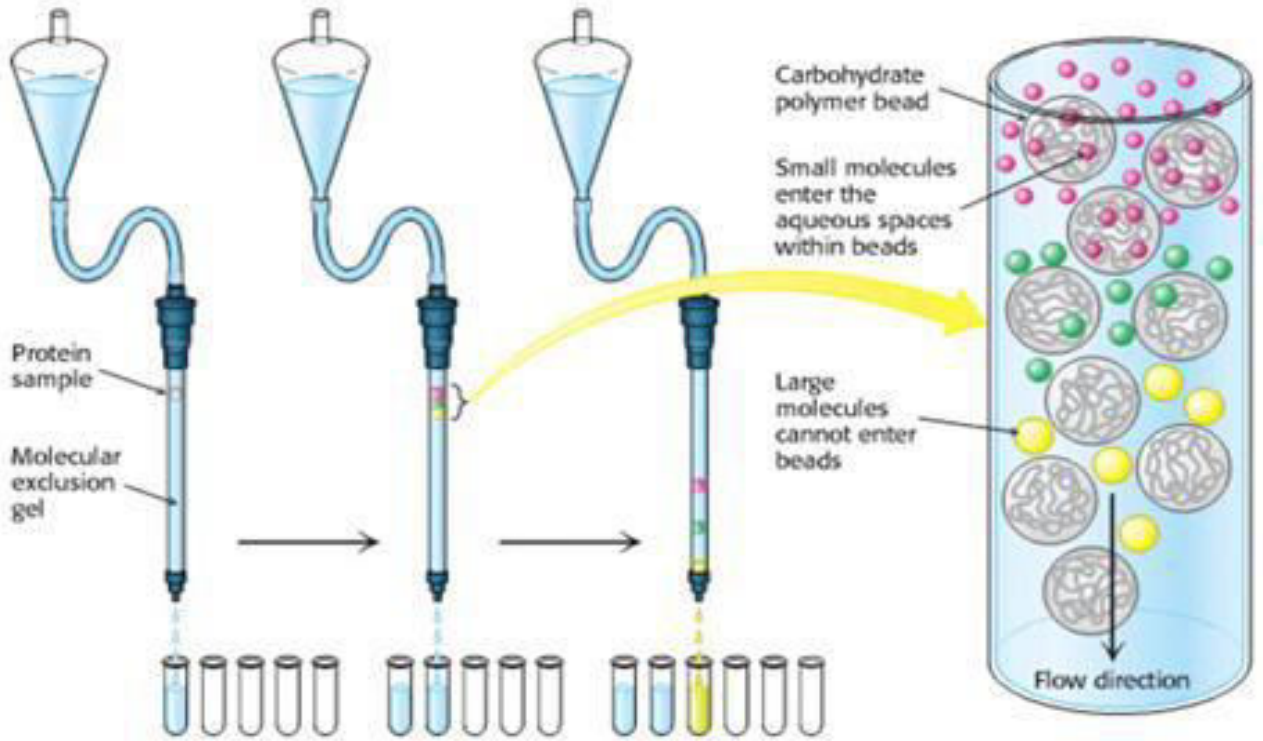
- More discriminating separations on the basis of size can be achieved by the technique of gel-filtration chromatography. The sample is applied to the top of a column consisting of porous beads made of an insoluble but highly hydrated polymer such as dextran or agarose (which are carbohydrates) or polyacrylamide.

- Sephadex, Sepharose, and Bio-gel are commonly used commercial preparations of these beads, which are typically 100 mm (0.1 mm) in diameter.

- Small molecules can enter these beads, but large ones cannot. The result is that small molecules are distributed in the aqueous solution both inside the beads and between them, whereas large molecules are located only in the solution between the beads.

- Large molecules flow more rapidly through this column and emerge first because a smaller volume is accessible to them. Molecules that are of a size to occasionally enter a bead will flow from the column at an intermediate position, and small molecules, which take a longer, tortuous path, will exit last.

c. Gel Filtration Chromatography:



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PROTEIN ISOLATION AND PURIFICATION

4. PURIFICATION:

As a protein sample becomes progressively more pure, more sophisticated methods are used to achieve further purification. The most commonly used techniques include chromatography and electrophoresis. Of these methods mainly two dimensional gel electrophoresis and affinity chromatography mostly used.

1.PROTEIN ESTIMATION:

2.TWO DIMENSIONAL GEL ELECTROPHORESIS (2DGE)

3.AFFINITY CHROMATOGRAPHY:

PROTEIN ESTIMATION:

- To determine the success of a protein purification scheme, monitor the procedure at each step by determining specific activity and by performing an SDS-PAGE analysis. Consider the results for the purification of a fictitious protein. At each step, the following parameters are measured:
- **Total protein:** The quantity of protein present in a fraction is obtained by determining the protein concentration of a part of each fraction and multiplying by the fraction's total volume.
- **Total activity:** The enzyme activity for the fraction is obtained by measuring the enzyme activity in the volume of fraction used in the assay and multiplying by the fraction's total volume.

PROTEIN ESTIMATION:

- **Specific activity:** This parameter is obtained by dividing total activity by total protein.
- **Yield:** This parameter is a measure of the activity retained after each purification step as a percentage of the activity in the crude extract. The amount of activity in the initial extract is taken to be 100%.
- **Purification level:** This parameter is a measure of the increase in purity and is obtained by dividing the specific activity, calculated after each purification step, by the specific activity of the initial extract.

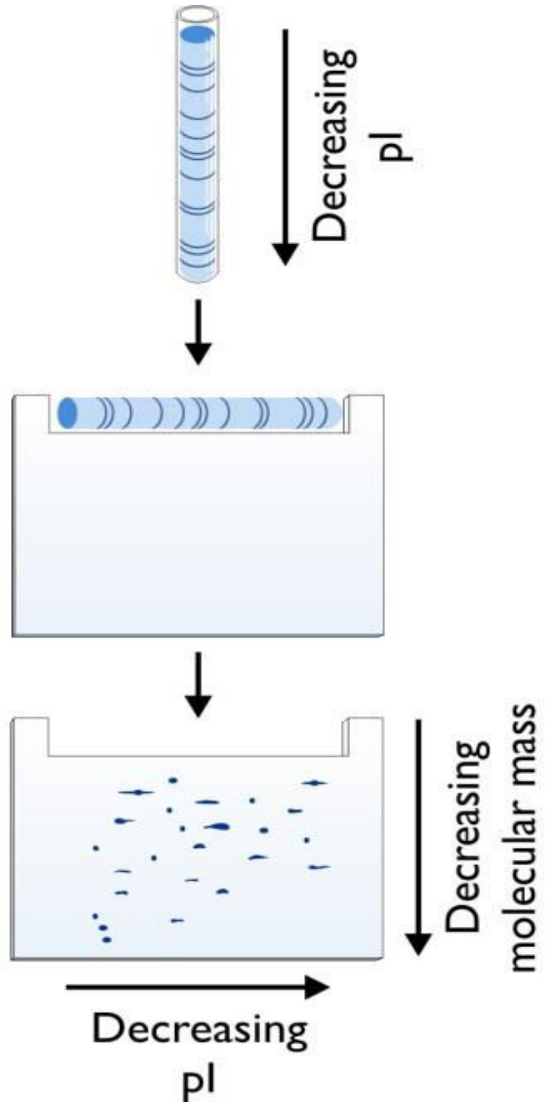
TWO DIMENSIONAL GEL ELECTROPHORESIS (2DGE)

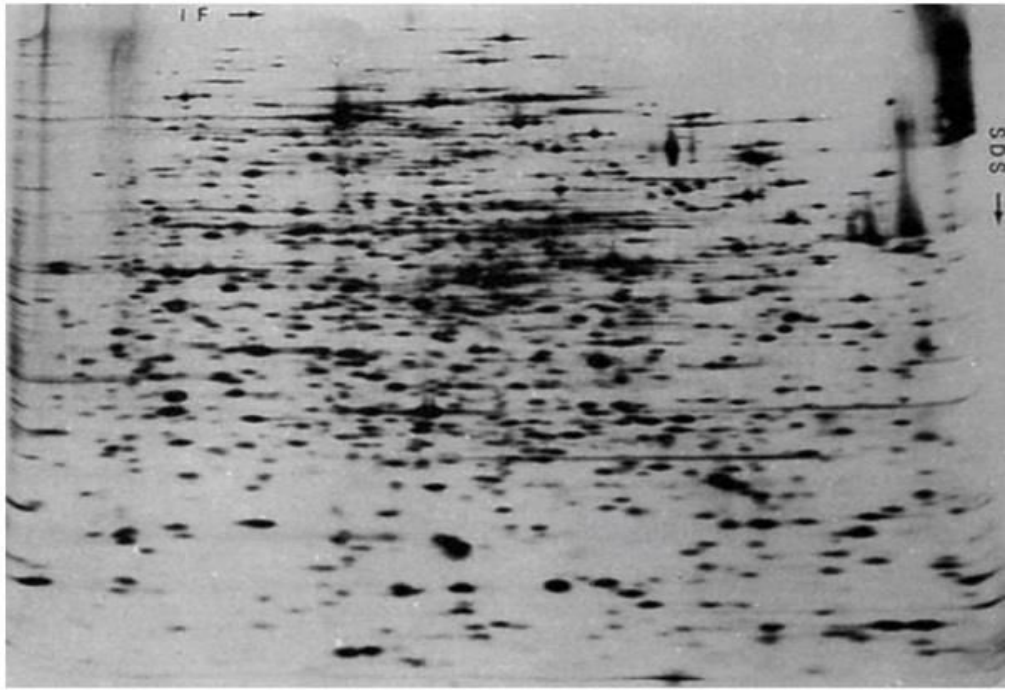
- Two-dimensional gel electrophoresis (2-D electrophoresis) is a powerful and widely used method for the analysis of complex protein mixtures extracted from cells, tissues, or other biological samples.
- This technique separate proteins in two steps, according to two independent properties: the first-dimension is isoelectric focusing (IEF), which separates proteins according to their isoelectric points (pI); the second-dimension is SDS-polyacrylamide gel electrophoresis (SDS-PAGE), which separates proteins according to their molecular weights (MW).
- In this way, complex mixtures consisted of thousands of different proteins can be resolved and the relative amount of each protein can be determined.

1. Separation of proteins by pI value

2. Soaking the gel in SDS solution and fitting it on an SDS PA gel

3. Separating the proteins by molecular mass with SDS PAGE

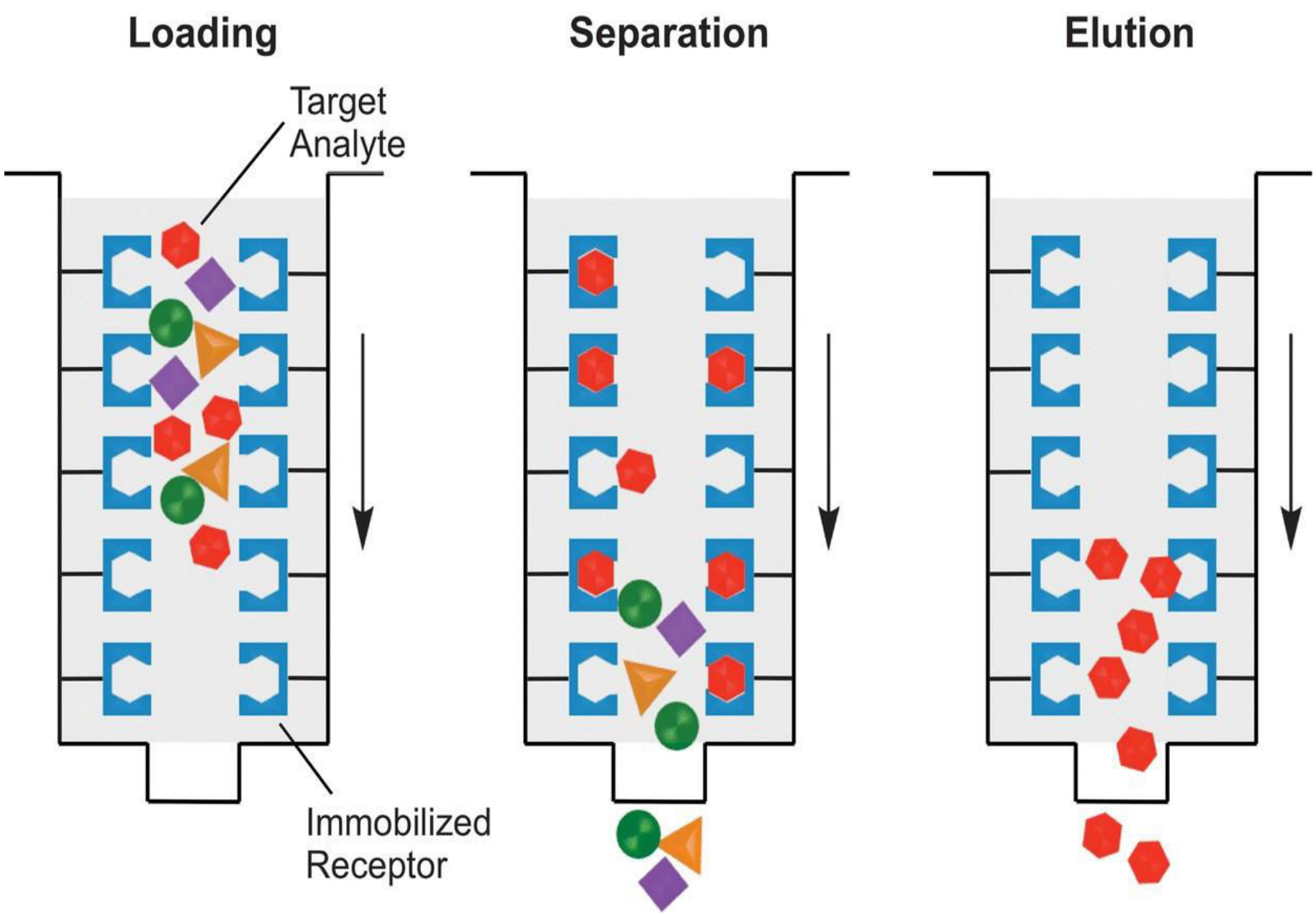




Approximately 1000 *E. coli* proteins on a single 2D gel

Affinity chromatography

- By affinity chromatography, high-selectivity separation of biomolecules can be achieved through their specific interactions. This separation technique is special because it is based on the biological function or the unique chemical structure of a given biomolecule.
- During affinity chromatography, the interacting partner of the biomolecule is immobilized on a chromatographic resin. This ligand, fixed to the stationary phase, reversibly binds the desired biomolecule present in the multi-component mobile phase. The materials can be eluted from the column by changing the composition of the mobile phase.
- The technique provides high selectivity, high resolution and generally high capacity for the desired protein. The degree of purification can be thousands of times, and the achieved yield can also be usually very good.


Loading
Separation
Elution
Target
Analyte
Immobilized
Receptor

Protein Visualization and Assessment of Purification

Reverse-phase chromatography

- **Reverse-phase chromatography** (RPC) separates proteins based on their relative **hydrophobicities**.
- This technique is highly selective but requires the use of organic solvents.
- Some proteins are permanently denatured by solvents and will lose functionality during RPC.
- Therefore this method is not recommended for all applications, particularly if it is necessary for the target protein to retain activity.

Immunoblotting

- **Immunoblotting** is a protein visualization technique applied in combination with affinity chromatography.
- Antibodies for a specific protein are used as ligands on an affinity chromatography column.
- The target protein is retained on the column, then removed by rinsing the column with a salt solution or other agents.
- Antibodies linked to radioactive or dye labels aid in the detection of the target protein once it is separated from the rest of the mixture.

Non-specific methods that detect total protein only

- **Absorbance:** Read at 280 or 205 nm. Can be very inaccurate. Detection in the range of 100 µg/mL to 1 mg/mL. Ratio of absorbance readings taken at 260/280 can indicate purity/contamination of the sample (pure samples have a ratio <0.8)
- **Bradford protein assay**: Detection in the range of ~1 mg/mL
- **Lowry Protein assay:** Detection in the range of 0.01–1.0 mg/mL
- **Fluorescamine:** Quantifies proteins and peptides in solution if primary amine are present in the amino acids
- **Amido black:** Detection in the range of 1-12 µg/mL
- **Colloidal gold:** Detection in the range of 20 - 640 ng/mL
- **Nitrogen detection**
- Kjeldahl method: used primarily for food and requires oxidation of material
- Dumas method: used primarily for food and requires combustion of material

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Specific methods which can detect amount of a single protein

- **Spectrometry methods:**
- High-performance liquid chromatography (HPLC): Chromatography method to detect proteins or peptides

- Liquid chromatography–mass spectrometry (LC/MS): Can detect proteins at low concentrations (ng/mL to pg/mL) in blood and body fluids, such as for Pharmacokinetics.

- **Antibody dependent methods:**

- **Enzyme-linked immunosorbent assay (ELISA)**: Specifically can detect protein down to pg/mL.
- **Protein immune-precipitation**: technique of precipitating a protein antigen out of solution using an antibody that specifically binds to that particular protein.
- **Immuno-electrophoresis:** separation and characterization of proteins based on electrophoresis and reaction with antibodies.
- **Western blot:** couples gel electrophoresis and incubation with antibodies to detect specific proteins in a sample of tissue homogenate or extract (a type of Immuno-electrophoresis technique).
- **Protein immune-staining**

Computational methods

- Molecular dynamics
- Protein structure prediction
- Protein sequence alignment (sequence comparison, including BLAST)
- Protein structural alignment
- Protein ontology

Interactions involving proteins

Protein–protein interactions

- (Yeast) two-hybrid system
- Protein-fragment complementation assay
- Co-immunoprecipitation
- Affinity purification and mass spectrometry
- Proximity ligation assay

Protein–DNA interactions

- ChIP-on-chip
- Chip-sequencing
- Microscale thermophoresis

Protein–RNA interactions

- Toeprinting assay
- TCP-seq

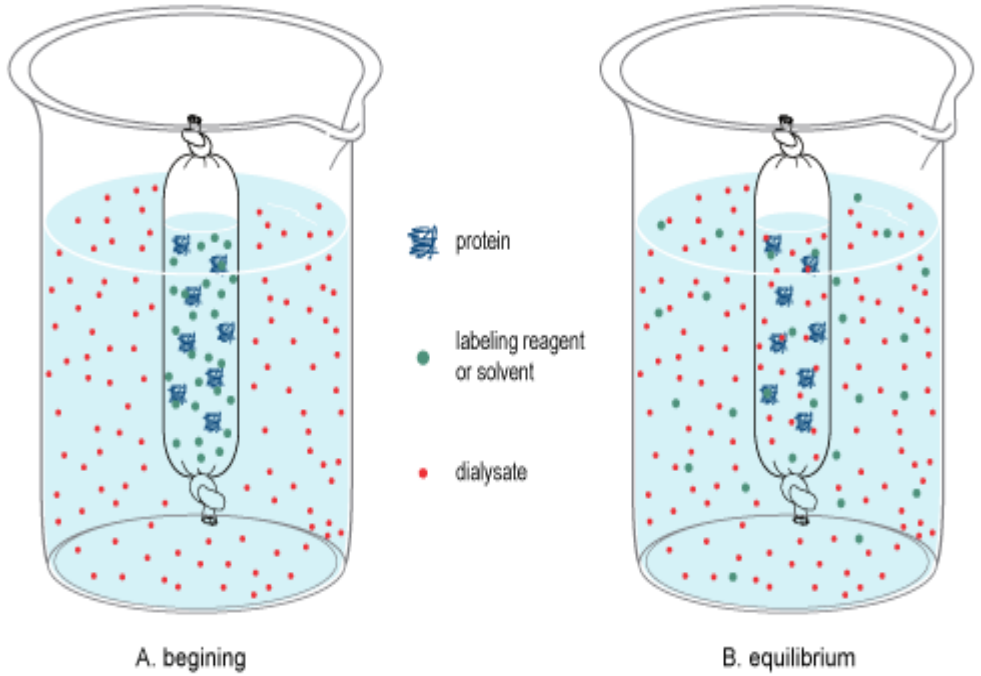
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Dialysis

Dialysis:

- **Dialysis** is the process of separating molecules in solution by the difference in their rates of diffusion through a semipermeable membrane



Principles of Dialysis

Diffusion is the random, thermal movement of molecules in solution (Brownian motion) that leads to the net movement of molecules from an area of higher concentration to a lower concentration until equilibrium is reached.

In dialysis, a sample and a buffer solution (called the dialysate) are separated by a semi-permeable membrane that causes differential diffusion patterns, thereby permitting the separation of molecules in both the sample and dialysate.

Principles of Dialysis

- Due to the pore size of the membrane, large molecules in the sample cannot pass through the membrane, thereby restricting their diffusion from the sample chamber.

- By contrast, small molecules will freely diffuse across the membrane and obtain equilibrium across the entire solution volume, thereby changing the overall concentration of these molecules in the sample and dialysate.

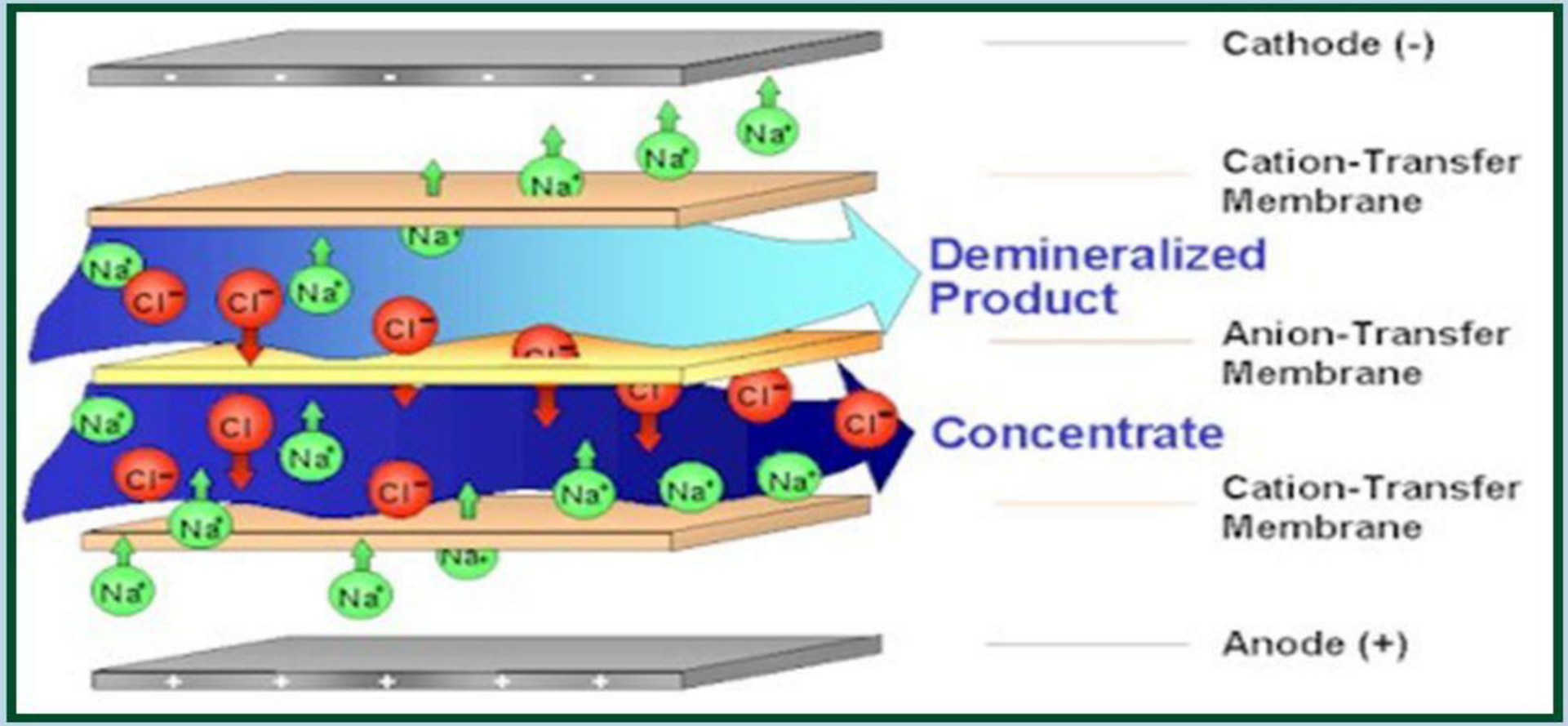
- Once equilibrium is reached, the final concentration of molecules is dependent on the volumes of the solutions involved, and if the equilibrated dialysate is replaced (or exchanged) with fresh dialysate, diffusion will further reduce the concentration of the small molecules in the sample.

ELECTRODIALYSIS

- Electro dialysis (ED) is used to transport salt ions from one solution through ion-exchange membranes to another solution under the influence of an applied electric potential difference.
- This is done in a configuration called an electro dialysis cell. The cell consists of a feed (dilute) compartment and a concentrate (brine) compartment formed by an anion exchange membrane and a cation exchange membrane placed between two electrodes.
- In almost all practical electro-dialysis processes, multiple electro-dialysis cells are arranged into a configuration called an electro-dialysis stack, with alternating anion and cation exchange membranes forming the multiple electro-dialysis cells.

Electrodialysis

- Uses electrical power to draw ions from product water to the concentrate stream.



- Electro dialysis processes are different than distillation techniques and other membrane based processes (such as reverse osmosis(RO)) in that dissolved species are moved away from the feed stream rather than the reverse.

- Because the quantity of dissolved species in the feed stream is far less than that of the fluid, electro-dialysis offers the practical advantage of much higher feed recovery in many applications.

- Electro dialysis reversal (EDR) is an advanced electro-dialysis process which utilizes a flow and polarity reversal to de-scale membrane surfaces and enable high concentration operation.

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Dialysis - Process

Dialysis:

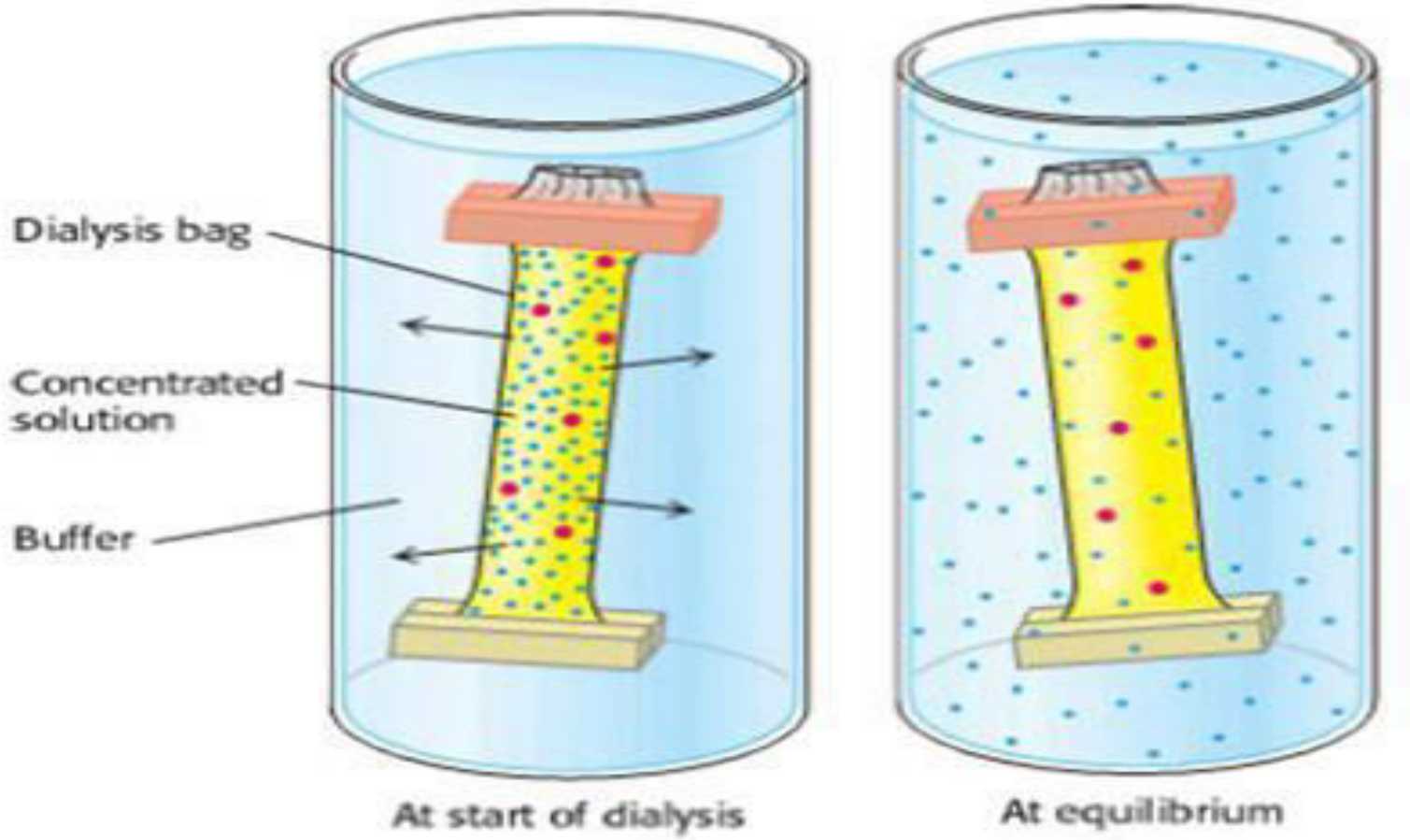
- Proteins can be separated from small molecules by dialysis through a semipermeable membrane, such as a cellulose membrane with pores.

- Molecules having dimensions significantly greater than the pore diameter are retained inside the dialysis bag, whereas smaller molecules and ions traverse the pores of such a membrane and emerge in the dialysate outside the bag. This technique is useful for removing a salt or other small molecule, but it will not distinguish between proteins effectively.

- The speed of dialysis might be increased by using hot water and electrical field. Ultrafiltration is an improvement on the dialysis principle. Filters having pore sizes over the range of bio molecular dimensions are used to filter solutions to select for molecules in a particular size range.

- Because the pore sizes in these filters are microscopic, high pressures are often required to force the solution through the filter. This technique is useful for concentrating dilute solutions of macromolecules. The concentrated protein can then be diluted into the solution of choice.

Dialysis:


Dialysis bag
Concentrated solution
Buffer
At start of dialysis
At equilibrium

General Protocol

A typical dialysis procedure for protein samples is as follows:

- Prepare the membrane
- Load the sample into dialysis tubing, cassette or device
- Place sample into an external chamber of dialysis buffer (with gentle stirring of the buffer)
- Dialyze for 2 hours (at room temperature or 4 °C)
- Change the dialysis buffer and dialyze for another 2 hours
- Change the dialysis buffer and dialyze for 2 hours or overnight
- The total volume of sample and dialysate determine the final equilibrium concentration of the small molecules on both sides of the membrane

Protein concentration using dialysis membranes

- Many samples will take on water or buffer during the dialysis process due to osmotic pressure.
- This occurs frequently with samples that have a high starting salt concentration or if a component of the sample is hygroscopic.
- In the case of high starting salt concentration, osmosis causes water to enter the sample faster than buffer salts within the sample are able to diffuse out, resulting in the swelling of the sample within the dialysis sample compartment.
- When this occurs, it may be desirable to return the sample to its original concentration, or to decrease the sample volume even further.

- To concentrate the sample, dialysis membrane containing the sample is placed in a small plastic bag containing a solution of hygroscopic compound instead of ordinary dialysate.

- To avoid contamination of the sample, the hygroscopic compound must be composed of molecules that are larger than the pore size of the dialysis tubing (e.g., high-molecular weight polyethylene glycol).

- With this set-up, concentration occurs upon diffusion of the water (osmosis) and other small molecules out of the sample and into the hygroscopic solution.

Forced dialysis

- Another method to concentrate samples is through forced dialysis.
- Vacuum is applied to a sample contained within a dialysis membrane; this effectively "pulls" water, buffer salts and other low-MW compounds out of the dialysis sample-chamber.
- Another form of dia-filtration involves "pushing" samples through a dialysis membrane by centrifugal force; this is the basis for protein concentrators, which have become popular in recent years.

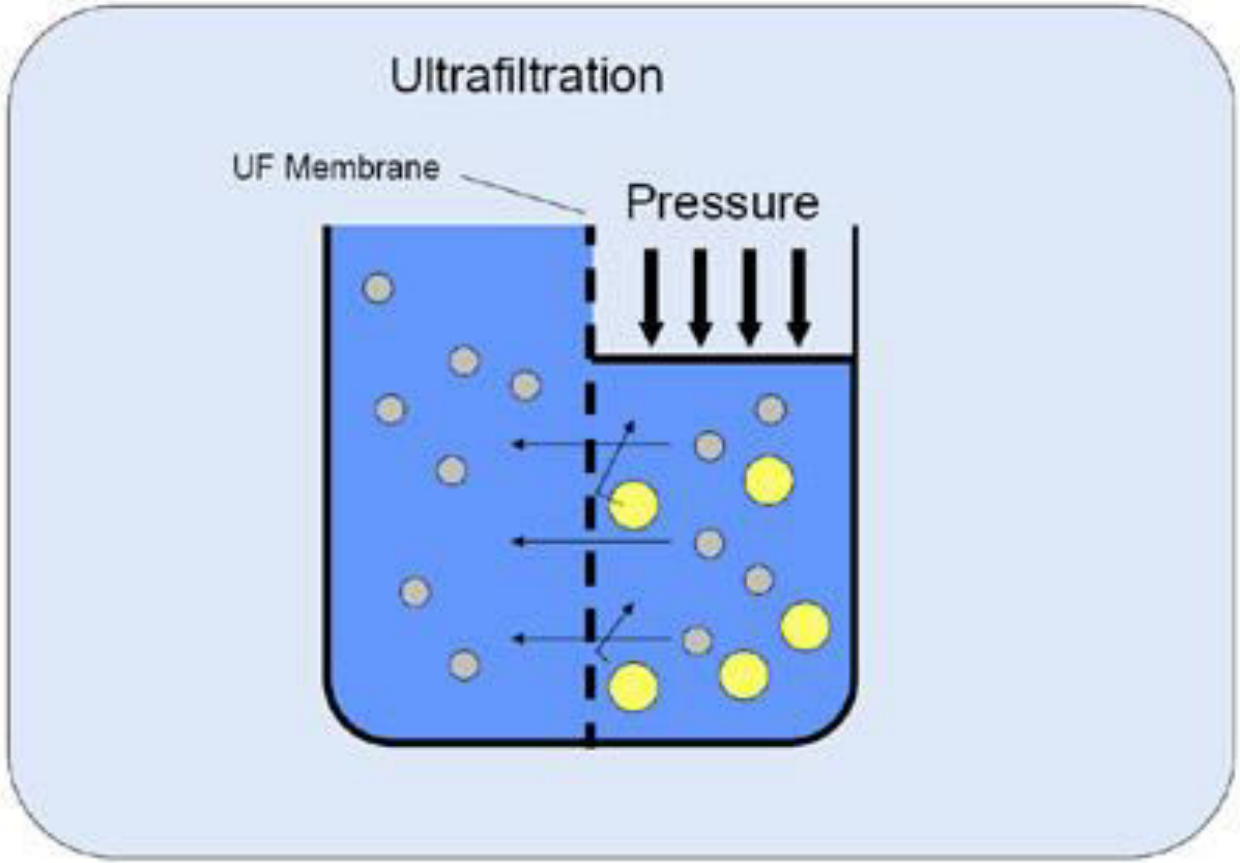
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Ultrafiltration

Ultrafiltration

- Ultrafiltration (UF) is a variety of membrane filtration in which forces like pressure or concentration gradients lead to a separation through a semipermeable membrane.
- Suspended solids and solutes of high molecular weight are retained in the so-called retentate, while water and low molecular weight solutes pass through the membrane in the permeate (filtrate).
- This separation process is used in industry and research for purifying and concentrating macromolecular (103 - 106 Da) solutions, especially protein solutions.

Ultrafiltration



Ultrafiltration

Ultrafiltration is typically used to:

- Separate proteins from buffer components for buffer exchange, desalting, or concentration
- Remove or exchange of sugars
- Remove or exchange non-aqueous solvents
- Separate free from protein-bound ligands
- Remove materials of low molecular weight
- Rapidly change ionic and/or pH environment

Principle

- The basic operating principle of ultrafiltration uses a pressure induced separation of solutes from a solvent through a semi permeable membrane.

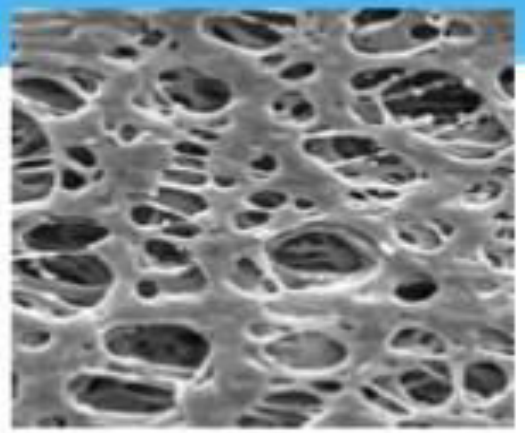
- The relationship between the applied pressure on the solution to be separated and the flux through the membrane is most commonly described by the **Darcy equation:**

$$J = TMP / \mu Rt$$

Where,

- **J** is the flux (flow rate per membrane area)
- **TMP** is the trans membrane pressure (pressure difference between feed and permeate stream)
- **μ** is solvent viscosity
- **Rt** is the total resistance (sum of membrane and fouling resistance).

Membrane Options



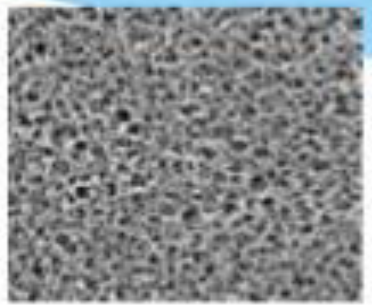
Polyethersulfone (Polymeric)

Excellent hydrolytic stability-

High flux, high retention-

Wide range of operating pH-

Durable and thus - economically efficient

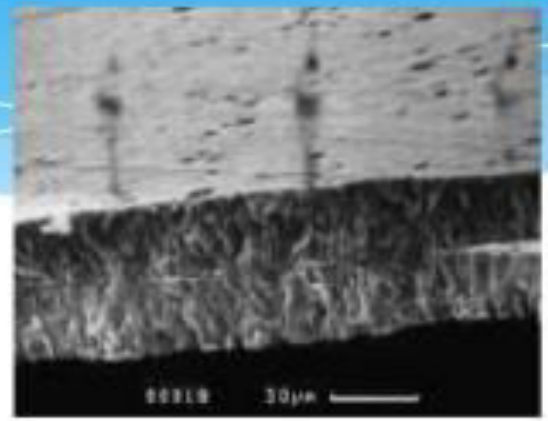


Regenerated Cellulose

-Expensive

-Susceptible to fouling and deformation

-Good flux and retention



Ceramic

-Comparatively fragile

-Narrow operating pH range

-Lower flux than polymer membranes

Applications

- Filtration of effluent from paper pulp mill
- Cheese manufacture,
- Removal of pathogens from milk
- Process and waste water treatment
- Enzyme recovery
- Fruit juice concentration and clarification
- Dialysis and other blood treatments
- Desalting and solvent-exchange of proteins (via diafiltration)
- Laboratory grade manufacturing

Thank you

LYOPHILIZATION
(Freeze-Drying)

LYOPHILIZATION

- Lyophilisation, or freeze drying, is a process in which the solvent (usually water) is:
- First frozen and then -removed by sublimation in a vacuum environment.
- **Freeze-drying**—technically known as **lyophilisation,** or **cryodesiccation**—is a dehydration process typically used to preserve a perishable material or make the material more convenient for transport.
- Freeze-drying works by freezing the material and then reducing the surrounding pressure to allow the frozen water in the material to sublimate directly from the solid phase to the gas phase.

PRINCIPLES INVOLVED IN FREEZE DRYING

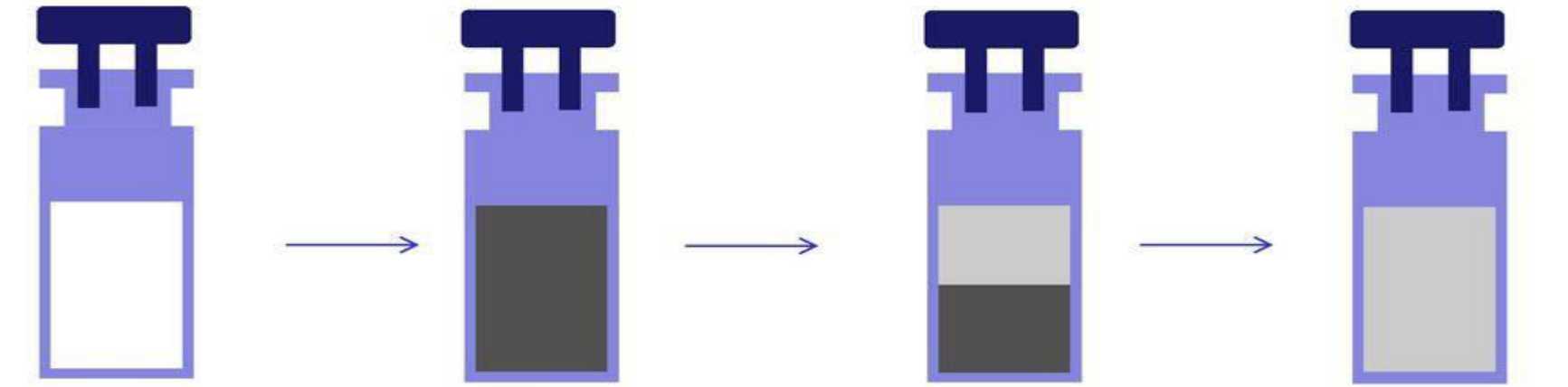
- Water is removed from frozen state by sublimation.
- Drying is achieved by subjecting material to temperature and pressures below triple point.
- The major factors that determine the phase which substance takes place depends on

1. Temperature

2. Pressure

- If temperature is b/w sea level freeze point (32°F/ 0°C) and the sea level BP (212°F/100°C), the water takes a liquid form.
- If the temperature increases above 32°F while keeping the pressure below 1 atm, the water is warm enough, but there is no enough pressure for a liquid to form, It become a gas.

Fundamentals of Lyophilization Process



PROTOCOL

1.Prepare ice box (dry ice)

2.Add chilled ethanol to reduce mist.

3.Transfer samples into ice box.

4.Remove cap and seal with parafilm.

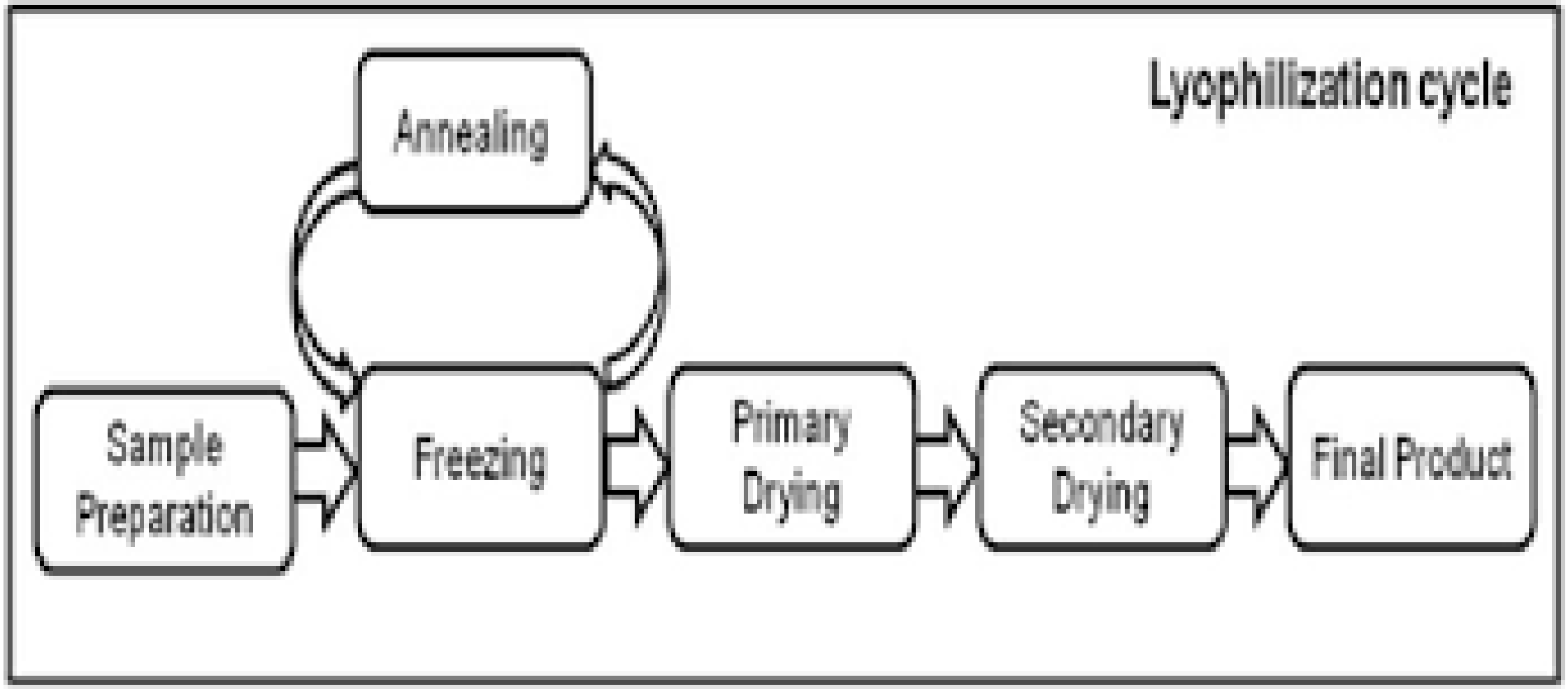
5. Puncture holes with syringe needles.

6.Turn on lyophizer and close ballast.

7.Wait for vacuum to reach <100 mT, and condenser temperature should be at least -40˚c

8.Load the sample.

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Lyophilisation cycle

Lyophilisation cycle is divided in three phases:

1. An initial freezing process, carried out in such a way that:

The product exhibits the desired crystalline structure. The product is frozen below its eutectic temperature.

2. A primary drying (sublimation) phase during which:

The partial pressure of the vapor surrounding the product must be lower than the pressure of the vapor from the ice, at the same temperature. The energy supplied in the form of heat must remain lower than the product's eutectic temperature (the highest allowable product temperature during the conditions of sublimation.)

Lyophilisation cycle

3. A secondary drying aimed at eliminating the final traces of water which remain due to absorption, and where:

The partial pressure of the vapor rising from the product will be at its lowest levels.

At the completion of the process, the treated product will have retained its form, volume and original structure-as well as all its physical, chemical and biological properties.

It can then be stored (provided packaging is effective to the reduction of moisture migration) for an almost indefinite period of time. As the product is porous, it can be re-dissolved by the simple addition of a proper solvent.

Applications of freeze-drying

- Pharmaceutical and biotechnology -shelf life of the products -easily stored, shipped -to produce tablets or wafers
- Food and agriculturally-based industries -freeze-dried ice cream -remains in good condition, longer than wet food -Instant coffee.
- Advanced ceramics processes sometimes use freeze-drying to create a formable powder from a sprayed slurry mist.
- Freeze-drying creates softer particles with a more homogeneous chemical composition than traditional hot spray drying, but it is also more expensive.

Applications of freeze-drying

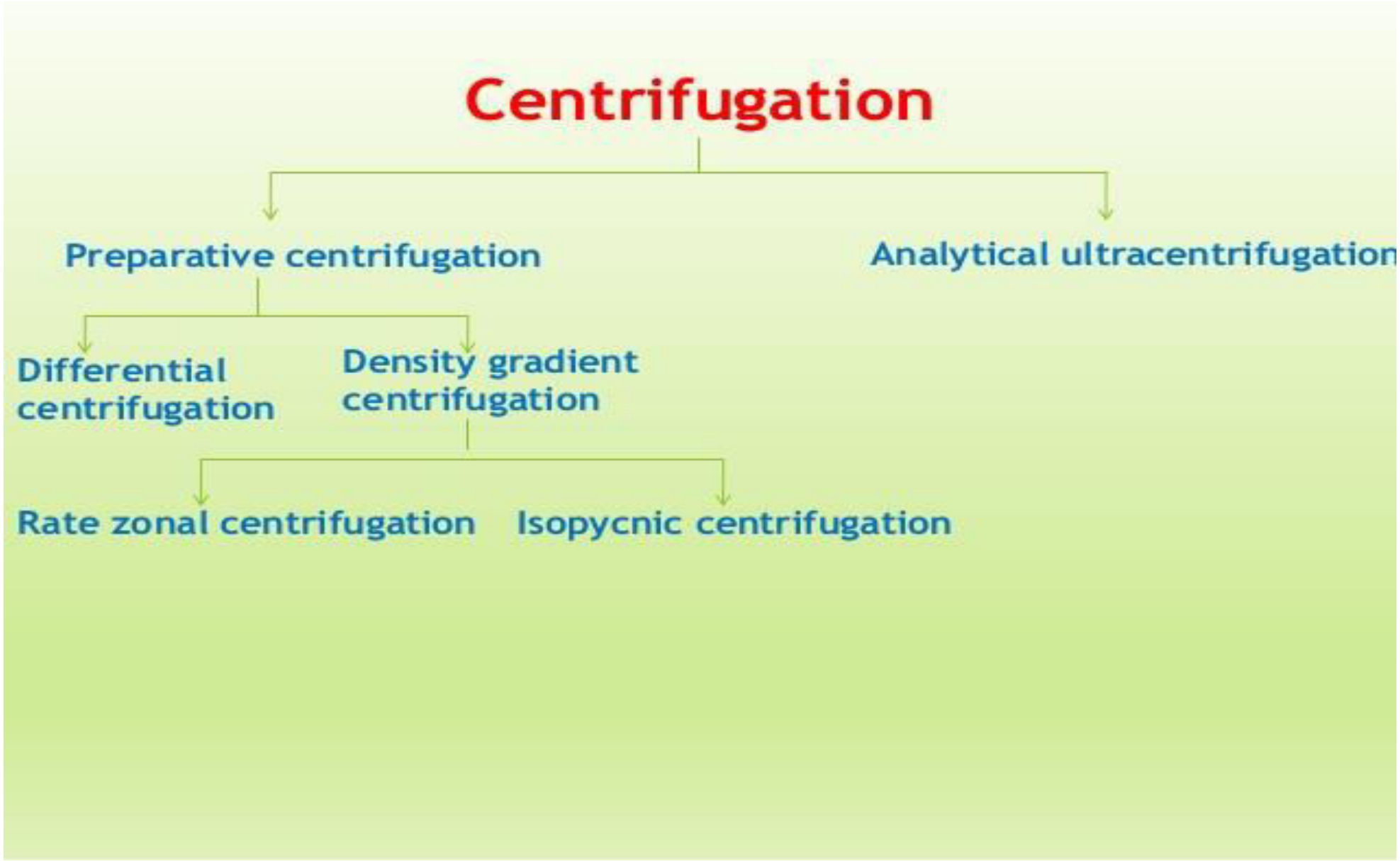
- Freeze drying is also used for floral preservation.
- Wedding bouquet preservation has become very popular with brides who want to preserve their wedding day flowers.
- A new form of burial which previously freeze-dries the body with liquid nitrogen has been developed by the **Swedish company Promessa Organic AB**, which puts it forward as an environmentally friendly alternative to traditional casket and cremation burials.

Thank you

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ULTRACENTRIFUGATION

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Types

Types of centrifugation techniques

1. **Density gradient centrifugation**
- Rate zonal centrifugation
- Isopycnic or sedimentation equilibrium centrifugation

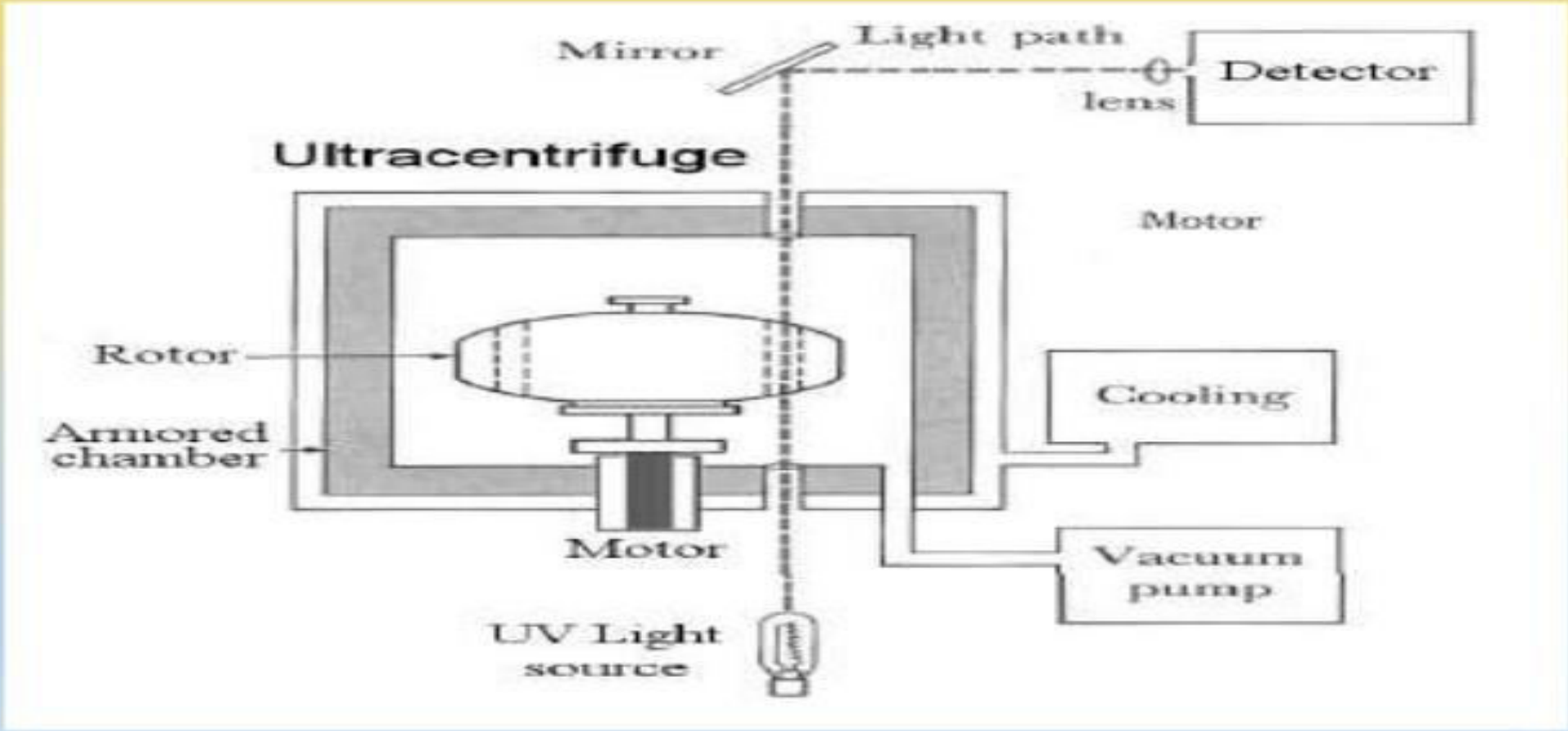
2. Differential centrifugation

3. Ultra centrifugation

ULTRACENTRIFUGATION

- It is an important tool in biochemical research. Which through rapid spinning imposes high centrifugal forces on suspended particles, or even molecules in solution, and causes separations of such matter on the basis of differences in weight.
- Example; Red cells separated from plasma of blood, nuclei from mitochondria in cell homogenates, one protein from another in complex mixtures.
- And also isolation of macromolecules such as DNA, RNA, Lipids etc.
- Its rotational speed up to 150,000 rpm.
- It is creating a centrifugal force up to 900,000 x g

Schematic presentation of a Ultracentrifuge:



Fig; A Beckman Ultracentrifugation.

TYPES

- There are two types of ultracentrifugation:

1. **Analytical ultracentrifugation:-** The aim of Analytical ultracentrifugation is use to study molecular interactions between macromolecules or to analyze the properties of sedimenting particles such as their apparent molecular weight.

2. **Preparative ultracentrifugation:-** The aim of Preparative ultracentrifugation to isolate and purify specific particles such as subcellular organelles.

Density gradient centrifugation

It allow separation of many or all components in a mixture and allows for measurement to be made. There are two forms of Density gradient centrifugation :

1. **Rate zonal centrifugation**
2. **Isopynic or sedimentation equilibrium centrifugation**

Rate zonal centrifugation

- In Rate zonal centrifugation the solution have a density gradient. The sample has a density i.e. greater than all the layers in the solution.
- The sample is applied in a thin zone at the top of the centrifuge tube on a density gradient.
- Under centrifugal force, the particles will begin sedimenting through the gradient.
- The particles will begin sedimenting in separate zones according to their size shape and density.

Isopycnic or sedimentation equilibrium centrifugation

- In this type of centrifugation , the solution contains a greater range of densities.
- The density gradient contains the whole range of densities of the particles in the sample.
- Each particle will sediment only to the position in the centrifuge tube at which the gradient density is equal to its own density.
- In Isopycnic centrifugation separation of particles occurs into zones on the basis of their density differences, independent of time.

Differential centrifugation

- Differential centrifugation is a common procedure in microbiology and cytology used to separate certain organelles from whole cells for further analysis of specific parts of cells.

- In the process, a tissue sample is first homogenized to break the cell membranes and mix up the cell contents.

- The homogenate is then subjected to repeated centrifugations, each time removing the pellet and increasing the centrifugal force.

Thank you

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Amino Acid Analysis

Types of Amino Acid Analysis

The following groups of tests are important

- Screening tests
- Quantitative tests to monitor treatment or confirm an initial diagnosis
- Specific tests that identify an unknown amino acid or metabolite
- The Amino Acid Analyzer perform the Quantitative tests

Amino Acid Analysis

The Amino Acid analyzer analysis all known amino acids beside it also analyses the following

- Taurine
- α-Aminoadipicacid
- Urea
- α-Aminobytyricacid
- Asparagine
- Hydroxyproline
- Citrulline
- 1-methyl histidine
- Ornithine
- Ammonia
- Homocysteine

Sample preparation

In physiological fluids, usually, the unbound amino acids are analyzed. Peptides, proteins or other highly molecular compounds have to be removed so as not to obstruct separation column and this could be done by

- Acid precipitation (by using Sulfosalisilic acid)
- Ultra filtration
- Ultracentrifugation

Steps of sample analysis

The sample go through 5 steps

1. Auto sampler
2. Separation column reaction
3. Coil reaction
4. Photometer
5. Recorder

2. Separation column reaction

- In this situation the separation column is the stationary phase and the buffers are the mobile phase (in comparison with thin layer chromatography where the slide is the stationary phase and the buffers are the mobile phase)
- A special pump pumps the buffer to the separation column and since amino acids are eluted at different pH medium different pH values buffers are used
- In the separation column ion exchange reaction take place and the positively charged amino acids are bound to the negatively charged sites in the separation column

1. Auto sampler

- Before samples are analyzed the commercially available standard solution have to be analyzed to calibrate the instrument

- The amino acid glutamine is not present in the standard since it decomposes quickly. Therefore when required a fresh solution of glutamine standard is prepared (1.0 µ mol / ml) and injected then the freshly prepared sample is injected and glutamine peak is identified from retention time
- Following sample preparation the sample is inserted in the specified place then the auto sampler inject 130 µl of the sample and pass it to separation column

3. Coil reaction

- After separation the instrument add the ninhydrin solution that react with the products at 130° C

- All products give purple color and are estimated at 570 nm except proline and hydroxyproline they give yellow color and are estimated at 440 nm

4. Photometer

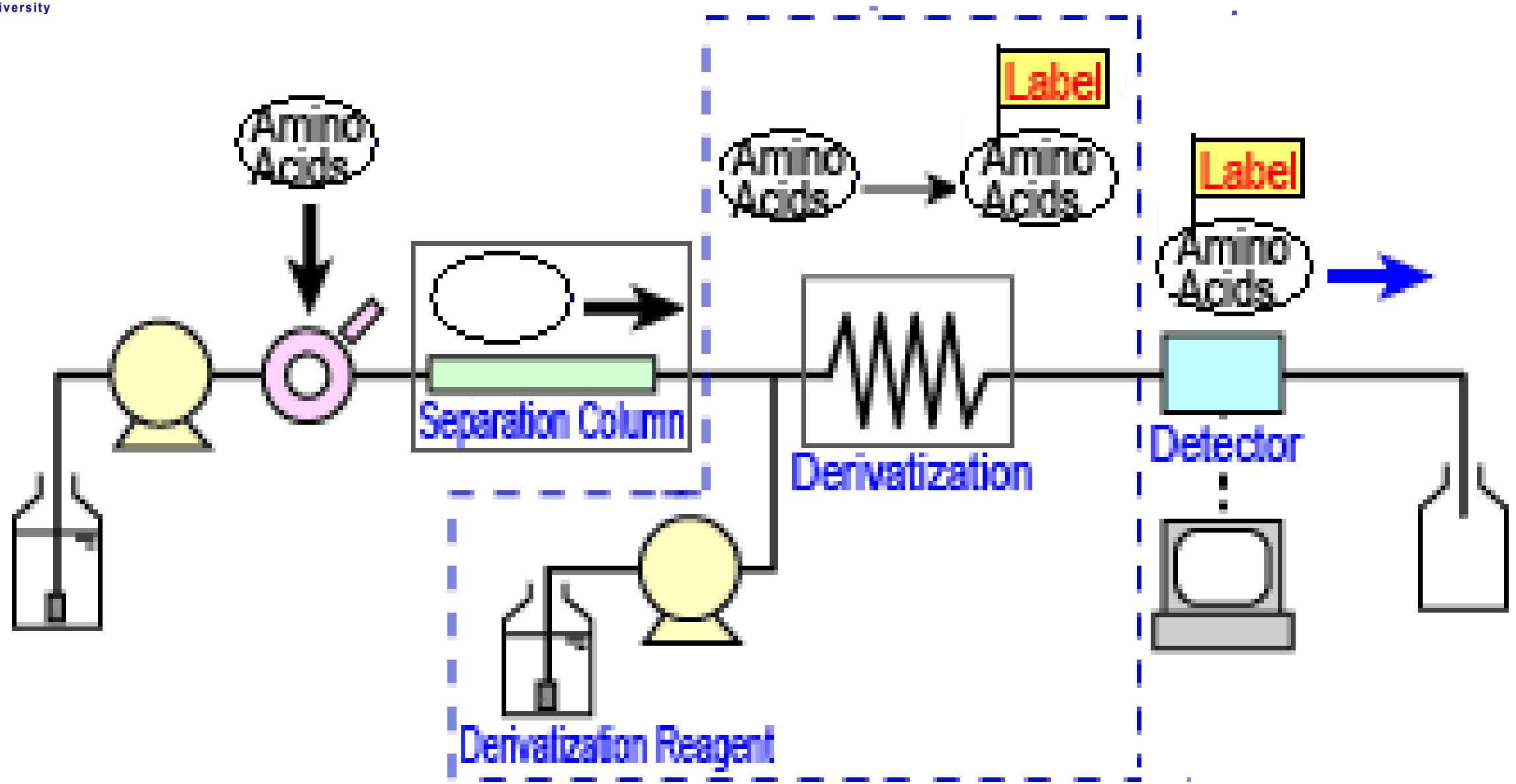
- After the ninhydrin reaction, The resultant colored species then are detected with a spectrophotometer where the absorption of the colored complex is measured at two wavelength length 570 and 440 nm

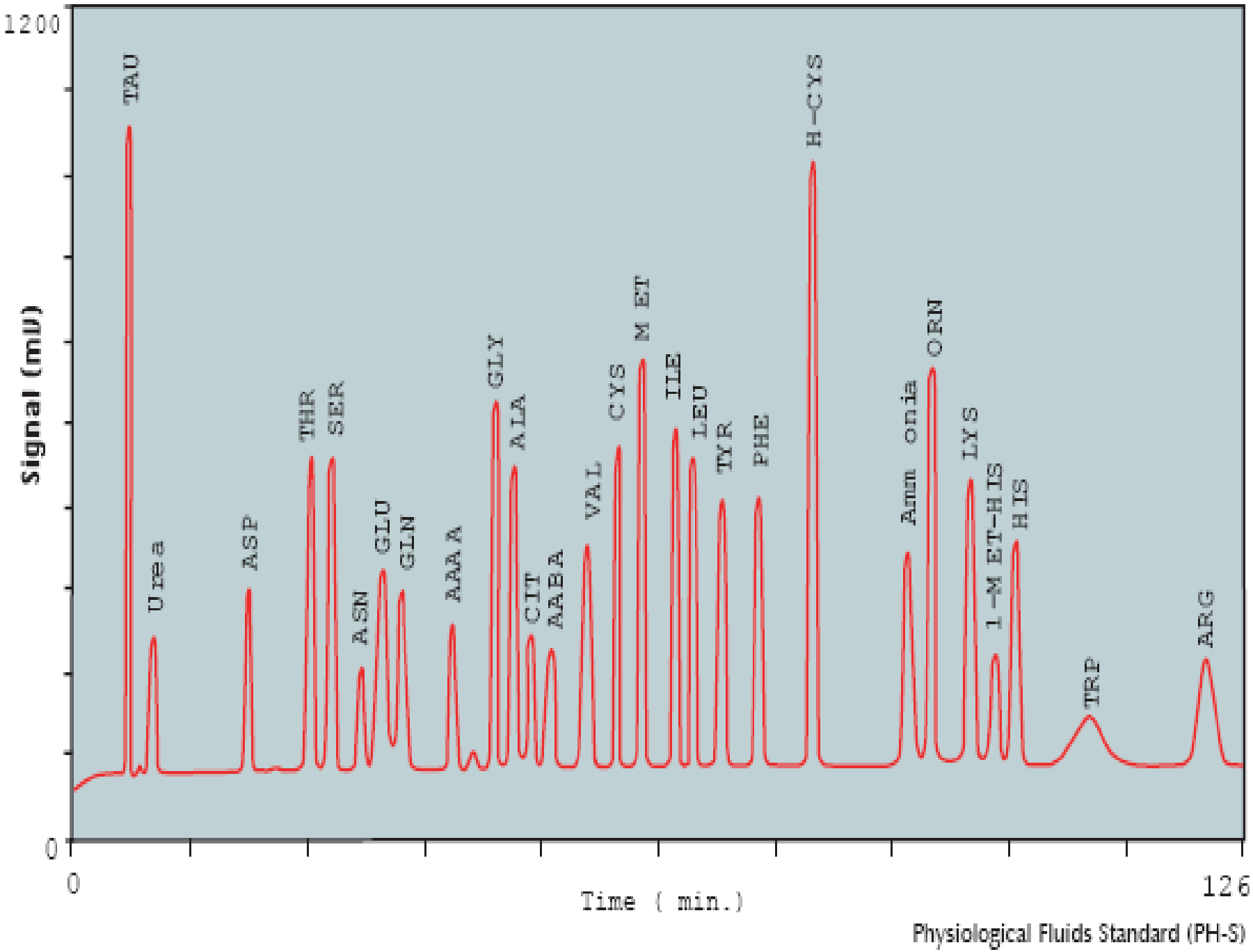
- The quantity of the colored complex produced is directly proportional to the concentration of the particular amino acid present in the sample

5. Recorder

- Photometer is linked to a two channel recorder where a series of peaks representing the amino acids are recorded

- The amino acids are identified by the comparison of the retention times of the components in the specimen to those of reference compounds


Amino Acids
Label
Amino Acids
Amino Acids
Label
Amino Acids
Separation Column
Derivatization
Detector
Derivatization Reagent


1200
0
Signal (mV)
TAU
Urea
ASP
THR
SER
ASN
GLU
GLN
AAAA
GLY
ALA
CIT
AABA
VAL
CYS
MET
ILE
LEU
TYR
PHE
H-CYS
Amm onia
ORN
LYS
1-MET-HIS
HIS
TRP
ARG
0
126
Time (min.)
Physiological Fluids Standard (PH-S)

Amino Acid Analyzer



Thank you

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Molecular forces stabilizing ligand binding

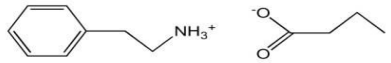
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Molecular forces stabilizing ligand binding

- The strength of protein-ligand binding is usually determined by a variety of non-covalent interactions
- These interactions differ from one another in their strength (binding energy) and their range. These interactions are as follows:

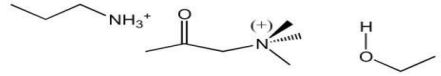
- Covalent bonding. The strongest interaction resulting irreversible binding. Energy: ~ -500
- Ionic (Coulomb) interaction: -20 – -40 kJ/mol, distance dependence: 1/*r*.

$$E = \frac{q_1 q_2}{4\pi\varepsilon_o r_{1,2}}$$



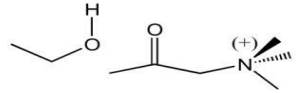
- Charge-dipole interaction: -10 – -30 kJ/mol, $1/r^2$

$$E = \frac{q_1 \mu}{4\pi\varepsilon_o r_{1,2}^2}$$



- Dipole-dipole interaction: -5 – -20 kJ/mol, $1/r^3$

$$E = \frac{\mu_1 \mu_2}{4\pi\varepsilon_o r_{1,2}^3}$$

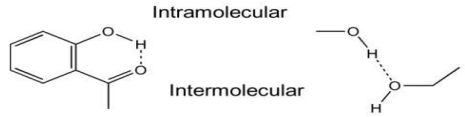


q, charge; *μ*, dipole moment; *r*, distance, ε_0, dielectric constant

- Hydrogen bond: -4 – -30 kJ/mol

H-bond donor ... acceptor

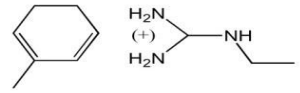
N-H ... O
N-H ... N
O-H ... N
O-H ... O



Hydrogen bond energy is largely dependent on the donor-acceptor distance, which is shorter 3,5 Å, orientation

- Charge-induced dipole interaction: -20 – -40 kJ/mol, $1/r^4$

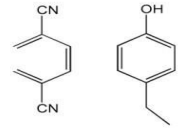
$$E = -\frac{1}{2}\frac{q_1^2 \alpha_2}{r_{1,2}^4}$$



q_1, charge; α_2, polarizability; $r_{1,2}$, distance

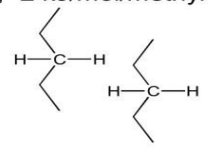
- Dipole-induced dipole interaction: -5 – -30 kJ/mol, $1/r^6$

$$E = -\frac{1}{2}\frac{\mu_1^2 \alpha_2}{r_{1,2}^6}$$



- London-dispersion forces, -2 kJ/mol/methylene group

$$E = -\frac{3 I_1 \alpha_2^2}{4 r_{1,2}^6}$$



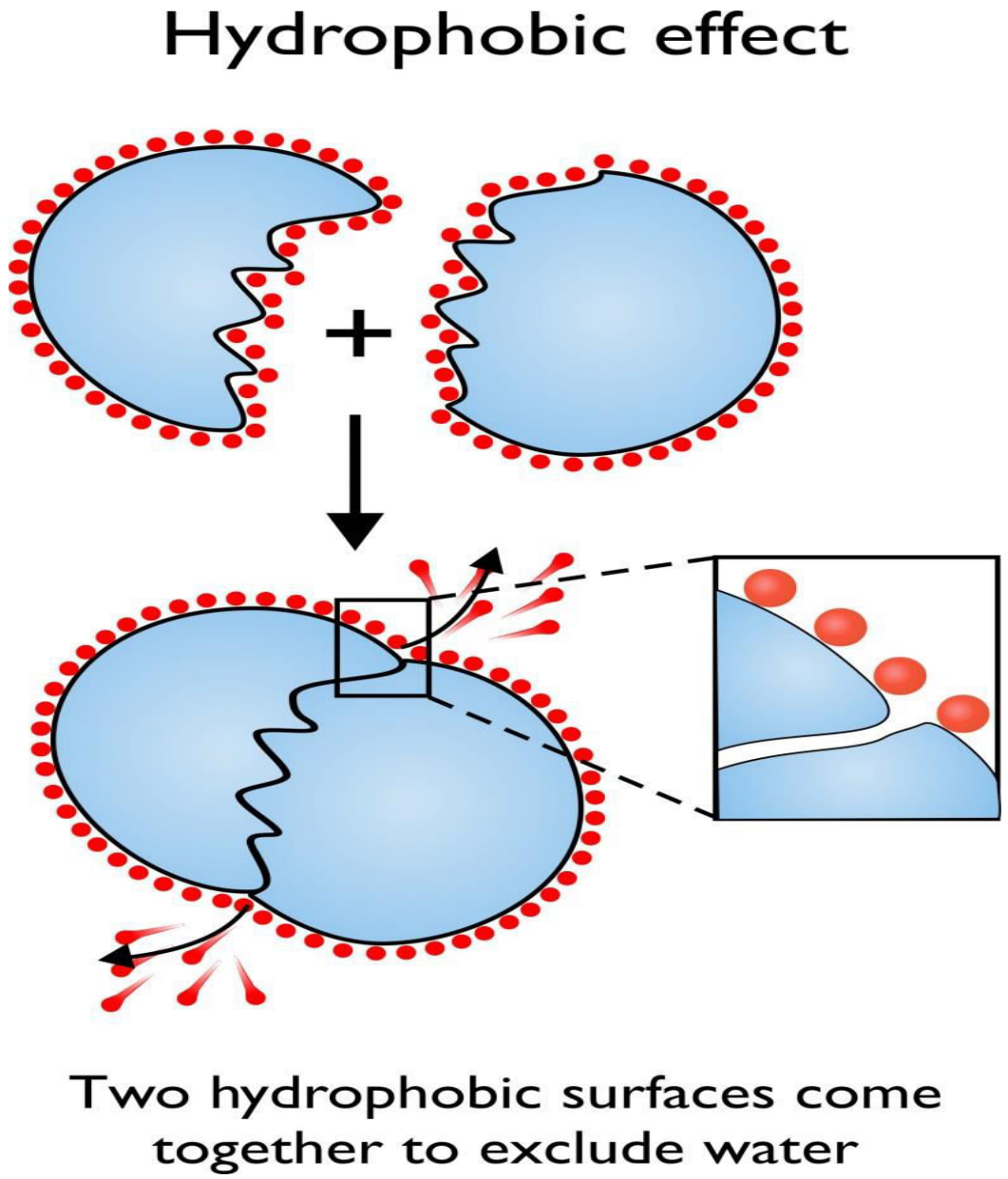
μ, dipole moment; *α*, polarizability; *I*, ionizability; *r*, distance

In the case of clashing atoms a strong repulsion is observable, which is explained by the Pauli e principle and is usually taken into account with $1/r^{12}$ distance dependence.

Beyond the previous ones, the hydrophobic effect is a complex, noncovalent interaction in which the hydrophobic part of the molecule becomes buried from water and thus the water-accessible polar surface area of the macromolecule decreases.

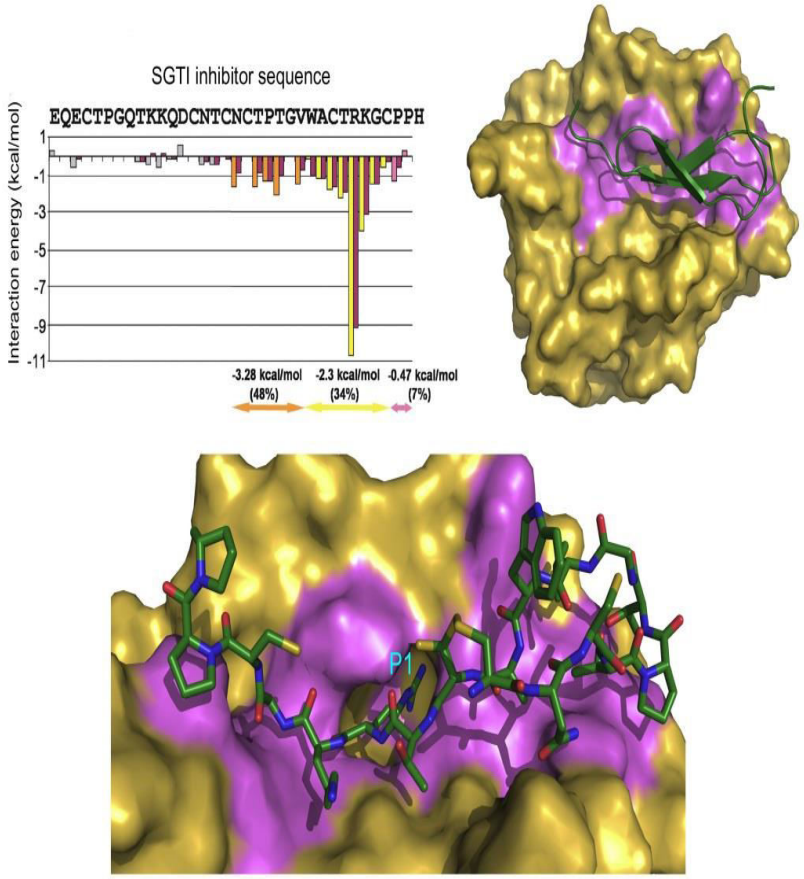
The interaction is additive; its energy is roughly proportional to the buried surface area: ~ -80-100 J/mol/Å^2.

This represents approximately -3 kJ/mol energy for the burial of a methylene group.



- The strong binding affinity and high specificity of protein-ligand interactions result from a high amount of weak noncovalent interactions.

- Figure shows an enzyme-inhibitor complex with a dissociation constant of 10-13 M.



The interaction of a trypsin inhibitor with crayfish trypsin

Methods for the experimental determination of the binding constant

- Understanding the molecular mechanisms and interactions that govern various processes and reactions in living systems is a central issue in molecular biology, biochemistry, medicine and in the new fields of proteomics and genomics.
- To reach this goal, it is indispensable to identify the interacting partners and characterize their interactions.

- A variety of techniques are available for screening and measuring protein-ligand interactions.
- Methods include affinity chromatography, cross-linking, gel filtration, co-localisation, two-hybrid methods, spectroscopic methods, 3D structure determination, equilibrium dialysis, radioactive labelling, sedimentation velocity measurements, isothermal titration calorimetry, surface plasmon resonance, microarray, immunoblotting and ELISA.

- The **radioimmunoassay (RIA)** is suitable for the determination of the concentration of various antigens, hormones and drugs in different body fluids such as the human blood.

- During the experiment, a known amount of a radioactively-labelled antigen is mixed into the solution of its corresponding antibody.

- Usually, gamma-radiating isotopes such as tyrosine-bound iodine are used.

- Subsequently, the unlabeled (“cold”) antigen sample is added to the solution and the liberation of the labelled antigen is measured.

- The unlabeled material competes with the labelled one and may “chase it off” from the antibody, depending on the concentrations.

- The material bound in complex is separated from the free antigen.

- This is a technique with high sensitivity; however, the handling of radioactive material needs special care. Nowadays, the ELISA technique is more popular, which measures the antigen-antibody interactions by applying colour reactions.

- In ELISA (enzyme-linked immunosorbent assay), the antigen is bound to a surface and then recognized by a specific antibody.

- Subsequently, a second antibody is applied that is specific to the first one and usually carries a covalently-bound enzyme molecule.

- By the addition of the substrate of the enzyme, a colorful and/or fluorescent product will form.

- This way, the presence of the antigen—or even its concentration—can be determined. A more sophisticated variant of ELISA is the “sandwich” ELISA.

- In this case, a “capture” antibody is immobilized on the membrane to which the antigen binds in the second step.

- Then a second antibody is added that also binds the antigen but—similarly to the above procedure—it carries a covalently-bound enzyme.
- Thus a “sandwich” is being formed, and then detection is carried out by adding substrate.

- The sandwich ELISA is much more sensitive than the common ELISA technique.

- Pregnancy tests are usually based on this method.

Thank You

Dissociation Constant

Dissociation Constant

- The dissociation constant is an immediate consequence of the law of mass action which describes equilibria in a more general way.
- The dissociation constant is also sometimes called ionization constant when applied to salts.

- The inverse of the dissociation constant is called association constant.

Dissociation Constant K_d



$$K_d = \frac{1}{K_a} = \frac{[\mathrm{P}] \cdot [\mathrm{L}]}{[\mathrm{PL}]}$$

the higher the binding affinity, the smaller K_d

Example: $K_a = 10^6\ M^{-1} \Leftrightarrow K_d = 10^{-6}\ M$

Statement: the rate of any chemical reaction is proportional to the product of the **masses** of the reacting substances, with each **mass** raised to a power equal to the coefficient that occurs in the chemical equation.

Law of Mass Action

- Derived from rate laws by Guldberg and Waage (1864)
  - For a balanced chemical reaction in equilibrium:

$$a\,A + b\,B \leftrightarrow c\,C + d\,D$$

  - Equilibrium constant expression (K_{eq}):



Cato Guldberg (1836-1902) Peter Waage (1833-1900)

$$K_c = \frac{[C]^c\,[D]^d}{[A]^a\,[B]^b} \quad \text{or} \quad K_p = \frac{(P_C)^c\,(P_D)^d}{(P_A)^a\,(P_B)^b}$$

- K_{eq} is strictly based on stoichiometry of the reaction (is independent of the mechanism).
- Units: K_{eq} is considered dimensionless (no units)

Dissociation constant of water

- Pure water dissociates according to the following reaction:

$$H_2O_{(l)} \Leftrightarrow H^+_{(aq)} + OH^-_{(aq)}$$

- There is an equal amount of H^+ and OH^- ions in solution (neutral, pH = 7)
- at 25°C $[H^+] = [OH^-] = 1x10^{-7}$ mol/L
- equilibrium constant for the dissociation of water: K_w

$$K_w = [H^+][OH^-]$$
$$= (1x10^{-7})(1x10^{-7})$$
$$= 1x10^{-14}$$

* small k, reactants are favoured (does not go to completion)

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Effect of temperature on K_w

$$H_2O_{(l)} \rightleftharpoons H^+_{(aq)} + OH^-_{(aq)}$$

Dissociation of water is endothermic

Increasing temp increases K_w

Increasing temp increases $[H^+]$

At higher temps pH of water will decrease

Water Temperature [°C]	K_w $[10^{-14}]$	pK_w
0	0.1	14.92
10	0.3	14.52
18	0.7	14.16
25	1.2	13.92
30	1.8	13.75
50	8.0	13.10
60	12.6	12.90
70	21.2	12.67
80	35	12.46
90	53	12.28
100	73	12.14

- The concept of the dissociation constant is applied in various fields of chemistry and pharmacology.

- In protein-ligand binding the dissociation constant describes the affinity between a protein and a ligand.

- A small dissociation constant indicates a more tightly bound the ligand.

- In the case of antibody-antigen binding the inverted dissociation constant is used and is called affinity constant.

Thank you

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Acid dissociation constant

Acid dissociation constant

- An acid dissociation constant, Ka, (also known as acidity constant, or acid-ionization constant) is a quantitative measure of the strength of an acid in solution.

- It is the equilibrium constant for a chemical reaction known as dissociation in the context of acid–base reactions.

In aqueous solution, the equilibrium of acid dissociation can be written symbolically as:

$$HA_{(aq)} \longrightarrow H^+_{(aq)} + A^-_{(aq)}$$

Strong Acid

$$HA_{(aq)} + H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + A^-_{(aq)}$$

Weak Acid

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

K_a - acid dissociation constant

Larger K_a : stronger acid : more product : more H^+

Base Ionization Constant

- **Base Ionization Constant** (K_b): the equilibrium constant for the ionization reaction of a base with water
  - $B + H_2O \rightleftharpoons OH^- + BH^+$

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

- Large K_b = Strong base
- Small K_b = Weak base

Ionic Product of Water

$$H_2O_{(l)} \rightleftarrows H^+_{(aq)} + OH^-_{(aq)}$$

$$Kc = \frac{[H^+][OH^-]}{[H_2O]}$$

In the above expression, the value of $[H_2O]$ may be taken as having a constant value because the degree of ionisation is so small.

$$Kc = \frac{[H^+][OH^-]}{[H_2O]}$$

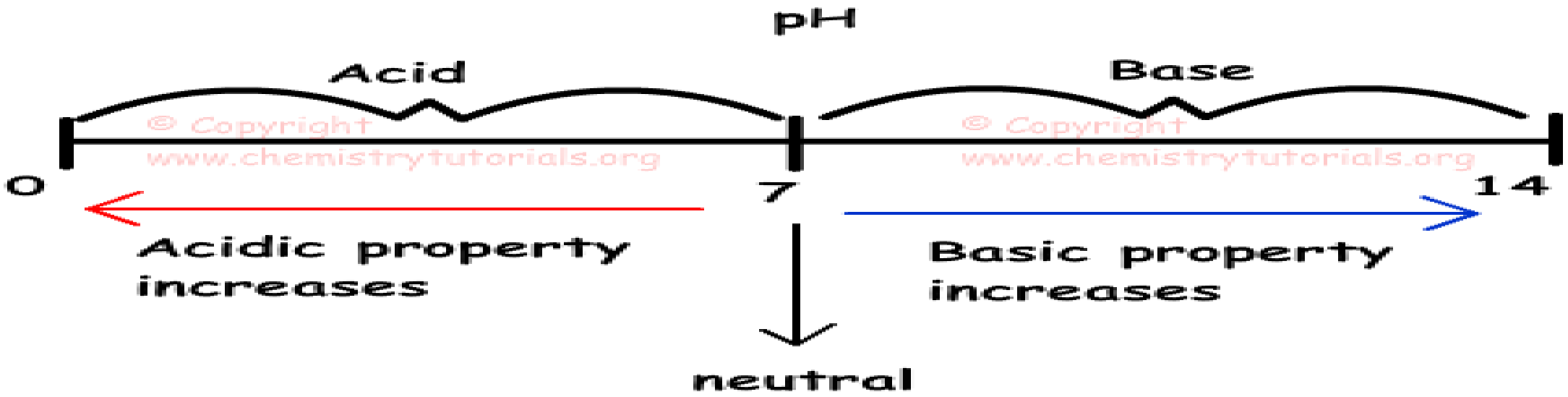
$$Kc\,[H_2O] = [H^+]\,[OH^-]$$

Both Kc and $[H_2O]$ are constant values so

$$Kw = Kc\,[H_2O] = [H^+]\,[OH^-]$$

$Kw = [H^+]\,[OH^-]$ is the ionic product of water


pH Scale
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Acid
Alkaline


pH
Acid
Base
© Copyright
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www.chemistrytutorials.org
0
7
14
Acidic property
increases
Basic property
increases
neutral

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Calculations and practice

- You will need to memorize the following:

$$pH = -\log_{10}[H^+]$$

$$pOH = -\log_{10}[OH^-]$$

$$pH + pOH = 14$$

Equilibria Involving A Weak Base

You have 0.010 M NH_3. Calc. the pH.

$NH_3 + H_2O \rightarrow NH_4^+ + OH^-$

$K_b = 1.8 \times 10^{-5}$

Step 3. Calculate pH

$[OH^-] = 4.2 \times 10^{-4}$ M

so pOH = - log $[OH^-]$ = 3.37

Because pH + pOH = 14,

pH = 10.63



Practice Test

- Assuming complete dissociation, calculate the pH of the following solutions:
- (i) 0.003 M HCl
- (ii) 0.005 M NaOH
- (iii) 0.002 M HBr
- (iv) 0.002 M KOH

(i) 0.003MHCl:

$$H_2O + HCl \longleftrightarrow H_3O^+ + Cl^-$$

Since HCl is completely ionized,

$$[H_3O^+] = [HCl].$$

$$\Rightarrow [H_3O^+] = 0.003$$

Now,

$$pH = -\log[H_3O^+] = -\log(.003)$$
$$= 2.52$$

Hence, the pH of the solution is 2.52.

(ii) 0.005MNaOH:

$$NaOH_{(aq)} \longleftrightarrow Na^{+}_{(aq)} + HO^{-}_{(aq)}$$

$$[HO^{-}] = [NaOH]$$

$$\Rightarrow [HO^{-}] = .005$$

$$pOH = -\log[HO^{-}] = -\log(.005)$$

$$pOH = 2.30$$

$$\therefore pH = 14 - 2.30$$

$$= 11.70$$

Hence, the pH of the solution is 11.70.

(iii) 0.002 HBr:

$$HBr + H_2O \longleftrightarrow H_3O^+ + Br^-$$

$$\left[H_3O^+\right] = \left[HBr\right]$$

$$\Rightarrow \left[H_3O^+\right] = .002$$

$$\therefore pH = -\log\left[H_3O^+\right]$$

$$= -\log(0.002)$$

$$= 2.69$$

Hence, the pH of the solution is 2.69.

(iv) 0.002 M KOH:

$$KOH_{(aq)} \longleftrightarrow K^{+}_{(aq)} + OH^{-}_{(aq)}$$

$$[OH^{-}] = [KOH]$$

$$\Rightarrow [OH^{-}] = .002$$

$$\text{Now, } pOH = -\log[OH^{-}]$$

$$= 2.69$$

$$\therefore pH = 14 - 2.69$$

$$= 11.31$$

Hence, the pH of the solution is 11.31.

Thank You

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Interactions of molecules in 3-D space

Molecular interactions

- Molecular Interactions are attractive or repulsive forces *between molecules* and between non-bonded atoms.
- Molecular interactions are important in diverse fields of protein folding, drug design, material science, sensors, nanotechnology, separations, and origin of life.
- Molecular interactions are also known as noncovalent interactions or intermolecular interactions.
- Molecular interactions are not bonds.

- Bonds hold atoms together within molecules.
- A molecule is a set of atoms that associates tightly enough that it does not dissociate or lose its structure when it interacts with its environment.
- At room temperature two nitrogen atoms form a bond. Two argon atoms do not.
- The breaking of chemical bonds never releases energy to the external environment. Energy is only released when chemical bonds are *formed*.
- In general, a chemical reaction involves two steps:
- 1) the original chemical bonds between the atoms are broken
- 2) new bonds are formed.
- These two steps are sometimes lumped into one event for simplicity, but they are really two separate events. For instance, when you burn methane (natural gas) in your stove, the methane is reacting with oxygen to form carbon dioxide and water. Chemists often write this as:

$$CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O + \text{energy}$$

By contrast, covalent bonds remain intact and unchanged when

- (a) ice melts
- (b) water boils
- (c) proteins unfold
- (d) RNA unfolds
- (e) DNA strands separate
- (f) membranes disassemble.

These processes are not chemical reactions. The processes (a-f) of melting, boiling, unfolding, strand separation and disassembly involve changes in molecular interactions.

Boiling Points.

- When a molecule transitions from the liquid to the gas phase (as during boiling), ideally all molecular interactions are disrupted.
- Differences in boiling temperatures give good qualitative indications of strengths of molecular interactions.
- High boiling liquids have strong molecular interactions.
- The boiling point of H2O is hundreds of degrees greater than the boiling point of N2 because of stronger molecular interactions in H2O(liq) than in N2(liq).
- The forces between molecules in H2O(liq) are greater than those in N2(liq).

Folding and assembly of biological macromolecules.

Native states. In biological systems

(i) proteins fold into globular structures called native states

(ii) ribosomal and transfer RNAs also fold into native globular structures

(iii) DNA forms double stranded helices

(iv) phospholipids form membranes

(v) proteins assemble with DNA, RNA, membranes and with other proteins.

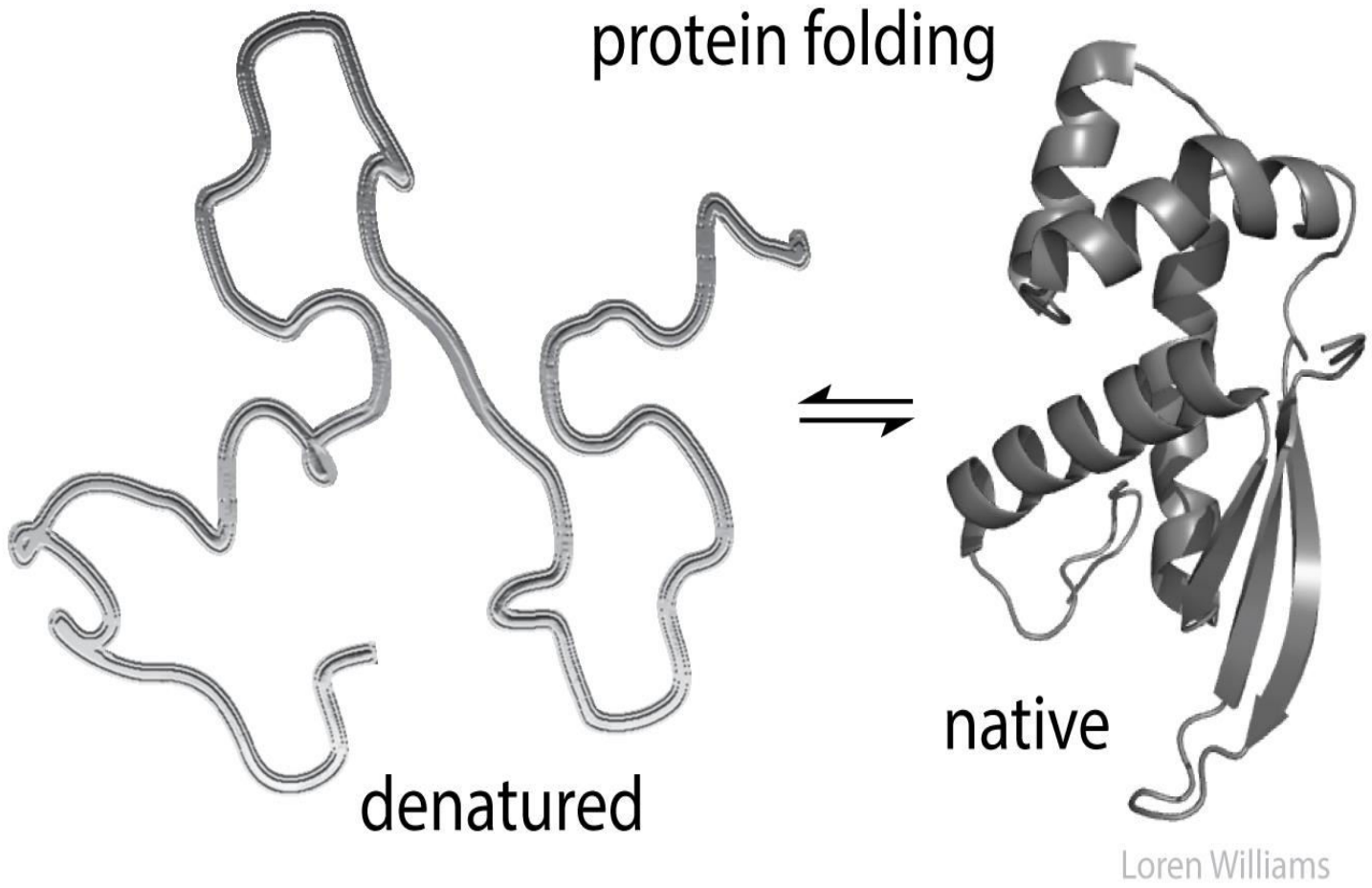
These native states and assemblies are stabilized by molecular interactions of enormous number and complexity. Native states are destabilized by their low conformational entropy.

Folding and assembly of biological macromolecules.

- *Denatured states.* When you unfold a protein or an RNA (denature them) or separate two strands of DNA (melt it), or disassemble and melt the ribosome, then interior regions become exposed to the surroundings, which are mostly water plus some ions.

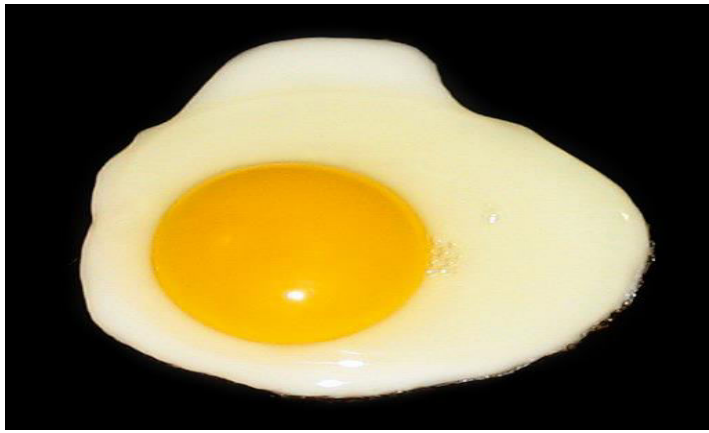
- Molecular interactions within the native state or assembly are replaced by molecular interactions with aqueous surroundings.

- Figure shows the folding of a protein. Protein folding is the conversion from a denatured state (a random coil) to a native state. On the right-hand side the arrows are β-strands and the coils are α-helices.



- Near the tipping point. Molecular interactions stabilize both native and denatured states.
- Huge numbers of intramolecular interactions within a protein native state are opposed by huge numbers of intermolecular interactions in the denatured state (with surrounding water molecules).
- On balance, native biological macromolecules and assemblies are marginally stable. Biological systems in general are held near the tipping point, pushed in opposing directions by powerful forces.
- A small perturbation can tip the balance from folded state to unfolded state. A small change in pH or temperature or a single mutation can unfold a protein.

- Have you ever denatured a protein (converted it from the native state to denatured state)? Yes.
- When you heat an egg to around 60° C, the albumin proteins denature and aggregate. You are not breaking bonds when you boil an egg - you are changing and rearranging molecular interactions.
- The aggregated protein forms large assemblies that scatter light, giving the egg a white appearance.
- When you add lemon juice to milk, the pH drops and the proteins denature and aggregate. Have you ever melted DNA? Yes, if you have run a PCR reaction.



Thank you

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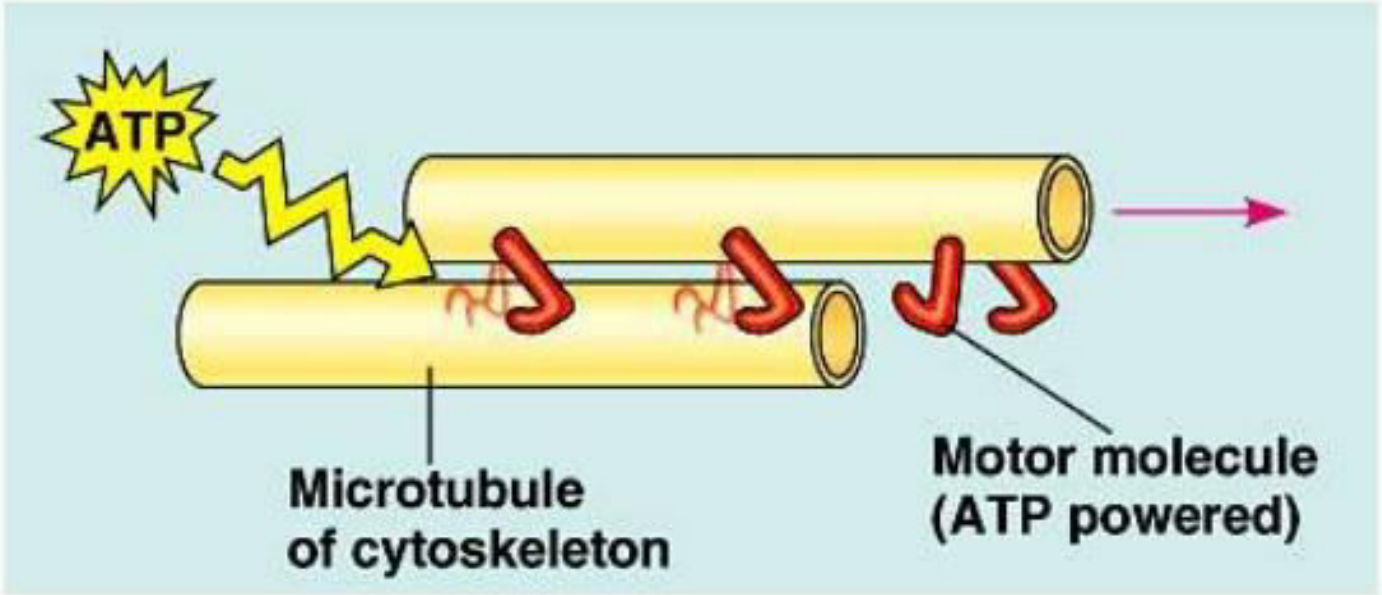
Molecular motors

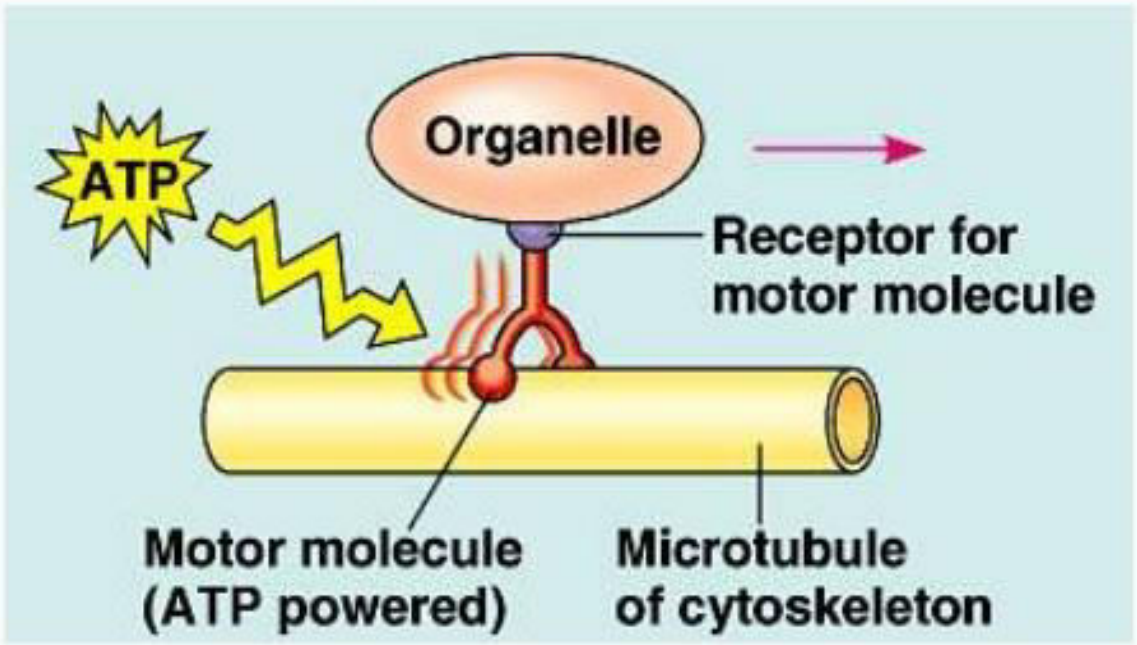
Molecular machines and motors

- Molecular motors, also called molecular machines, are either natural or synthetic molecules that convert chemical energy into mechanical forces and motion.

- An example of a biological motor is the protein kinesin, which uses the hydrolysis of adenosine triphosphate to move along microtubule filaments.

Motor proteins pull components of the cyto-skeleton past each other.





Motor molecules also carry vesicles or organelles to various destinations along "monorails' provided by the cytoskeleton.

Molecular motors

- **Molecular motors** are enzyme protein devices that move along cytoskeletal filaments and the proteins are run by ATP hydrolysis.

- The main system of Biological motility is driven by the factor of intracellular motility.

- For example, the microtubules and the actin filaments in cells are *tracks*, and the *molecular motors* have two parts the *tail domain* that are carriers of cargo (organelle or protein) and the *motor domain*, which work along the track causing motion and thus causes movement.

- Molecular motors convert the chemical energy derived from ATP hydrolysis into mechanical action.

- Molecular motors are generally of 10^{-8} m size and are driven by the Hydrolysis of ATP and have an average speed of $4*10^{-3}$ m/hr

Molecular Motors consist of different varieties:

·**Cytoskeletal** – Kinesin, Dynein and Myosin motor groups. These are of large of groups and there are many motors from these groups.

·**DNA Motors** - Helicases and polymerases.

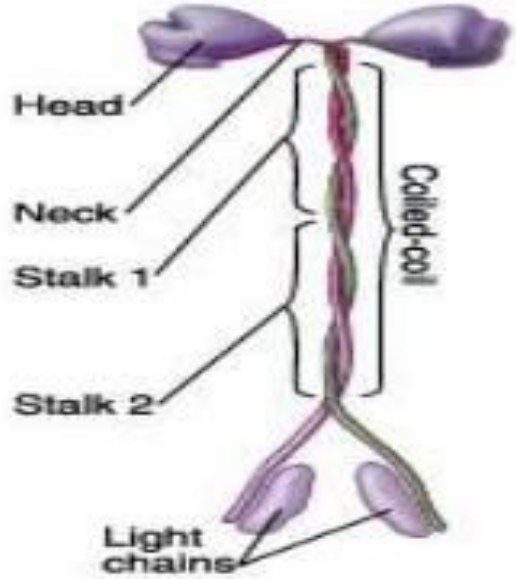
·**Rotatory motors** – Bacterial flagellum, ATPase

for example the aerobic respiration between the mitochondrial matrix and the membrane of the mitochondria.

Molecular Motors, Kinesin

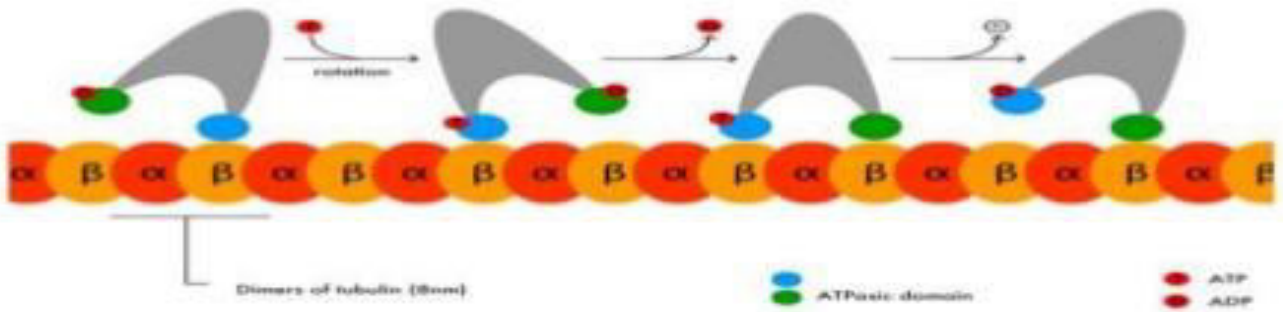
- The conventional form of kinesin is a doubleheaded molecule that tends to move its cargo toward the "+" ends of microtubules.
- Dimer of two heavy chains
- Each heavy chain complexes with a light chain
- Three domains
  - Two globular head domains
  - Long central coiled-coil stalk
  - Two small globular tail domains

  (contain light chains



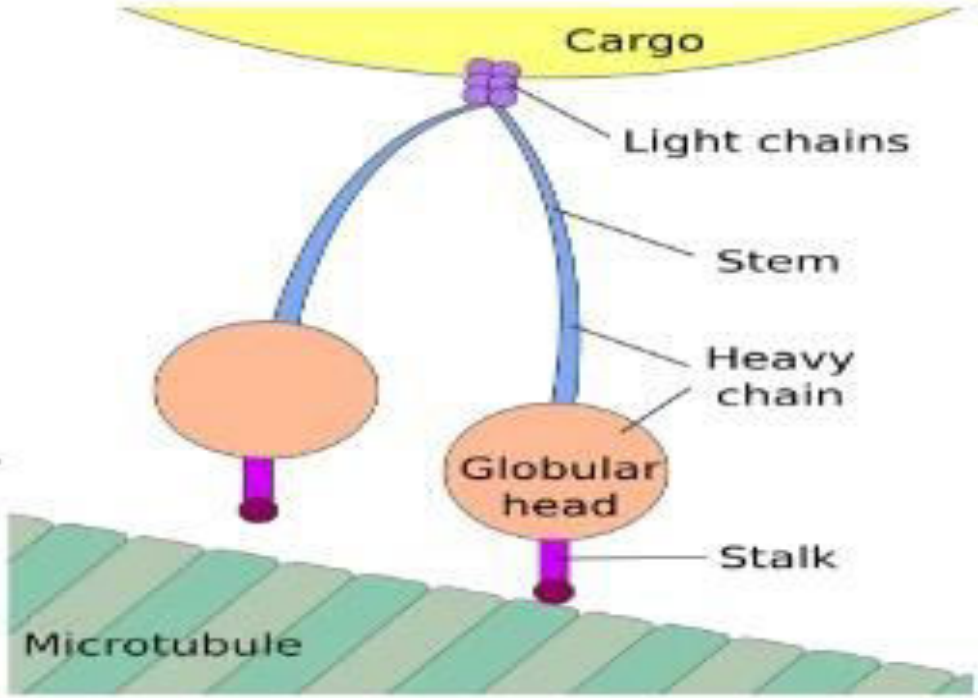
Molecular Motors, Kinesin

- Kinesin accomplishes transport by "walking" along a microtubule. Two mechanisms have been proposed to account for this movement.
  - In the "hand-over-hand" mechanism, the kinesin heads step past one another, alternating the lead position.
  - One head binds to the microtubule and then bends its neck while the other head swings forward and binds, producing almost continuous movement
  - In the "inchworm" mechanism, one kinesin head always leads, moving forward a step before the trailing head catches up.



Molecular Motors, Dyenin

- Dynein transports various cellular cargo by "walking" along cytoskeletal microtubules towards the minus-end of the microtubule.
- Composed of two or three heavy chains (that include the motor domain) and a large and variable number of associated light chains.
- Dyneins can be divided into two groups:
  - cytoplasmic dyneins and
  - axonemal dyneins, which are also called ciliary or flagellar dyneins.



Molecular Motors, Myosin

- Myosins comprise a family of ATP-dependent motor proteins and are best known for their role in muscle contraction and their involvement in a wide range of other eukaryotic motility processes.
- Most myosin molecules are composed of a head, neck, and tail domain.
  - The head domain binds the filamentous actin, and uses ATP hydrolysis to generate force and to "walk" along the filament towards the barbed (+) end (with the exception of myosin VI, which moves towards the pointed (-) end).
  - the neck domain acts as a lever arm for transducing force generated by the catalytic motor domain.
  - The tail domain generally mediates interaction with cargo molecules and/or other myosin subunits. In some cases, the tail domain may play a role in regulating motor activity

Molecular Motors, Myosin

- 18 different families (identified by genetic analysis)
- Have different functions
  - Myosin II powers muscle contraction and cytokinesis
  - Myosins I transport of endocytic vesicles
  - Myosin V phagocytosis and transport of cellular elements
  - Myosins VI and VII – transport endocytic vesicles into the cell. Found in the inner ear and mutations in the gene coding for myosin VII cause deafness in mice and humans.

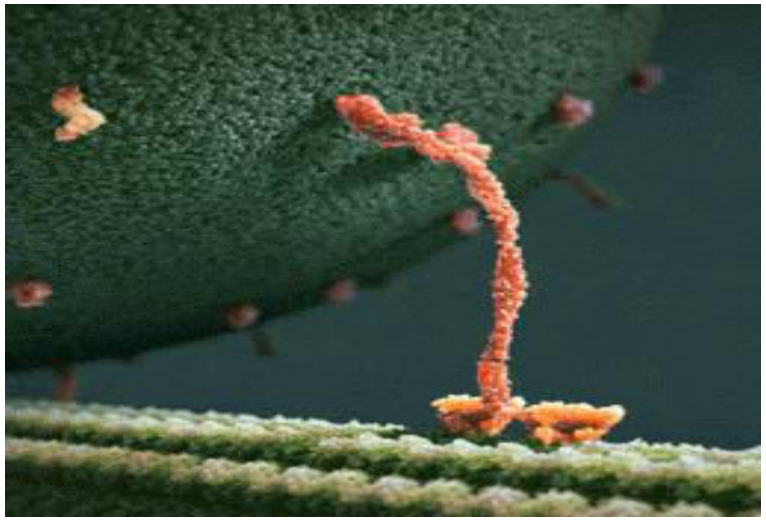
There are 2 types of general tracks in the cytoskeletal system

Microtubules – Cylindrical Polymer and made of α and β tubulents.

- Mainly driven motors are Kinesin and Dynein

Actin filaments – Helical polymer

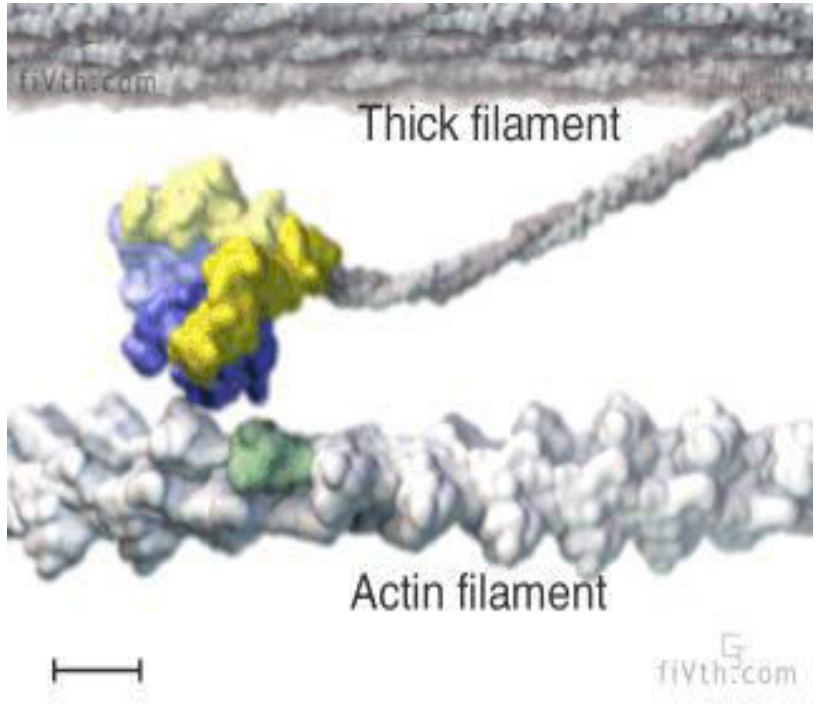
- Mainly driven motor is Myosin



The molecule motor drives on a Track.
The track is a polar structure and motors can go only in a unidirectional way

The Process

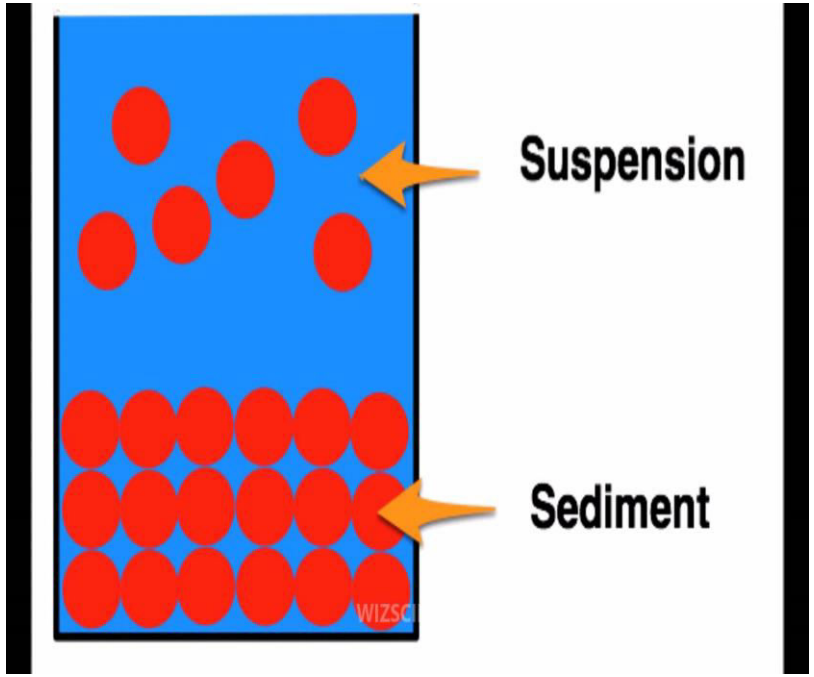
- The actual motion as shown in the picture follows a chemical cycle.
- The Picture 3 is an example of muscle contraction of a Myosin motor on a actin filament.
- This movement is the motor to bind on the actin track and slide the actin filament so that it can contract the sarcomere thus making a contraction.
- Firstly in the animation you can see the myosin motor binding to the track and the actin filament causes a phosphate release and that causes a big rotation of 10nm of one part of the motor domain(highlighted red).
- Then the ATP comes in to the motor which causes the bonded motor to release from the track. Then the hydrolysis of ATP causes a movement or repositioning of motor domain in the actin filament. Thus, the cycle repeats again: the phosphate release and then a stroke, ATP hydrolysis occurs and reposting of motor.



Movement of Myosin motor on Actin filament

Thank You

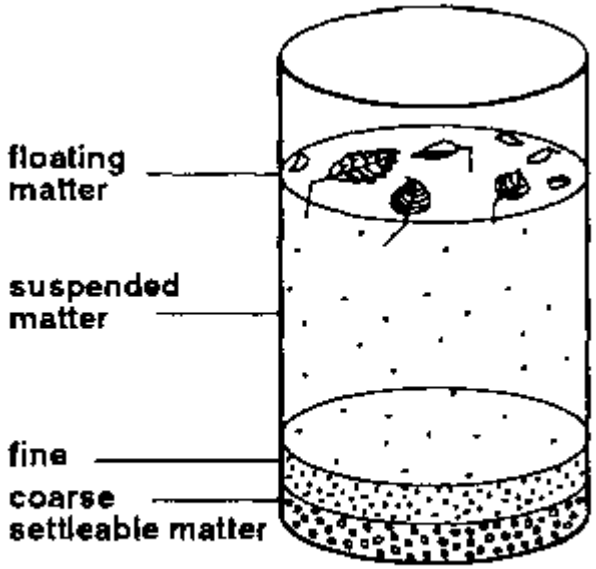
Sedimentation





Sedimentation

- Sedimentation is the tendency for particles in suspension to settle out of the fluid in which they are entrained and come to rest against a barrier.
- This is due to their motion through the fluid in response to the forces acting on them: these forces can be due to gravity, centrifugal acceleration, or electromagnetism.
- In geology, sedimentation is often used as the opposite of erosion, i.e., the terminal end of sediment transport. In that sense, it includes the termination of transport by saltation or true bedload transport.
- Settling is the falling of suspended particles through the liquid, where as sedimentation is the termination of the settling process.



- Sedimentation may pertain to objects of various sizes, ranging from large rocks in flowing water to suspensions of dust and pollen particles to cellular suspensions to solutions of single molecules such as proteins and peptides.

- Even small molecules supply a sufficiently strong force to produce significant sedimentation.

Theory for sedimentation

- Whenever a particle is moving through a fluid, a number of forces will be acting on the particle.
- First, a density difference is needed between the particle and the fluid.
- If the densities of the fluid and particle are equal, the buoyant force on the particle will counterbalance the external force and the particle will not move relative to the fluid.
- There are three forces acting on the body:-

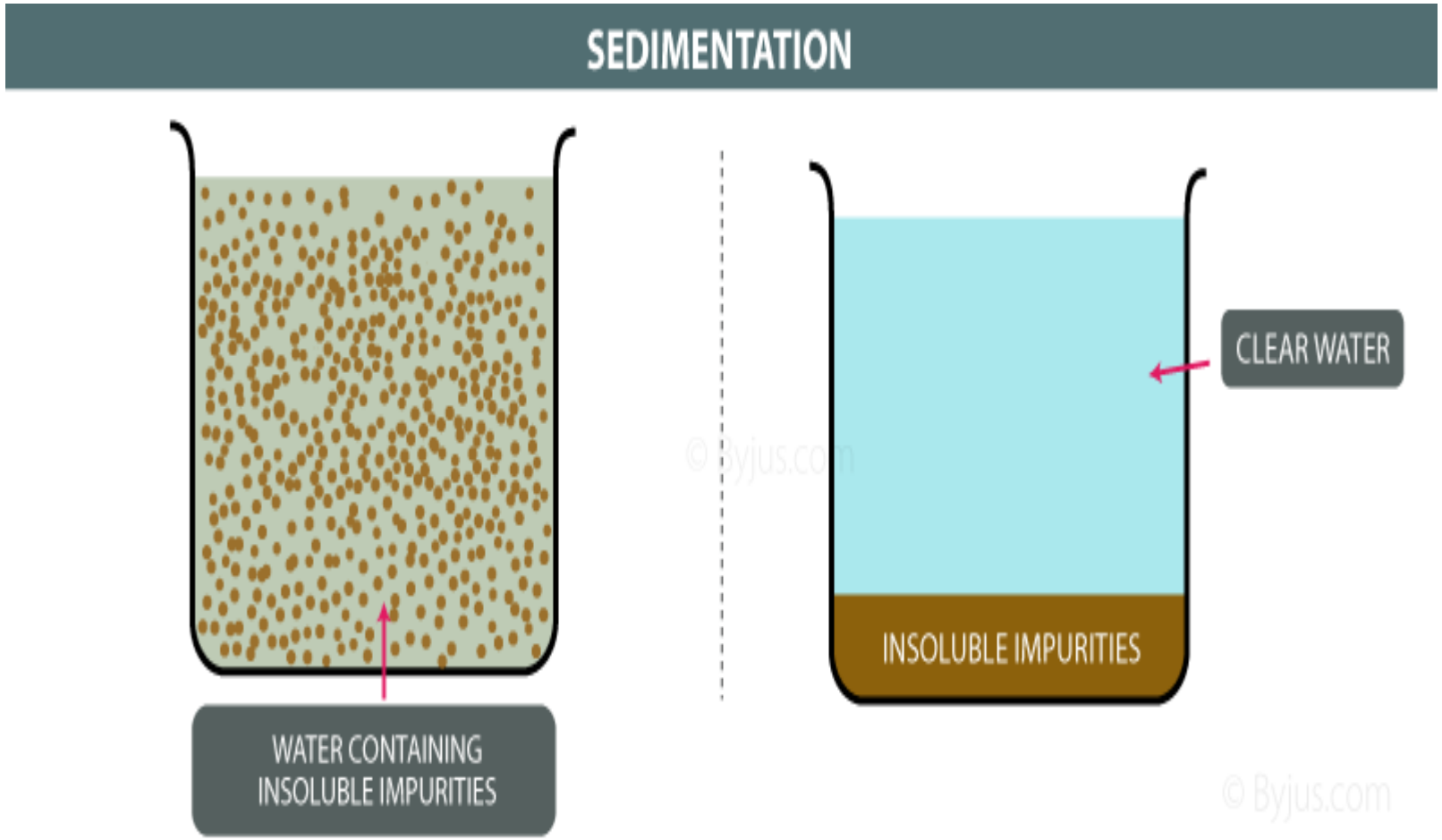
1. Gravity Force
2. Buoyant Force
3. Drag Force

- **Gravity, or gravitation**, is a natural phenomenon by which all things with mass are brought toward (or gravitate toward) one another, including objects ranging from atoms and photons, to planets and stars. Since energy and mass are equivalent, all forms of energy (including light) cause gravitation and are under the influence of it. On Earth, gravity gives weight to physical objects.

- **Buoyancy** is, basically, the ability of something to float or rise in a fluid. Simple as that. More scientifically, buoyancy refers to a force that arises from the pressure exerted on an object by a fluid (a liquid or a gas). Since it's a force, we call it the **buoyant force**.

- **Drag** (sometimes called air resistance, a type of friction, or fluid resistance, another type of friction or fluid friction) is a force acting opposite to the relative motion of any object moving with respect to a surrounding fluid. This can exist between two fluid layers (or surfaces) or a fluid and a solid surface.

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How Does Sedimentation Work?

Removing sediment from turbid water.

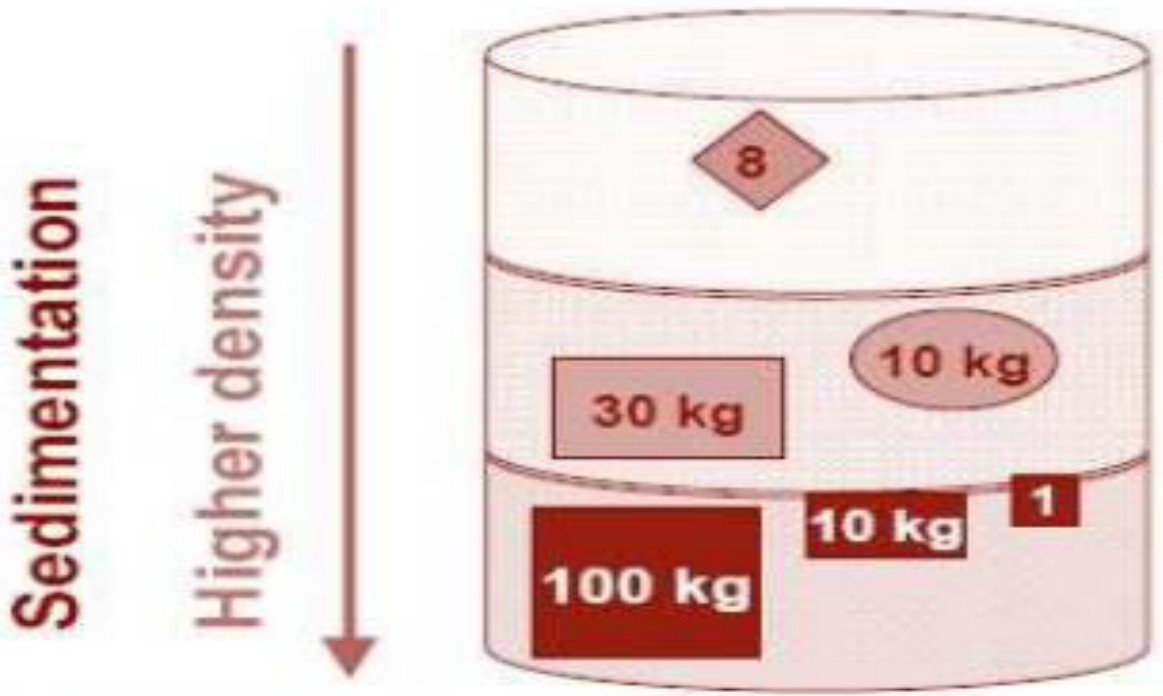
Let a container of water sit without moving for 24 hours

pour the clear water into a clean container.

repeated 2 to 3 times as needed.

Separation by Sedimentation

Weight	100 kg	30 kg	10 kg	10 kg	8	1
Material	Iron	Stone	Iron	Stone	Cotton	Iron



- Mass
- Density
- Shape

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Reynold's number

Reynold's number

- The Reynolds number, referred to as Re, is used to determine whether the fluid flow is laminar or turbulent.
- It is one of the main controlling parameters in all viscous flows where a numerical model is selected according to pre-calculated Reynolds number.
- The dimensionless Reynolds number plays a prominent role in foreseeing the patterns in a fluid's behavior.

Difference between laminar and turbulent flow?

- **Laminar Flow**: the flow of a fluid when each particle of the fluid follows a smooth path, paths which never interfere with one another. One result of laminar flow is that the velocity of the fluid is constant at any point in the fluid.

  Turbulent Flow: irregular flow that is characterized by tiny whirlpool regions. The velocity of this fluid is definitely **not** constant at every point.

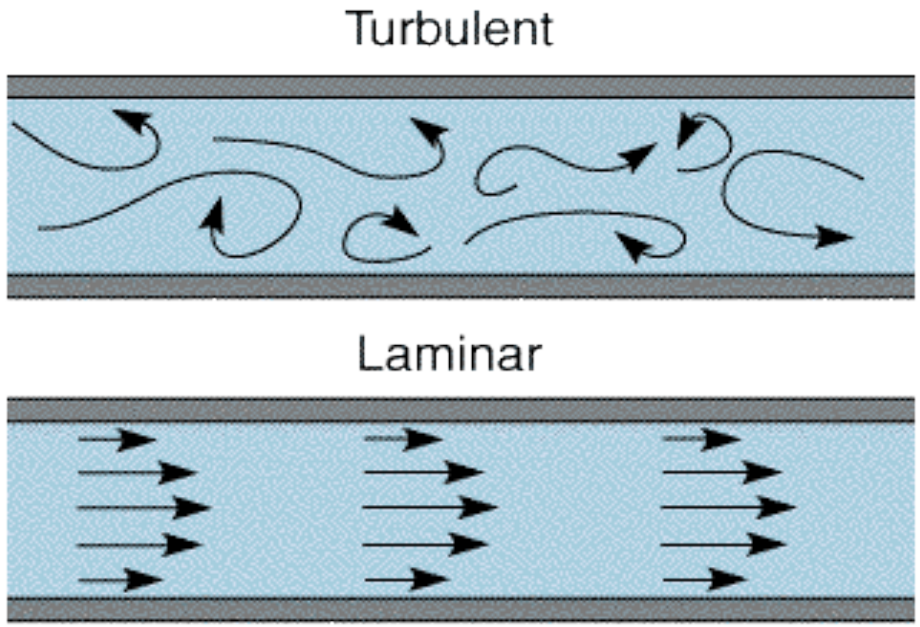
  Typical examples of both Laminar and Turbulent flows can be observed in the cigarette smoke (see image).



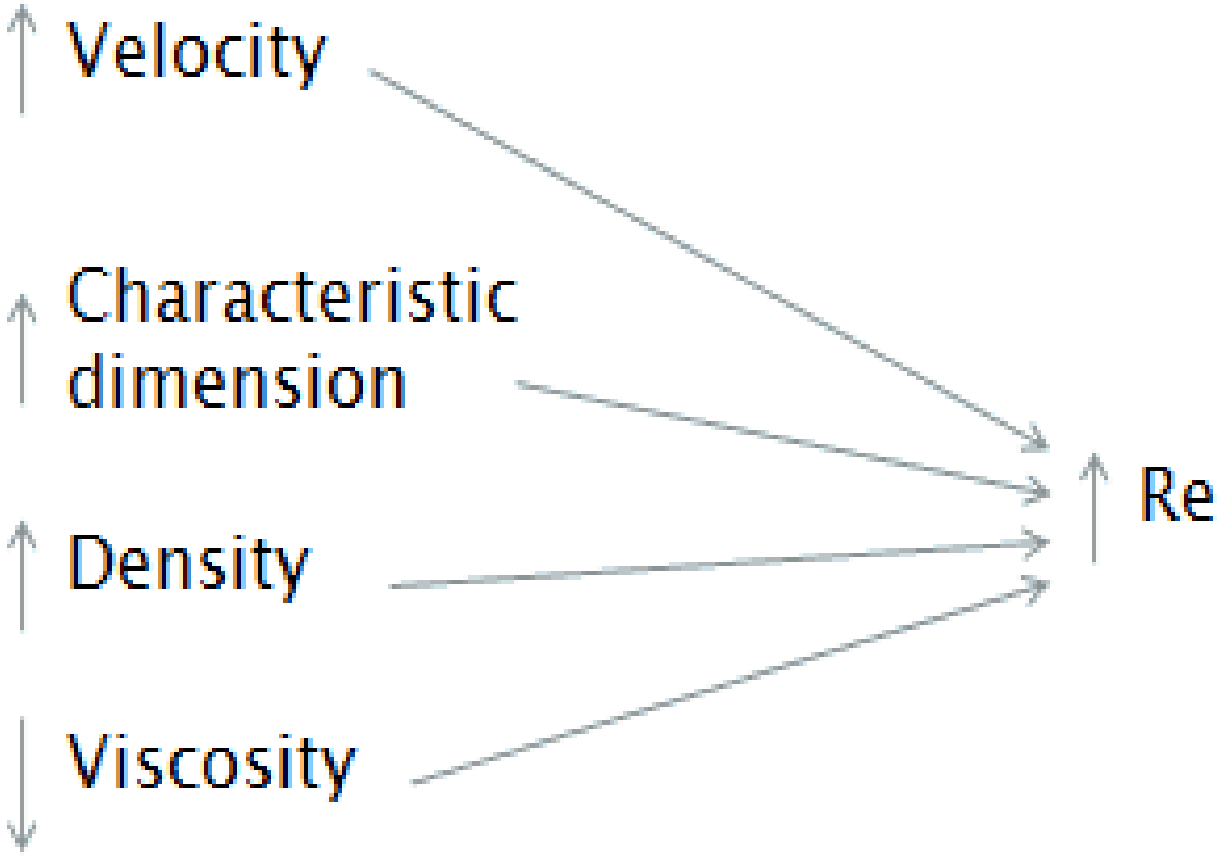
- Although the Reynolds number comprises both static and kinematic properties of fluids, it is specified as a flow property since dynamic conditions are investigated.

- Technically speaking, the Reynolds number is the ratio of the inertial forces and the viscous forces.

- In practice, the Reynolds number is used to predict if the flow will be laminar or turbulent.



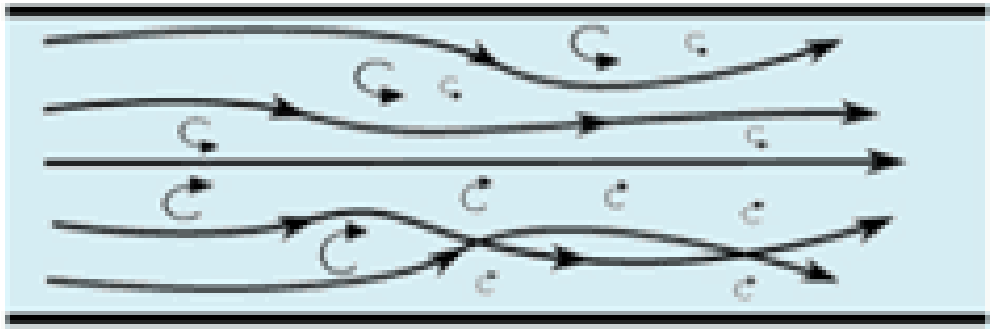
$$Re = \frac{\text{inertia forces}}{\text{viscous forces}} = \frac{\rho \cdot V \cdot D}{\mu}$$



laminar flow



turbulent flow



With respect to laminar and turbulent flow regimes:

•Laminar flow occurs at low Reynolds numbers, where viscous forces are dominant, and is characterized by smooth, constant fluid motion.

•Turbulent flow occurs at high Reynolds numbers and is dominated by inertial forces, which tend to produce chaotic eddies, vortices and other flow instabilities.

•In fluid dynamics, an eddy is the swirling of a fluid and the reverse current created when the fluid is in a turbulent flow regime

REYNOLDS NUMBER

1. Laminar Flow — $Re < 2000$
2. Unstable Flow — $2000 < Re < 4000$
3. Turblent Flow — $Re > 4000$

Normally Re > 4000 for flow in most piping systems.

$$Re = \frac{V\ D}{\gamma}$$

Where

D = Inside diameter (ft)

V = Velocity (ft/sec)

γ = Kinematics viscosity (ft^2/sec)

Re = Reynolds Number (dimensionless)

History

- The concept was introduced by **Sir George Stokes in 1851**, but the Reynolds number was named by **Arnold Sommerfeld in 1908 after Osborne Reynolds** (1842–1912), who popularized its use in 1883.

- Osborne Reynolds famously studied the conditions in which the flow of fluid in pipes transitioned from laminar flow to turbulent flow.

- In his 1883 paper Reynolds described the transition from laminar to turbulent flow in a classic experiment in which he examined the behavior of water flow under different flow velocities using a small stream of dyed water introduced into the centre of clear water flow in a larger pipe.

Applications

- Reynolds number has wide applications, ranging from liquid flow in a pipe to the passage of air over an aircraft wing.
- It is used to predict the transition from laminar to turbulent flow, and used in the scaling of similar but different-sized flow situations, such as between an aircraft model in a wind tunnel and the full size version.
- The predictions of the onset of turbulence and the ability to calculate scaling effects can be used to help predict fluid behavior on a larger scale, such as in local or global air or water movement and thereby the associated meteorological and climatological effects.

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Molecular self-assembly

Molecular self-assembly

Molecular self-assembly is the process by which molecules adopt a defined arrangement without guidance or management from an outside source.

There are two types of self-assembly. These are

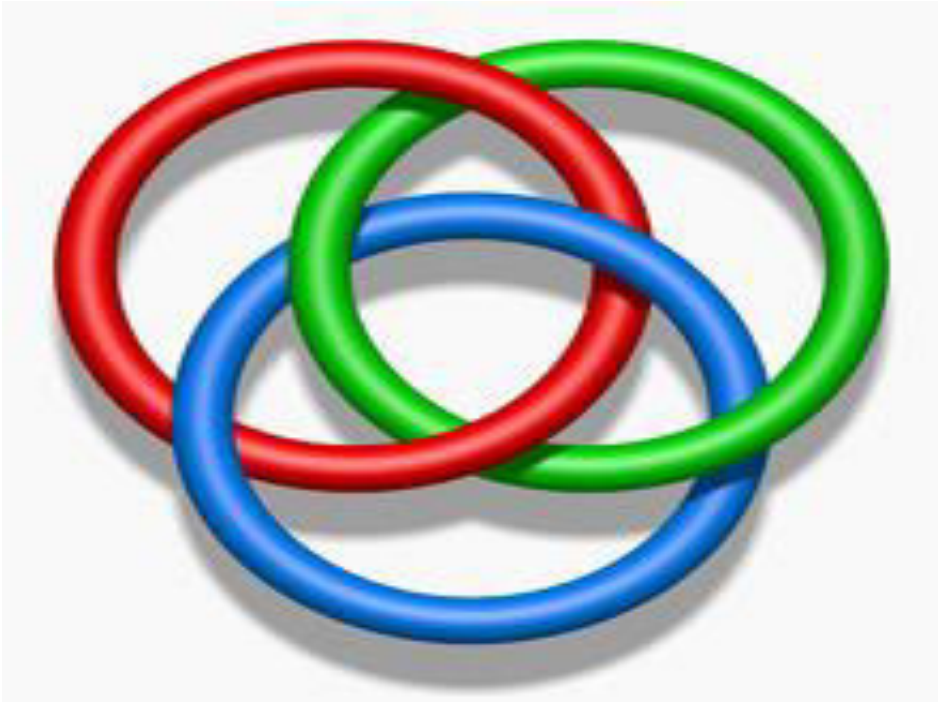
1. Intra-molecular self-assembly
2. Intermolecular self-assembly.

Commonly, the term molecular self-assembly refers to intermolecular self-assembly, while the intra-molecular analog is more commonly called folding.

Supra-molecular systems

- Molecular self-assembly is a key concept in supramolecular chemistry.
- This is because assembly of molecules in such systems is directed through non-covalent interactions (e.g., hydrogen bonding, metal coordination, hydrophobic forces, van der Waals forces, π-π interactions, and/or electrostatic) as well as electromagnetic interactions.
- Common examples include the formation of micelles, vesicles, liquid crystal phases, and Langmuir monolayers by surfactant molecules.
- Further examples of supramolecular assemblies demonstrate that a variety of different shapes and sizes can be obtained using molecular self-assembly

- Molecular self-assembly allows the construction of challenging molecular topologies.
- One example is Borromean rings, interlocking rings wherein removal of one ring unlocks each of the other rings.
- DNA has been used to prepare a molecular analog of Borromean rings.
- More recently, a similar structure has been prepared using non-biological building blocks.



The **Borromean rings** consist of three topological circles which are linked.

Biological systems

- Molecular self-assembly underlies the construction of biologic macromolecular assemblies in living organisms, and so is crucial to the function of cells.
- It is exhibited in the self-assembly of lipids to form the membrane, the formation of double helical DNA through hydrogen bonding of the individual strands, and the assembly of proteins to form quaternary structures.
- Molecular self-assembly of incorrectly folded proteins into insoluble amyloid fibers is responsible for infectious prion-related neurodegenerative diseases.

Nanotechnology

- Molecular self-assembly is an important aspect of bottom-up approaches to nanotechnology.
- Using molecular self-assembly the final (desired) structure is programmed in the shape and functional groups of the molecules.
- Self-assembly is referred to as a 'bottom-up' manufacturing technique in contrast to a 'top-down' technique such as lithography where the desired final structure is carved from a larger block of matter.
- In the theoretical vision of molecular nanotechnology, microchips of the future might be made by molecular self-assembly.
- An advantage to constructing nanostructure using molecular self-assembly for biological materials is that they will degrade back into individual molecules that can be broken down by the body.

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Self-assembly

- Self-assembly is the formation of a well-defined, complex structure under equilibrium conditions spontaneously from non-covalent interactions between two or more molecular components.

- Self-assembly encompasses the self-association of proteins and nucleic acids, such as the formation of different forms of a double-stranded DNA double helix from two complementary poly-deoxynucleotide strands, the association of multiple copies of the same polypeptide chain, the formation of complexes comprised of more than one type of polypeptide chain (ie, heteromultimeric proteins), and the reconstitution of an infectious viral particle, comprised of one or more proteins, plus nucleic acid.

Self-assembly can also be viewed as part of a more general topic of supramolecular assembly or self-organization, which encompasses the formation of a wide variety of structures with masses greater than 1 MDa and sizes greater than 10 nm.

These include pathogens such as:

- Viruses
- large multi-enzyme complexes involved in biosynthetic pathways (eg, pyruvate dehydrogenase) and in protein degradation (eg, the proteasome)
- multiprotein complexes that bind to nucleic acids and are active in DNA replication
- RNA biosynthesis and protein biosynthesis (eg, nucleosomes, polymerases, spliceosomes, ribosomes)
- transmembrane protein complexes (eg, acetylcholine receptor, respiratory enzyme complexes, photosynthetic reaction centers)
- and the protein complexes encaging transport vesicles (clathrin-coated vesicles and circulating lipoproteins).

- The foregoing definition of self-assembly implies that all the necessary information is encoded within the covalent structures of the components and hence requires complete reversibility of the assembly process.

- Reversibility is demonstrated in vitro by the reassembly of the completely dissociated structure upon return to the initial solution conditions.

Dissociation into individual molecules can be effected in many different ways:

- (a) increased hydrostatic pressure
- (b) addition or removal of specific ligands, such as coenzymes, substrates, allosteric effectors or other prosthetic groups
- (c) decrease in concentration
- (d) alteration of pH, temperature, or ionic strength
- (e) the addition of dissociating and denaturing chaotropic agents such as urea, guanidinium salts, and detergents (neutral, cationic, or anionic).

At present, overexpression in Escherichia coli of the individual wild-type or mutagenized subunits makes it possible to elucidate the pathways of self-assembly and the roles in the self-assembly process of various protein domains within each subunit.

Intermolecular Forces Responsible for Self-Assembly

- Although the covalent structure of each component encodes all the information necessary for its three-dimensional structure and its interactions with the other components in the self-assembly process, the intermolecular forces involved in both processes are all noncovalent, being energetically weaker than covalent bonds by more than an order of magnitude (<10kcal/mol vs ~100kcal/mol).
- These intermolecular forces represent both specific, geometrically directed interactions, like hydrogen bonds and salt bridges (which may be regarded as hydrogen-bonded ion pairs), and relatively nonspecific electrostatic and van der Waals interactions, including the hydrophobic effect.
- The resulting energetic stabilization depends on the amount of accessible surface area buried in the contact between the interacting components and, most critically, on the complementarity in the chemical nature and shapes of the interacting surfaces.

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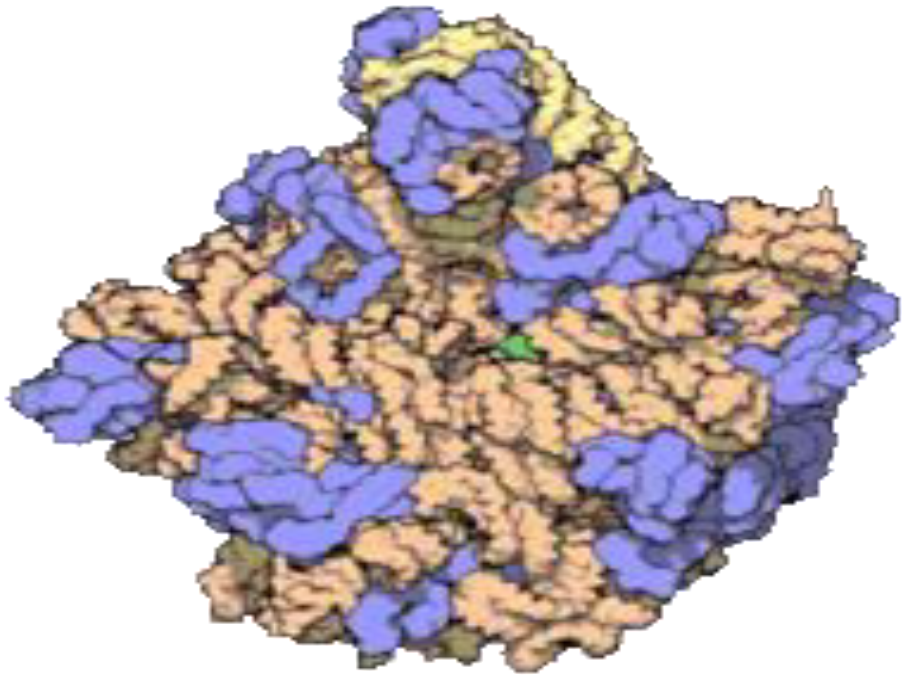
Macromolecular assembly

Macromolecular assembly

- The term macromolecular assembly (MA) refers to massive chemical structures such as viruses and non-biologic nanoparticles, cellular organelles and membranes and ribosomes, etc. that are complex mixtures of polypeptide, polynucleotide, polysaccharide or other polymeric macromolecules.
- They are generally of more than one of these types, and the mixtures are defined spatially (i.e., with regard to their chemical shape), and with regard to their underlying chemical composition and structure.
- Macromolecules are found in living and nonliving things, and are composed of many hundreds or thousands of atoms held together by covalent bonds; they are often characterized by repeating units (i.e., they are polymers).

Structure of a macromolecular assembly (MA), the 50S ribosomal subunit from *H. marismortui*, based on X-ray crystallography.

Of the 31 component proteins, 27 are shown (blue), along with its 2 RNA strands (orange, yellow).



- MAs of macromolecules are held in their defined forms by non-covalent intermolecular interactions (rather than covalent bonds), and can be in either non-repeating structures, or in repeating linear, circular, spiral, or other patterns (e.g., as in actin filaments and the flagellar motor, image).

- The process by which MAs are formed has been termed molecular self-assembly, a term especially applied in non-biologic contexts.

- A wide variety of physical/biophysical, chemical/biochemical, and computational methods exist for the study of MA; given the scale (molecular dimensions) of MAs, efforts to elaborate their composition and structure and discern mechanisms underlying their functions are at the forefront of modern structure science.

Roles

- The complexes of macromolecules that are referred to as MAs occur ubiquitously in nature, where they are involved in the construction of viruses and all living cells.
- In addition, they play fundamental roles in all basic life processes (protein translation, cell division, vesicle trafficking, intra- and inter-cellular exchange of material between compartments, etc.).
- In each of these roles, complex mixtures of become organized in specific structural and spatial ways.
- While the individual macromolecules are held together by a combination of covalent bonds and intra-molecular non-covalent forces (i.e., associations between parts within each molecule, via charge-charge interactions, van der Waals forces, and dipole-dipole interactions such as hydrogen bonds), by definition MAs themselves are held together solely via the non-covalent forces, except now exerted between molecules (i.e., intermolecular interactions).

Research into MAs

- The study of MA structure and function is challenging, in particular because of their mega dalton size, but also because of their complex compositions and varying dynamic natures.
- Most have had standard chemical and biochemical methods applied (methods of protein purification and centrifugation, chemical and electrochemical characterization, etc.).
- In addition, their methods of study include modern proteomic approaches, computational and atomic-resolution structural methods (e.g., X-ray crystallography), small-angle X-ray scattering (SAXS) and small-angle neutron scattering (SANS), force spectroscopy, and transmission electron microscopy and cryo-electron microscopy.

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Ion channels

Ion channels

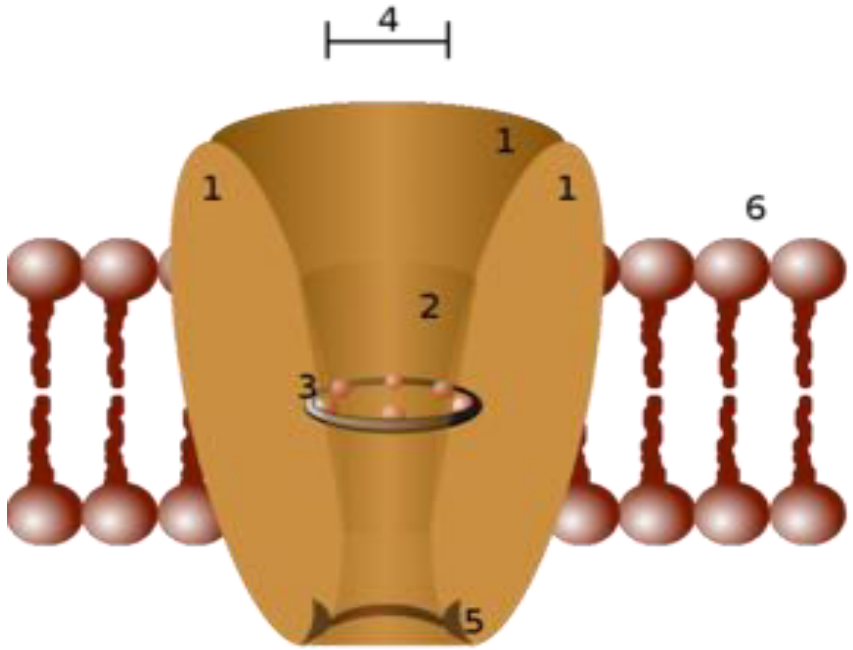
Ion channels are pore-forming membrane proteins that allow ions to pass through the channel pore.

Their functions include:

1. Establishing a resting membrane potential
2. Shaping action potentials and other electrical signals by gating the flow of ions across the cell membrane
3. Controlling the flow of ions across secretory and epithelial cells
4. Regulating cell volume.

- Ion channels are present in the membranes of all excitable cells.

- Ion channels are one of the two classes of ionophoric proteins, along with ion transporters (including the sodium-potassium pump, sodium-calcium exchanger, and sodium-glucose transport proteins).



Schematic diagram of an ion channel.
1 - channel domains (typically four per channel)
2 - outer vestibule
3 - selectivity filter
4 - diameter of selectivity filter
5 - phosphorylation site
6 - cell membrane.

- There are two distinctive features of ion channels that differentiate them from other types of ion transporter proteins:

- The rate of ion transport through the channel is very high (often 106 ions per second or greater).

- Ions pass through channels down their electrochemical gradient, which is a function of ion concentration and membrane potential, "downhill", without the input (help) of metabolic energy (e.g. ATP, co-transport mechanisms, or active transport mechanisms).

- Ion channel, protein expressed by virtually all living cells that creates a pathway for charged ions from dissolved salts, including sodium, potassium, calcium, and chloride ions, to pass through the otherwise impermeant lipid cell membrane.

- Operation of cells in the nervous system, contraction of the heart and of skeletal muscle, and secretion in the pancreas are examples of physiological processes that require ion channels.

- In addition, ion channels in the membranes of intracellular organelles are important for regulating cytoplasmic calcium concentration and acidification of specific subcellular compartments (e.g., lysosomes).

Thank you

Ion Channels

- Ion channels are membrane protein complexes and their function is to facilitate the diffusion of ions across biological membranes.

- Membranes, or phospholipid bilayers, build a hydrophobic, low dielectric barrier to hydrophilic and charged molecules.

- They are electrical insulators. Ion channels provide a high conducting, hydrophilic pathway across the hydrophobic interior of the membrane.

- The channel, or pore structure, is said to catalyze the 'reaction' of transporting charged molecules across a low dielectric medium.

- The 'catalytic site', the central channel, is either open or closed.

- The open channel conformation can be compared to the transition state of the enzyme-substrate complex, where ions are tightly associated to the catalytic site.

- The conformational change between closed and open state is called gating, as in opening and closing a gate.

- Channel gating is controlled by external factors like enzymes are controlled by modulators and effectors.

- Ion channels can be classified according to which chemical or physical modulator controls their gating activity.

History

- In the late1800s, the chemical mechanism underlying nerve and muscle tissue messaging was not known.

- **Ludimar Hermann** was able to conclude that nerve and muscle cells were capable of exhibiting a "self- propagating wave of negative charge which advances in steps along the tissue ".

- In 1880s, **Sidney Ringer** used a solution of water and ran it through the vessels of an isolated heart from a frog and discovered that in order for the heart to continue beating salts needed to be present in the water.

- Sodium, calcium, and potassium salts were important and had to be in specific concentration relative to each other.

- In 1970s, the existence of ion channels was confirmed by the invention of ‘patch clamp’ technique by **Erwin Neher and Bert Sakmann** who won a Nobel Prize for it.

- In 2003, the Nobel Prize was awarded to American scientists, **Roderick MacKinon** and **Peter Agre** for their x-ray crystallographic structure studies on ion channels.

Evolution And Selectivity

- Ions flow passively through channels toward equilibrium. This movement may be driven by electrical (voltage) or chemical (concentration) gradients.
- The ability to alter ion flow as a result of the development of ion channels may have provided an evolutionary advantage by allowing single-celled organisms to regulate their volume in the face of environmental changes.
- Through subsequent evolution, ion channels have come to play essential roles in cellular secretion and electrical signaling.

Groups of channels

We have different groups of channels as summarized below:

1. Ligand gated channels (neurotransmitters)
2. Voltage gated channels trans-membrane potential (electric field)
3. Second messenger gated channels nucleotides, (G-proteins)
4. Mechano-sensitive channels osmotic pressure (membrane curvature)
5. Gap junctions, (porins not gated)

Thank you

Types of Ion Channels

VOLTAGE GATED ION CHANNEL

TRANSMEMBRANE ION CHANNEL that are activated by changes in electrical membrane potential.

- Voltage-gated ion channels are a class of trans membrane proteins that form ion channels that are activated by changes in the electrical membrane potential near the channel.
- The membrane potential alters the conformation of the channel proteins, regulating their opening and closing.
- Cell membranes are generally impermeable to ions, thus they must diffuse through the membrane through trans membrane protein channels.
- They have a crucial role in excitable cells such as neuronal and muscle tissues, allowing a rapid and co-ordinated depolarization in response to triggering voltage change.
- Found along the axon and at the synapse, voltage-gated ion channels directionally propagate electrical signals.

- Voltage-gated ion-channels are usually ion-specific, and channels specific to sodium (Na+), potassium (K+), calcium (Ca2+), and chloride (Cl–) ions have been identified.

- The opening and closing of the channels are triggered by changing ion concentration, and hence charge gradient, between the sides of the cell membrane

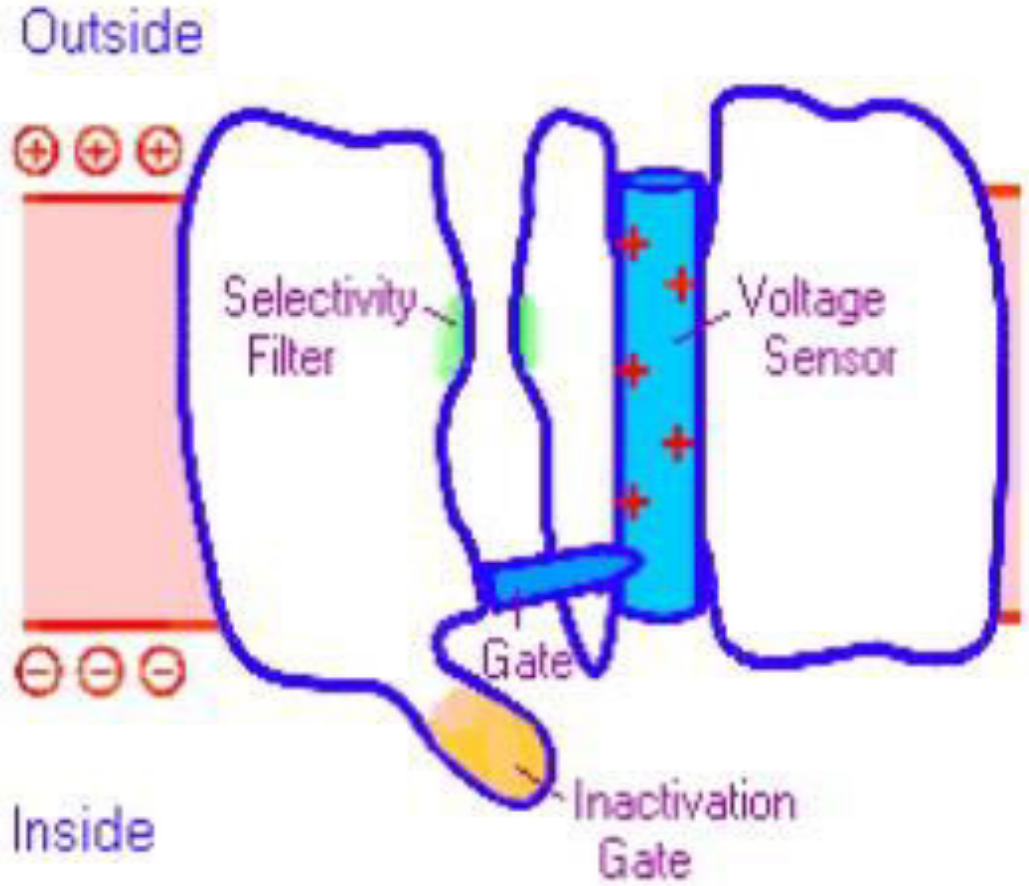
STRUCTURE

- Voltage Gated ion channels are made of <u>three basic parts:</u>

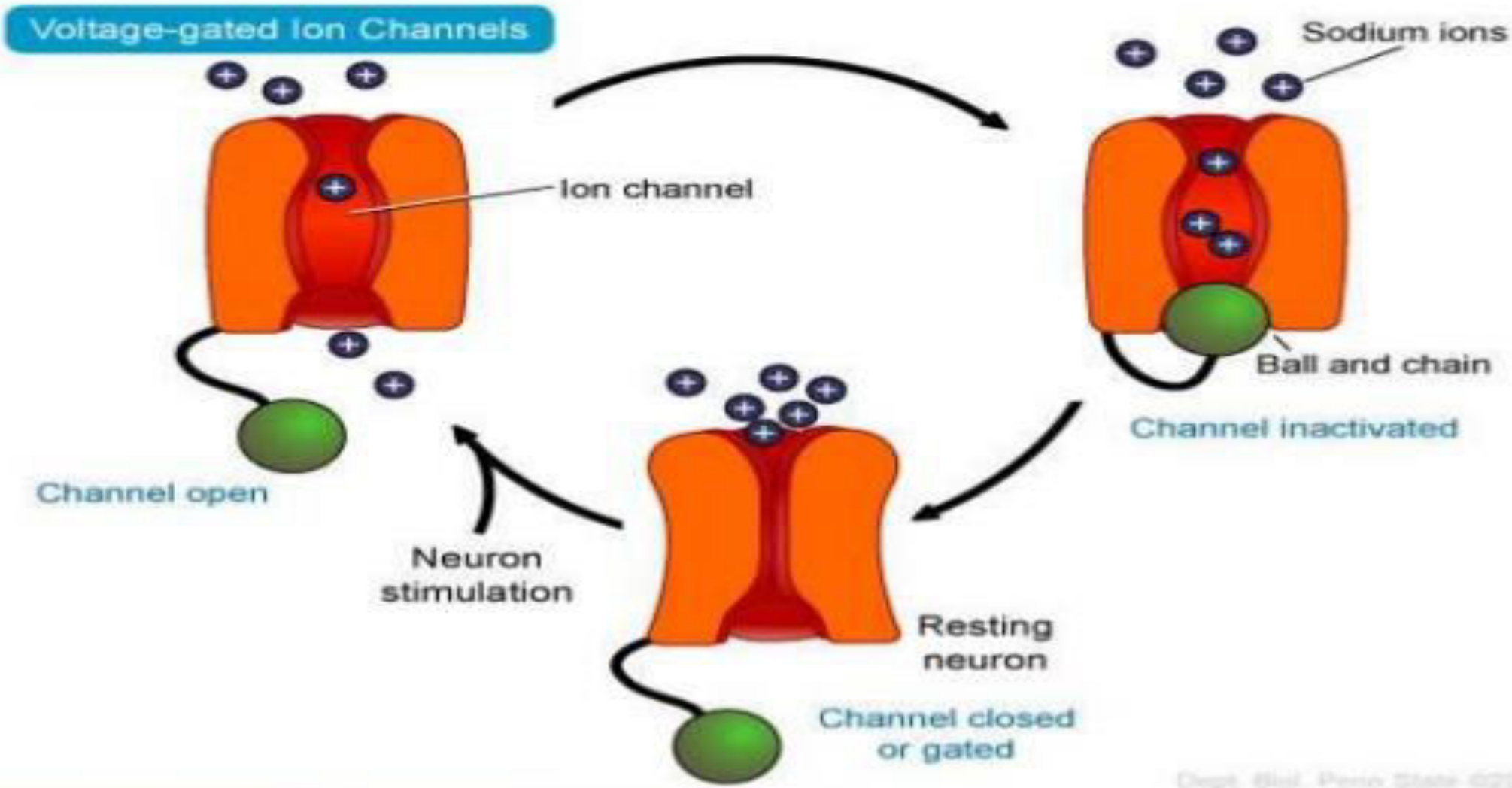
1) **The transmembrane pore**
2) **Voltage sensor**
3) **Selectivity filter**

- Contains **different subunits:**

α subunit and other auxillary Subunits.



Mechanism of voltage gated ion channel:



LIGAND GATED ION CHANNELS

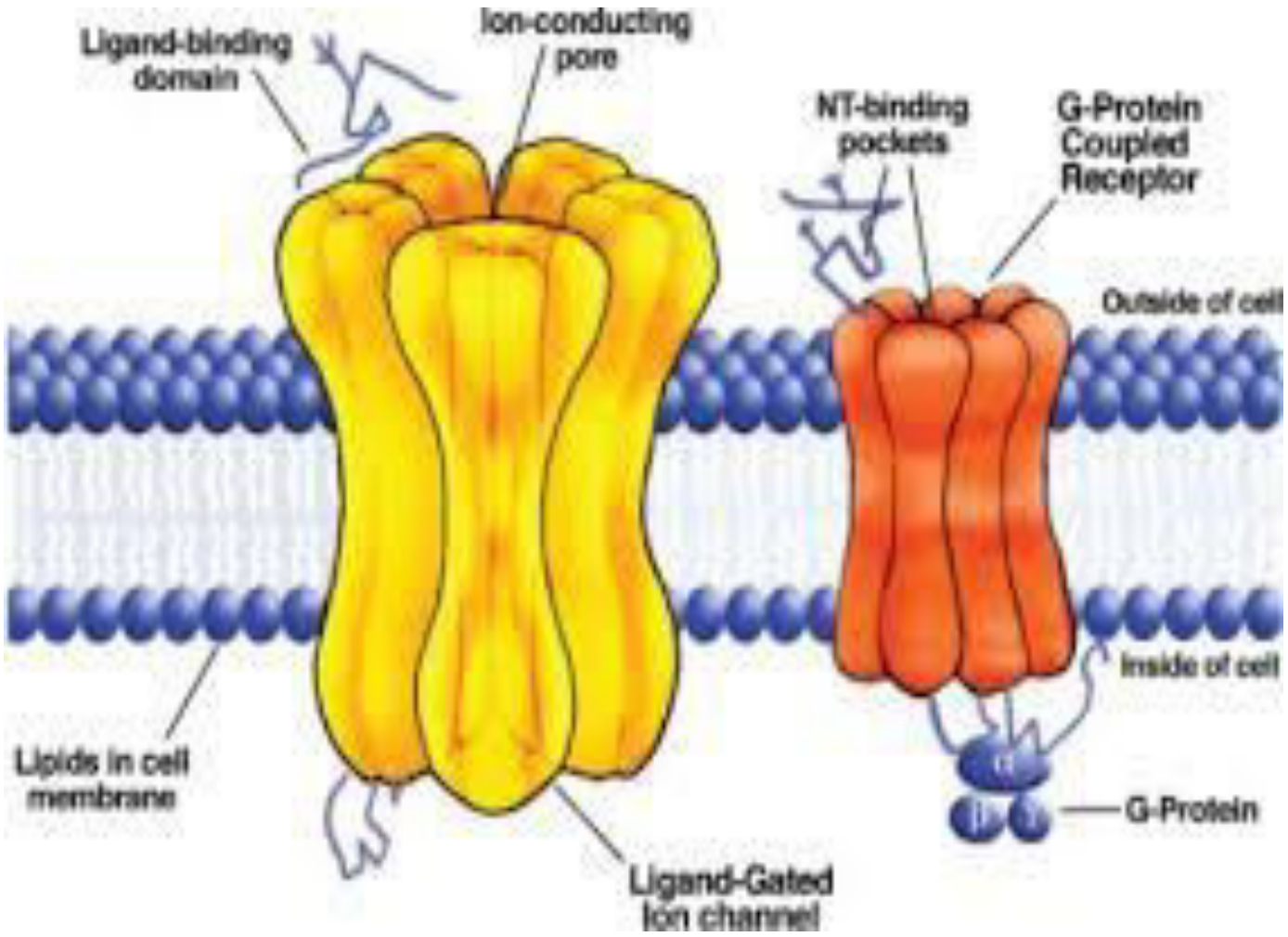
TRANSMEMBRANE PROTEIN which open to allow ions such as Na+, K+, Ca2+, or Cl– to pass through the membrane in response to the binding of a chemical messenger (i.e. a ligand).

They are all receptors

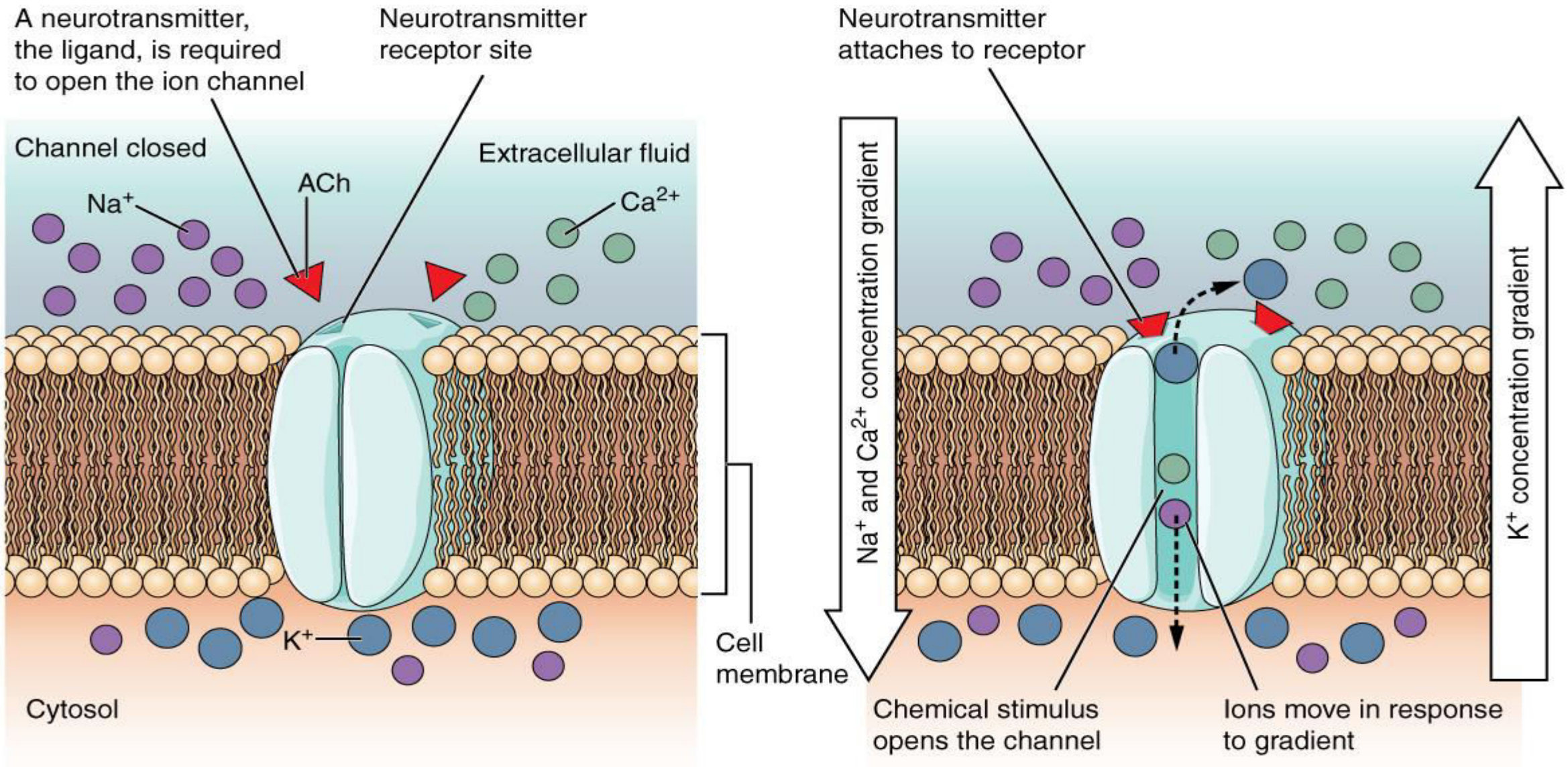
These proteins are typically composed of at least two different domains:

1. A trans membrane domain which includes the ion pore.

2. An extracellular domain which includes the ligand binding location (an allosteric binding site).



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Different types of voltage gated ion channels

- **Sodium (Na+) channels**
- Sodium channels have similar functional properties across many different cell types. While ten human genes encoding for sodium channels have been identified, their function is typically conserved between species and different cell types.
- **Calcium (Ca2+) channels**
- With sixteen different identified genes for human calcium channels, this type of channel differs in function between cell types. Ca2+ channels produce action potentials similarly to Na+ channels in some neurons. They also play a role in neurotransmitter release in pre-synaptic nerve endings. In most cells, Ca2+ channels regulate a wide variety of biochemical processes due to their role in controlling intracellular Ca2+ concentrations.

- **Potassium (K+) channels**
- Potassium channels are the largest and most diverse class of voltage-gated channels, with over 100 encoding human genes.
- These types of channels differ significantly in their gating properties; some inactivating extremely slowly and others inactivating extremely quickly.
- This difference in activation time influences the duration and rate of action potential firing, which has a significant effect on electrical conduction along an axon as well as synaptic transmission.
- Potassium channels differ in structure from the other channels in that they contain four separate polypeptide subunits, while the other channels contain four homologous domain but on a single polypeptide unit.

- **Chloride (Cl–) channels**
- Chloride channels are present in all types of neurons. With the chief responsibility of controlling excitability, chloride channels contribute to the maintenance of cell resting potential and help to regulate cell volume.
- **Proton (H+) channels**
- Voltage-gated proton channels carry currents mediated by hydrogen ions in the form of hydronium, and are activated by depolarization in a pH-dependent manner. They function to remove acid from cells.

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Toxins And Disease

Toxins And Disease

A variety of inorganic and organic molecules can modulate ion channel activity and conductance. Some commonly used blockers include:

- **Tetrodotoxin (TTX),** used by puffer fish and some types of newts for defense. It blocks sodium channels.

- **Saxitoxin** is produced by a dinoflagellate also known as "red tide". It blocks voltage-dependent sodium channels.

- **Conotoxin** is used by cone snails to hunt prey.

- **Lidocaine and Novocaine** belong to a class of local anesthetics which block sodium ion channels.

- **Dendrotoxin** is produced by mamba snakes, and blocks potassium channels.

- **Iberiotoxin** is produced by the **Buthus tamulus** (Eastern Indian scorpion) and blocks potassium channels.

- **Heteropodatoxin** is produced by **Heteropoda venatoria** (brown huntsman spider or laya) and blocks potassium channels.

- Inherited mutations in ion channel genes and in genes encoding proteins that regulate ion channel activity have been implicated in a number of diseases, including ataxia (the inability to coordinate voluntary muscle movements), diabetes mellitus, certain types of epilepsy, and cardiac arrhythmias (irregularities in heartbeat).

- For example, genetic variations in sodium-selective and potassium-selective channels, or in their associated regulatory subunits, underlie some forms of long-QT syndrome.

- This syndrome is characterized by a prolongation in the depolarization time-course of cardiac myocyte action potentials, which can lead to fatal arrhythmias.

- In addition, mutations in adenosine triphosphate (ATP)-sensitive potassium channels that control insulin secretion from cells in the pancreas underlie some forms of diabetes mellitus.

Diseases

- **Shaker gene** mutations cause a defect in the voltage gated ion channels.

- **Cystic fibrosis** is caused by mutations in the CFTR gene, which is a chloride channel.

- Mutations in and overexpression of ion channels are important events in cancer cells. In **Glioblastoma multiforme**, upregulation of gBK potassium channels and ClC-3 chloride channels enables glioblastoma cells to migrate within the brain, which may lead to the diffuse growth patterns of these tumors.

Role In Research

- Ongoing basic research on ion channels seeks to understand the structural basis for permeability, ion selectivity, and gating at the molecular level.
- Research efforts also attempt to answer questions about the cellular regulation of ion channel protein synthesis and about the subcellular distribution and ultimate degradation of channels.
- In addition, compounds with greater specificity and potency for channels involved in pain sensation, cardiovascular disease, and other pathological conditions are potential sources for drug development.

Thank you

Facilitated Diffusion of Ions

Facilitated Diffusion of Ions

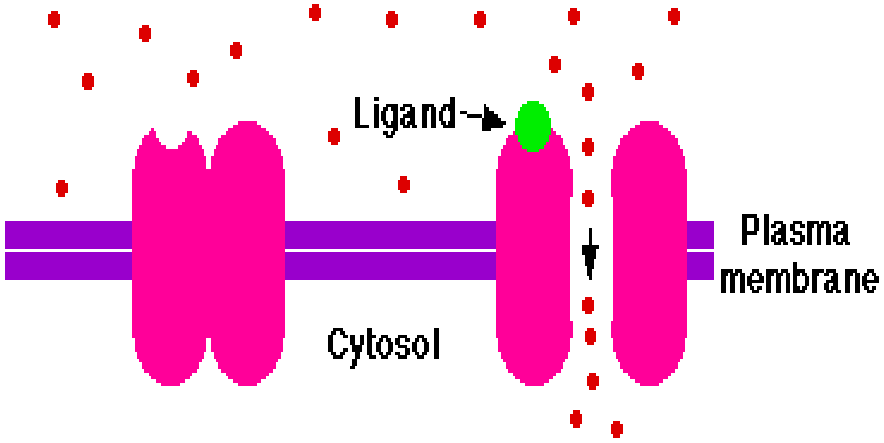
Facilitated diffusion of ions takes place through proteins, or assemblies of proteins, embedded in the plasma membrane.

These trans-membrane proteins form a water-filled channel through which the ion can pass down its concentration gradient.

- The trans-membrane channels that permit facilitated diffusion can be opened or closed. They are said to be "gated".
- Some types of gated ion channels:
- Ligand-gated
- Mechanically-gated
- Voltage-gated
- Light-gated

Ligand-gated ion channels.

- Many ion channels open or close in response to binding a small signaling molecule or "ligand". Some ion channels are gated by extracellular ligands; some by intracellular ligands. In both cases, the ligand is not the substance that is transported when the channel opens.
- **External ligands**
- External ligands (shown here in green) bind to a site on the extracellular side of the channel.
- **Examples:**
- Acetylcholine (ACh). The binding of the neurotransmitter acetylcholine at certain synapses opens channels that admit Na+ and initiate a nerve impulse or muscle contraction.
- Gamma amino butyric acid (GABA). Binding of GABA at certain synapses — designated GABAA — in the central nervous system admits Cl- ions into the cell and inhibits the creation of a nerve impulse.



Facilitated diffusion through a ligand-gated channel

- **Internal ligands**
- Internal ligands bind to a site on the channel protein exposed to the cytosol.
- Examples:
- "Second messengers", like cyclic AMP (cAMP) and cyclic GMP (cGMP), regulate channels involved in the initiation of impulses in neurons responding to odors and light respectively.
- ATP is needed to open the channel that allows chloride (Cl-) and bicarbonate (HCO3-) ions out of the cell.
- This channel is defective in patients with cystic fibrosis. Although the energy liberated by the hydrolysis of ATP is needed to open the channel, this is not an example of active transport; the ions diffuse through the open channel following their concentration gradient.

Mechanically-gated ion channels

- **Examples:**
- Sound waves bending the cilia-like projections on the hair cells of the inner ear open up ion channels leading to the creation of nerve impulses that the brain interprets as sound.

- Mechanical deformation of the cells of stretch receptors opens ion channels leading to the creation of nerve impulses.

Facilitated Diffusion of Molecules

- Some small, hydrophilic organic molecules, like sugars, can pass through cell membranes by facilitated diffusion.
- Once again, the process requires transmembrane proteins. In some cases, these — like ion channels — form water-filled pores that enable the molecule to pass in (or out) of the membrane following its concentration gradient.
- **Example: Maltoporin**. This homotrimer in the outer membrane of E. coli forms pores that allow the disaccharide maltose and a few related molecules to diffuse into the cell.
- **Another example**: The plasma membrane of human red blood cells contain transmembrane proteins that permit the diffusion of glucose from the blood into the cell.
- Note that in all cases of facilitated diffusion through channels, the channels are selective; that is, the structure of the protein admits only certain types of molecules through.
- Whether all cases of facilitated diffusion of small molecules use channels is yet to be proven. Perhaps some molecules are passed through the membrane by a conformational change in the shape of the transmembrane protein when it binds the molecule to be transported.

Thank you

Ion Channels – Direct Active Transport

- **Learning Objectives:**
- The Na+/K+ ATPase
- The H+/K+ ATPase
- The Ca2+ ATPases
- ABC Transporters

1. The Na+/K+ ATPase

The cytosol of animal cells contains a concentration of potassium ions (K+) as much as 20 times higher than that in the extracellular fluid.

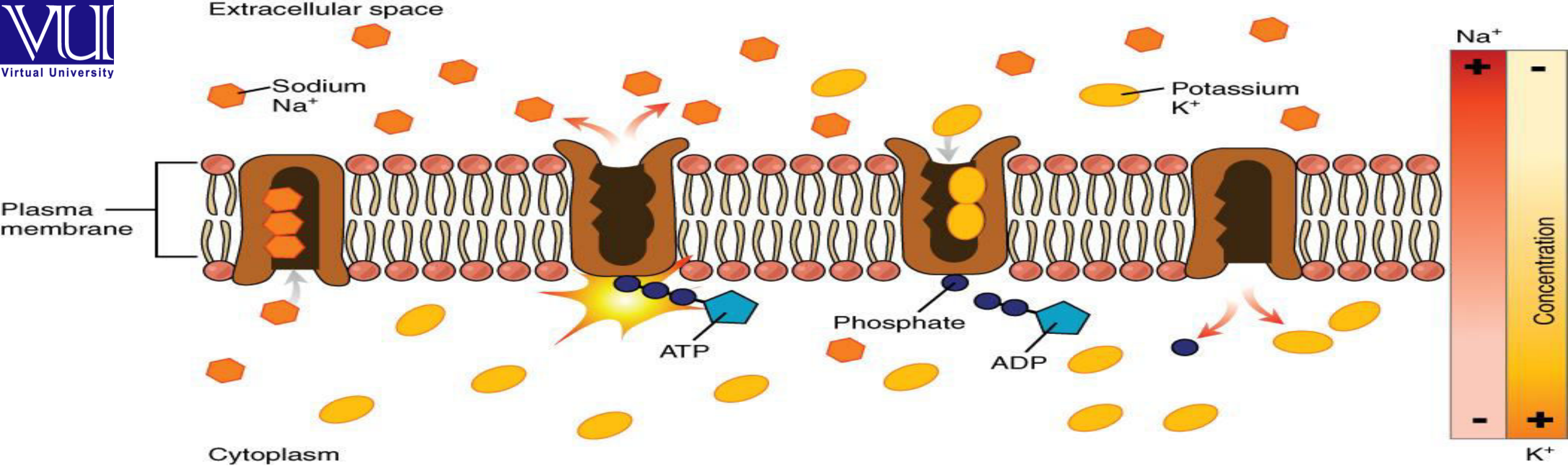
Conversely, the extracellular fluid contains a concentration of sodium ions (Na+) as much as 10 times greater than that within the cell.

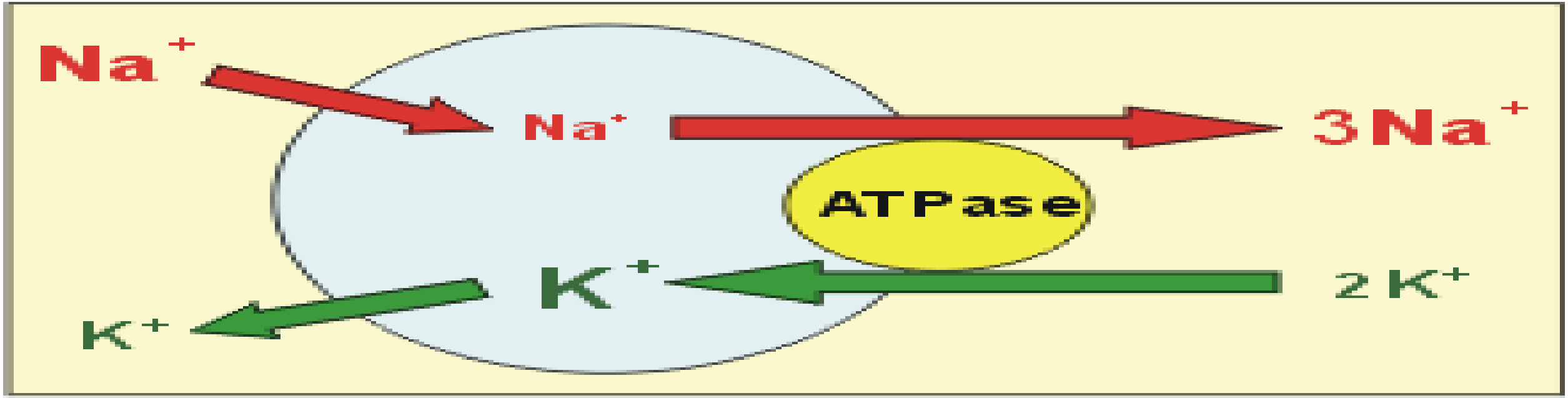
These concentration gradients are established by the active transport of both ions.

And, in fact, the same transporter, called the Na+/K+ ATPase, does both jobs. It uses the energy from the hydrolysis of ATP to

- Actively transport 3 Na+ ions out of the cell
- For each 2 K+ ions pumped into the cell.

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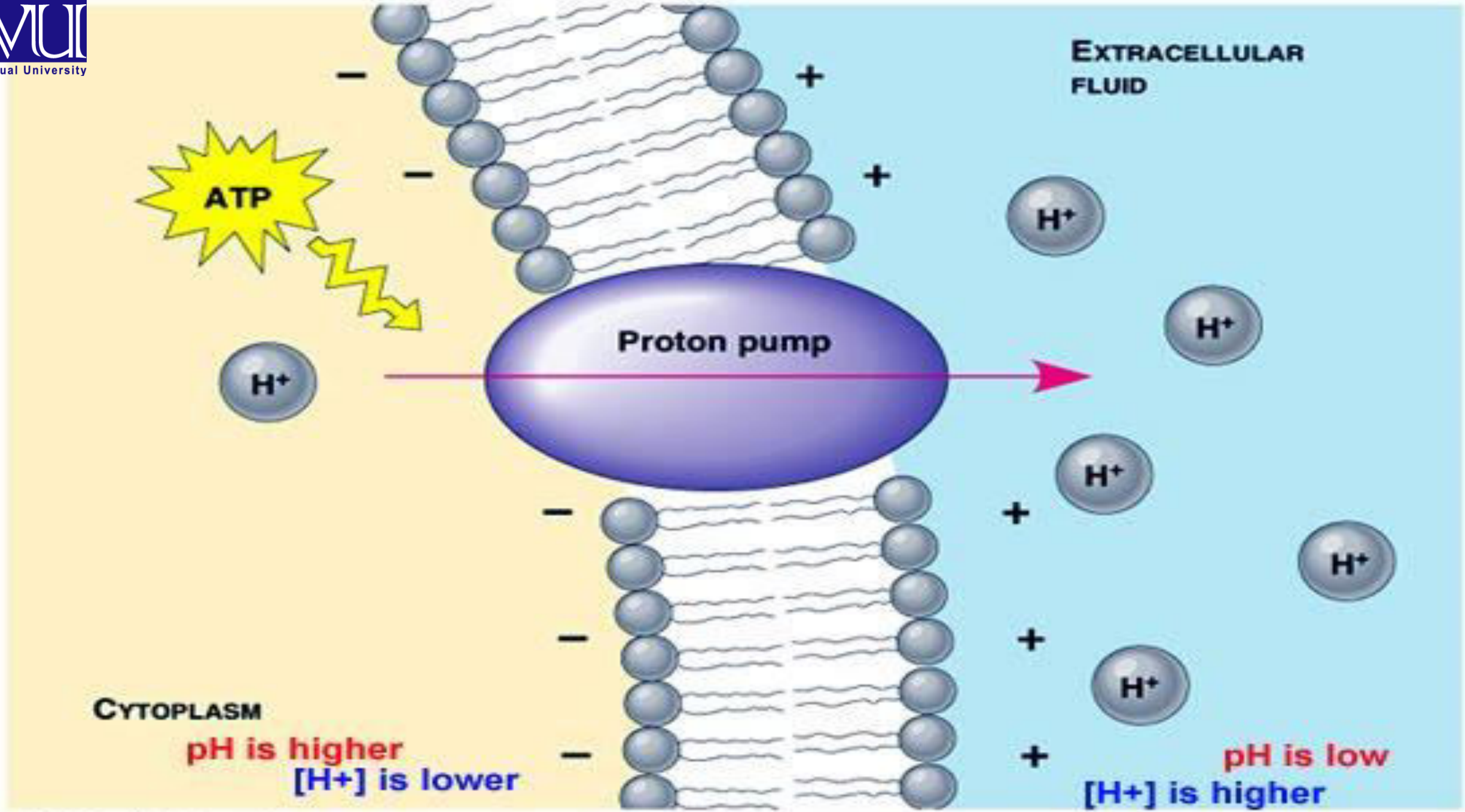


This accomplishes several vital functions:

- It helps establish a net charge across the plasma membrane with the interior of the cell being negatively charged with respect to the exterior.
- This resting potential prepares nerve and muscle cells for the propagation of action potentials leading to nerve impulses and muscle contraction.
- The accumulation of sodium ions outside of the cell draws water out of the cell and thus enables it to maintain osmotic balance (otherwise it would swell and burst from the inward diffusion of water).
- The gradient of sodium ions is harnessed to provide the energy to run several types of indirect pumps.
- The crucial roles of the Na+/K+ ATPase are reflected in the fact that almost one-third of all the energy generated by the mitochondria in animal cells is used just to run this pump.

2. The H+/K+ ATPase

- The parietal cells of your stomach use this pump to secrete gastric juice.
- These cells transport protons (H+) from a concentration of about 4 x 10-8 M within the cell to a concentration of about 0.15 M in the gastric juice (giving it a pH close to 1).
- Small wonder that parietal cells are stuffed with mitochondria and uses huge amounts of ATP as they carry out this three-million fold concentration of protons.

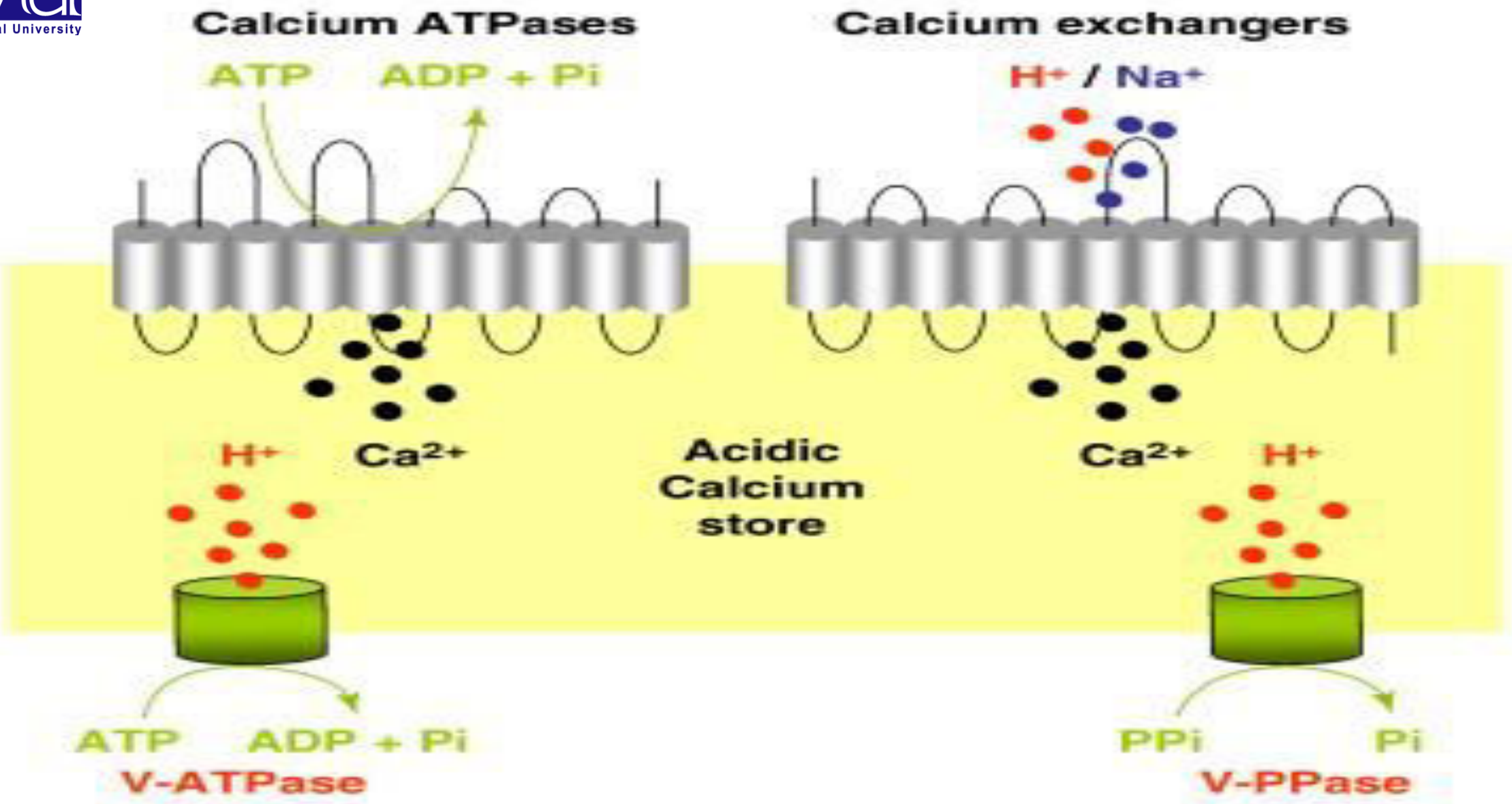


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3. The Ca2+ ATPases

- A Ca2+ ATPase is located in the plasma membrane of all eukaryotic cells.
- It uses the energy provided by one molecule of ATP to pump one Ca2+ ion out of the cell.
- The activity of these pumps helps to maintain the ~20,000-fold concentration gradient of Ca2+ between the cytosol (~ 100 nM) and the ECF (~ 20 mM).
- In resting skeletal muscle, there is a much higher concentration of calcium ions (Ca2+) in the sarcoplasmic reticulum than in the cytosol.
- Activation of the muscle fiber allows some of this Ca2+ to pass by facilitated diffusion into the cytosol where it triggers contraction.
- After contraction, this Ca2+ is pumped back into the sarcoplasmic reticulum. This is done by another Ca2+ ATPase that uses the energy from each molecule of ATP to pump 2 Ca2+ ions.

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TRENDS in Cell Biology

4. ABC Transporters

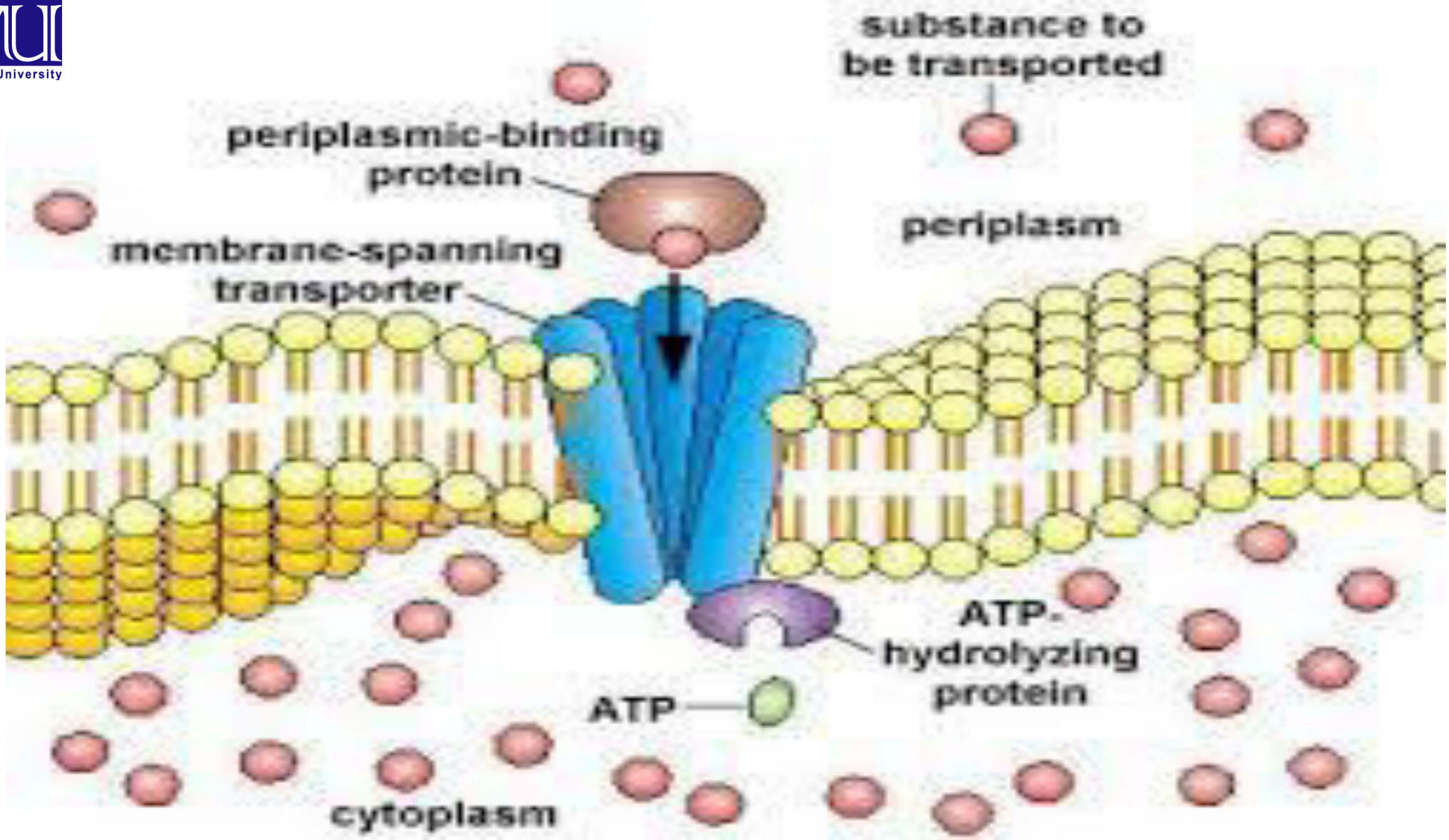
ABC ("ATP-Binding Cassette") transporters are transmembrane proteins that

- Expose a ligand-binding domain at one surface and a
- ATP-binding domain at the other surface.
- The ligand-binding domain is usually restricted to a single type of molecule.
- The ATP bound to its domain provides the energy to pump the ligand across the membrane.

The human genome contains 48 genes for ABC transporters. Some examples:

- CFTR — the cystic fibrosis transmembrane conductance regulator
- TAP, the transporter associated with antigen processing.
- The transporter that liver cells use to pump the salts of bile acids out into the bile.
- ABC transporters that pump chemotherapeutic drugs out of cancer cells thus reducing their effectiveness.

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Channelopathies

Channelopathies

- Channelopathies are diseases caused by disturbed function of ion channel subunits or the proteins that regulate them.

- These diseases may be either congenital (often resulting from a mutation or mutations in the encoding genes) or acquired (often resulting from autoimmune attack on an ion channel).

Channelopathies

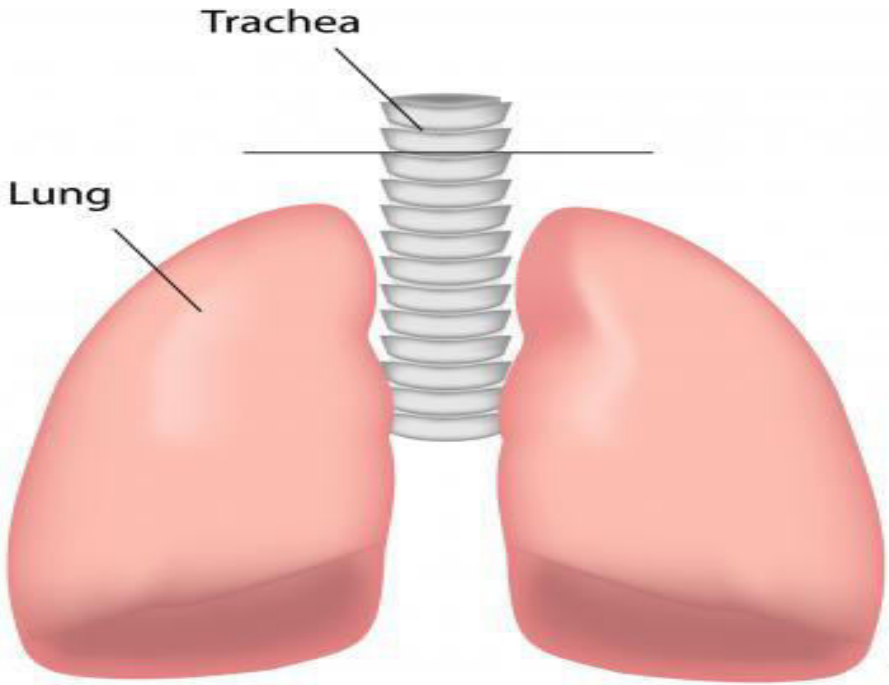
Definition: A disease caused by mutations of ion channels.

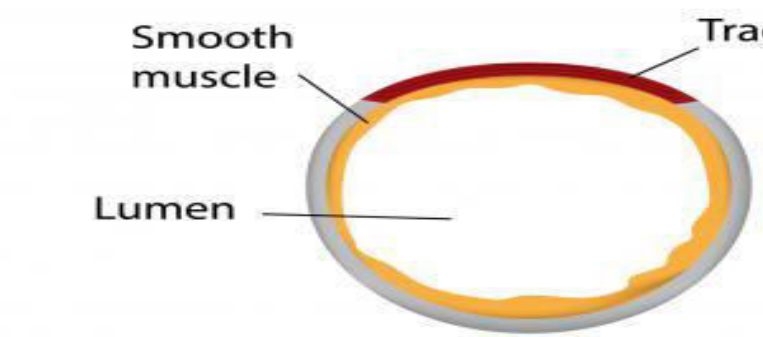
Increasingly recognized as important cause of disease (>30 diseases).

Numerous mutation sites may cause similar channelopathy e.g. cystic fibrosis where

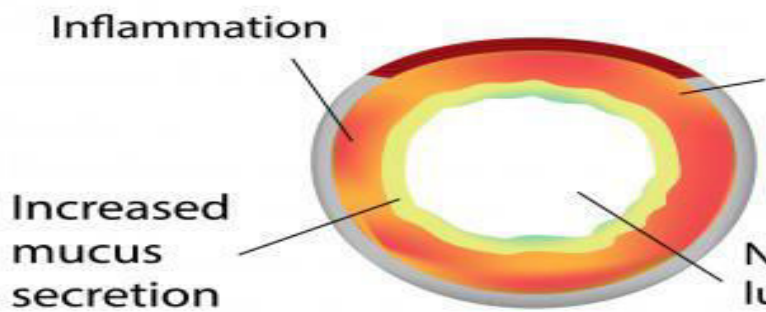
>1000 different mutations of CFTR described

Cystic Fibrosis





Healthy



Cystic Fibrosis

wiseGEEK

Channelopathies

- Extensive research has been done and is continued on ion channels.
- Ion channels are a favorite site for invention of new drugs.
- Moreover many genetic disorders are found to be caused by defective channel proteins.
- In addition to that, many toxins and venoms produced by spiders, snakes, scorpion, bees, fish, snails and others act by incapacitating ion channels.

Paramyotonia Congenita (PMC)

- Paramyotonia Congenita (PMC) is one of the periodic paralyses caused by mutations in (alpha-1 subunit) the sodium channel.

- PMC causes muscle stiffness (myotonia) which is made worse by chilling or activity.

Mechanism:

- In Paramyotonia the sodium channels fail to regulate the flow of ions properly.
- At first this imbalance causes the muscle fiber to contract uncontrollably, but as the imbalance worsens the muscle stops responding to nerve signals and becomes weak or paralyzed.
- This can clearly be seen in an EMG measurement of muscle from a PMC patient.
- **Treatment:** Tocainide is a new antiarrhythmic agent which seems to reduce effectively sodium conductance.

Brugada syndrome

- The Brugada syndrome is a genetic disease that is characterized by abnormal electrocardiogram (ECG) findings and an increased risk of sudden cardiac death.

- It is the major cause of sudden death in young's and termed as sudden unexplained/ adult death syndrome (SUDS) or (SADS).

Genetics:

- The cases of Brugada syndrome have been shown to be associated with mutations (loss of function mutation) in the gene (named SCN5A of alpha subunit) that encodes for the sodium ion channel in the cell membranes of the muscle cells of the heart (the myocytes).

Treatment:

- The cause of death in Brugada syndrome is ventricular fibrillation. These arrhythmias appear with no warning.

- While there is no exact treatment modality, treatment lies in termination of this lethal arrhythmia before it causes death.

- This is done via implantation of an implantable cardioverter-defibrillator(ICD), which continuously monitors the heart rhythm and will defibrillate an individual if ventricular fibrillation is noted.

Malignant Hyperthermia

- MH is a life-threatening clinical syndrome of hypermetabolism involving the skeletal muscle.

- It is triggered in susceptible individuals primarily by the volatile inhalational anesthetic agents (Halothane) and the muscle relaxant succinylcholine.

Genetics:

- The defect is typically located on chromosome 19 (involving the ryanodine receptor) located in the N-terminus of the protein, which interacts with L-type calcium channels.

- This region is important for allowing Ca2+ passage through the protein following opening.

Treatment:

- During an episode of malignant hyperthermia, wrapping the patient in a cooling blanket can help reduce fever and the risk of serious complications.

- The current treatment of choice is the intravenous administration of Dantrolene, the only known antidote, and supportive therapy directed at correcting hyperthermi

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Biological Effects of Ionizing Radiation

Biological Effects of Ionizing Radiation.

- A dose of ionizing radiation absorbed in a biological environment will initiate a complex process of various events.
- The physical events start with energy transfer from the ionizing particles to the atoms and molecules of surrounding tissues.
- This process takes about 10-13 seconds.
- Following this, physicochemical processes take place, such as intra-molecular energy transfers, excitation and ionization of atoms.
- The time length of this is about 10-10 s. Then the chemical processes start. At this point primary damage to biological structures begins at the time length of about 10-6 seconds.

- Following the chemical processes, biological processes starts to occur. In this phase the damaged biological structures can be either repaired or irreversibly damaged.

- Biological damage can lead to cellular death, which can occur over time periods from seconds and up to years.

- All these processes lead to the final radiobiological effect of the radiation on the human body. Ionizing radiation can, apart from causing damage, also lead to activation of adaptive and protective mechanisms.

The pathogenesis of ionizing radiation damage starts at with the physicochemical processes. These processes give origin to reactive compounds that act on molecular level and cause radiation cellular changes.

This causes cells to losing their specific properties. At the subcellular level there are irregularities happening to the biochemical processes:

- The activity of enzymes changes
- Phosphorylation mechanisms are disrupted
- The synthesis of nucleic acid
- Specific proteins do no longer work.

- The cellular level of damage first manifests itself with a decrease in the number of proliferating cellular populations.

- The loss of specialized cells causes the biochemical changes to intensify.

- The changes starts to interfere with the functions of vitally important organs such as hematopoietic tissues, intestinal epithelium etc.

- The resulting organ and systemic changes to the whole organism initiates the development of radiation sickness.

- There are also latent somatic and genetic damages, which can be seen on a population level.

- Biological effects of the same doses of ionizing radiation can differ greatly.

- These effects depend on the linear energy transfer and on the spatial distribution of a given dose.

- DNA molecules are the most critical cellular structures that can be affected by ionizing radiation.

- Ionizing radiation can induce a number of changes in the size, structure and shape of these molecules.

There are two different types of damaging biological effects due to ionizing radiation on the cellular level.

1. Exposure can cause either cell death, or
2. It can change its genetic information through mutations, leaving the cell able to reproduce itself.

In an irradiated population, both of these processes generally work simultaneously, i.e. part of the cells dies and another part is a subject to mutations or malignant transformation.

The origin of malignancies lies in changes of the functional parts of chromosomes, the genes. These mutations can originate spontaneously, but the effects of ionizing radiation can significantly increase the number of mutations.

Even under normal circumstances, there are about 10^10 spontaneous genome changes during the human lifetime.

- The final result of the exposure to ionizing radiation is defined by the extent of reparation processes.

- These processes are able to repair a part of the radiation damage, but the ability to correct mistakes is depends on the time factor.

- Molecular level or reparation includes enzymatic activity which repairs the damaged DNA structures.

- In order to restore the organ functions, the most important aspect is the restoration via proliferation of new cells, which originates from the remaining fraction of healthy stem cells.

Thank you

Basic mechanism of the biological effects of the ionizing radiation

Basic mechanism of the biological effects of the ionizing radiation

- When the ionizing electromagnetic radiation passes through a biological environment, they induce ionization and excitation of the atoms and molecules located near their trajectory.

- Each ionizing reaction causes 2 to 4 excitations.
-
- The electromagnetic radiation and neutrons cause indirect ionization.

- The ions, excited atoms, and molecules give origin to chemically reactive compounds, which are mainly free radicals.

Effect of ionizing radiation

The effect of ionizing radiation can be either direct or indirect:

- The **direct effect** of ionizing radiation and the radiant energy is immediately absorbed inside the cellular nucleus.
- The **indirect effect** induces radiolysis of water, which produces very aggressive free radicals responsible for radiation damage of important molecules. The absorption of radiation in biological environment depends on high water content.

Indirect effect

- Up to 50% of the ionizing radiation energy can be absorbed in free water.
- During radiolysis of water, production of free radicals such as H. and OH occurs.
- The resultant radicals can interact with DNA molecules and cause single- or double-strand DNA breaks.
- This may have lethal consequences. Other products of this reaction are so called molecular products of radiolysis (H2, O2, H2O2).
- The lifetime of these products is about 10-6 s.
- They undergo fast recombination or energy migration and transform into different reactive products.
- The resulting products can cause various redox reactions.

- In all of the chemical reactions following radiolysis, the major role is played by the presence of oxygen.

- Oxygen is one of the main factors influencing the radio sensitivity of cells because the oxygen can act as a radiation sensitizer.

- This means that tissues with lower oxygen content, for example hypoxic areas inside solid tumors can be more radio resistant.

- The influence of oxygen presence is most significant in the case of radiation with low linear energy transfer (X-rays, gamma radiation, electrons).
- Due to the significance of the oxygen presence, a new term is established, the oxygen ratio.
- Oxygen ratio is defined as the ratio between doses causing the same effect in hypoxic cells and in cells that are sufficiently saturated with oxygen.
- The presence of oxygen increases the production of free radicals and interferes with the reparation processes within the cell.
- Cells that are sufficiently saturated with oxygen are also 2-3 times more sensitive to radiation.
- The number of cells with insufficient oxygen saturation within tumours is generally between 10-15%.
- In the final phase of radiotherapy, these cells decide the long-term positive effect of the radiation therapy

- The ionizing radiation's ability to ionize atoms and molecules, including biologically important macromolecules, is the main factor of biological effects to the exposed organism.

- The resulting products are chemically highly reactive compounds, called free radicals.

- The primary physical and physicochemical processes are followed by other reactions, which leads to the damage of the whole irradiated organism.

- This damage gradually manifests itself by morphological and functional changes, and can be proved on molecular, cellular, organ and systemic level (i.e. the damage affects all of the levels of internal organisation of the organism).

- The amount of ions generated along the trajectory of ionizing particles per unit of track is called ion density.

- The higher the ion density, the higher the probability of hitting some of the important molecules and thus achieving greater biological effect.

- The final effects of ionizing radiation depend mainly on the radiation dose, the dose rate, way of exposure, the spatial distribution, and type of radiation.

- Metabolic condition of the organism at the time of exposure also influences the effects on an organism.

Characteristics

- One characteristic of ionizing radiation on human body is that the energy absorbed is low but the biological effects are serious.

- For example after receiving a lethal dose of 10 Gy, the body temperature will only increase by 0.02 ^{o}C but the dose may lead to death of all the exposed entities.

[gray (symbol: Gy) is a derived unit of ionizing radiation dose in the International System of Units (SI)]

- The second characteristic is the latent biological effects of radiation. Acute biological effects can occur within several hours to several days while the long term effects usually appear several years after the exposure.

Type of effects

Generally speaking, the biological effects of ionizing radiation can be classified according to the characteristics of effects, occurring times and the object that shows the effects.

Characteristic of effects	Occurring time	Object	Effects on organs
Deterministic Effects	Acute Effects	Somatic Effects	Skin damage Damage of reproductive system Damage of blood forming system Damage of digestive system Damage of central nervous system
	Latent Effects		Cataract Damage of immunization system
Stochastic Effects			Cancer
		Genetic Effects	Heredity effects

Sensitivity of body organs to radiation

Gonad:	The probability of cancer induction in this organ is comparatively low. The main concern is the genetic effects due to irradiation of this organ and the probability is proportional to the absorbed dose.
Red bone marrow:	The effect on bone marrow plays an important role in the induction of leukemia. Based on statistical data from radiotherapy patients and atomic bomb survivors in Japan, the probability of leukemia is maximum at several years after the irradiation and will return to the normal incident level after about 25 years.
Bone:	The radiosensitive cells in bone are endosteal cells and epithelial cells on the bone surfaces. The sensitivity of bone is less than that of the breast, red bone marrow, lung and thyroid.
Lung:	Cancer of the lung has been observed in miners exposed to high concentration of radon and its decay products. External irradiation can also induce lung cancer and the risk is about the same as that for the development of leukemia.
Thyroid:	Radiation can induce thyroid cancer. However, the mortality of this type of cancer is much lower than that of leukemia primarily because of the success in the treatment of thyroid cancer.
Breast:	During reproductive life, the female breast may be one of the more radiosensitive tissues of the human body. The risk for breast cancer is about one half of that of leukemia.
Skin:	The effects on skin depend on absorbed dose and the area of irradiation. Major effects are erythema and ulceration of the skin. The probability of induction of skin cancer is less than the above organs.
Lens:	The effects of radiation on the eyes are causing cataract, which is the opacity of the lens. The lens is 5 to 10 times more sensitive to neutrons than to x-ray. Thresholds exist with lower values for children than those for the adults.
Fetus:	Effects depend on the stage of pregnancy at the time of exposure. At the early stage of pregnancy when the cells are still not differentiated, it usually causes death of the embryo. It is generally believed that exposure of fetus at the first 3 weeks of pregnancy should not cause any deterministic and stochastic effect to the embryo. Later stage irradiation of the fetus may leads to malformation of body organs and increase in cancer risk of the child.

Thank you

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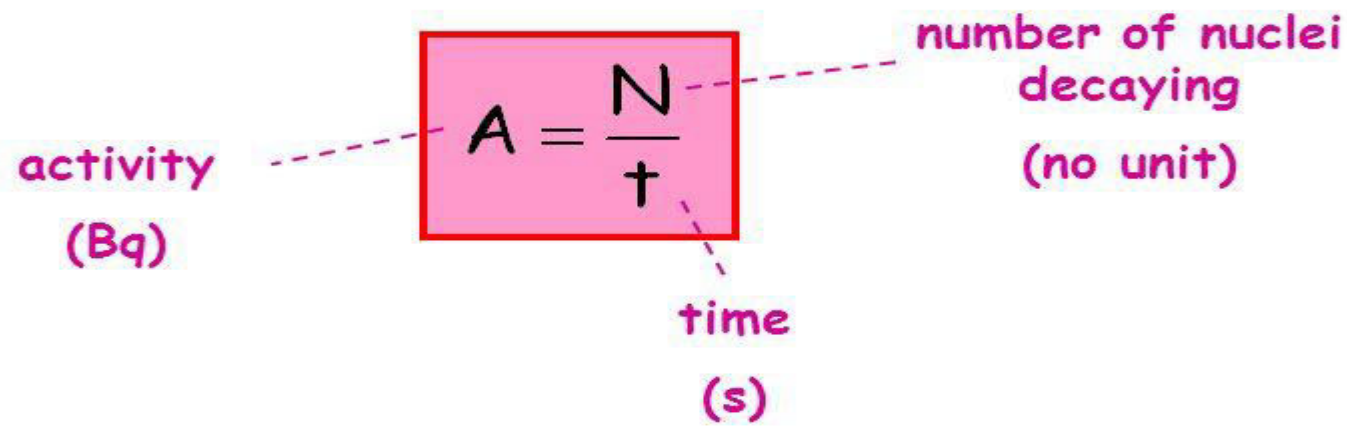
Units of Ionizing radiation measurement

Radiation Activity

- Radiation activity is measured in an international (SI) unit called a becquerel (Bq).
- The becquerel counts how many particles or photons (in the case of wave radiation) are emitted per second by a source.
- The device used for measurement is often the familiar Geiger counter.
- If you put a Geiger counter over a gram of substance and count 3 clicks per second, the radioactivity of that substance would be 3 becquerel.

Activity of Radiation

The activity of a radioactive source is the average number of nuclei decaying per unit time.



1 becquerel (Bq) = 1 decay per second

UNITS OF RADIOACTIVITY

- Unit of activity is curie (Ci)
- 1 Curie = 3.7×10^{10} disintegration/sec (or Bq)
- SI unit is Becquerel
- 1 Bq = 1 radioactive decay per second
  = 2.703×10^{-11} Ci

Question 2

A radioactive source has an activity of 25 Bq.

How many decays would be expected over a 3 hour period?

Activity is 25 Bq

= 25 decays per second

= (25 x 60) = 1500 decays in one minute

= (1500 x 60) = 90 000 decays in one hour

= (90 000 x 3) decays in 3 hours

Number of decays in 3 hours = 270 000

Question 1

A radioactive source undergoes 72 000 decays over a ten minute period.

What is its average activity in becquerels?

Activity in becquerels equals decays per second.

72 000 per 10 minutes

= 72 000 / 10 per minute

= 72 000 / (10 x 60) per second

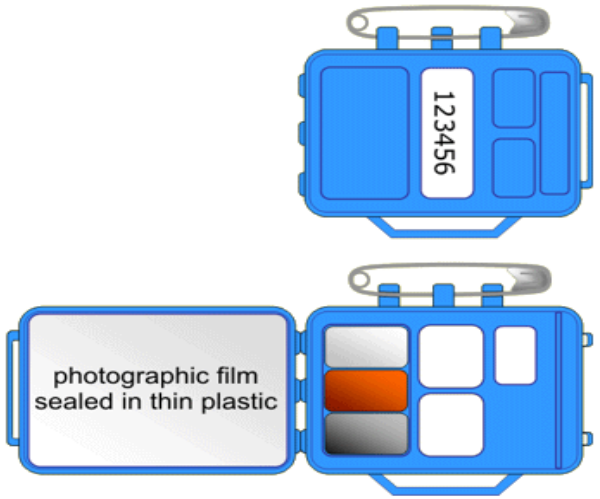
= 72 000 / 600

= 120 per second

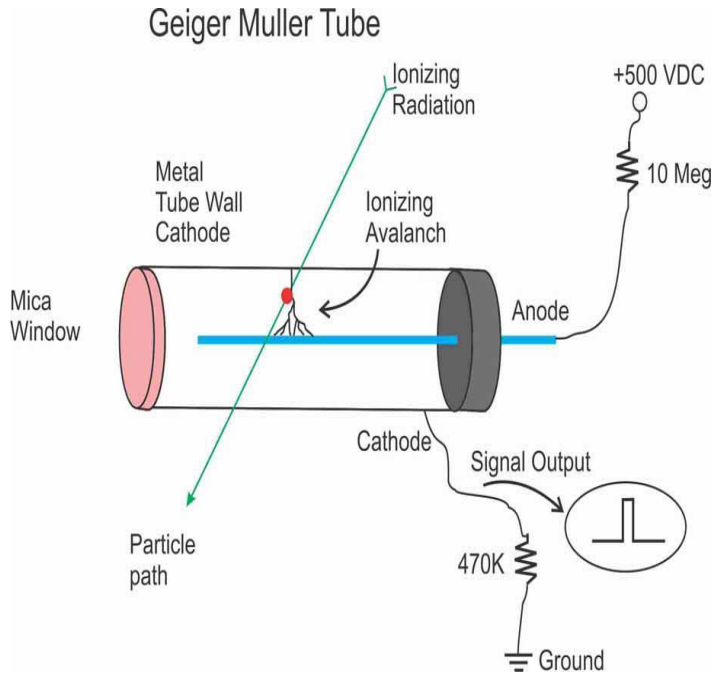
Activity = 120 becquerel

Detecting radioactivity

- Radioactivity can be detected using photographic film or a Geiger counter.



Radiation badge containing photographic film



Geiger tube and counter

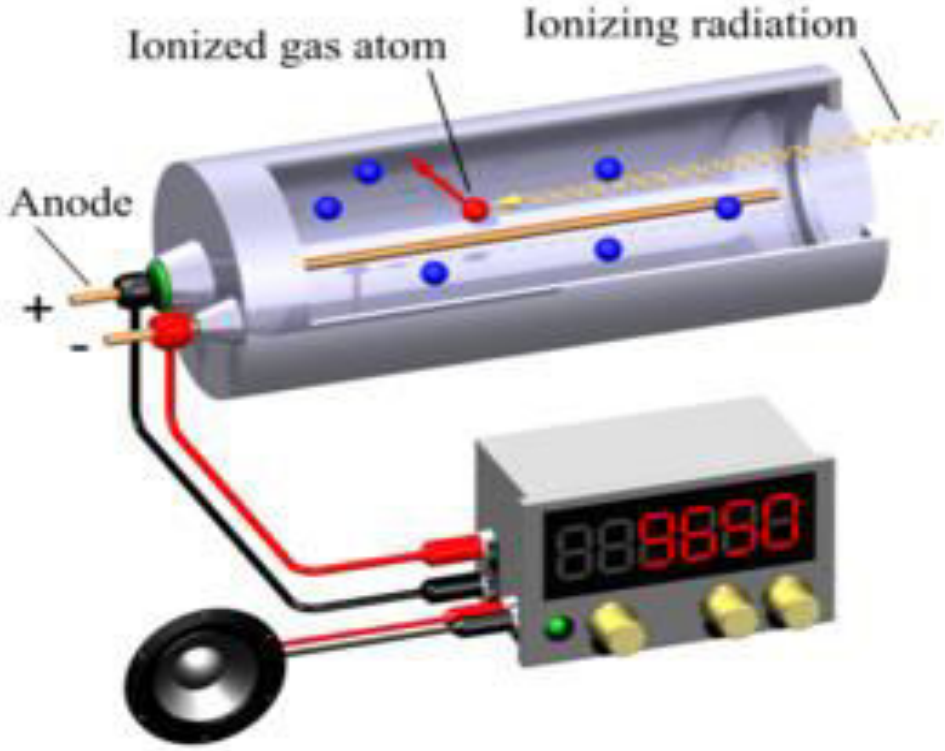
Radiation badges

- Photographic film darkens on exposure to radiation and light.
- Light cannot penetrate the badge but ionizi radiation can.
- Darkening of the film indicates that a person h been exposed to too much radiation.



The Geiger tube

- Radiation produces ions in a low pressure gas between a central positively charged electrode and the outer negatively charged tube.
- A pulse of current then flows that is registered by the counter.
- The thin mica window allows the least penetrating radiation (alpha) to enter the tube.
- Gamma radiation and most beta can enter through the sides of the metal tube.



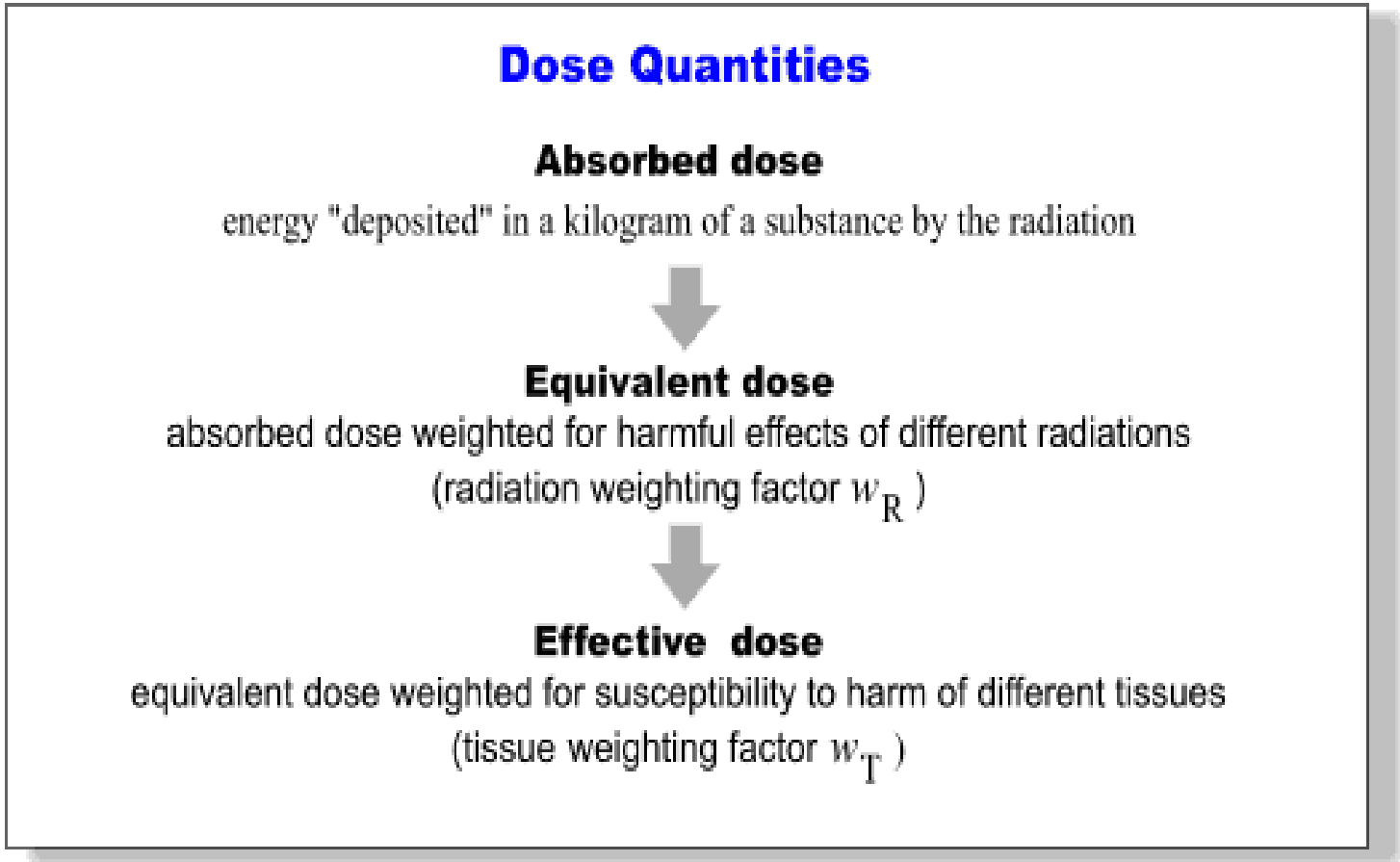
Thank you

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Radiation Exposure

Radiation Exposure

Radiation exposure is expressed in several ways to account for the different levels of harm caused by different forms of radiation and the different sensitivity of body tissues.



Absorbed Dose

- Radiation exposure is measured in an international (SI) unit called the gray (Gy).
- The radiation exposure is equivalent to the energy "deposited" in a kilogram of a substance by the radiation.
- Exposure is also referred to as absorbed dose.
- The important concept is that exposure is measured by what radiation does to substances, not anything particular about the radiation itself.
- This allows us to unify the measurement of different types of radiation (i.e., particles and wave) by measuring what they do to materials.

- The gray is a large unit and for normal radiation protection levels a series of prefixes are used:
- nanogray (nGy) is one thousand millionth of a gray (1/1,000,000,000)
- microgray (µGy) is one millionth of a gray (1/1,000,000)
- milligray (mGy) is one thousandth of a gray (1/1,000)

Equivalent Dose

- Often we are interested in the effect of radiation exposure on human tissue.
- Enter a quantity called equivalent dose. This relates the absorbed dose in human tissue to the effective biological damage of the radiation.
- Not all radiation has the same biological effect, even for the same amount of absorbed dose.

Equivalent dose is measured in an international (SI) unit called the sievert (Sv).

Like the gray, the sievert is a large unit and for normal radiation protection levels a series of prefixes are used:

- nanosievert (nSv) is one thousand millionth of a sievert (1/1,000,000,000)
- microsievert (µSv) is one millionth of a sievert (1/1,000,000)
- millisievert (mSv) is one thousandth of a sievert (1/1,000)

- To determine equivalent dose (Sv), you multiply absorbed dose (Gy) by a radiation weighting factor that is unique to the type of radiation.

- The radiation weighting factor (WR) takes into account that some kinds of radiation are inherently more dangerous to biological tissue, even if their "energy deposition" levels are the same.

- For x-rays and gamma rays and electrons absorbed by human tissue, WR is 1. For alpha particles it is 20.

- To compute sieverts from grays, simply multiply by WR. This is obviously a simplification.

- The radiation weighting factor WR approximates what otherwise would be very complicated computations.

- The values for WR change periodically as new research refines the approximations.

Effective Dose

- The probability of a harmful effect from radiation exposure depends on what part or parts of the body are exposed.
- Some organs are more sensitive to radiation than others.
- A tissue weighting factor (W_T) is used to take this into account.
- When an equivalent dose to an organ is multiplied by the tissue weighting factor for that organ the result is the effective dose to that organ.
- The unit of effective dose is the sievert (Sv).

- If more than one organ is exposed then the effective dose, E, is the sum of the effective doses to all exposed organs.

Tissue	**Tissue weighting factor W_T**	**Sum of Tissue weighting factors**
Bone-marrow (red), colon, lung, stomach, breast, remaining tissues(*)	0.12	0.72
Gonads	0.08	0.08
Bladder, oesophagus, liver, thyroid	0.04	0.16
Bone surface, brain, salivary glands, skin	0.01	0.04
	Total	**1.00**

Thank you

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Measuring Radiation

- Ionizing radiation is emitted when radioactive substances decay.

- Radioactive decay occurs when the nucleus of an atom spontaneously decays by emitting a particle (an alpha particle, an electron, or one or more neutrons).

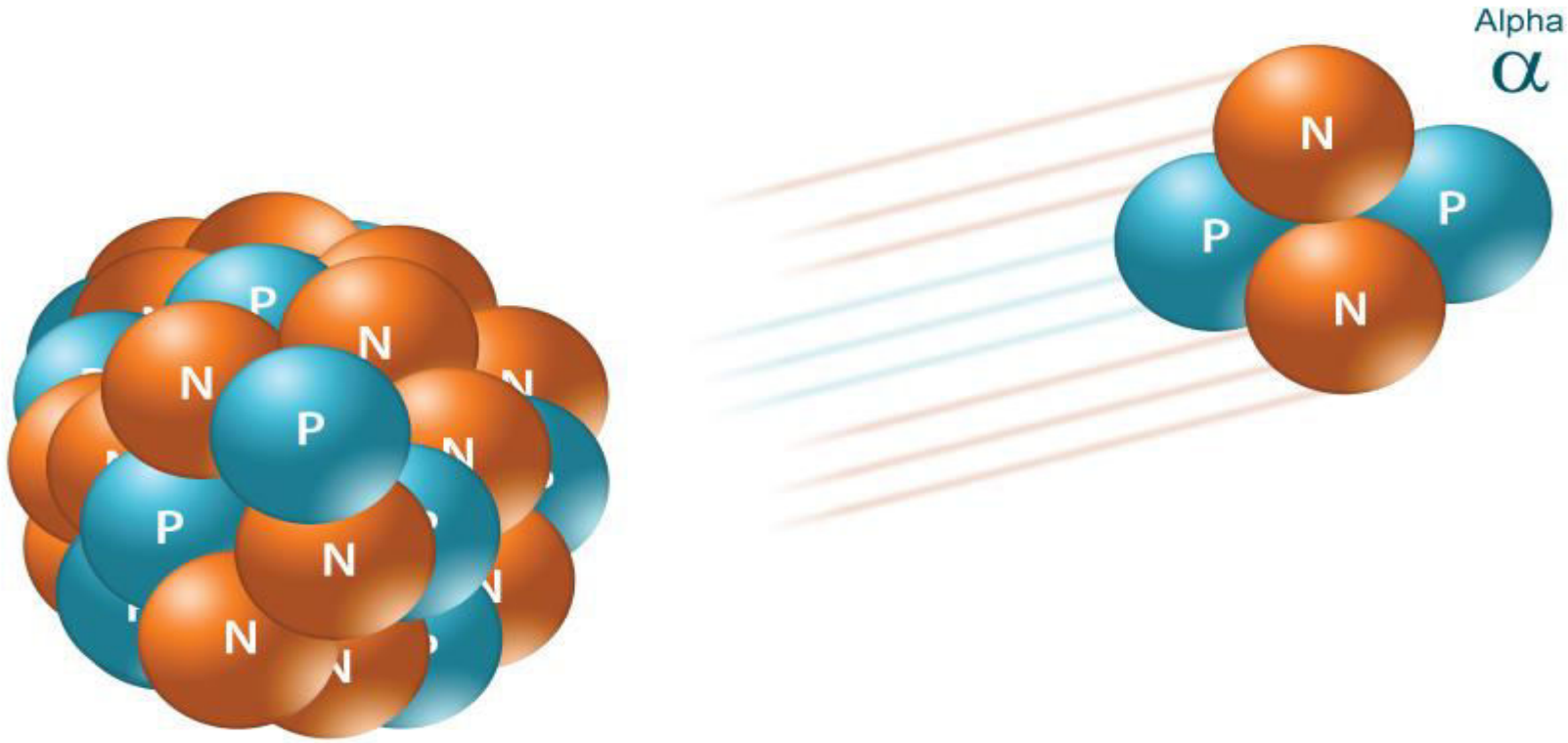
- The four forms of ionizing radiation are:

1. alpha particles
2. beta particles
3. gamma rays
4. and, indirectly, neutrons.

alpha particle

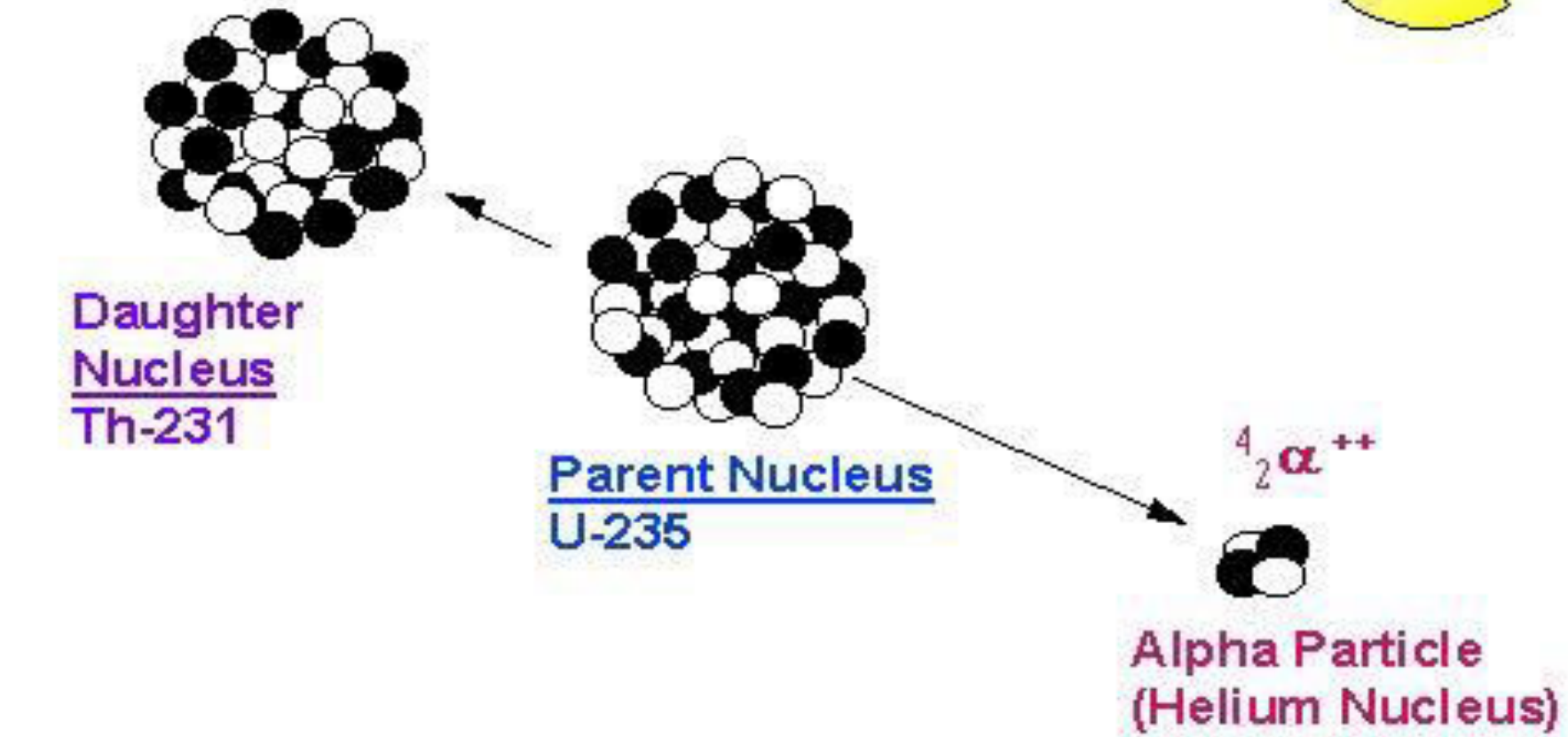
- An alpha particle consists of two protons and two neutrons, the equivalent of the nucleus of a helium atom.
- Alpha particles readily ionize material they contact and transfer energy to that material's electrons.
- An alpha particle can travel several millimeters in air, but in general its range decreases with increasing density of the medium.
- For example, alpha particles do not penetrate the outer layer of human skin, but if inhaled, alpha particles can damage lung tissue.

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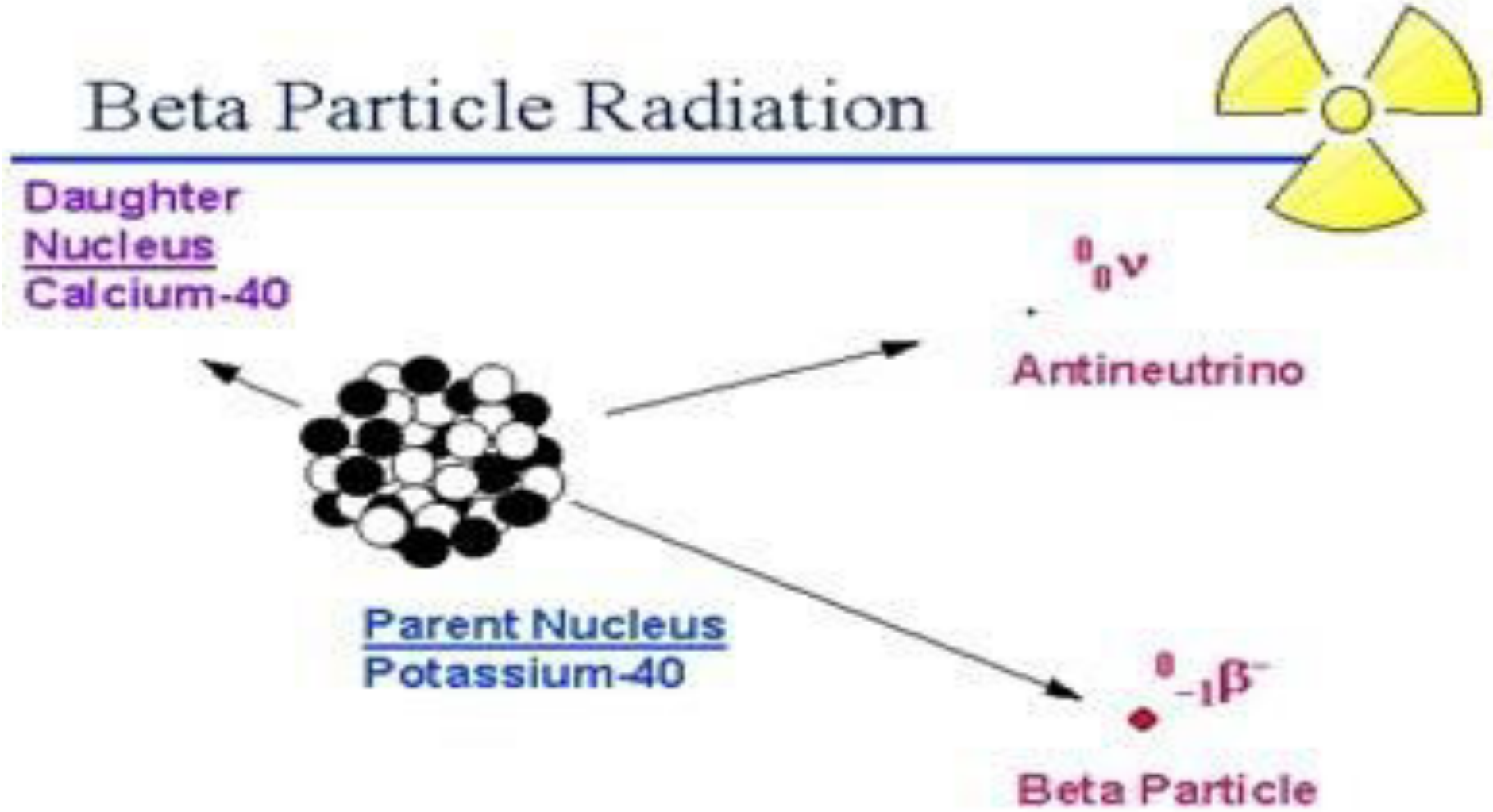
Alpha Particle Radiation



beta particle

- A beta particle is an electron or a positron and is much lighter than an alpha particle.
- Thus, it takes beta particles a longer distance than alpha particles to lose energy.
- A medium-energy beta particle travels about one meter in air and one millimeter in body tissue.

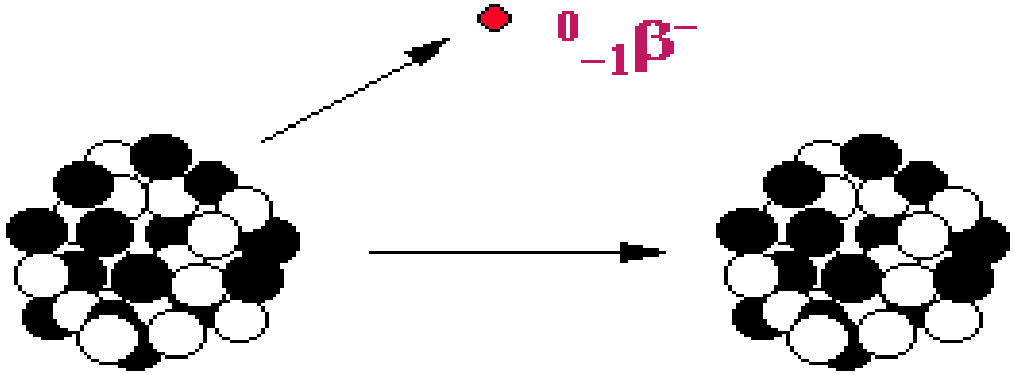
Beta Particle Radiation



Gamma rays

- Gamma rays are electromagnetic radiation.
- A radioactive element may emit gamma rays (in discrete bundles, or quanta, called photons) if the nucleus remaining after alpha or beta decay is in an excited state.
- Gamma rays can penetrate much more deeply than alpha or beta particles; a high-energy gamma ray photon may pass through a person without interacting with tissue at all.
- When gamma rays interact with tissue, they ionize atoms.
- The term “X rays” is also sometimes used for the gamma rays emitted in the process of radioactive decay that are at the lower end of the energy spectrum of electromagnetic radiation resulting from radioactive decay.

Gamma-Ray Radiation





Parent Nucleus
Cobalt-60

Daughter Nucleus
Ni-60

Neutrons

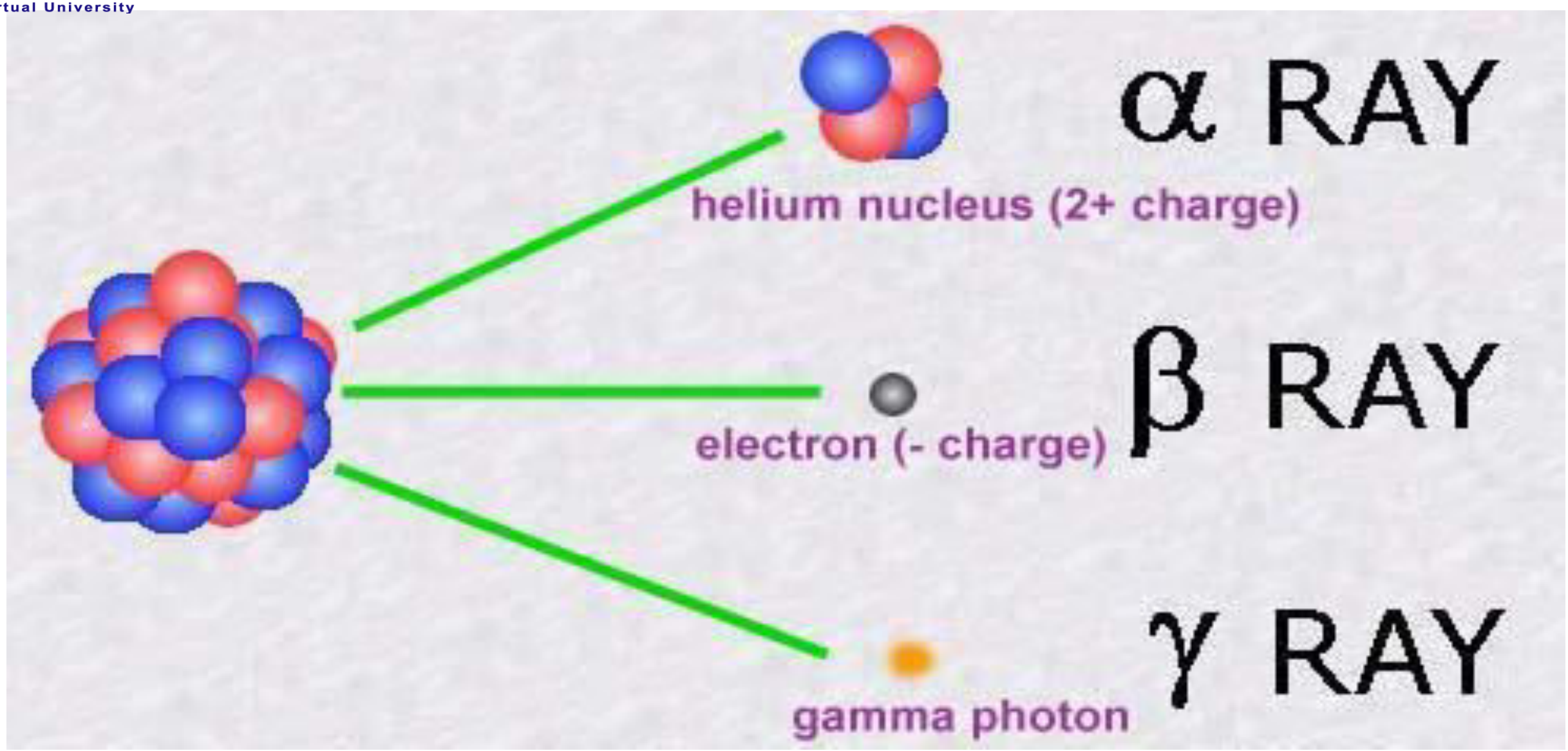
- **Neutrons** are neutral particles that have no electric charge.
- Unlike alpha and beta particles, they do not interact with electrons or cause ionization directly.
- Neutrons can, however, ionize indirectly in a variety of ways: elastic collisions, inelastic scattering, nonelastic scattering, capture reactions, or spallation processes.
- These processes variously result in the emission of gamma rays, beta radiation, and, in the case of spallation, more neutrons.

Thank you

Radioactive decay

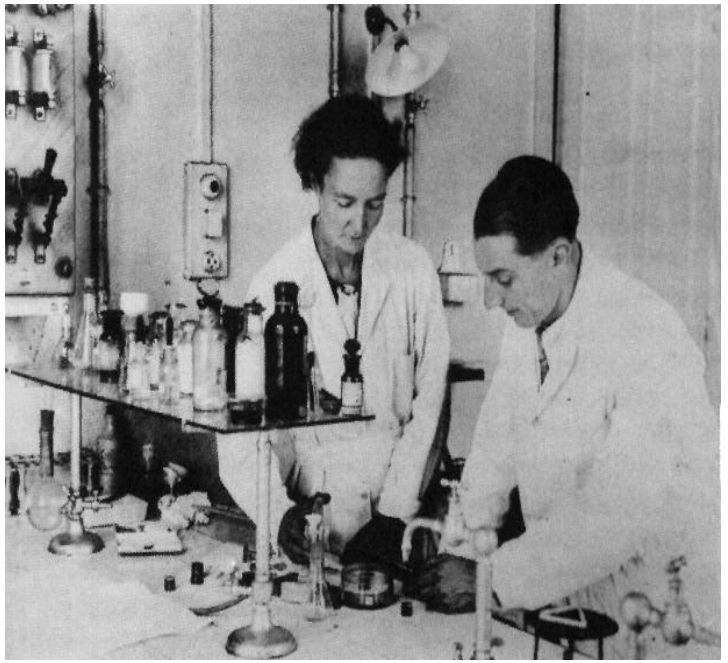
Radioactive decay

- Radioactive decay (also known as nuclear decay or radioactivity) is the process by which an unstable atomic nucleus loses energy (in terms of mass in its rest frame) by emitting radiation, such as an alpha particle, beta particle with neutrino or only a neutrino in the case of electron capture, gamma ray, or electron in the case of internal conversion.
- A material containing such unstable nuclei is considered radioactive.
- Certain highly excited short-lived nuclear states can decay through neutron emission, or more rarely, proton emission.


α RAY
helium nucleus (2+ charge)
β RAY
electron (- charge)
γ RAY
gamma photon

History of discovery

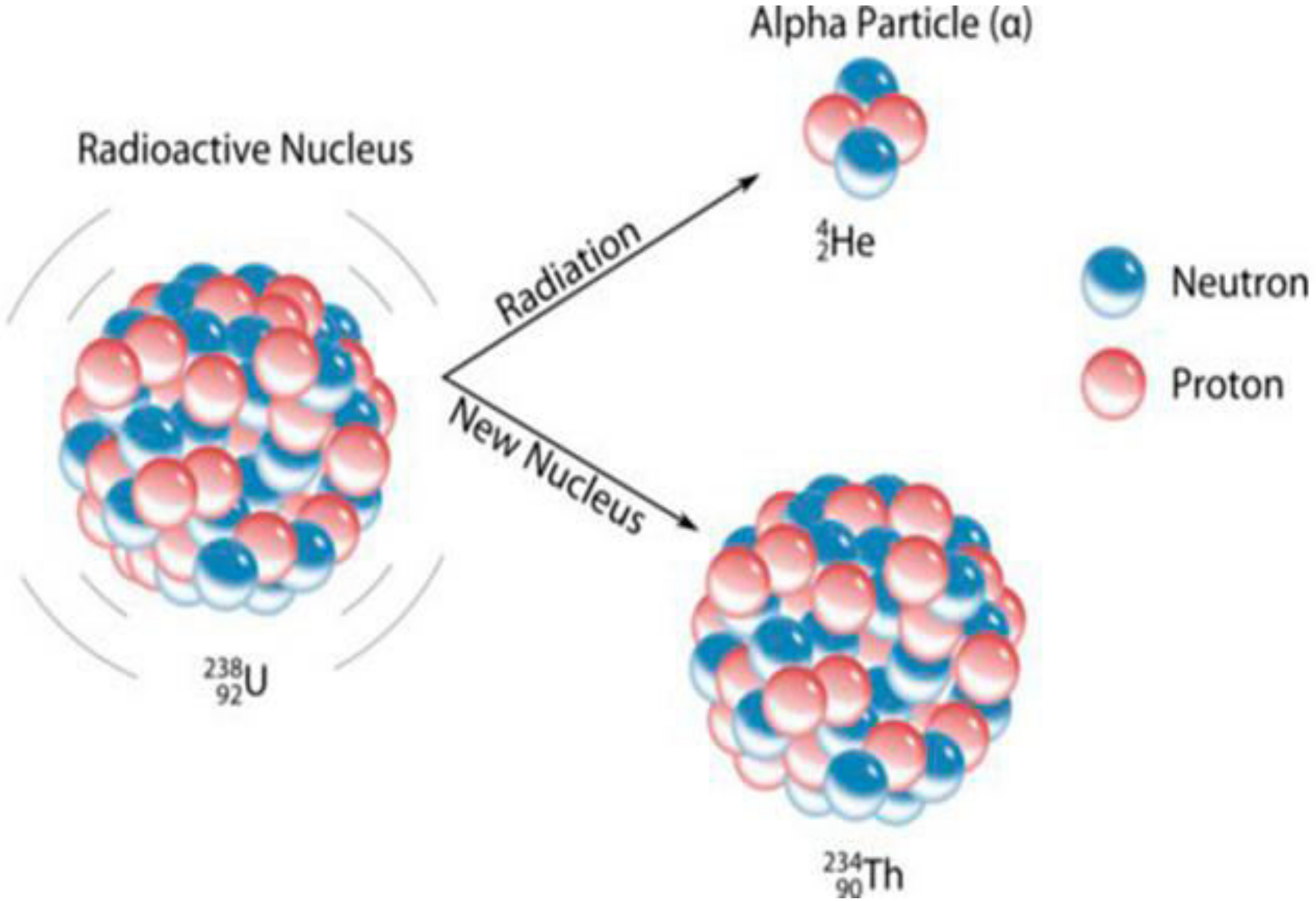
- Radioactivity was discovered in 1896 by the French scientist Henri Becquerel, while working with phosphorescent materials.
- These materials glow in the dark after exposure to light, and he suspected that the glow produced in cathode ray tubes by X-rays might be associated with phosphorescence.
- He wrapped a photographic plate in black paper and placed various phosphorescent salts on it. All results were negative until he used uranium salts.
- The uranium salts caused a blackening of the plate in spite of the plate being wrapped in black paper. These radiations were given the name "Becquerel Rays".



- Radioactive decay occurs in unstable atomic nuclei – that is, ones that don't have enough binding energy to hold the nucleus together due to an excess of either protons or neutrons.
- It comes in three main types – named alpha, beta and gamma for the first three letters of the Greek alphabet.

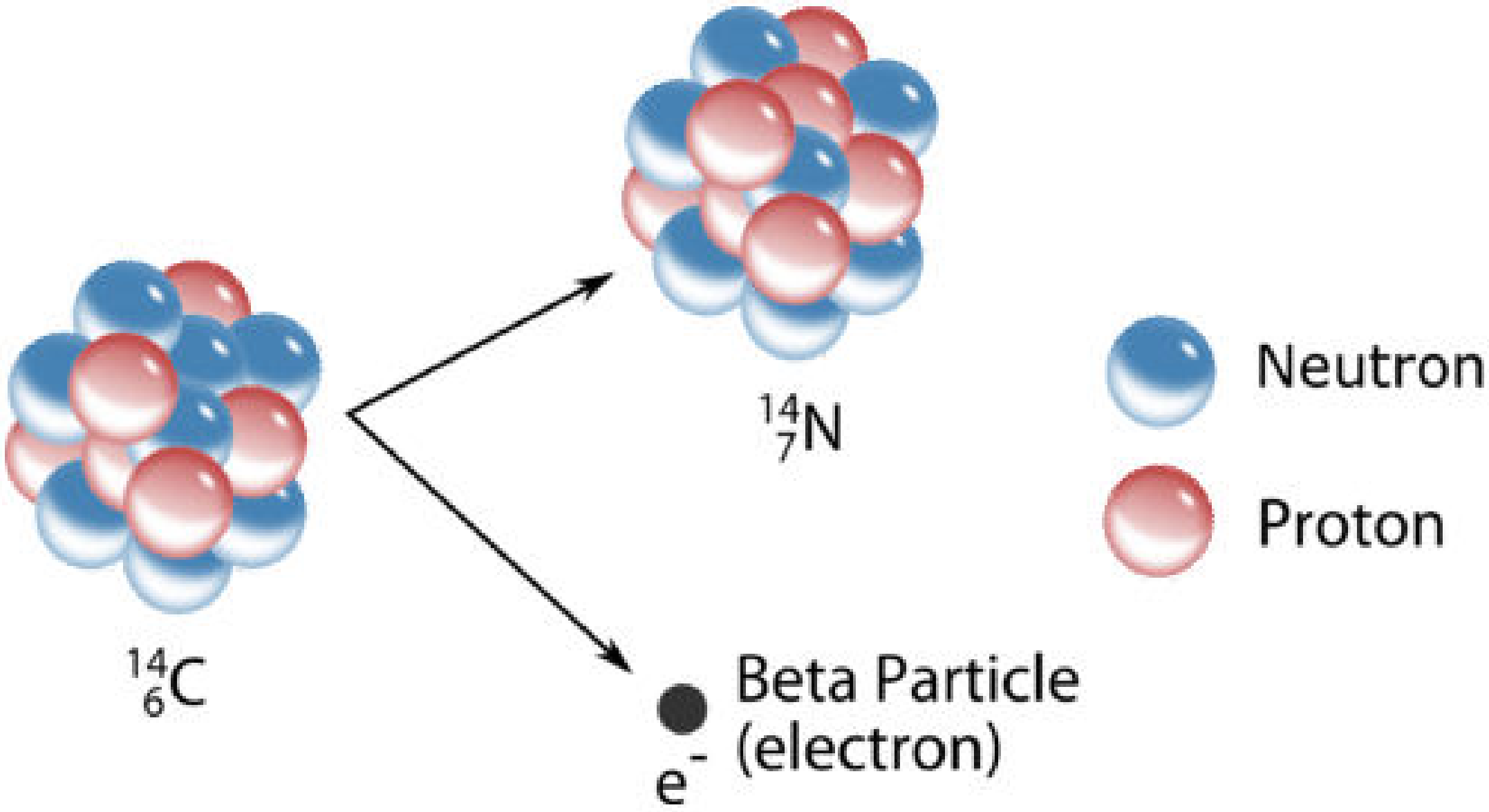
Alpha decay

- An alpha particle is identical to a helium nucleus, being made up of two protons and two neutrons bound together.
- It initially escapes from the nucleus of its parent atom, invariably one of the heaviest elements, by quantum mechanical processes and is repelled further from it by electromagnetism, as both the alpha particle and the nucleus are positively charged.
- The process changes the original atom from which the alpha particle is emitted into a different element.
- Its mass number decreases by four and its atomic number by two. For example, uranium-238 will decay to thorium-234.


Alpha Particle (α)
Radioactive Nucleus
Radiation
New Nucleus
$^{4}_{2}He$
$^{238}_{92}U$
$^{234}_{90}Th$
Neutron
Proton

Beta decay

- Beta decay itself comes in two kinds: β+ and β-.
- β- emission occurs by the transformation of one of the nucleus's neutrons into a proton, an electron and an antineutrino. Byproducts of fission from nuclear reactors often undergo β- decay as they are likely to have an excess of neutrons.
- β+ decays is a similar process, but involves a proton changing into a neutron, a positron and a neutrino.


$^{14}_{6}C$
$^{14}_{7}N$
Neutron
Proton
Beta Particle
(electron)
e^-

Gamma decay

- After a nucleus undergoes alpha or beta decay, it is often left in an excited state with excess energy.
- Just as an electron can move to a lower energy state by emitting a photon somewhere in the ultraviolet to infrared range, an atomic nucleus loses energy by emitting a gamma ray.
- Gamma radiation is the most penetrating of the three, and will travel through several centimetres of lead.
- Beta particles will be absorbed by a few millimetres of aluminium, while alpha particles will be stopped in their tracks be a few centimetres of air, or a sheet of paper – although this type of radiation does the most damage to materials it hits.

Gamma (γ) decay



$$^{12}_{6}C \longrightarrow {}^{12}_{6}C + \gamma$$

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$$^{235}_{92}U \xrightarrow{\text{decay by releasing an alpha particle}} {}^{4}_{2}\alpha + {}^{231}_{90}Th$$

$$^{14}_{6}C \xrightarrow{\text{decay by releasing an beta particle}} {}^{0}_{-1}\beta + {}^{14}_{7}N$$

$$^{235}_{92}U \xrightarrow{\text{decay by releasing a gamma wave}} {}^{0}_{0}\gamma + {}^{235}_{92}U$$

Thank you

Common sources of radiation

Types of radiation

- Radiation can be classified as non-ionizing and ionizing:
- **non-ionizing radiation** – e.g. visible light, signals from mobile phones, and radio waves; and
- **ionizing radiation** – e.g. radiation emissions from uranium ore, and high frequency waves in the electromagnetic spectrum such as x-rays.

Common sources of radiation

- Man made sources radiation
- Natural sources of radiation

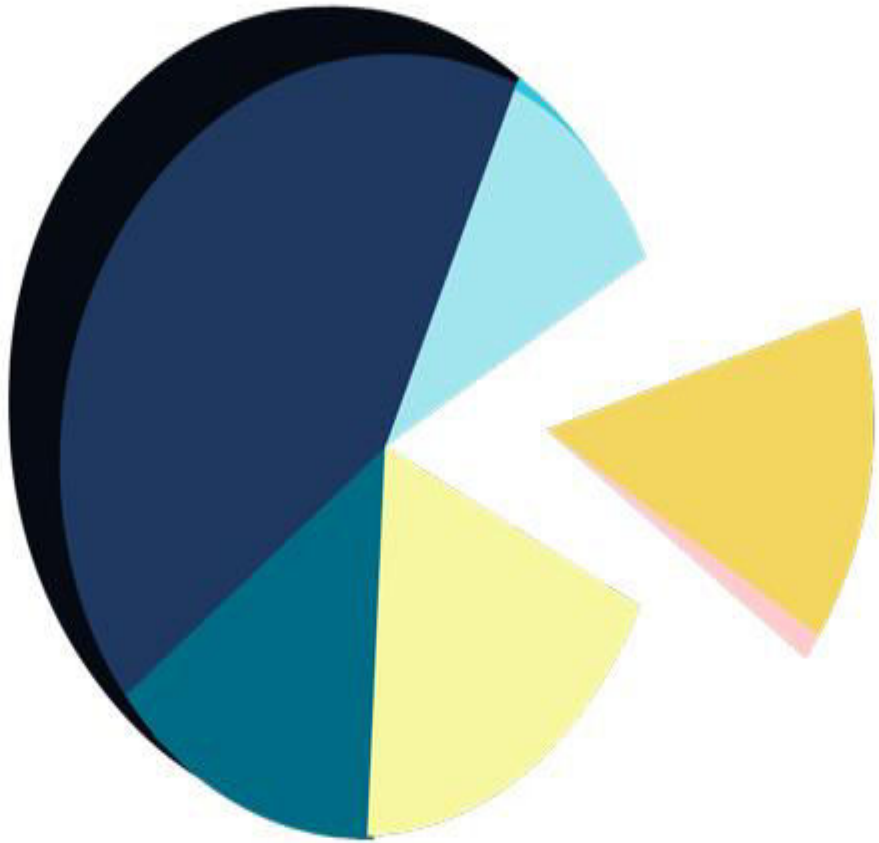
- **Natural radiation**
  Natural radiation includes radioactivity in the rocks and soil of the Earth's crust; radon, a radioactive gas given out by many volcanic rocks and uranium ore; cosmic radiation; and radioactivity in food and drinks.
- Natural radiation accounts for about 80% of the radiation doses to which we are subjected. It may vary from place to place.

- **Artificial (man-made) radiation**
  Medical use of radiation is the most significant source of man-made radiation. This includes x-ray radiology, nuclear medicine imaging and radiation therapy.
- Radiation arising from human activities typically accounts for about 20% of public exposure.
- Exposure due to fallout from past testing of nuclear weapons and generation of electricity in nuclear power plants constitutes less than 0.3%.

Sources of Radiation

Natural Radiation 80%

Food/drinks	9 %
Radon gas from ground	42 %
Cosmic rays	13 %
Buildings/soil	16 %



Man-made Radiation 20%

Medicine	Almost 20 %
Others	Below 1 %

(Including occupational exposure, fallout, products and nuclear discharges)

Reference: Sources and Effects of Ionizing Radiation, UNSCEAR 2008 Report

Man-made radiation

Smoke detectors

- Smoke detectors make use of the isotope Americium-241. This isotope emits alpha-particles at energies up to 5.4 MeV.
- The energetic alpha particles are used to ionize air.
- Once the air is ionized, a small current runs through it. When smoke enters the chamber, the current experiences an increase in resistance and a circuit sounds the alarm.

Coal-burning power plants

- Coal is an impure fuel, and it usually contains 1.3 ppm of uranium and 3.4 ppm of thorium (not to mention arsenic, mercury, and sulfur).
- When coal burns, these isotopes are emitted into the atmosphere, where they enter our ecosystem.
- This leads to the surprising fact that the population effective dose equivalent from coal plants is 100 times that from nuclear plants.

Nuclear weapon detonations

The hundreds of atmospheric nuclear weapons tests that occurred before they were banned by the 1963 Limited Test Ban Treaty left long-lived radioisotopes in the atmosphere.

Some of these are still in the atmosphere and account for some of our daily dose.

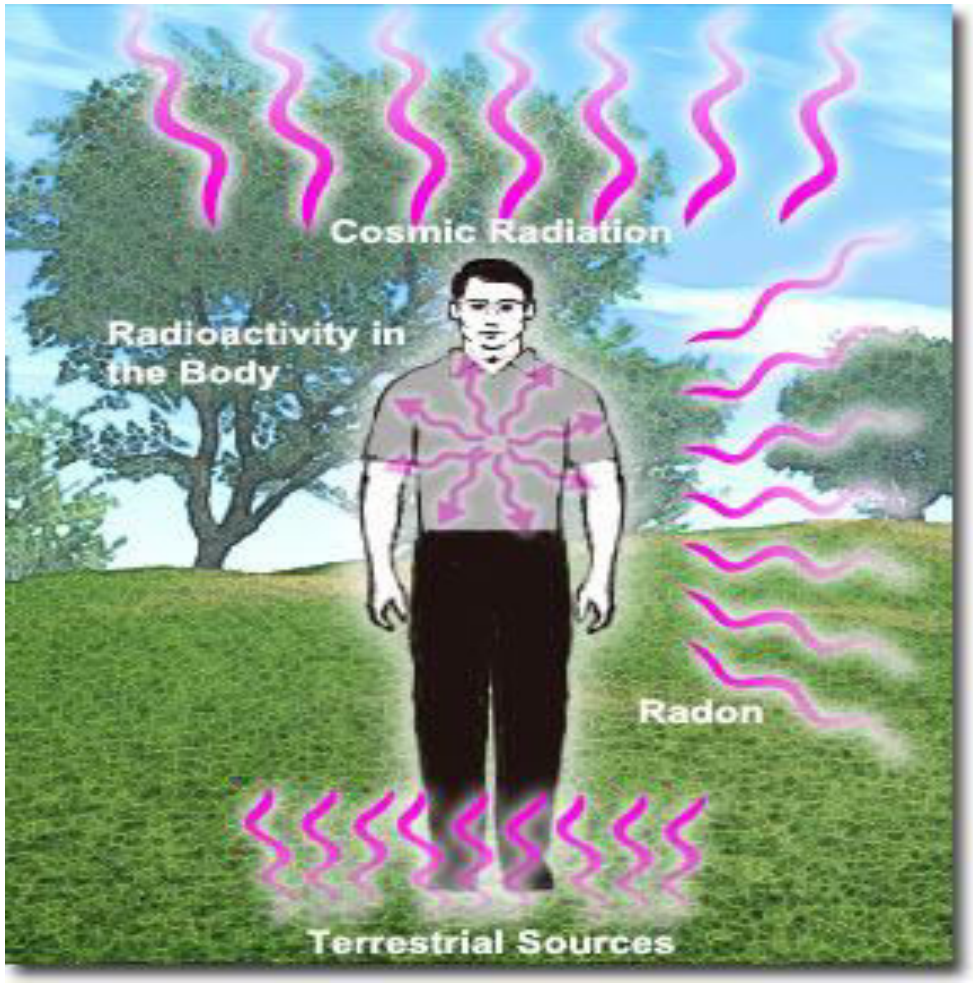
- In general, the following man-made sources expose the public to radiation (the significant radioactive isotopes are indicated in parentheses):
- **Medical Sources (by far, the most significant man-made source)**
  - Diagnostic x-rays
  - Nuclear medicine procedures (iodine-131, cesium-137, and others)
- **Consumer Products**
  - Building and road construction materials
  - Combustible fuels, including gas and coal
  - X-ray security systems
  - Televisions
  - Fluorescent lamp starters
  - Smoke detectors (americium)
  - Luminous watches (tritium)
  - Lantern mantles (thorium)
  - Tobacco (polonium-210)
  - Ophthalmic glass used in eyeglasses
  - Some ceramics

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Natural Background Radiation: Radioactivity in the Body

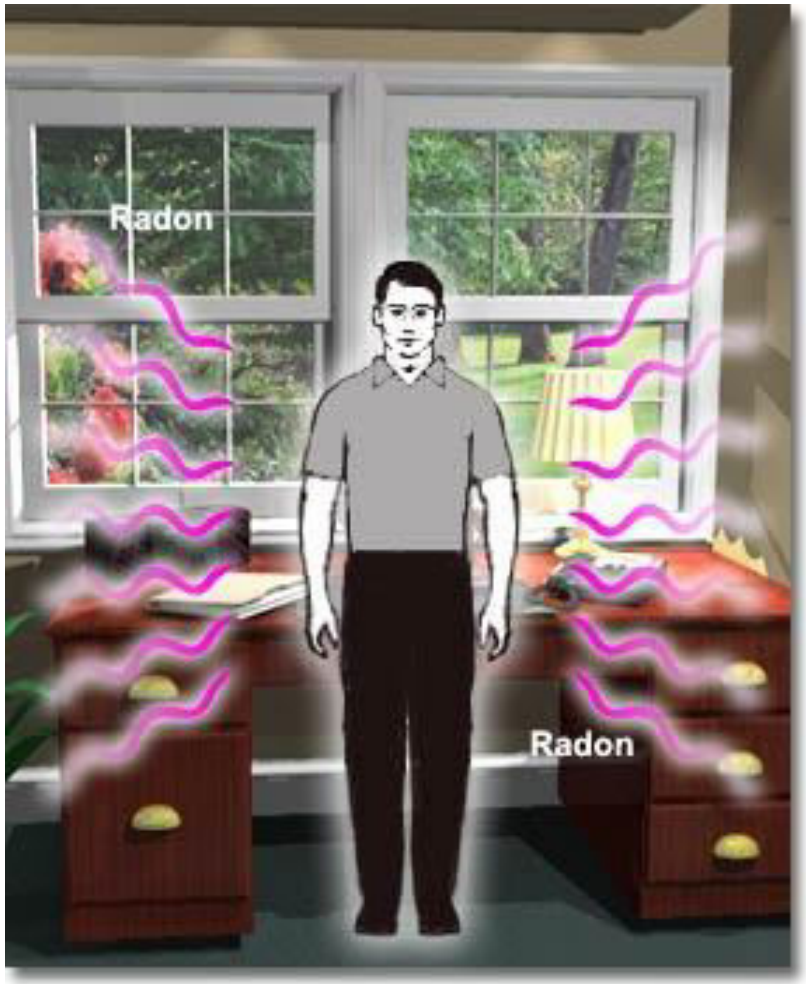
Natural Background Radiation

- Natural background radiation comes from four main sources:
- Radon
- Cosmic radiation
- Terrestrial sources
- Radioactivity in the body



Natural Background Radiation: Radon

- Radon seeps out of the earth all over the world. Radon concentrates in indoor air when buildings are, by and large, closed spaces.
- The doses from radon, however, are contributed by the radon decay products (daughters) rather than by the gas itself.
- The decay products of the gas are solid, and they attach themselves to dust particles in the air which when inhaled, irradiate the lung.
- Radon contributes, on average, about 2 mSv/year (200 mrem/year), which is one-half of the dose received by individuals from all natural and man-made sources.

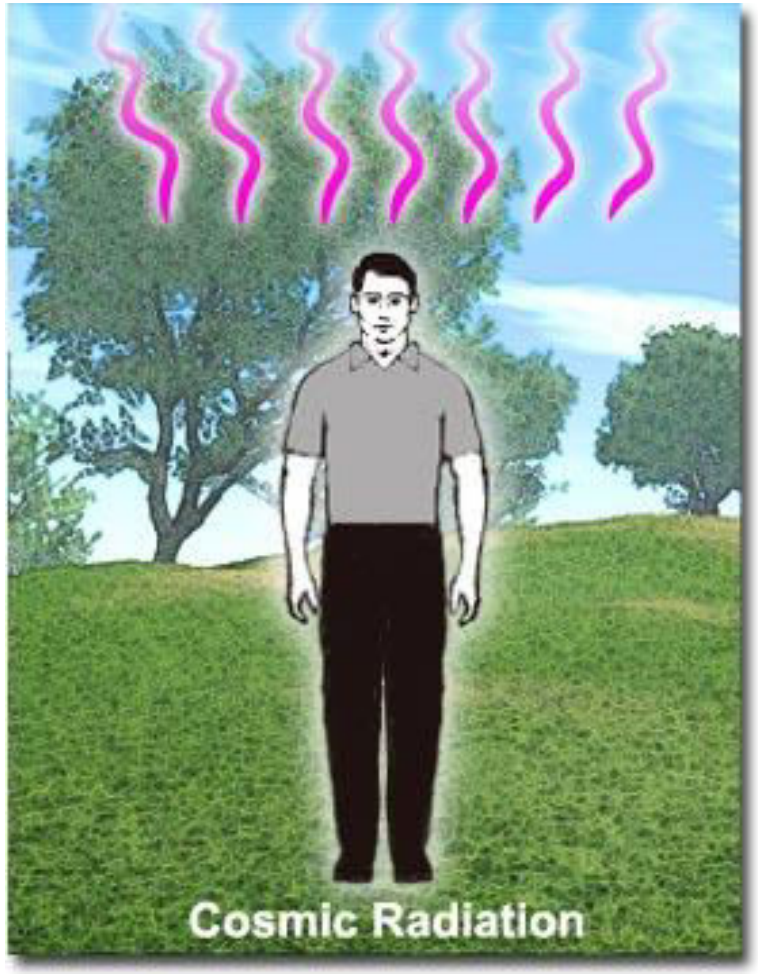


- A rem is a large dose of **radiation**, so the **millirem** (**mrem**), which is one thousandth of a rem, is often used for the dosages commonly encountered, such as the amount of **radiation** received from medical x-rays and background sources.

- The millisievert and milligray as measures of radiation dose and exposure. In the SI system, a millisievert (**mSv**) is defined as "the average accumulated background radiation dose to an individual for 1 year, exclusive of radon, in the United States."

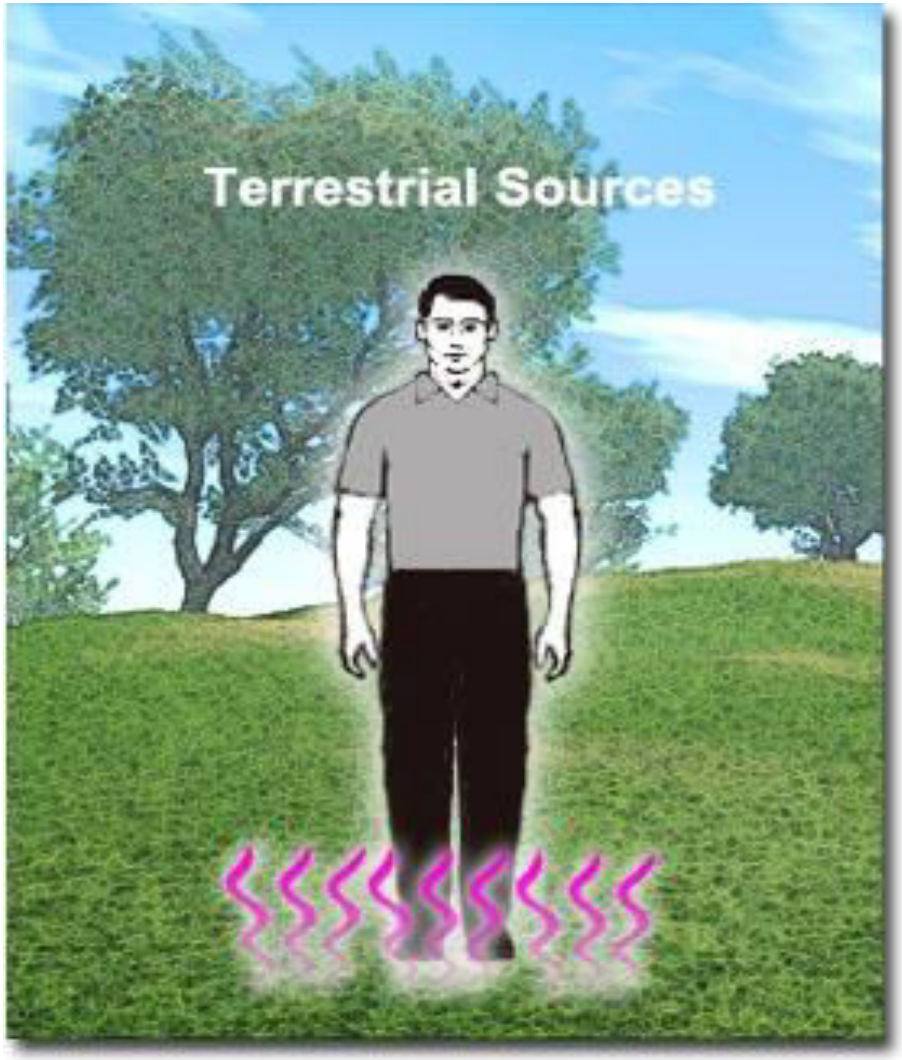
Natural Background Radiation: Cosmic radiation

- Cosmic radiation reaches the earth primarily from the sun.
- It is composed of a very wide range of penetrating radiations which undergo many types of reactions with the elements they encounter in the atmosphere.
- The atmosphere acts as a shield and considerably reduces the amount of cosmic radiation reaching the earth's surface.
- The average dose rate in the U.S. from cosmic radiation is approximately 0.3 mSv/year or 30 mrem/year.
- Much higher doses (up to 140 mrem/year or 1.4 mSv/year) are received by individuals living in higher elevation areas such as Denver.



Natural Background Radiation: Terrestrial sources

- The environment we live in is filled with radioactive materials.
- For example, the rocks and soil of the earth contain small quantities of the natural radioactive elements uranium and thorium.
- The concentration of these and other radioactive elements varies considerably depending on the type of rock formation. Thus, the dose rate from this source depends on the geographical location.
- In the U.S., the dose rate to the body may vary between approximately 15 and 140 mrem/year (0.15 and 1.4 mSv/year).
- The average is approximately 30 mrem/year (0.3 mSv/year).

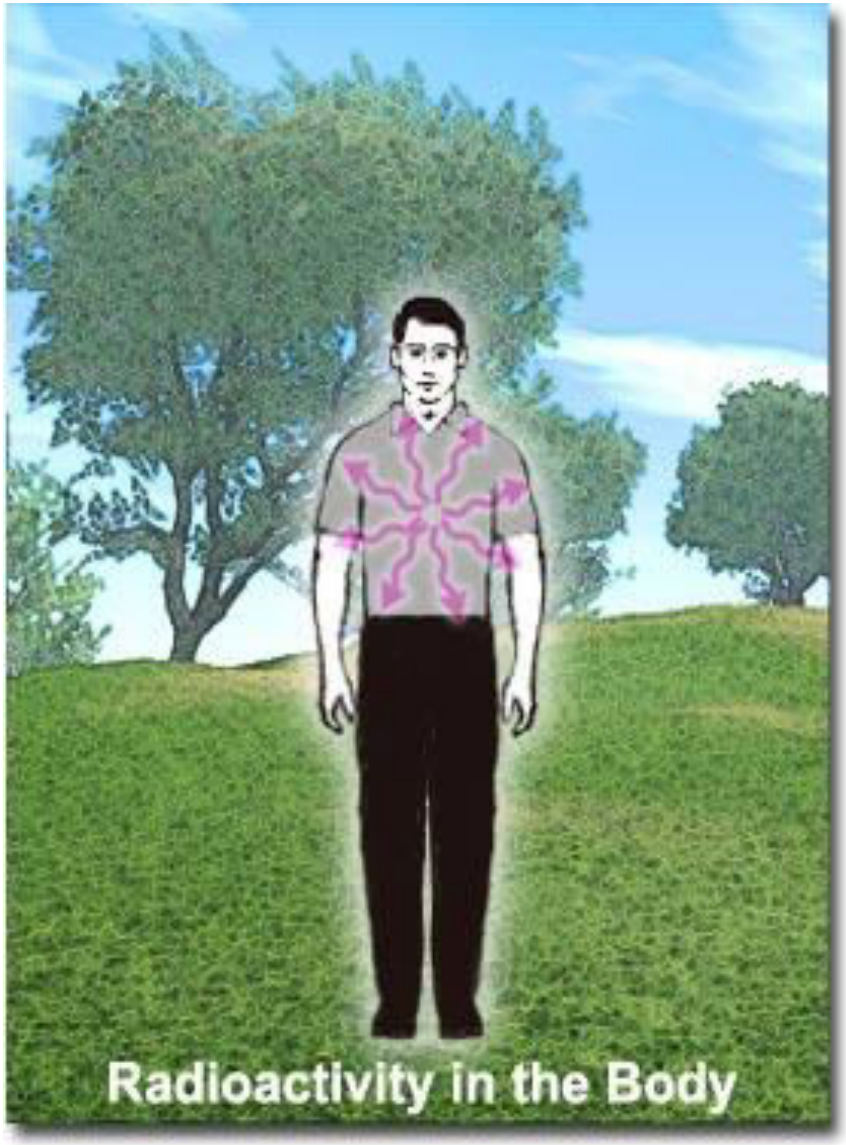


Natural Background Radiation: Radioactivity in the Body

- The human body contains very small quantities of radioactive carbon and potassium. This radioactive material gets incorporated into tissues and organs throughout the body.

- The radioactive carbon originates in the atmosphere and results in a dose of approximately 1mrem/year (0.01 mSv/year) in the soft tissue.

- Radioactive potassium is naturally occurring and contributes approximately 20 mrem/year (0.2 mSv/year) to the testes or ovaries.



Thank you

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Radiation Tidbits

Where does man-made radiation come from ?

- Man-made radiation sources include X-ray machines, nuclear power plants, nuclear industrial incidents, nuclear weapon tests, as well as consumer products which produce radiation.

- Radiation is widely used for medical diagnosis and therapy. This includes the familiar body and dental X-ray examination, as well as computer tomography.

- In radiotherapy, radioactive material is used to kill cancer tumors.

- The vacuum tubes of television and video display units used in our daily life emit X-ray.

- Other consumer products, such as radio-luminous watches and smoke detectors, consist of radioactive material, and are another source of man-made radiation.

What is the main source of man-made radiation? What is its radiation level ?

- More than 90% of man-made radiation received by human comes from medical diagnoses. The annual dose of man-made radiation received by people arising from medical diagnoses ranges from 0.04 mSv to 1.0 mSv.

- This is only a fraction when compared with the annual dose of natural background radiation received by people, which is in the range of 1 mSv to 10 mSv.

What is X-ray ? Is it safe to take X-ray examination ?

- X-ray is high-energy electromagnetic wave. Its wavelength is shorter than those of visible light and is not visible to the human eye.
- X-ray is widely used for medical diagnosis since its discovery in 1895.
- X-ray is highly penetrating. The higher the density of the material it acts on, the lower will be the degree of penetration.
- Since human tissue, organs and bone have different densities, the X-ray passing through the body will be attenuated differently thus producing images very useful in medical diagnosis.



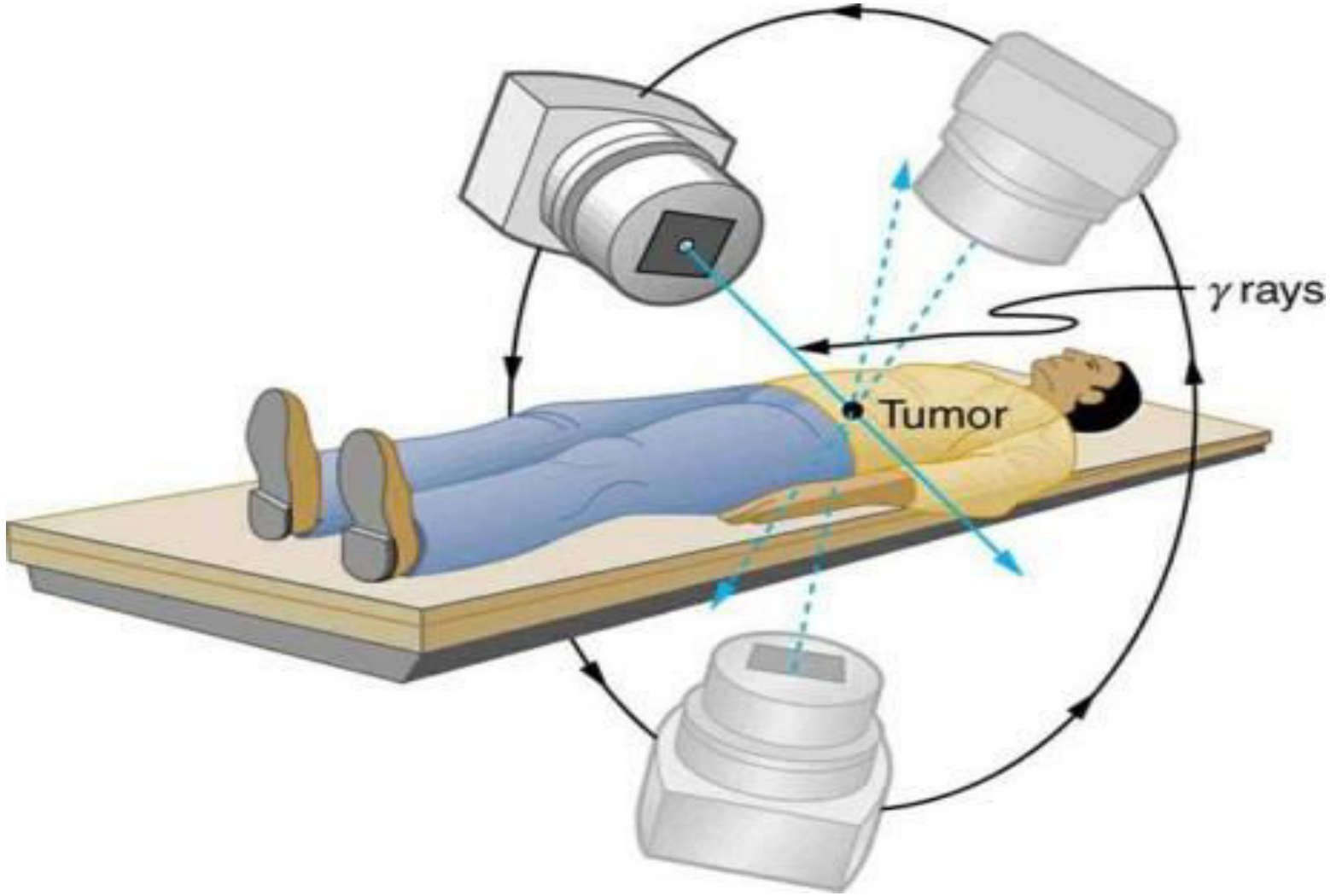
Chest X-ray examination before (left) and after (right) surgical abortion. The diameter of the largest pulmonary manifestation decreased significantly

- The X-ray dosage in medical diagnosis is very low each time and is unlikely to cause any harm to the patient.

- Furthermore, the radiographer will shield those parts of the body not requiring medical diagnosis using lead sheets to reduce the effect of scattered radiation to the patient.

How is radiotherapy performed ?

- Radiotherapy is the use of high-energy radiation to treat cancer cells.
- Usually an external, collimated beam is used to irradiate the parts requiring treatment, and in some cases radioactive material is placed near the tumor so that the radiation will act on the tumor more directly.


γ rays
Tumor

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Thank you

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Radiation Tidbits

How is nuclear energy generated ?

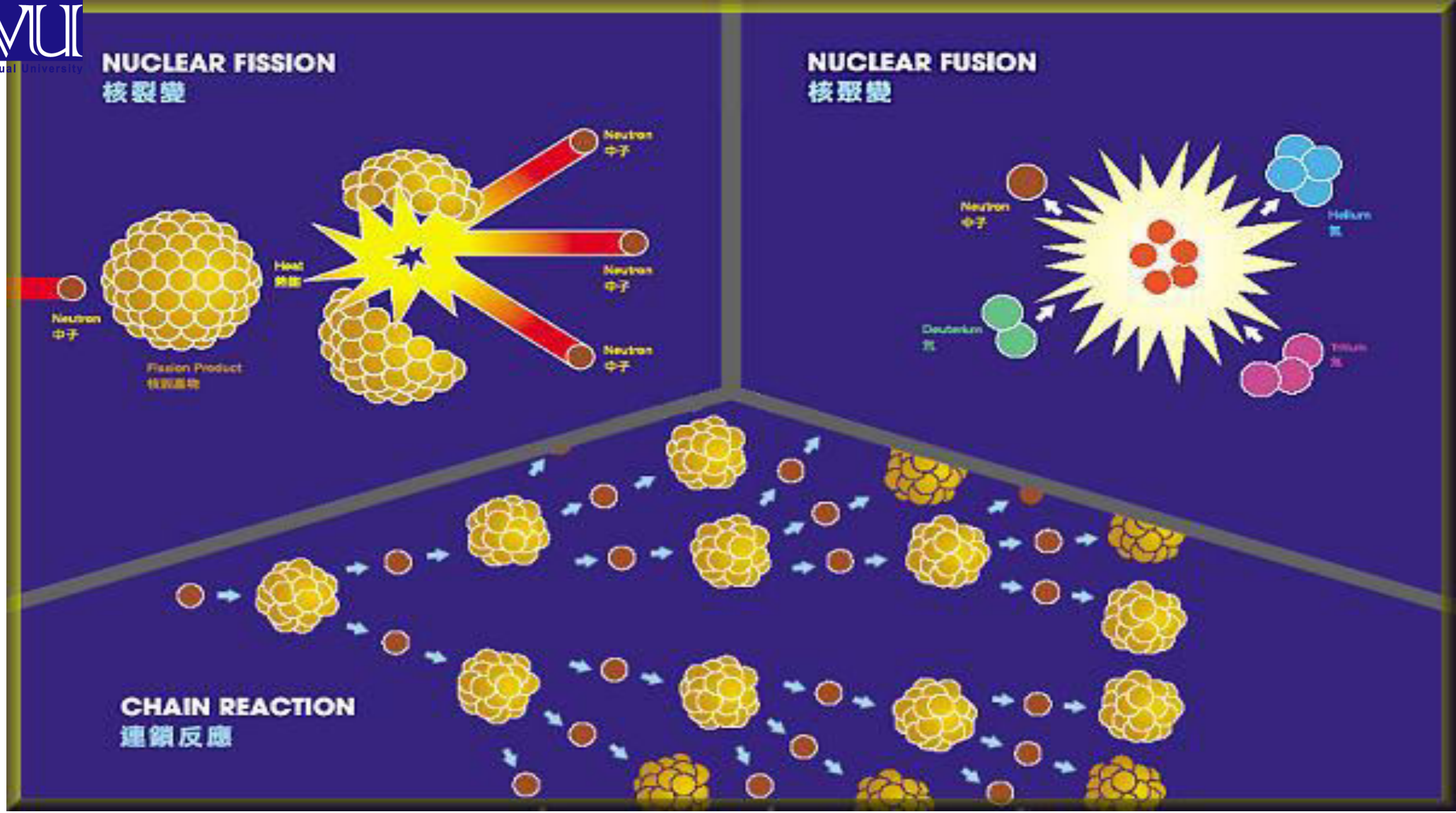
- Nuclear energy is the energy released when the structure of nuclei inside an atom changes. This is released in two ways, nuclear fission and fusion.

- In nuclear fission, relatively large nuclei of, say, uranium or plutonium, are split, while in nuclear fusion, smaller nuclei of deuterium and tritium, for example, are combined.

- In both cases, a large and highly concentrated amount of energy is released. For instance, 1 gram of uranium-235 can produce as much energy as that generated by burning 2.5 tons of coal.

What is nuclear fission and nuclear fusion ?

- Nuclear fission is a nuclear reaction in which a relatively large nucleus such as uranium splits into two smaller nuclei and releases energy in the process.
- Nuclear fission can be spontaneous or induced. Very large nuclei are unstable normally so some of them have a chance of splitting up spontaneously.
- Nuclear fission can also be initiated when the nucleus of an atom is hit by other particles. For example, uranium may split up when hit by a neutron.
- Nuclear fusion is a nuclear reaction in which two nuclei join to form a larger nucleus and release energy.
- The energy released in nuclear fusion is usually larger than that in nuclear fission. It is hydrogen fusion (producing helium) in the Sun that results in light and heat coming out of it.


NUCLEAR FISSION
核裂變
Neutron
中子
Heat
熱能
Neutron
中子
Neutron
中子
Neutron
中子
Fission Product
NUCLEAR FUSION
核聚變
Neutron
中子
Helium
氦
Deuterium
氘
CHAIN REACTION
連鎖反應

What are the common consumer products that produce radiation ?

- Common consumer products in our daily life such as smoke detectors, luminous signs, radioactive lightning conductors, contain radioactive material.
- It is the presence of radioactive material in these products that make them work.
- They are properly designed and their use according to instructions will not endanger our health.

How does radon affect the human body ?

- Radon enters our body through respiration. Its decay gives off alpha particles, which results in internal exposure to our body and may cause damage to the respiratory system.
- Long-term exposure to alpha particles may cause damage to tissues and increase the incidence of lung cancer.

- Relevant guidelines should be given to professionals to protect building occupants from excessive radon exposure.

Where can I get more information on radiation ?

- To get more information on man-made radiation, please visit the Observatory's webpage 'Radiation Monitoring, Assessment and Protection' at http://www.weather.gov.hk/education/dbcp/menu/eng/emenu.htm

Where does radiation in the earth crust originate from ?

- Natural radiation has been part of the earth since it came into existence. Primordial (i.e. since the beginning of the world) radionuclides include uranium-235, uranium-238, thorium-232, potassium-40 and neptunium-237 in rock and soil.
- They are unstable and undergo radioactive decay until they reach a stable state. As the decay of uranium-235, uranium-238, thorium-232 and potassium-40 is very slow, these radionuclides are still found today.
- Neptunium-237, on the other hand, decays fast and is no longer naturally present on earth nowadays.

What is cosmic radiation ?

- Cosmic radiation is natural-occurring and arises from cosmic rays.
- Cosmic rays consist largely of streams of charged particles and carry high energy.
- Their origin is still not well understood, although it is generally believed that cosmic rays come from supernova explosions or from activities of distant galaxies.

What are the factors affecting the radiation level in soil ?

- Soil is formed from the erosion and ablation of rock.
- The speed of soil formation varies with different rock types (and the weathering processes).
- Therefore, the composition of natural radionuclides in soil differs in accordance with the rock type they originated from.
- The amount of natural radionuclides in soil and their radiation levels depend on the geologic origin, geographic location, hydrologic and climatic conditions of the soil.

How does the strength of cosmic radiation vary with height ?

- The Earth's atmosphere protects us from exposure to cosmic rays.
- Before they reach the earth's surface, the cosmic rays are blocked by the atmosphere and lost some of their energy.
- As a result, the strength of cosmic radiation at high altitudes is many times that on the surface.

Thank you

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Radiation Tidbits

How does natural radiation enter our body through the biosphere ?

- Natural radiation enters the human body through three pathways.

- First, some natural radionuclides in the earth crust can be washed away and carried in underground water. They then enter into the human body through water intake.

- Secondly, the radionuclides on the soil surface can become airborne, in the form of dust or particulates, by wind action. They then enter into the human body through respiration.
- Thirdly, soluble radionuclides in deep soil can be absorbed by plant root and transported up the stem. The edible portion of the plant then becomes food for human or herbivore. Radiation present in the latter may eventually enter our body through ingestion.

How does rain affect the ambient radiation level ?

- Initially, rain will wash down natural radioactive material in the air to the ground, raising the ambient radiation level in a very short time.
- However, after prolonged rainfall, there will be less natural radioactive material in the air.
- This lowers the ambient radiation from its peak level for some time before it returns to normal.

How does wind affect the ambient radiation level ?

- Air from the continent contains more natural radioactive material (which may be gaseous) released from rock and soil when compared to that from the ocean.

- When the wind is northerly, it brings the natural radioactive material from upwind land areas, raising the ambient radiation level.

- On the contrary, when the wind is from the south, there is much less natural radioactive material over the ocean and the ambient radiation level is lower.

What do radioactive sources look like?

- Radioactive sources are typically “sealed,” or encased in metal or foil.

- They are usually very small; their size can vary from tiny "seeds" used in cancer treatment, to the size of the tip of a ballpoint pen or a pencil eraser, to rods up to several inches in length, depending on the material and its configuration.

How can I tell if something is radioactive?

- You can't, without the help of a radiation detector.

- In addition, it is important to know what type of detector you have and the type of radiation — alpha, beta, gamma, x-ray, and/or neutron — that it can detect. Scanning an object with a typical gamma/x-ray radiation detector will not detect alpha particles, for example.

What is tritium, and what can it do to you?

- Tritium (hydrogen-3 or 3H) is a weakly radioactive isotope of the element hydrogen, which occurs both naturally and during the operation of nuclear power plants.
- Tritium is one of the least dangerous radioisotopes because it emits very weak radiation and leaves the body relatively quickly.
- Nonetheless, exposure to very small amounts of ionizing radiation is thought to minimally increase the risk of developing cancer, and that risk increases as exposure increases.
- Since tritium is almost always found as water, if ingested, it goes directly into soft tissues and organs. The dose to these tissues is generally uniform and depends on the tissue's water content.

What kind and how much radiation is produced by a nuclear power plant?

- An operating nuclear power plant produces very small amounts of radioactive gases and liquids, as well as small amounts of direct radiation.

- If you lived within 50 miles of a nuclear power plant, you would receive an average radiation dose of about 0.01 millirem per year.

What happens to radiation produced by a plant?

- Nuclear power plants sometimes release radioactive gases and liquids into the environment under controlled, monitored conditions to ensure that they pose no danger to the public or the environment.

- These releases dissipate into the atmosphere or a large water source and, therefore, are diluted to the point where it becomes difficult to measure any radioactivity.

- By contrast, most of an operating nuclear power plant's direct radiation is blocked by the plant's steel and concrete structures. The remainder dissipates in an area of controlled, uninhabited space around the plant, ensuring that it does not affect any member of the public.

Thank you

Radiation protection

- **Learning Objectives**
- Radiation protection
- International Atomic Energy Agency (IAEA)
- Recommendations of **International Committee on Radiation Protection (ICRP) and International Commission on Radiation Units and Measurements (ICRU)**

Radiation protection

Radiation protection, sometimes known as radiological protection, is defined by the **International Atomic Energy Agency (IAEA)** as

"The protection of people from harmful effects of exposure to ionizing radiation, and the means for achieving this".

The IAEA also states

"The accepted understanding of the term radiation protection is restricted to protection of people. Suggestions to extend the definition to include the protection of non-human species or the protection of the environment are controversial".

Exposure can be from a radiation source external to the human body or due to the bodily intake of a radioactive material.

- Ionizing radiation is widely used in industry and medicine, and can present a significant health hazard by causing microscopic damage to living tissue.

- This can result in skin burns and radiation sickness at high exposures, known as "tissue" or "deterministic" effects (conventionally indicated by the gray), and statistically elevated risks of cancer at low exposures, known as "stochastic effects" (conventionally measured by the sievert).

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Exposure time

The less time you are exposed, the lower the dose of radiation you will receive

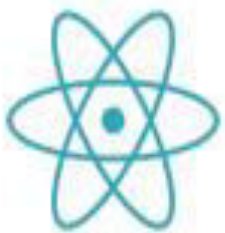




Distance

The further away you are from the source of the radiation, the less intense its effects will be





Shielding

Shield yourself behind a thick concrete wall or stay indoor. Protective covers made of thick concrete are very good at withstanding radiation penetration



- Fundamental to radiation protection is the reduction of expected dose and the measurement of dose uptake.

- For radiation protection and dosimeter assessment the **International Committee on Radiation Protection (ICRP) and International Commission on Radiation Units and Measurements (ICRU)** publish recommendations and data which is used to calculate the biological effects on the human body of certain levels of radiation, and thereby advise acceptable dose uptake limits.

- Supporting these are preventive dose reduction techniques such as radiation shielding, exposure planning and avoidance of ingestion of radioactive substances.

- Radiation protection instruments are used to indicate radiation hazards, and personal dosimeters and bioassay techniques are used to measure personal dose uptake.

Principles

- The ICRP recommends, develops and maintains the International System of Radiological Protection, based on evaluation of the large body of scientific studies available to equate risk to received dose levels.
- The system's health objectives are "to manage and control exposures to ionising radiation so that deterministic effects are prevented, and the risks of stochastic effects are reduced to the extent reasonably achievable".

- The ICRP's recommendations flow down to national and regional regulators, which have the opportunity to incorporate them into their own law; this process is shown in the accompanying block diagram.
- In most countries a national regulatory authority works towards ensuring a secure radiation environment in society by setting dose limitation requirements that are generally based on the recommendations of the ICRP.

Exposure situations

- The ICRP recognizes planned, emergency, and existing exposure situations, as described below;
- **Planned exposure** – defined as "...where radiological protection can be planned in advance, before exposures occur, and where the magnitude and extent of the exposures can be reasonably predicted." These are such as in occupational exposure situations, where it is necessary for personnel to work in a known radiation environment.
- **Emergency exposure** – defined as "...unexpected situations that may require urgent protective actions". This would be such as an emergency nuclear event.
- **Existing exposure** – defined as "...being those that already exist when a decision on control has to be taken". These can be such as from naturally occurring radioactive materials which exist in the environment.

Regulation of dose uptake

- The ICRP uses the following overall principles for all controllable exposure situations.

- **Justification:** No unnecessary use of radiation is permitted, which means that the advantages must outweigh the disadvantages.
- **Limitation:** Each individual must be protected against risks that are too great, through the application of individual radiation dose limits.
- **Optimization:** This process is intended for application to those situations that have been deemed to be justified. It means "the likelihood of incurring exposures, the number of people exposed, and the magnitude of their individual doses" should all be kept as Low As Reasonably Achievable (known as ALARA or ALARP). It takes into account economic and societal factors.

Thank you

Factors in dose uptake

- **Learning Objectives**
- Factors in external dose uptake
- Factors in Internal dose uptake
- Recommended limits on dose uptake

Factors in external dose uptake

- There are three factors that control the amount, or dose, of radiation received from a source. Radiation exposure can be managed by a combination of these factors:

- **Time:** Reducing the time of an exposure reduces the effective dose proportionally. An example of reducing radiation doses by reducing the time of exposures might be improving operator training to reduce the time they take to handle a radioactive source.
- **Distance:** Increasing distance reduces dose due to the inverse square law. Distance can be as simple as handling a source with forceps rather than fingers.
- **Shielding:** Sources of radiation can be shielded with solid or liquid material, which absorbs the energy of the radiation. The term 'biological shield' is used for absorbing material placed around a nuclear reactor, or other source of radiation, to reduce the radiation to a level safe for humans.

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Exposure time

The less time you are exposed, the lower the dose of radiation you will receive

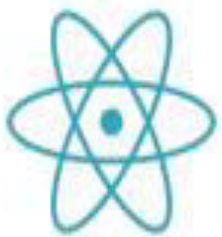




Distance

The further away you are from the source of the radiation, the less intense its effects will be





Shielding

Shield yourself behind a thick concrete wall or stay indoor. Protective covers made of thick concrete are very good at withstanding radiation penetration



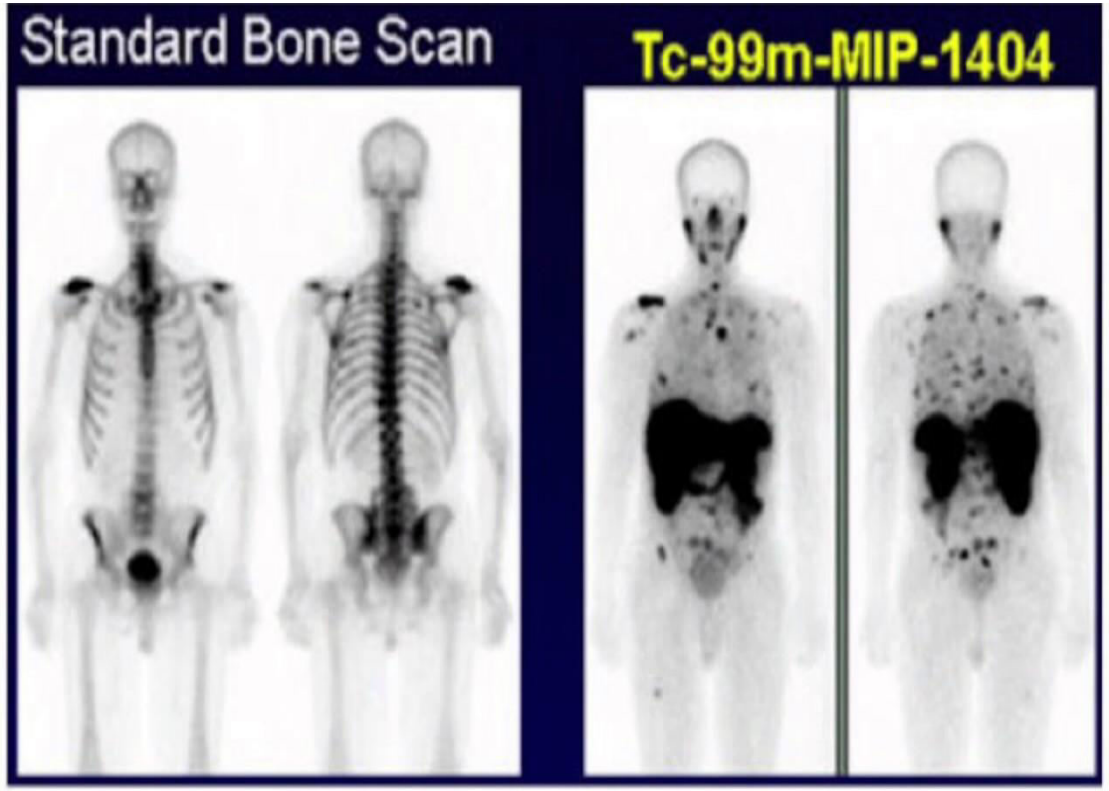
Internal dose uptake

Internal dose, due to the inhalation or ingestion of radioactive substances, can result in stochastic or deterministic effects, depending on the amount of radioactive material ingested and other bio-kinetic factors.

The risk from a low level internal source is represented by the dose quantity committed dose, which has the same risk as the same amount of external effective dose. The intake of radioactive material can occur through four pathways:

- Inhalation of airborne contaminants such as radon gas and radioactive particles
- Ingestion of radioactive contamination in food or liquids
- Absorption of vapors such as tritium oxide through the skin
- Injection of medical radioisotopes such as technetium-99m

Technetium-99m is a metastable nuclear isomer of technetium-99 (itself an isotope of technetium), symbolized as 99mTc, that is used in tens of millions of medical diagnostic procedures annually, making it the most commonly used medical radioisotope.



- The occupational hazards from airborne radioactive particles in nuclear and radio-chemical applications are greatly reduced by the extensive use of glove boxes to contain such material. To protect against breathing in radioactive particles in ambient air, respirators with particulate filters are worn.

- To monitor the concentration of radioactive particles in ambient air, radioactive particulate monitoring instruments measure the concentration or presence of airborne materials.

- For ingested radioactive materials in food and drink, specialist laboratory radiometric assay methods are used to measure the concentration of such materials.

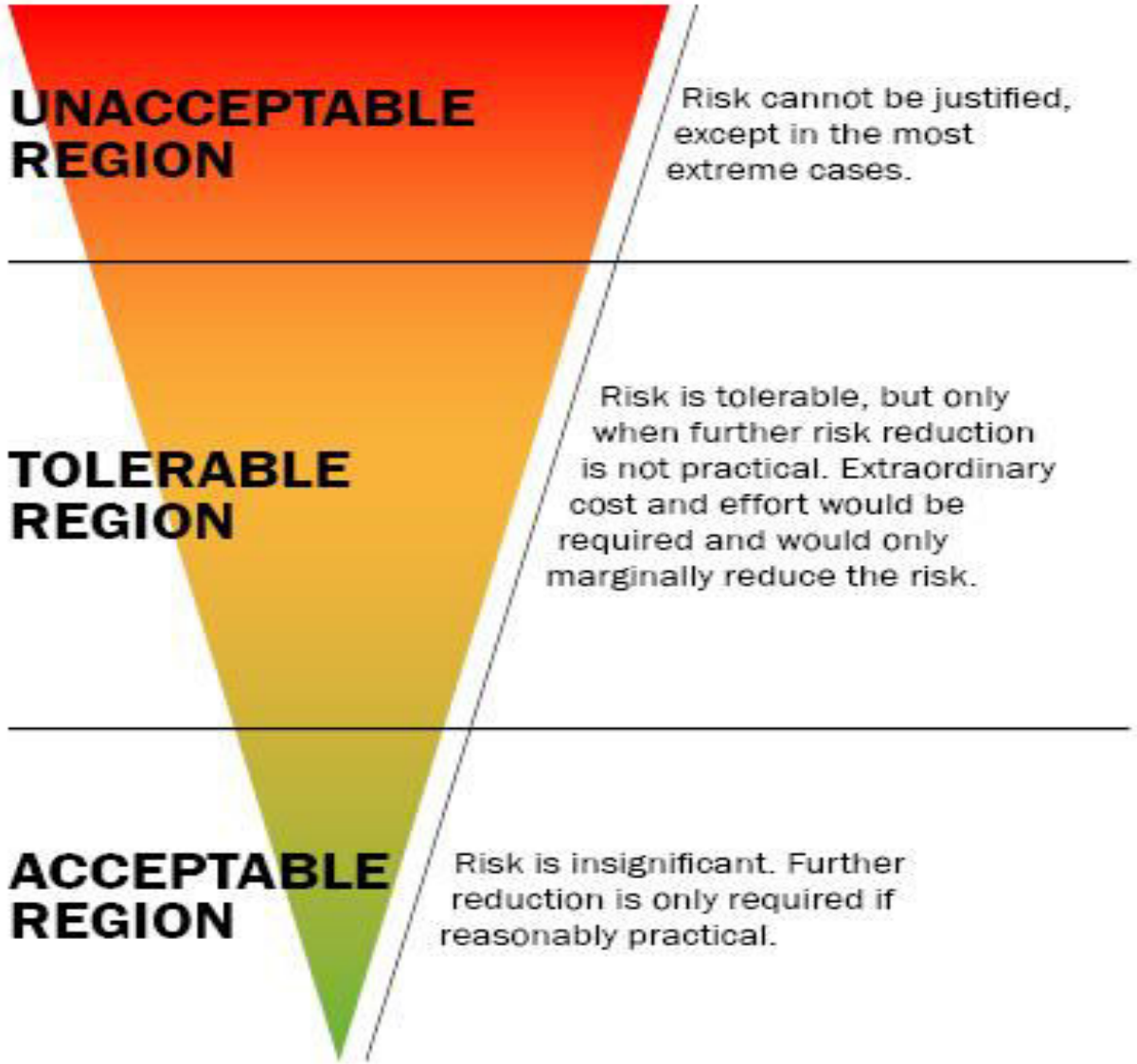
Recommended limits on dose uptake

- The ICRP recommends a number of limits for dose uptake in table 8 of ICRP report 103.
- These limits are "situational", for planned, emergency and existing situations. Within these situations, limits are given for certain exposed groups;
- **Planned exposure** – limits given for occupational, medical and public exposure. The occupational exposure limit of effective dose is 20 mSv per year, averaged over defined periods of 5 years, with no single year exceeding 50 mSv. The public exposure limit is 1 mSv in a year.
- **Emergency exposure** – limits given for occupational and public exposure
- **Existing exposure** – reference levels for all persons exposed

ALARP & ALARA

- ALARP is an acronym for an important principle in exposure to radiation and other occupational health risks and stands for **"As Low As Reasonably Practicable".**

- The aim is to minimize the risk of radioactive exposure or other hazard while keeping in mind that some exposure may be acceptable in order to further the task at hand. The equivalent term ALARA, **"As Low As Reasonably Achievable",** is more commonly used outside the UK.


UNACCEPTABLE REGION
Risk cannot be justified, except in the most extreme cases.
TOLERABLE REGION
Risk is tolerable, but only when further risk reduction is not practical. Extraordinary cost and effort would be required and would only marginally reduce the risk.
ACCEPTABLE REGION
Risk is insignificant. Further reduction is only required if reasonably practical.

- The application of radiation can aid the patient by providing doctors and other health care professionals with a medical diagnosis, but the exposure should be reasonably low enough to keep the statistical probability of cancers or sarcomas (stochastic effects) below an acceptable level, and to eliminate deterministic effects (e.g. skin reddening or cataracts).

- An acceptable level of incidence of stochastic effects is considered to be equal for a worker to the risk in another work generally considered to be safe.

Thank you

Personal radiation dosimeters

- **Learning Objectives**
- Film badge dosimeter
- Quartz fiber dosimeter
- Solid state (MOSFET or silicon diode) dosimeter
- Thermo-luminescent dosimeter

Personal radiation dosimeters

- The radiation dosimeter is an important personal dose measuring instrument. It is worn by the person being monitored and is used to estimate the radiation dose deposited in the individual wearing the device.
- They are used for Gamma, X-ray, beta and other strongly penetrating radiation, but not for weakly penetrating radiation such as alpha particles. Traditionally film badges were used for long term monitoring, and quartz fibre dosimeters for short term monitoring.
- However, these are being superseded by such as thermo-luminescent dosimetry badges and electronic dosimeters.
- Electronic dosimeters can give an alarm warning if a preset dose threshold has been reached, enabling safer working in potentially higher radiation levels, where the received dose must be continually monitored.

- Workers exposed to radiation, such as radiographers, nuclear power plant workers, doctors using radiotherapy, those in laboratories using radionuclides, and HAZMAT teams are required to wear dosimeters so a record of occupational exposure can be made.

- Such devices are generally termed "legal dosimeters" if they have been approved for use in recording personnel dose for regulatory purposes.

- Dosimeters can be worn to obtain a whole body dose and there are also specialist types that can be worn on the fingers or clipped to headgear, to measure the localized body irradiation for specific activities.

- A **Hazardous Materials** (**HAZMAT**) Response **team** is comprised of **hazardous material** experts who specialize in detecting, containing, and removing any release or potential release of hazardous substances in order to control or stabilize an incident.



Common types of wearable dosimeters for ionizing radiation include:

- Film badge dosimeter
- Quartz fiber dosimeter
- Solid state (MOSFET or silicon diode) dosimeter
- Thermo-luminescent dosimeter

Film Badges



- Most common type of radiation dosimeter used.
- Works by darkening of x-ray film in proportion to radiation absorbed.
- Cheap
- Not durable
- Short monitoring period per badge (6 months or less)





Quartz fiber dosimeter

- A quartz fiber dosimeter, sometimes called a self indicating pocket dosimeter (SIPD) or self reading pocket dosimeter (SRPD), is a type of radiation dosimeter, a pen-like device that measures the cumulative dose of ionizing radiation received by the device, usually over one work period.

- It is usually clipped to a person's clothing and worn to measure one's actual exposure to radiation.

- It is now being superseded by more modern dosimeter types such as the Electronic Personal Dosimeter (EPD).



Solid State

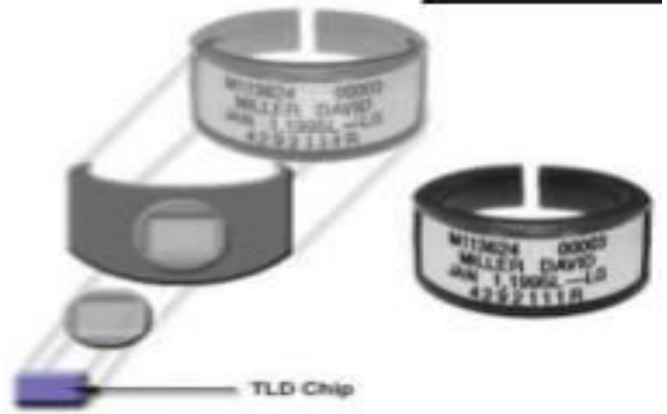


- Provides instantaneous information regarding dose accumulation
- Simple to use
- Not a “legal” record
- Dose range device dependent



THERMO LUMINESCENT DOSIMETERS(TLD)

- Looks like a film badge
- Contains a lithium fluoride crystal
- Responds to radiation similarly to skin
- Measured by a TLD analyzer
- Crystal will luminescence if exposed to radiation, then heated
- More accurate than a film badge









Thank you

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Radiation protection instruments

Radiation protection instruments

Practical radiation measurement using calibrated radiation protection instruments is essential in evaluating the effectiveness of protection measures, and in assessing the radiation dose likely to be received by individuals.

The measuring instruments for radiation protection are both:

1. “Installed" (in a fixed position)
2. Portable (hand-held or transportable).

DEPARTMENTAL SURVEYS

- The Exposure rates of the radiation leaking through barriers, or through cracks will usually be very small (of the order of mill roentgens per hour),

- So ionization chambers of quite large volume are usually used for their measurement.





Installed instruments



- Installed instruments are fixed in positions which are known to be important in assessing the general radiation hazard in an area.

- Examples are installed "area" radiation monitors, Gamma interlock monitors, personnel exit monitors, and airborne particulate monitors.

- The area radiation monitor will measure the ambient radiation, usually X-Ray, Gamma or neutrons; these are radiations which can have significant radiation levels over a range in excess of tens of metres from their source, and thereby cover a wide area.

- Gamma radiation "interlock monitors" are used in applications to prevent inadvertent exposure of workers to an excess dose by preventing personnel access to an area when a high radiation level is present. These interlock the process access directly.



- Airborne contamination monitors measure the concentration of radioactive particles in the ambient air to guard against radioactive particles being ingested, or deposited in the lungs of personnel.

- These instruments will normally give a local alarm, but are often connected to an integrated safety system so that areas of plant can be evacuated and personnel are prevented from entering an air of high airborne contamination.

- Personnel exit monitors (PEM) are used to monitor workers who are exiting a "contamination controlled" or potentially contaminated area. These can be in the form of hand monitors, clothing frisk probes, or whole body monitors.

- These monitor the surface of the workers body and clothing to check if any radioactive contamination has been deposited. These generally measure alpha or beta or gamma, or combinations of these.

Portable instruments

- Portable instruments are hand-held or transportable. The hand-held instrument is generally used as a survey meter to check an object or person in detail, or assess an area where no installed instrumentation exists.

- They can also be used for personnel exit monitoring or personnel contamination checks in the field. These generally measure alpha, beta or gamma, or combinations of these.

- Transportable instruments are generally instruments that would have been permanently installed, but are temporarily placed in an area to provide continuous monitoring where it is likely there will be a hazard.

- Such instruments are often installed on trolleys to allow easy deployment, and are associated with temporary operational situations.

Radiation Survey Instruments

- Area monitoring devices
- Detect and measure radiation
- Measures either quantity or rate
- Generally gas filled
- Major types of survey instruments

  - Ionization chamber - cutie pie
  - Proportional counter
  - Geiger-Müller detector
  - Calibration instruments

VU
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Ionization Chamber (Cutie Pie)

- Measures x or gamma radiation generally - can be equipped to measure beta
- Measures intensity from 1mR/hr to several thousand R/hr
- Most commonly used to measure patients receiving brachytherapy or diagnostic isotopes



Proportional Counter

- Generally used in laboratories to measure beta or alpha radiation
- Can discriminate between these particles
- Operator must hold the counter close to the object being surveyed to obtain accurate reading





Geiger-Müller Detector

- Generally used for nuclear medicine facilities
- Unit is sensitive enough to detect individual particles
- Can be used to locate a lost radioactive source
- Has an audible sound system
- Alerts to presence of radiation
- Meter readings are generally displayed in mR/hr



GAMMA AREA MONITOR

primarily meant to serve as a Gamma Zone Monitor to indicate dose rates and alarm status (visual and aural), once the dose rates exceed the preset level fixed by the user.



PERSONNEL MONITORING

The purpose of monitoring and exposure assessment is to gather and provide information on the actual exposure of workers and to confirm good working practices contributing to reassurance and motivation.

Radiation oncologists, Radiotherapy physicists, Radiation protection officers, Radiotherapy technologists, source handlers, maintenance staff and any nursing or other staff who must spend time with patients who contain radioactive sources are the ones who need personal monitorin

Monitoring includes not just measuring and determining the equivalent dose;

It includes interpretation and assessment.

Employers... and licensees should make arrangements for appropriate health surveillance in accordance with the rules established by the Regulatory Authority."

Thank you

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Bio-electromagnetics

Bio-electromagnetics

- Bio-electromagnetics, also known as bio-electromagnetism, is the study of the interaction between electromagnetic fields and biological entities.
- Areas of study include electrical or electromagnetic fields produced by living cells, tissues or organisms, including bioluminescent bacteria; for example, the cell membrane potential and the electric currents that flow in nerves and muscles, as a result of action potentials.
- Others include animal navigation utilizing the geomagnetic field; potential effects of man-made sources of electromagnetic fields like mobile phones; and developing new therapies to treat various conditions.
- The term can also refer to the ability of living cells, tissues, and organisms to produce electrical fields and the response of cells to electromagnetic fields

Medical Sensing and Bio Electromagnetics

- Electromagnetic energy is used extensively for many biomedical applications including:
- Imaging
- Electrical stimulation of the heart, brain, and other tissues
- Hyperthermia
- Sensing

Researchers are currently:

- Studying the microwave aspects of high field magnetic resonance imaging (MRI) and electron paramagnetic resonance (EPR) imaging.

- Developing game-changing:
- (a) wireless and fully-passive implants for continuous and unobtrusive monitoring of brain signals with minimum impact to the individual's activity, and
- (b) non-invasive and inconspicuous epidermal textile sensors for real-time monitoring of internal organs deep into the body in a carefree manner.

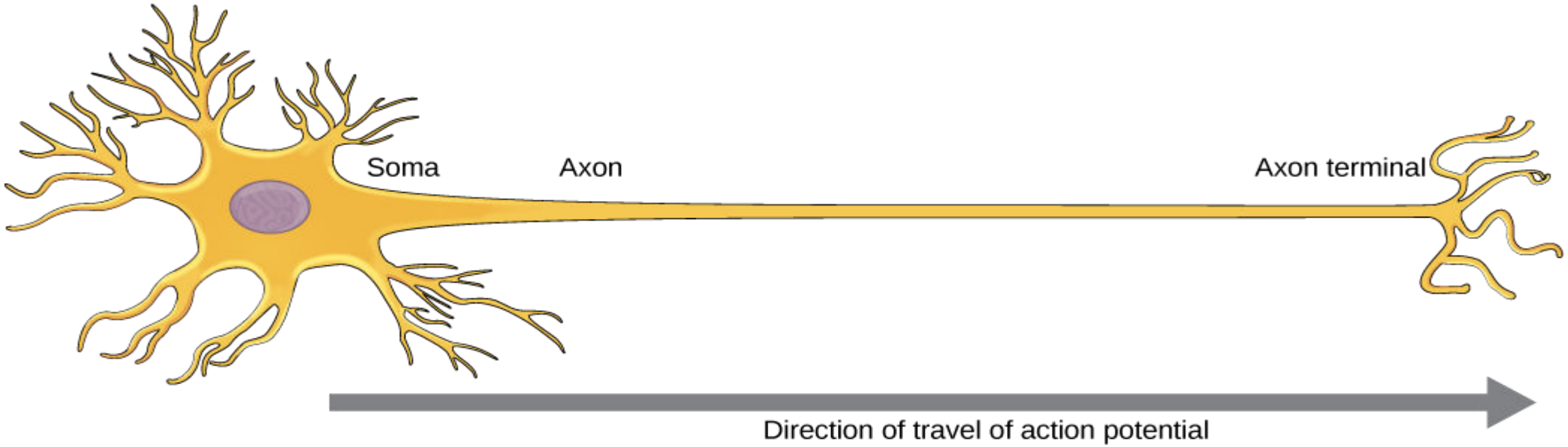
Biological phenomena

- Short-lived electrical events called action potentials occur in several types of animal cells which are called excitable cells, a category of cell include neurons, muscle cells, and endocrine cells, as well as in some plant cells.

- These action potentials are used to facilitate inter-cellular communication and activate intracellular processes.

- The physiological phenomena of action potentials are possible because voltage-gated ion channels allow the resting potential caused by electrochemical gradient on either side of a cell membrane to resolve.

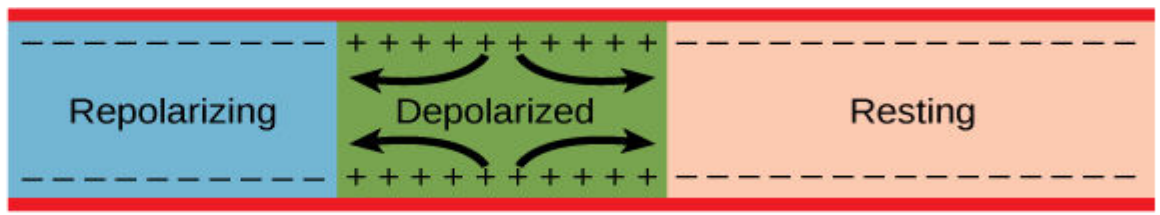
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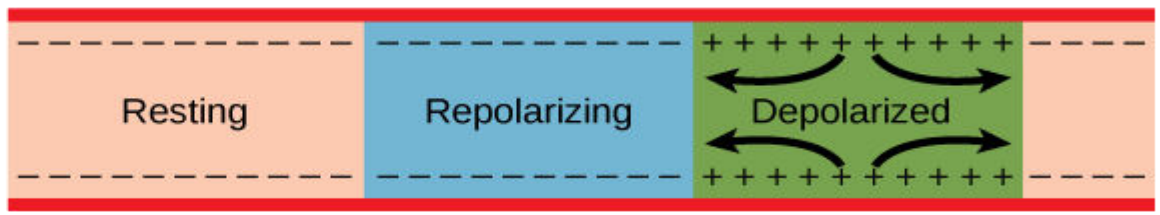
a. In response to a signal, the soma end of the axon becomes depolarized.



b. The depolarization spreads down the axon. Meanwhile, the first part of the membrane repolarizes. Because Na^+ channels are inactivated and additional K^+ channels have opened, the membrane cannot depolarize again.



c. The action potential continues to travel down the axon.



- Bio-electromagnetism is studied primarily through the techniques of electrophysiology. In the late eighteenth century, the Italian physician and physicist **Luigi Galvani** first recorded the phenomenon while dissecting a frog at a table where he had been conducting experiments with static electricity.

- Galvani coined the term animal electricity to describe the phenomenon, while contemporaries labeled it galvanism.

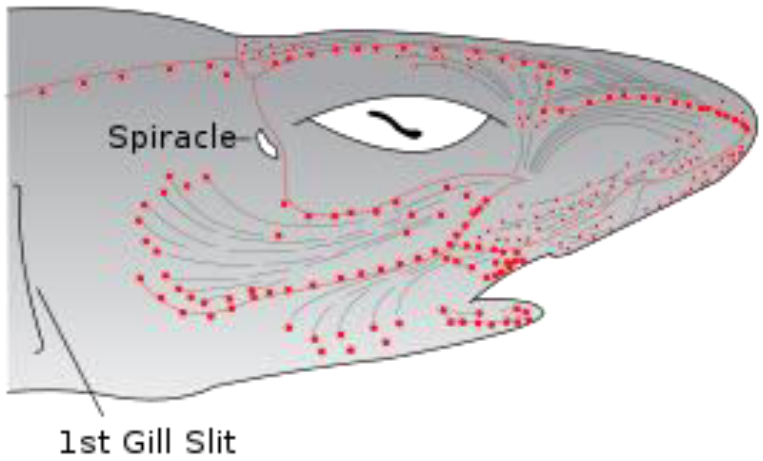
- Galvani and contemporaries regarded muscle activation as resulting from an electrical fluid or substance in the nerves.

- Some usually aquatic animals, such as sharks, have acute bioelectric sensors providing a sense known as electroreception, while migratory birds navigate in part by orienteering with respect to the Earth's magnetic field.

- In an extreme application of electromagnetism the electric eel is able to generate a large electric field outside its body used for hunting and self-defense through a dedicated electric organ.

Electroreception

- Electroreception or electroception is the biological ability to perceive natural electrical stimuli.
- It has been observed almost exclusively in aquatic or amphibious animals, since salt-water is a much better conductor than air; the currently known exceptions being the monotremes (echidnas and platypuses), cockroaches and bees.
- Electroreception is used in electrolocation (detecting objects) and for electrocommunication.



Electroreceptors (Ampullae of Lorenzini) and lateral line canals in the head of a shark.

Thank you

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Electrolocation

Electrolocation

- Electroreceptive animals use this sense to locate objects around them.
- This is important in ecological niches where the animal cannot depend on vision: for example in caves, in murky water and at night.
- Many fish use electric fields to detect buried prey.
- Some shark embryos and pups "freeze" when they detect the characteristic electric signal of their predators.

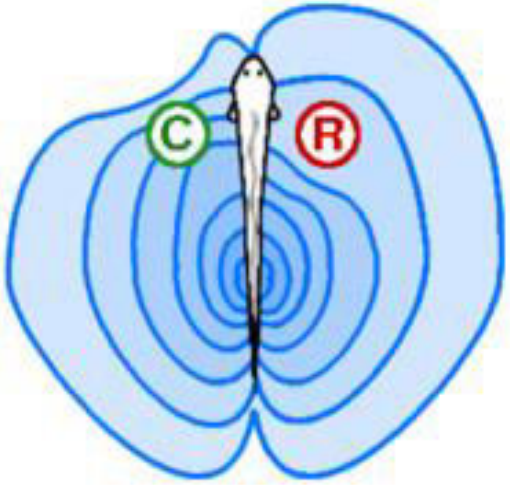
- It has been proposed that sharks can use their acute electric sense to detect the earth's magnetic field by detecting the weak electric currents induced by their swimming or by the flow of ocean currents.

- The walking behavior of cockroaches can be affected by the presence of a static electric field: they like to avoid the electric field.

- Cabbage loopers are also known to avoid electric fields

Active electrolocation

- In active electrolocation, the animal senses its surrounding environment by generating electric fields and detecting distortions in these fields using electroreceptor organs.
- This electric field is generated by means of a specialized electric organ consisting of modified muscle or nerves.
- This field may be modulated so that its frequency and wave form are unique to the species and sometimes, the individual.



Active electrolocation. Conductive objects concentrate the field and resistive objects spread the field.

- Animals that use active electroreception include the weakly electric fish, which either generate small electrical pulses ("pulse-type") or produce a quasi-sinusoidal discharge from the electric organ (termed "wave-type").

- These fish create a potential which is usually smaller than one volt.

- Weakly electric fish can discriminate between objects with different resistance and capacitance values, which may help in identifying the object.

- Active electroreception typically has a range of about one body length, though objects with an electrical impedance similar to that of the surrounding water are nearly undetectable.

Passive electrolocation

- In passive electrolocation, the animal senses the weak bioelectric fields generated by other animals and uses it to locate them.

- These electric fields are generated by all animals due to the activity of their nerves and muscles. A second source of electric fields in fish is the ion pumps associated with osmoregulation at the gill membrane. This field is modulated by the opening and closing of the mouth and gill slits.

- Many fish that prey on electrogenic fish use the discharges of their prey to detect them. This has driven the prey to evolve more complex or higher frequency signals that are harder to detect.

- Passive electroreception is carried out solely by ampullary electroreceptors in fish. It is tuned to low frequency signals (below one up to tens of Hertz)

- Fish use passive electroreception to supplement or replace their other senses when detecting prey and predators. In sharks, sensing an electric dipole alone is sufficient to cause them to try to eat it

Electro-communication

- Weakly electric fish can also communicate by modulating the electrical waveform they generate, an ability known as electro-communication.

- They may use this for mate attraction and territorial displays. Some species of catfish use their electric discharges only in agonistic displays.

Examples

- **Sharks and rays**
- Sharks and rays (members of the subclass Elasmobranchii), such as the lemon shark, rely heavily on electrolocation in the final stages of their attacks, as can be demonstrated by the robust feeding response elicited by electric fields similar to those of their prey.
- Sharks are the most electrically sensitive animals known, responding to DC fields as low as 5 nV/cm.



Bony fish

- The electric eel (actually a knifefish, not an eel), besides its ability to generate high voltage electric shocks, uses lower voltage pulses for navigation and prey detection in its turbid habitat.



Dolphins

- Dolphins have evolved electroreception in structures different from those of fish, amphibians and monotremes.



Bees

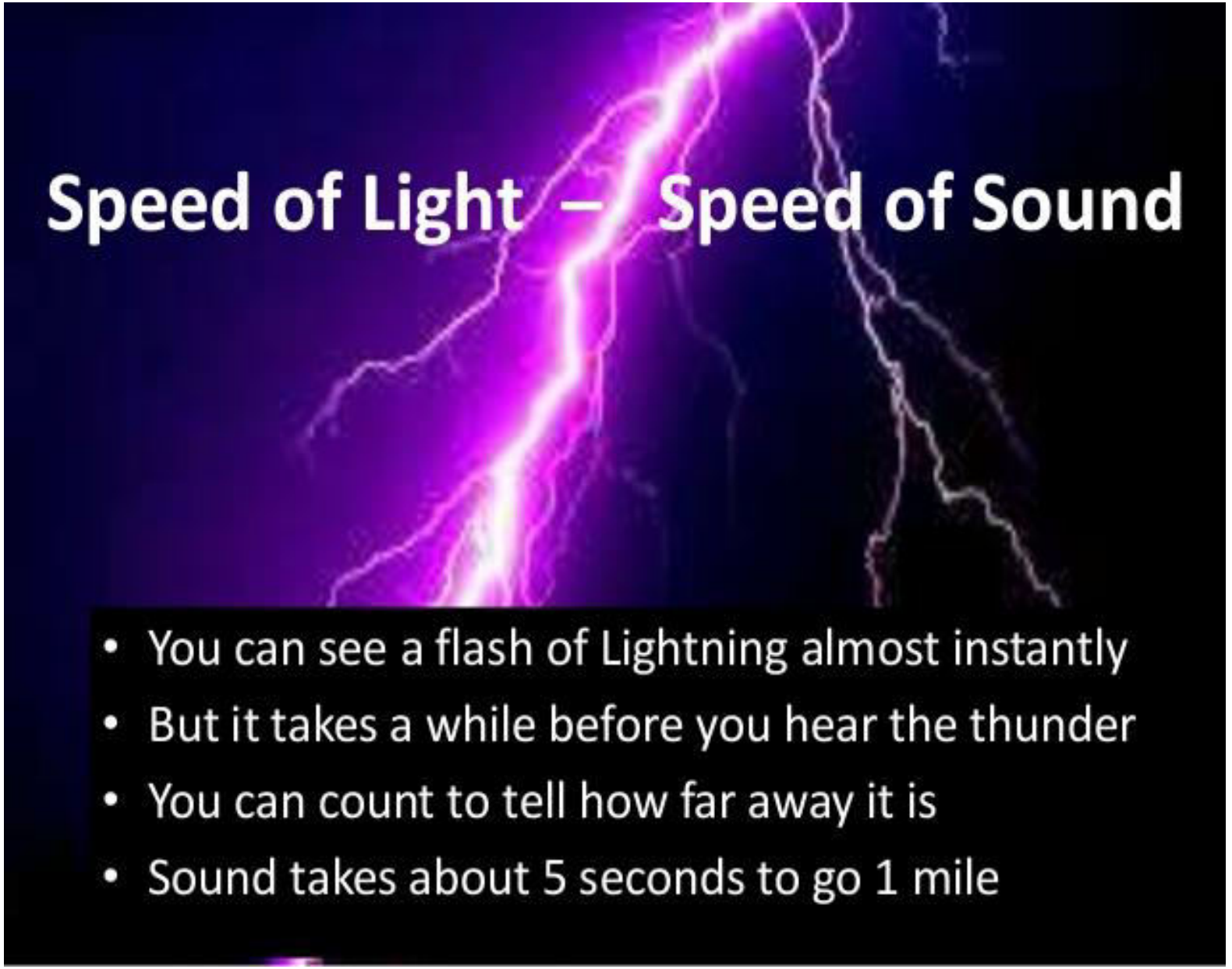
- Bees collect a positive static charge while flying through the air.
- When a bee visits a flower, the charge deposited on the flower takes a while to leak away into the ground. Bees can detect both the presence and the pattern of electric fields on flowers, and use this information to know if a flower has been recently visited by another bee and is therefore likely to have a reduced concentration of nectar.
- Animals detect electric fields through insulating air by mechano-reception, not electroreception. Bees sense the electric field changes via the Johnston's organs in their antennae and possibly other mechano-receptors.
- They distinguish different temporal patterns and learn them. During the waggle dance, honeybees appear to use the electric field emanating from the dancing bee for distance communication.

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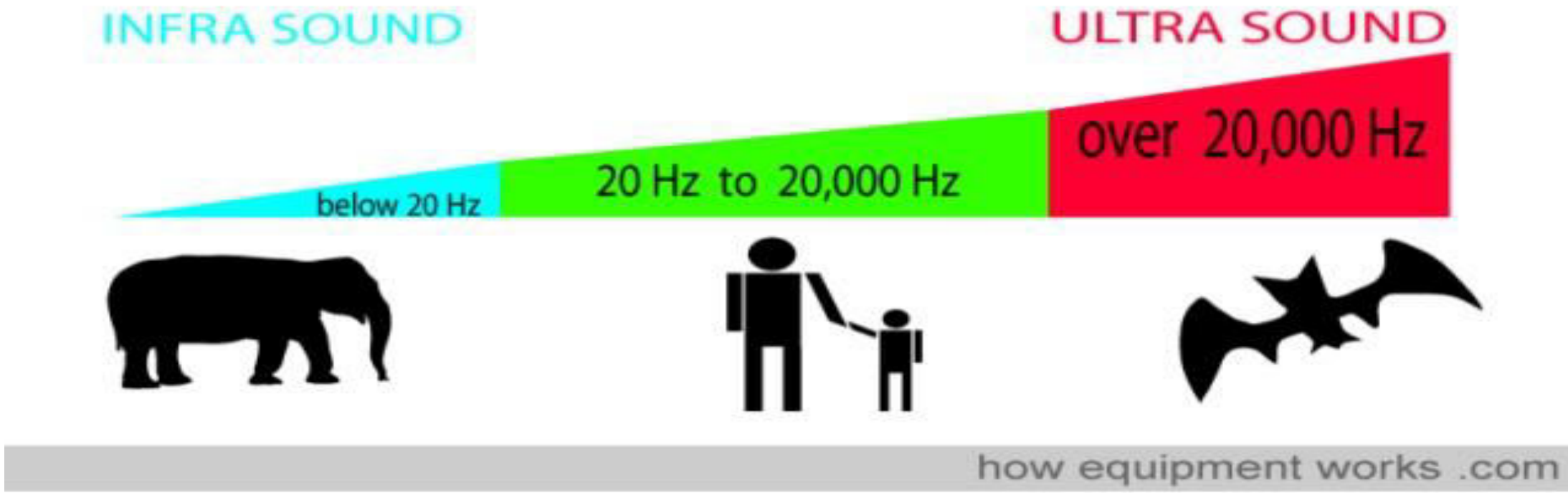
Echolocation



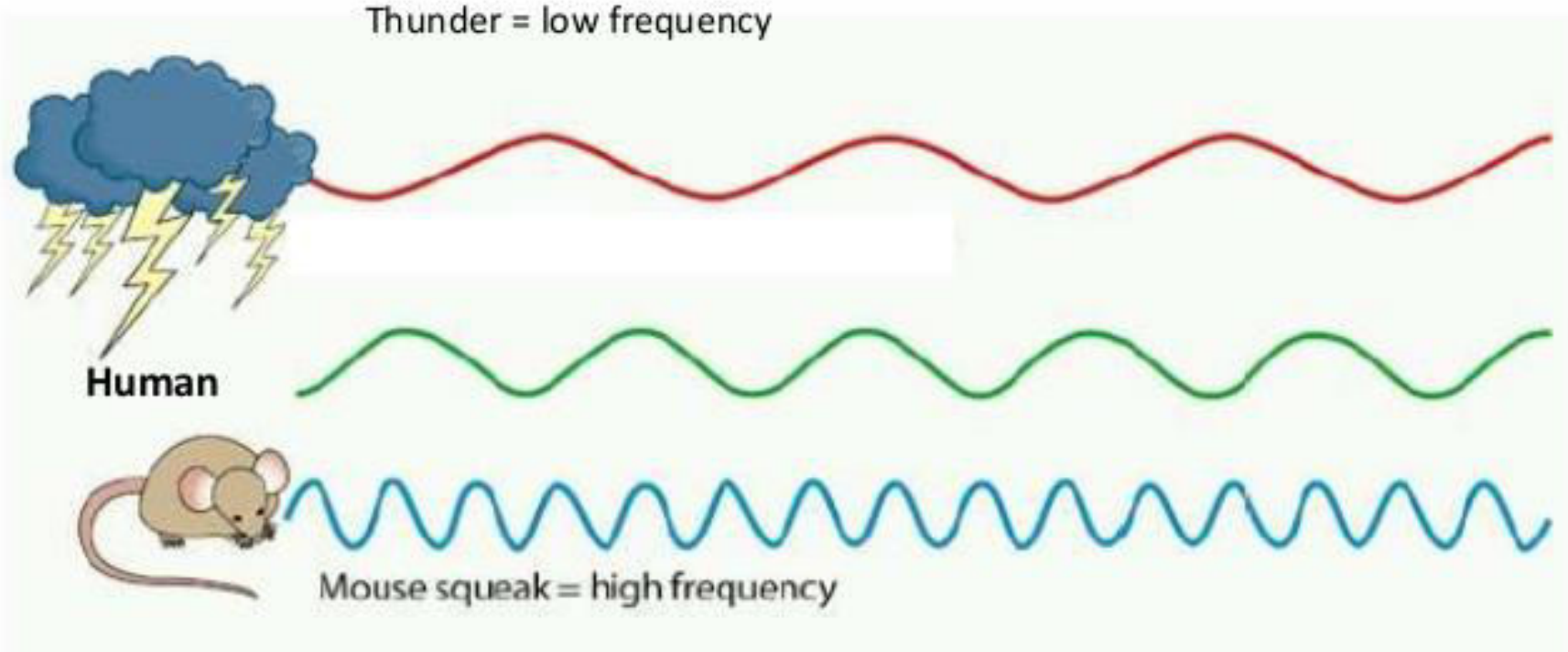
The speed of sound through air is about 340 meters per second. (The speed of sound in water is about 4 times faster than this).

The speed of light in air is about 300 million meters per second.

Sound Waves


INFRA SOUND
ULTRA SOUND
below 20 Hz
20 Hz to 20,000 Hz
over 20,000 Hz
how equipment works .com

What do sound waves look like?



Frequency is measured as the number of wave cycles that occur in **one second**. The unit of frequency measurement is Hertz (Hz for short).

What is Echolocation ?

"Echolocation is the use of sound waves and echoes to determine where objects are in space"

In other words, echoes help to find the location of an object.

http://askabiologist.asu.edu/echolocation

Echolocation

- Echolocation is the use of sound waves and echoes to determine where objects are in space.
- Echolocation, also called bio sonar, is the biological sonar used by several kinds of animals.
- Echo-locating animals emit calls out to the environment and listen to the echoes of those calls that return from various objects near them.
- They use these echoes to locate and identify the objects.
- Echolocation is used for navigation and for foraging (or hunting) in various environments.

Principle

- Echolocation is the same as active sonar, using sounds made by the animal itself.
- Ranging is done by measuring the time delay between the animal's own sound emission and any echoes that return from the environment.
- The relative intensity of sound received at each ear as well as the time delay between arrival at the two ears provide information about the horizontal angle from which the reflected sound waves arrive.

- Unlike some human-made sonars that rely on many extremely narrow beams and many receivers to localize a target (multi-beam sonar), animal echolocation has only one transmitter and two receivers (the ears) positioned slightly apart.

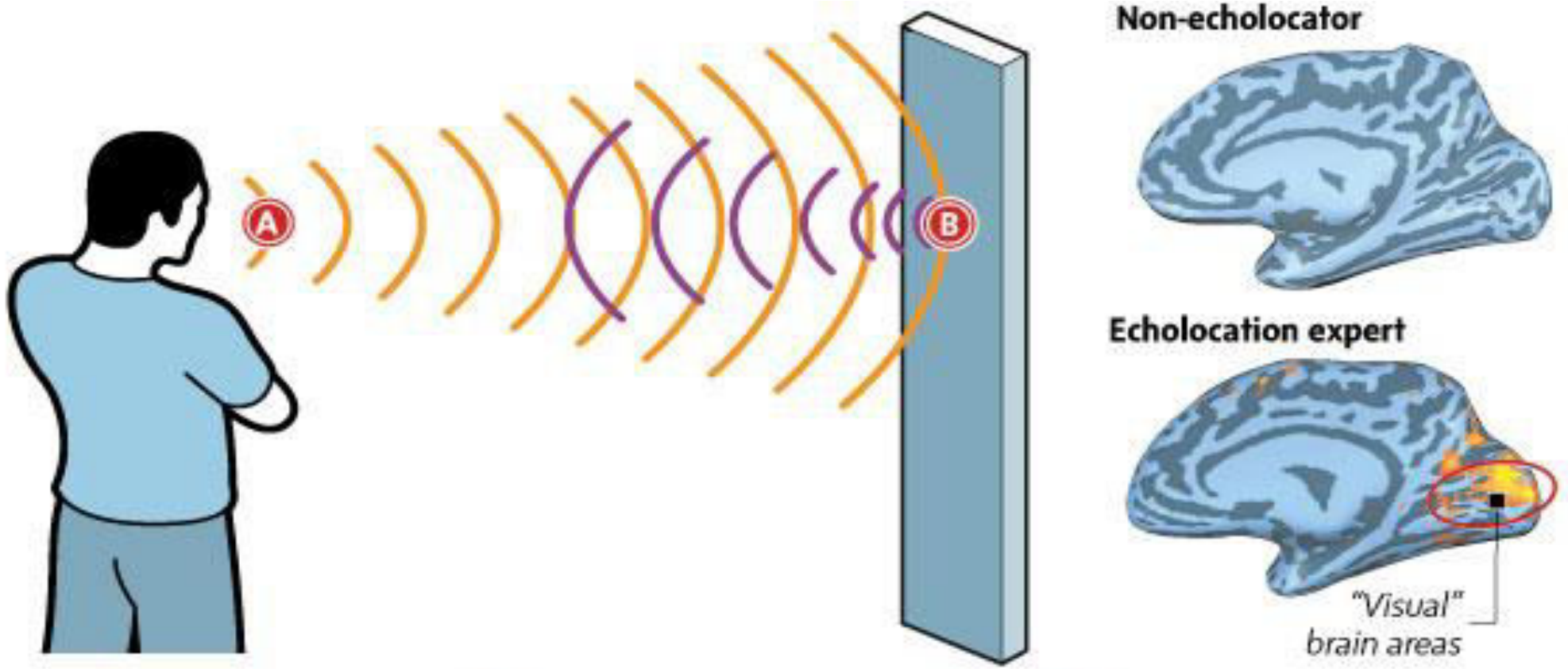
- The echoes returning to the ears arrive at different times and at different loudness levels, depending on the position of the object generating the echoes.

- The time and loudness differences are used by the animals to perceive distance and direction.

- With echolocation, the bat or other animal can see not only where it is going but also how big another animal is, what kind of animal it is, and other features.

- At the most basic level, echolocation is based on the neural anatomy of auditory brain circuitry.

- In essence, ascending brain pathways in the brain stem allow the brain to calculate the difference between the two ears to very small fractions of a second



A The echolocator uses his tongue and mouth to create clicking sounds.

B Sound bounces off object. A practiced echolocator can identify objects around him by the echo, much like bats do.

C Sounds are played back to participants later during MRI scan. These scans show activity in areas of the brain that normally process visual information.

MATTHEW BAMBACH/THE GLOBE AND MAIL » SOURCE: UNIVERSITY OF WESTERN ONTARIO

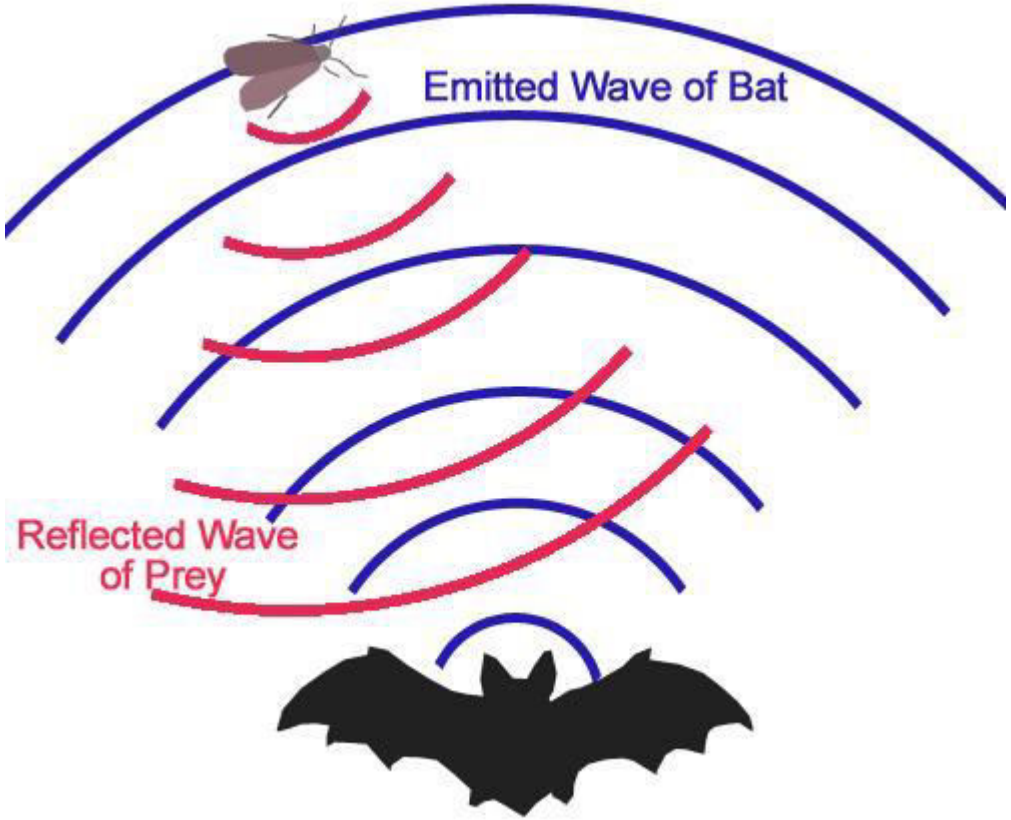
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Echolocation in bats

Echolocation In Bats

- Microbats use echolocation to navigate and forage, often in total darkness.
- They generally emerge from their roosts in caves, attics, or trees at dusk and hunt for insects into the night.
- Their use of echolocation allows them to occupy a niche where there are often many insects (that come out at night since there are fewer predators then), less competition for food, and fewer species that may prey on the bats themselves.



- Microbats generate ultrasound via the larynx and emit the sound through the open mouth or, much more rarely, the nose.

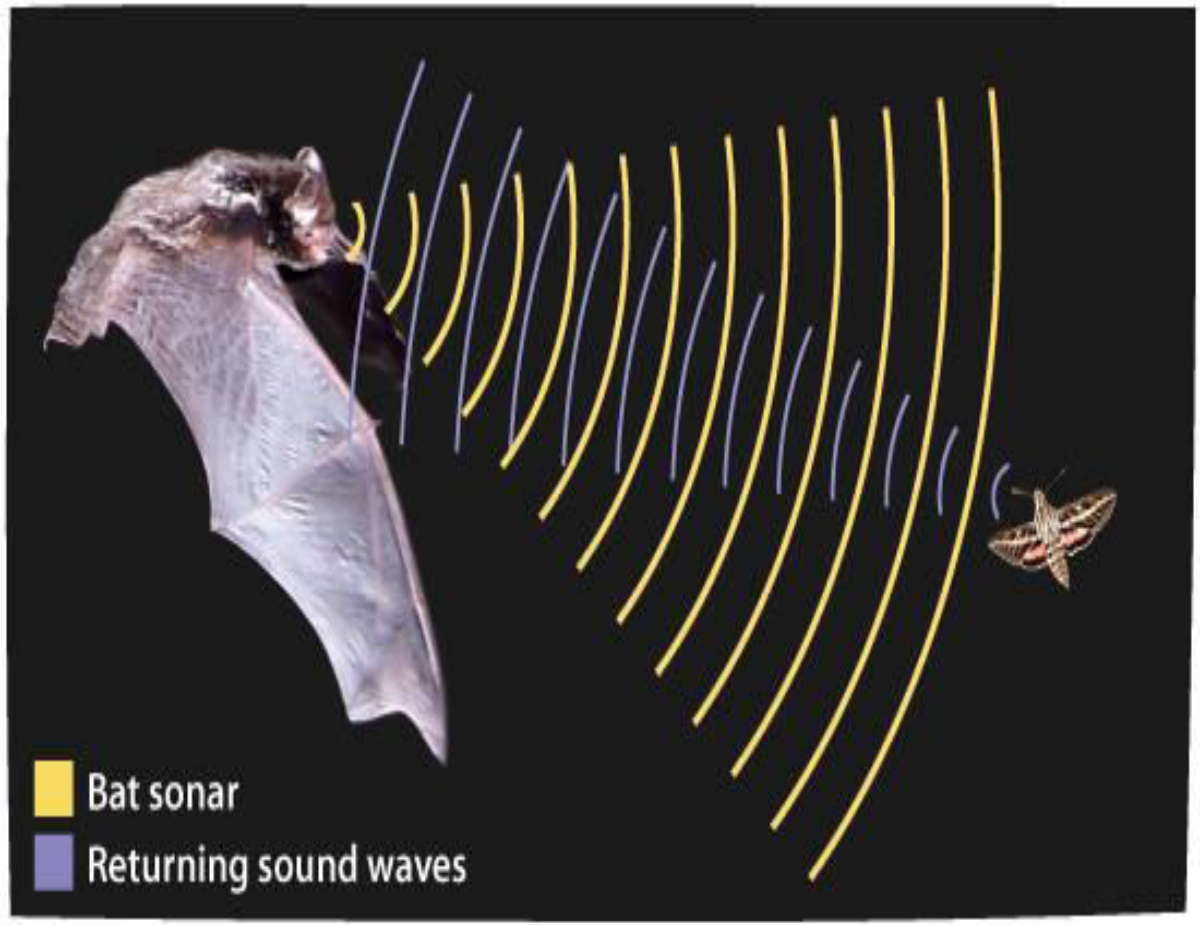
- About this sound calls range in frequency from 14,000 to well over 100,000 Hz, mostly beyond the range of the human ear (typical human hearing range is considered to be from 20 Hz to 20,000 Hz).
- Bats may estimate the elevation of targets by interpreting the interference patterns caused by the echoes reflecting from the tragus, a flap of skin in the external ear.

- Using echolocation, bats can detect objects as thin as a human hair in complete darkness.

- Echolocation allows bats to find insects the size of mosquitoes, which many bats like to eat.

- Bats aren't blind, but they can use echolocation to find their way around very quickly in total darkness.

- The image shows the sonogram of a Silver-haired bat screech from the Western Ecological Research Center.
- In this recording you can hear a standard repeated call that is for basic navigation.
- Bats use this to avoid flying into objects. The faster clicking is likely because the bat has detected an insect and the bat needs more accuracy to catch its prey.



Calls and ecology

- Bats belonging to the suborder Microchiroptera (microbats) occupy a diverse set of ecological conditions – they can be found living in environments as different as Europe and Madagascar, and hunting for food sources as different as insects, frogs, nectar, fruit, and blood.

- Additionally, the characteristics of an echolocation call are adapted to the particular environment, hunting behavior, and food source of the particular bat

Blind as a Bat?

- Most bats regularly eat flying insects.
- Bats can see as well as humans, but echolocation is much more important to them than their eyesight for finding food.
- Bat's Echolocation is so precise that it can detect an object the width of a human hair.
- Mother bats who are feeding their babies, may catch and eat up to 4500 insects in one night!

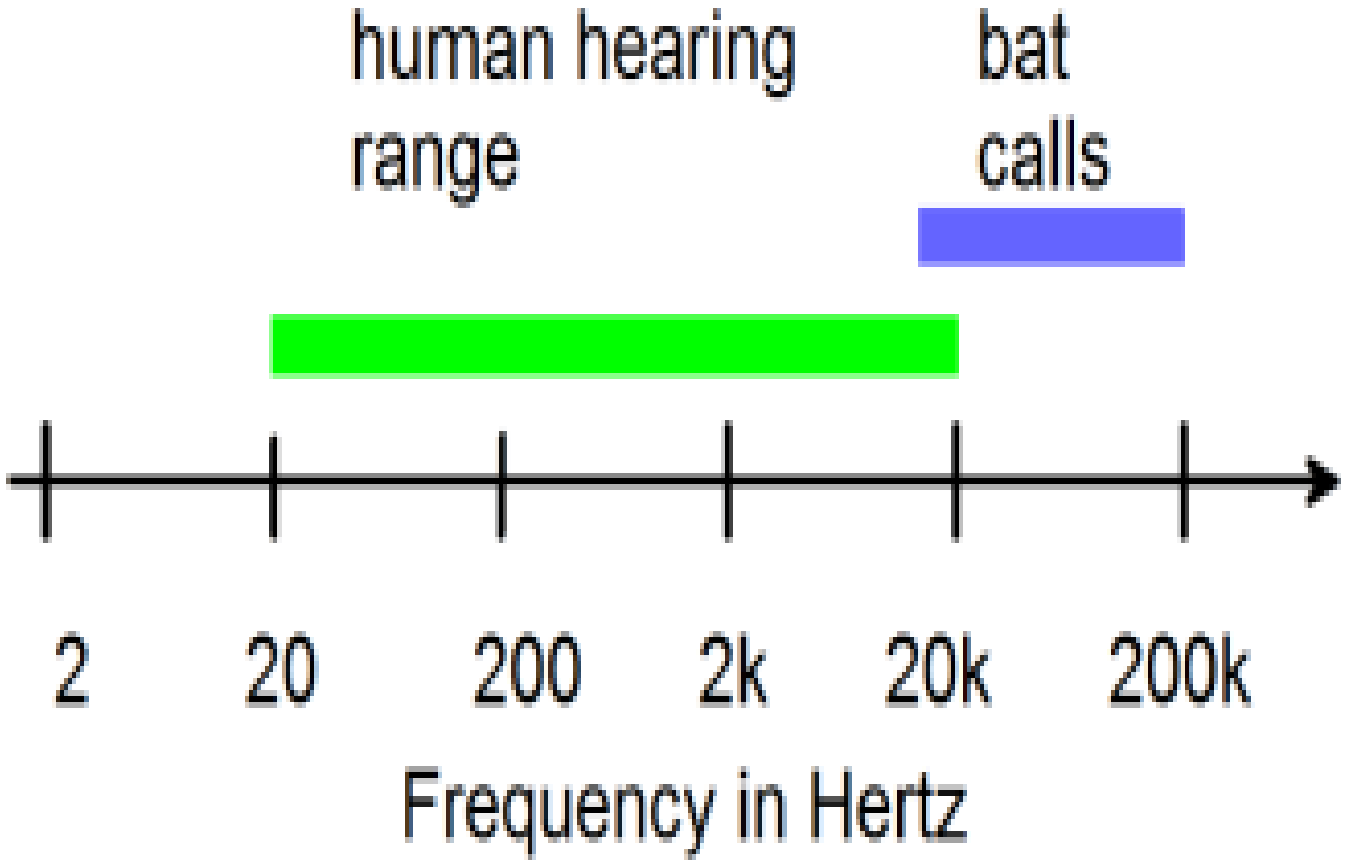
Acoustic features

- Describing the diversity of bat echolocation calls requires examination of the frequency and temporal features of the calls.

- It is the variations in these aspects that produce echolocation calls suited for different acoustic environments and hunting behaviors.

Frequency

- Bat call frequencies range from as low as 11 kHz to as high as 212 kHz.
- Insectivorous aerial-hawking bats have a call frequency between 20 kHz and 60 kHz because it is the frequency that gives the best range and imagea acuity and makes them less conspicuous to insects.
- However, low frequencies are adaptive for some species with different prey and environments.
- Euderma maculatum, a species that feeds on moths, uses a particularly low frequency of 12.7 kHz that cannot be heard by moths


human hearing range
bat calls
2
20
200
2k
20k
200k
Frequency in Hertz

Intensity

- Echolocation calls have been measured at intensities anywhere between 60 and 140 decibels.
- Certain microbat species can modify their call intensity mid-call, lowering the intensity as they approach objects that reflect sound strongly. This prevents the returning echo from deafening the bat.
- High-intensity calls such as those from aerial-hawking bats (133 dB) are adaptive to hunting in open skies. Their high intensity calls are necessary to even have moderate detection of surroundings because air has a high absorption of ultrasound and because insects' size only provide a small target for sound reflection.
- Additionally, the so-called "whispering bats" have adapted low-amplitude echolocation so that their prey, moths, which are able to hear echolocation calls, are less able to detect and avoid an oncoming bat.

Call duration

- A single echolocation call (a call being a single continuous trace on a sound spectrogram, and a series of calls comprising a sequence or pass) can last anywhere from 0.2 to 100 milliseconds in duration, depending on the stage of prey-catching behavior that the bat is engaged in.

- For example, the duration of a call usually decreases when the bat is in the final stages of prey capture – this enables the bat to call more rapidly without overlap of call and echo.

- Reducing duration comes at the cost of having less total sound available for reflecting off objects and being heard by the bat.

Thank you

Echolocation in Other Animals

Echolocation in Other Animals

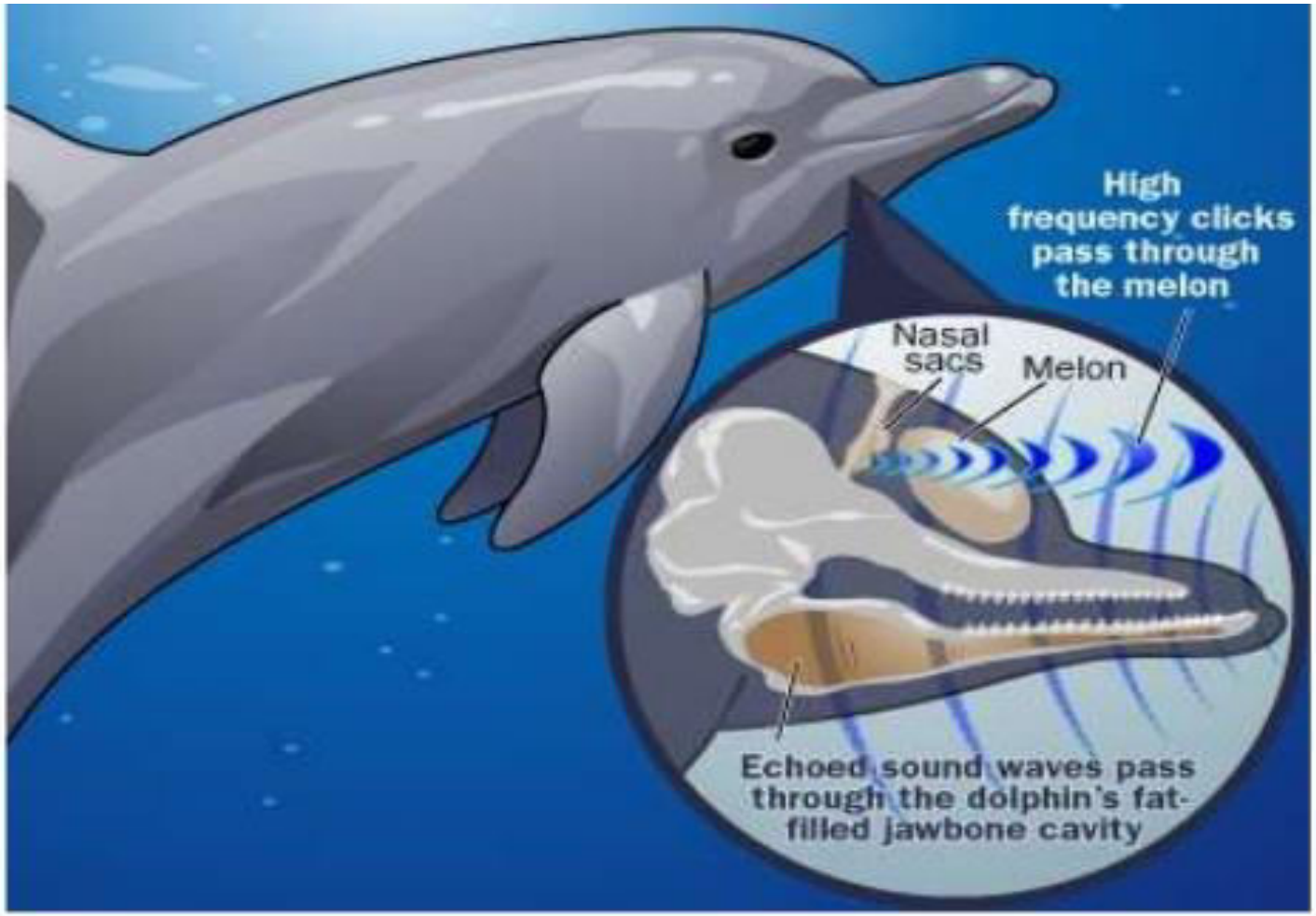
- Humans and most other animals see using light waves.
- Light reflects off the objects around you and reaches your eye, which provides information about the world around you.
- Sound waves can be used in exactly the same way to "see."
- Some animals use echos -- sound waves reflected off objects in their path -- to navigate and find food at night or in dark locations such as caves. This is known as echolocation.

Whales and Dolphins

- Marine mammals such as whales and dolphins also use echolocation to locate things at long distances, beyond the range of vision, and also in the depths of the ocean where it is very dark.

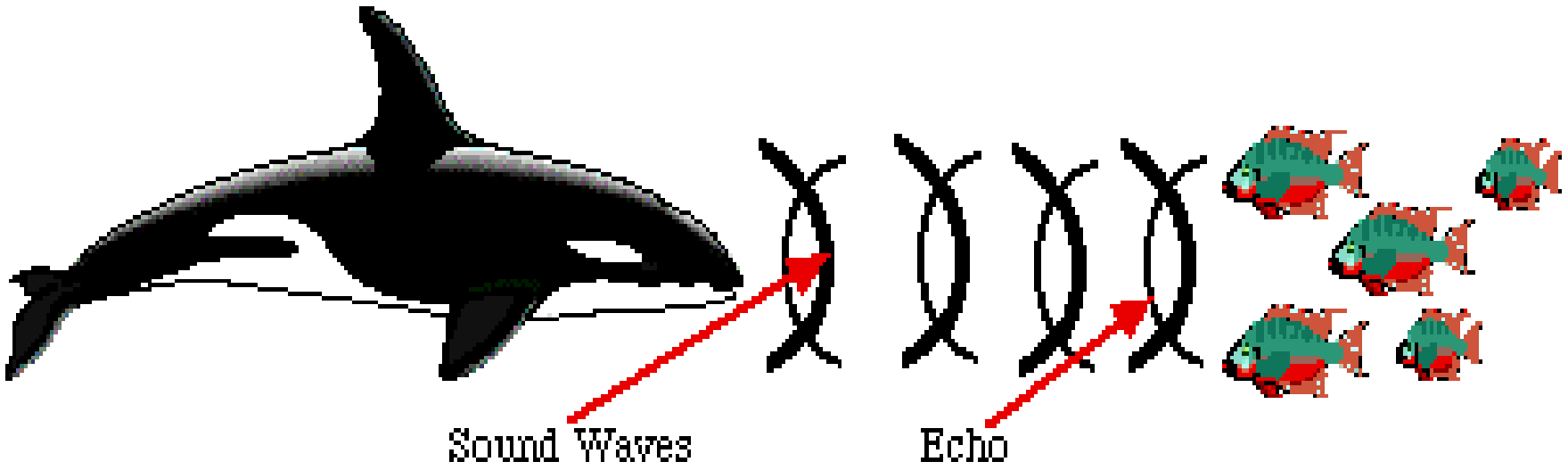
- Whales use echolocation for navigation and to locate food.

- Dolphins also emit clicks with their nasal tissue and use the echoes to find their way around and to hunt. They also use echolocation to communicate with other members of their group and to avoid predators.

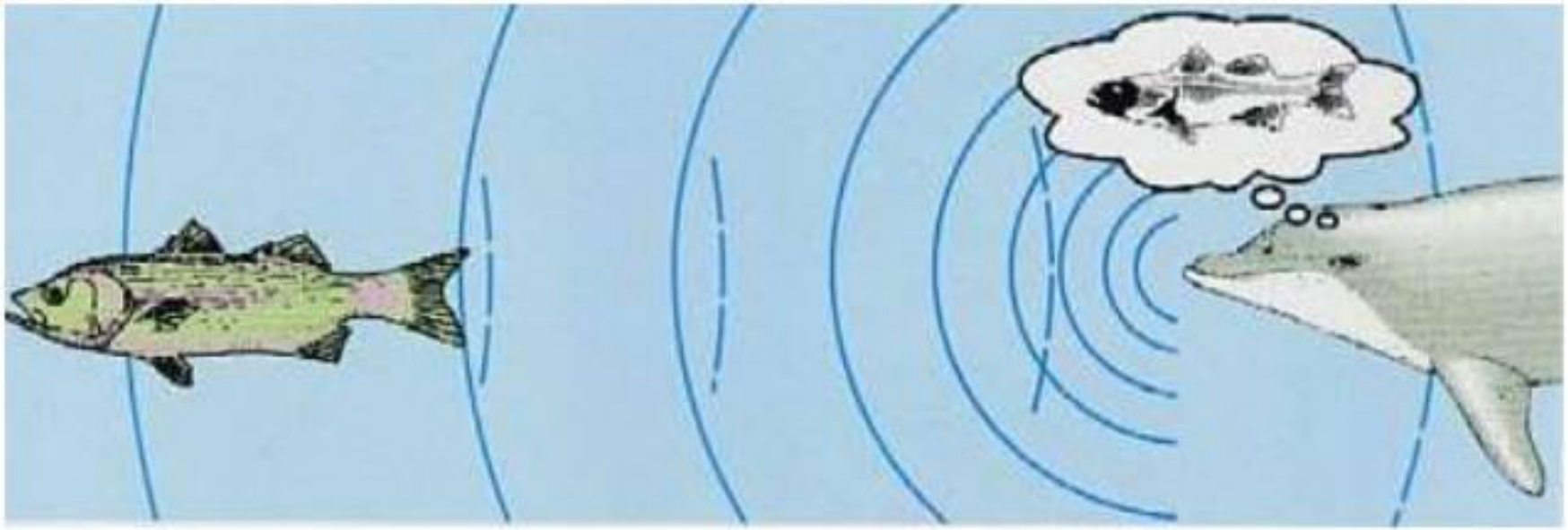


The **Dolphin** uses nasal passages to make a click and sends it through its forehead, which focuses the sounds together into a beam before sending it into the water.

Echolocation



Sound waves from the whale's song are reflected in an echo and travel back to the whale.



- When the sound hits an object in the water, it bounces back to the dolphin as an echo.
- The dolphin absorbs this returning echo through its jaw.
- A passage of fat from the jaw conducts the sound to the dolphin's inner ear
- The dolphin can tell things about the object, such as size, shape and material.

Oilbirds and Swiftlets

- Echolocation is rare among birds. Two species of birds that live in caves and are known to have developed echolocation are South American oilbirds and swiftlets.
- Oilbirds emit clicks and use the echoes to hep them navigate in total darkness.
- Swiftlets use echolocation for navigation in the dark and also for social purposes.
- The ears of these birds, unlike those of bats, don't show any modifications that make them particularly suited to echolocation.



House mice use Ultrasonic sounds to attract mates.

Rats and other rodents also use it to communicate.

Shrews

- Shrews are known to emit ultrasonic sound and use the echoes to locate insects and other prey.

- They open and close their mouths rapidly to emit quick pulses of low intensity sound as they get closer to their prey. Shrews also use echolocation for navigation.

- They find their way through leaf litter or in the darkness of tunnels under snow with the help of echoes produced by sounds they emit.

Humans

- Sonars and radars, used by people for navigation and to locate objects, are forms of echolocation.

- In fact, the development of these technologies was inspired by the work of zoologist **Donald Griffin** who discovered how bats navigate and coined the term "echolocation."

- Some blind people have developed the ability to locate obstacles by making clicking sounds with their tongues and listening for echoes.

Echolocation jamming

- Echolocation systems are susceptible to interference known as echolocation jamming or sonar jamming.

- Jamming occurs when non-target sounds interfere with target echoes. Jamming can be purposeful or inadvertent and can be caused by the echolocation system itself, other echolocating animals, prey, or humans.

- Echolocating animals have evolved to minimize jamming; however, echolocation avoidance behaviors are not always successful.

Thank you

Human echolocation

Human echolocation

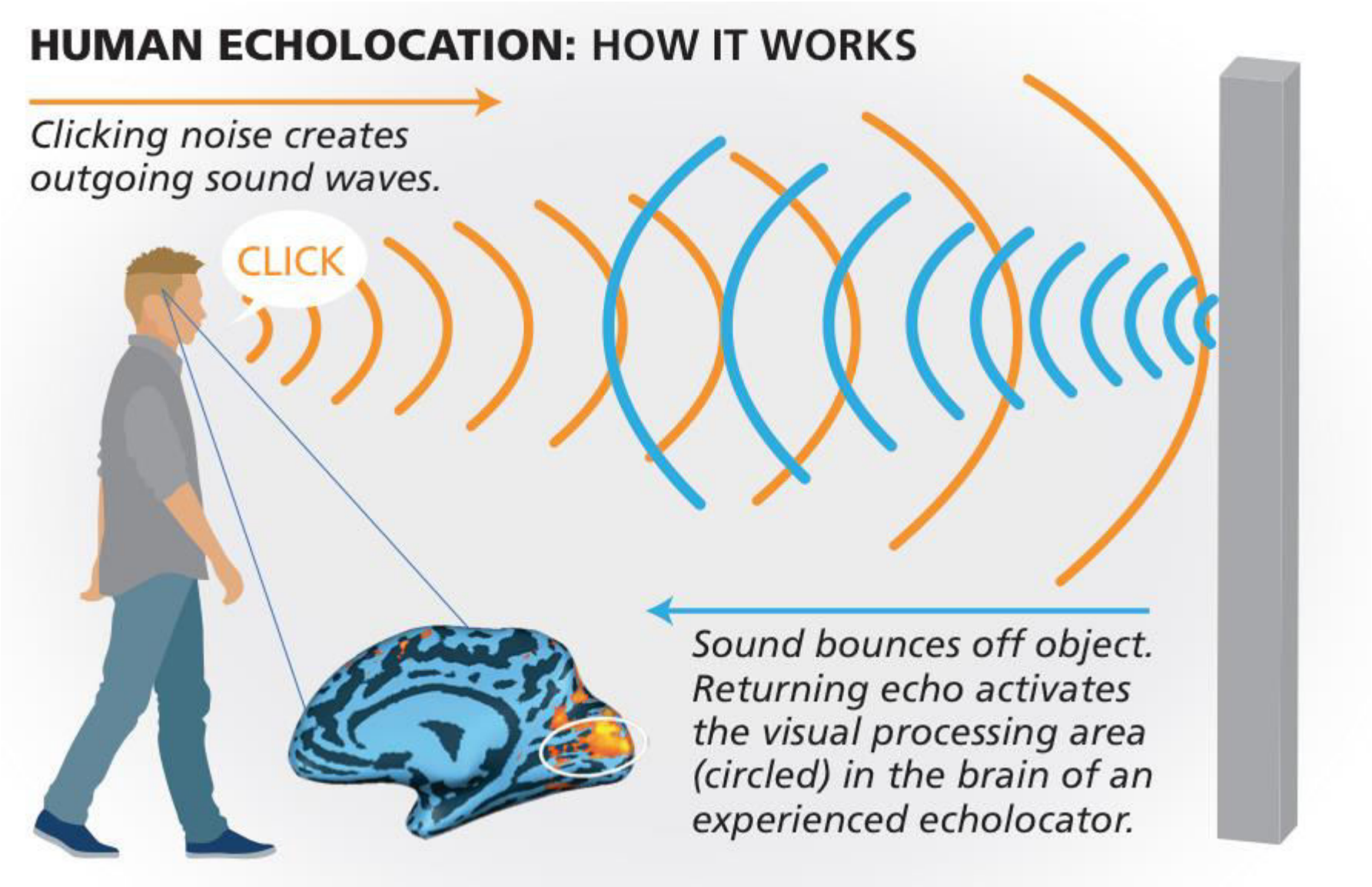
- Human echolocation is the ability of humans to detect objects in their environment by sensing echoes from those objects, by actively creating sounds – for example, by tapping their canes, lightly stomping their foot, snapping their fingers, or making clicking noises with their mouths – people trained to orient by echolocation can interpret the sound waves reflected by nearby objects, accurately identifying their location and size.

- Human Echolocation is the ability of humans to use sound echoes to help determine their immediate and nearby surroundings.

- Using methods such as tapping a cane on the ground, stamping their feet, or making clicking sounds with their mouths they are able to interpret the resulting echoes/reflections to provide them with an internal “map” of their surroundings.

- Similar in principle to sonar used by bats, toothed whales and dolphins, reflecting sound waves returning to the ears provide important clues such as proximity, density, shape and size of nearby objects.

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Mechanics

- Vision and hearing are closely related in that they can process reflected waves of energy.
- Vision processes light waves as they travel from their source, bounce off surfaces throughout the environment and enter the eyes.
- Similarly, the auditory system processes sound waves as they travel from their source, bounce off surfaces and enter the ears.
- Both systems can extract a great deal of information about the environment by interpreting the complex patterns of reflected energy that they receive. In the case of sound, these waves of reflected energy are called "echoes".

- Echoes and other sounds can convey spatial information that is comparable in many respects to that conveyed by light.

- With echoes, a blind traveler can perceive very complex, detailed, and specific information from distances far beyond the reach of the longest cane or arm.

- Echoes make information available about the nature and arrangement of objects and environmental features such as overhangs, walls, doorways and recesses, poles, ascending curbs and steps, planter boxes, pedestrians, fire hydrants, parked or moving vehicles, trees and other foliage, and much more.

- Echoes can give detailed information about location (where objects are), dimension (how big they are and their general shape), and density (how solid they are).

- Location is generally broken down into distance from the observer and direction (left/right, front/back, high/low).

- Dimension refers to the object's height (tall or short) and breadth (wide or narrow).

- Whether visually impaired or not, all humans (with operational auditory systems) use sound clues to understand their surroundings.

- Make a noise in a large enclosed space and you can hear the delay in reflections as the sound repeatedly bounces around the room, have a friend yell out your name from a distance and you can fairly easily determine the direction your friend is calling you from.

- However when it comes to the visually impaired, they seem to have the ability to tap into this phenomenon with much greater ability.

Thank you

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Molecular Imaging

Molecular Imaging

- Molecular imaging originated from the field of radio-pharmacology due to the need to better understand fundamental molecular pathways inside organisms in a noninvasive manner.
- It enables the visualization of the cellular function and the follow-up of the molecular process in living organisms without perturbing them.
- The multiple and numerous potentialities of this field are applicable to the diagnosis of diseases such as cancer, and neurological and cardiovascular diseases.
- This technique also contributes to improving the treatment of these disorders by optimizing the pre-clinical and clinical tests of new medication.

- Molecular imaging differs from traditional imaging in that probes known as biomarkers are used to help image particular targets or pathways.

- Biomarkers interact chemically with their surroundings and in turn alter the image according to molecular changes occurring within the area of interest.

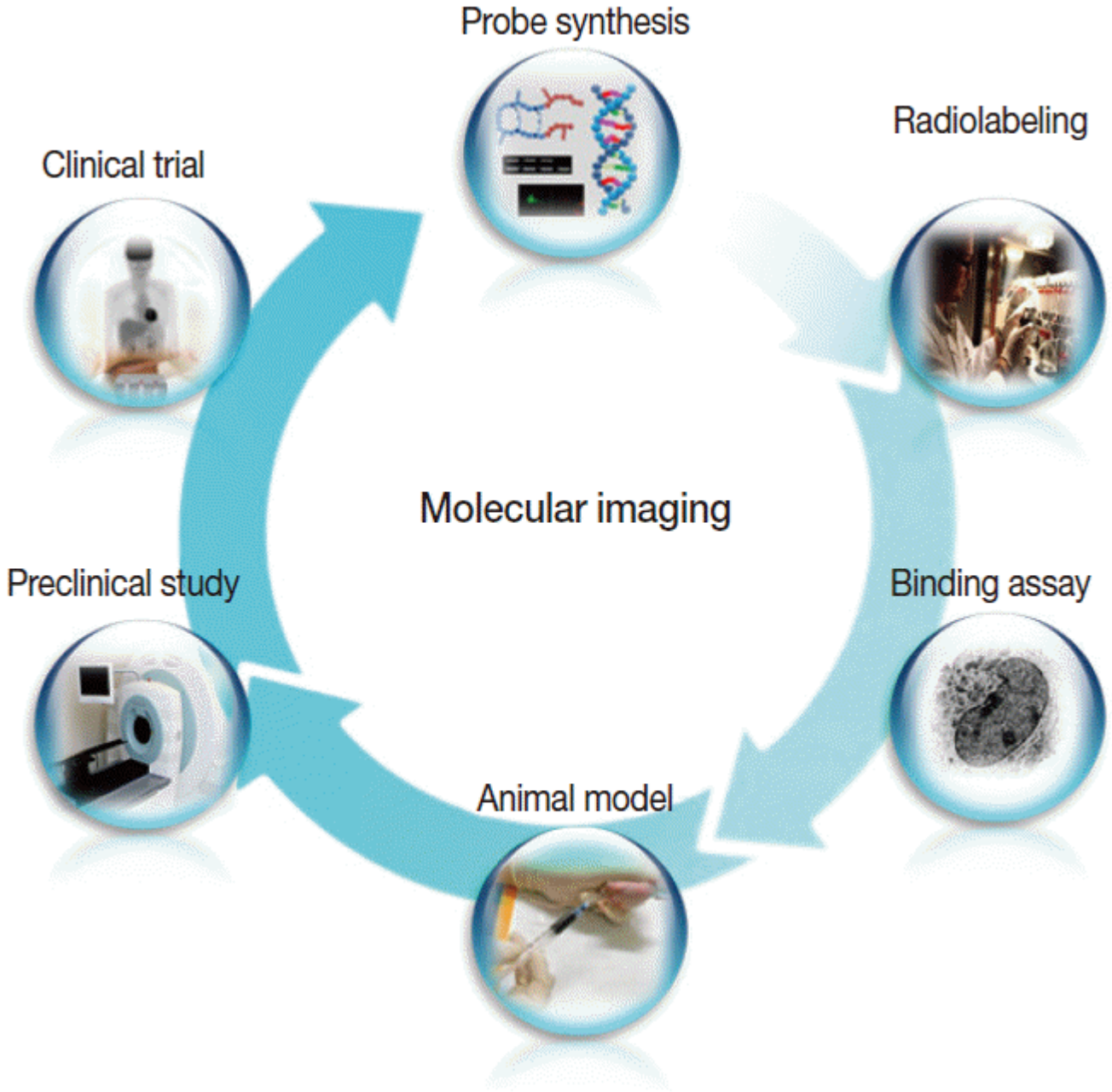
- This process is markedly different from previous methods of imaging which primarily imaged differences in qualities such as density or water content.

- This ability to image fine molecular changes opens up an incredible number of exciting possibilities for medical application, including early detection and treatment of disease and basic pharmaceutical development.

- Furthermore, molecular imaging allows for quantitative tests, imparting a greater degree of objectivity to the study of these areas

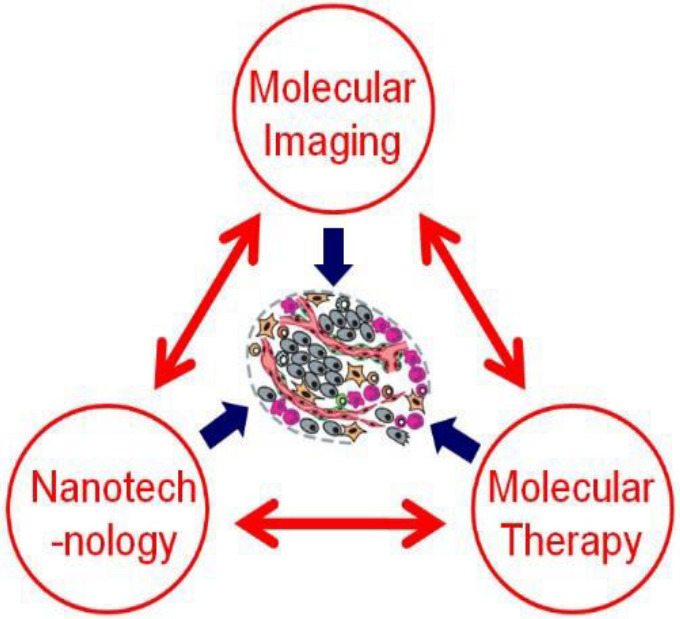
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The overall process of molecular imaging research, which is similar to that of new drug development.



- Molecular imaging, “the visualization, characterization and measurement of biological processes at the molecular and cellular levels in humans and other living systems”, can play pivotal roles in disease diagnosis, treatment efficacy assessment, drug discovery, and the understanding of fundamental biology.

- In addition, molecular imaging is also an extremely powerful tool in multiple disciplines of biomedical research (e.g. stem cell research, RNA interference, nanomedicine, gene/cancer therapy, etc.).



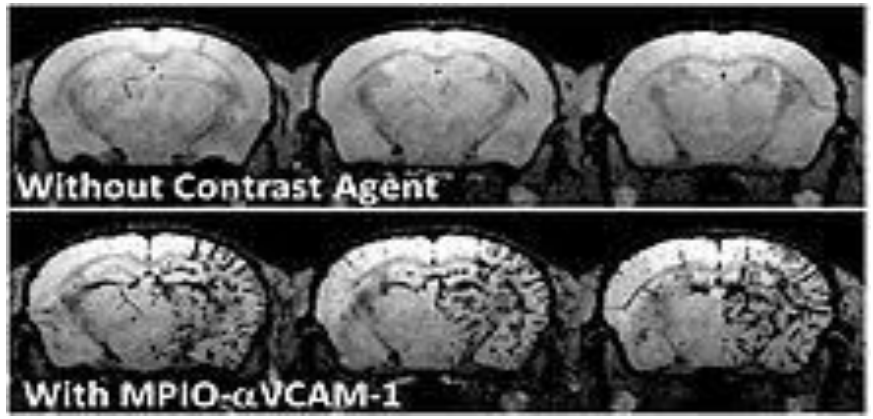
Imaging modalities

There are many different modalities that can be used for noninvasive molecular imaging. Each have their different strengths and weaknesses and some are more adept at imaging multiple targets than others.

1. **Magnetic resonance imaging**
2. **Optical imaging**
3. **Near Infrared imaging**
4. **Single photon emission computed tomography**
5. **Positron emission tomography**
6. **Probes and the imaging of molecular interactions**

Magnetic resonance imaging

- MRI has the advantages of having very high spatial resolution and is very adept at morphological imaging and functional imaging.
- MRI does have several disadvantages though. First, MRI has a sensitivity of around 10−3 mol/L to 10−5 mol/L which, compared to other types of imaging, can be very limiting.
- Improvements to increase MR sensitivity include increasing magnetic field strength, and hyperpolarization via optical pumping, dynamic nuclear polarization or para hydrogen induced polarization.



Molecular MRI of a mouse brain presenting acute inflammation in the right hemisphere. Whereas unenhanced MRI failed to reveal any difference between right en left hemispheres, injection of a contrast-agent targeted to inflamed vessels allows to reveal inflammation specifically in the right hemisphere.

Optical imaging

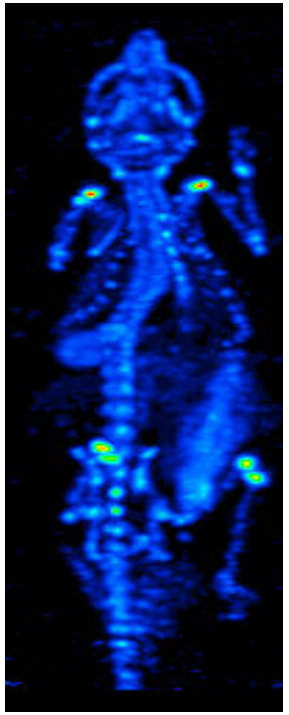
- There are a number of approaches used for optical imaging.
- The various methods depend upon fluorescence, bioluminescence, absorption or reflectance as the source of contrast.
- Optical imaging's most valuable attribute is that it and ultrasound do not have strong safety concerns like the other medical imaging modalities

Near Infrared imaging

- Fluorescent probes and labels are an important tool for optical imaging.
- Some researchers have applied NIR imaging in Rat model of acute myocardial infarction (AMI), using a peptide probe that can binds to apoptotic and necrotic cells.
- A number of near-infrared (NIR) fluorophores have been employed for in vivo imaging.

Single photon emission computed tomography

- The development of computed tomography in the 1970s allowed mapping of the distribution of the radioisotopes in the organ or tissue, and led to the technique now called single photon emission computed tomography (SPECT).



PECT image (bone tracer) of a mouse MIP

Positron emission tomography

- Positron emission tomography (PET) is a nuclear medicine imaging technique which produces a three-dimensional image or picture of functional processes in the body.

- The theory behind PET is simple enough. First a molecule is tagged with a positron emitting isotope. These positrons annihilate with nearby electrons, emitting two 511 keV photons, directed 180 degrees apart in opposite directions.

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Probes and the imaging of molecular interactions

- In order to image multiple targets you must first identify and develop the interactions of which you are trying to take advantage.

- Developing good probes is often difficult and is an area of intense research.

Thank you

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Molecular Imaging

Molecular Imaging

- Molecular imaging originated from the field of radio-pharmacology due to the need to better understand fundamental molecular pathways inside organisms in a noninvasive manner.
- It enables the visualization of the cellular function and the follow-up of the molecular process in living organisms without perturbing them.
- The multiple and numerous potentialities of this field are applicable to the diagnosis of diseases such as cancer, and neurological and cardiovascular diseases.
- This technique also contributes to improving the treatment of these disorders by optimizing the pre-clinical and clinical tests of new medication.

- Molecular imaging differs from traditional imaging in that probes known as biomarkers are used to help image particular targets or pathways.

- Biomarkers interact chemically with their surroundings and in turn alter the image according to molecular changes occurring within the area of interest.

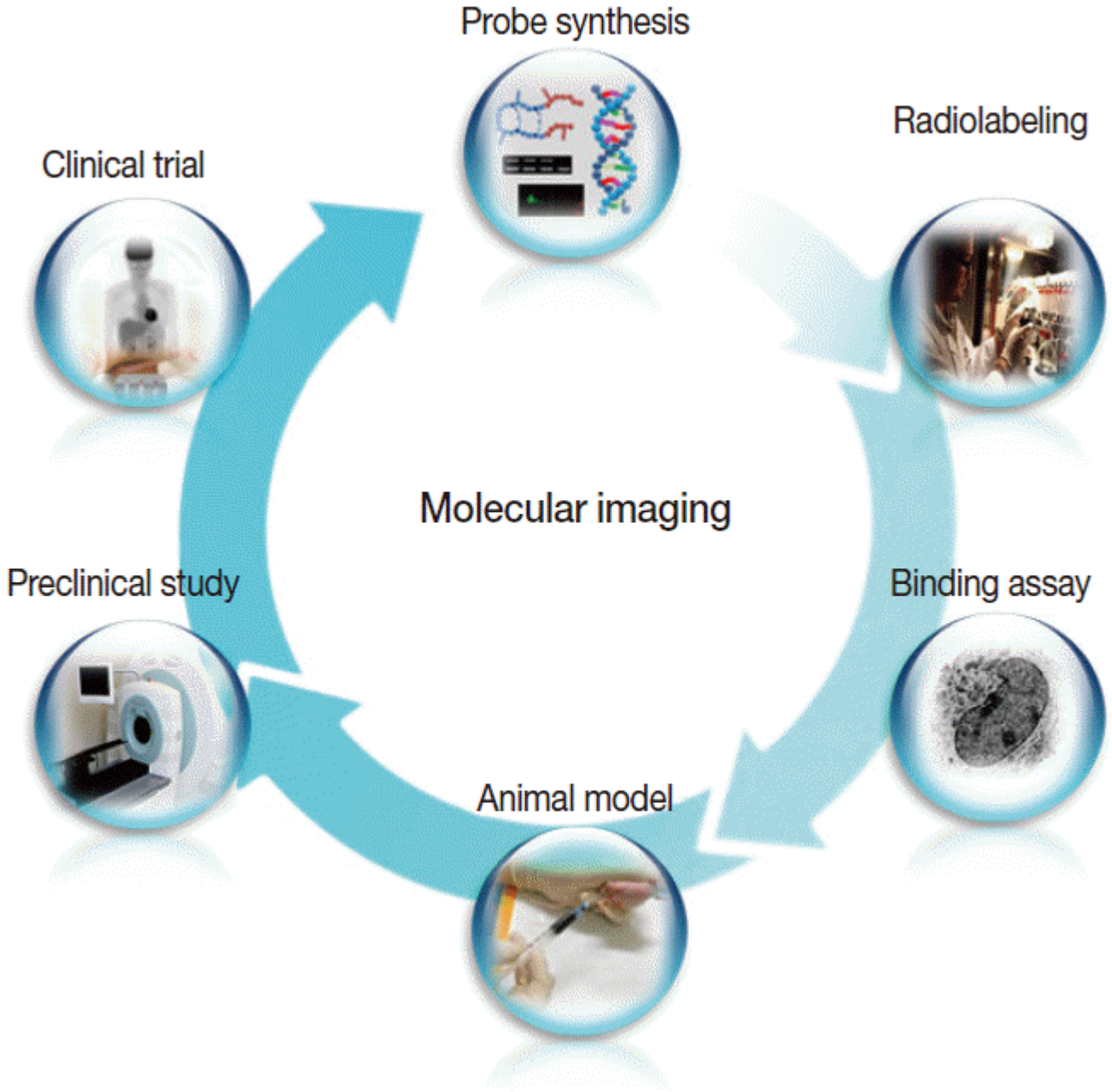
- This process is markedly different from previous methods of imaging which primarily imaged differences in qualities such as density or water content.

- This ability to image fine molecular changes opens up an incredible number of exciting possibilities for medical application, including early detection and treatment of disease and basic pharmaceutical development.

- Furthermore, molecular imaging allows for quantitative tests, imparting a greater degree of objectivity to the study of these areas

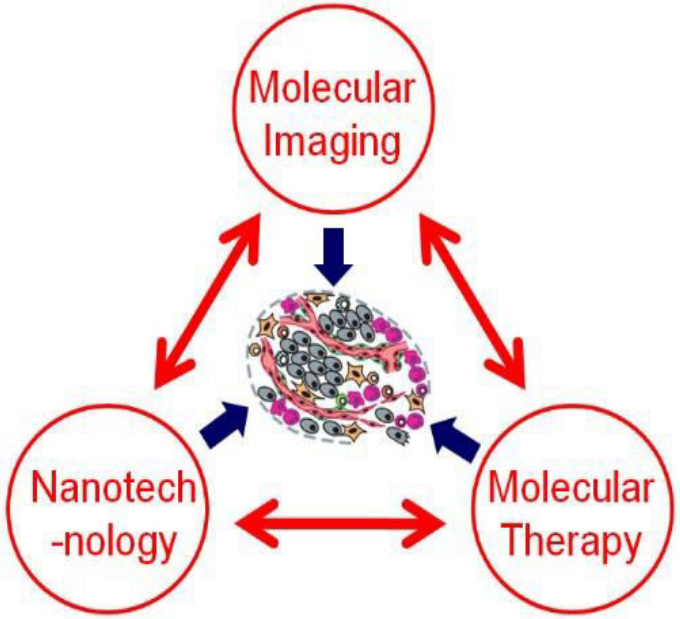
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The overall process of molecular imaging research, which is similar to that of new drug development.



- Molecular imaging, “the visualization, characterization and measurement of biological processes at the molecular and cellular levels in humans and other living systems”, can play pivotal roles in disease diagnosis, treatment efficacy assessment, drug discovery, and the understanding of fundamental biology.

- In addition, molecular imaging is also an extremely powerful tool in multiple disciplines of biomedical research (e.g. stem cell research, RNA interference, nanomedicine, gene/cancer therapy, etc.).



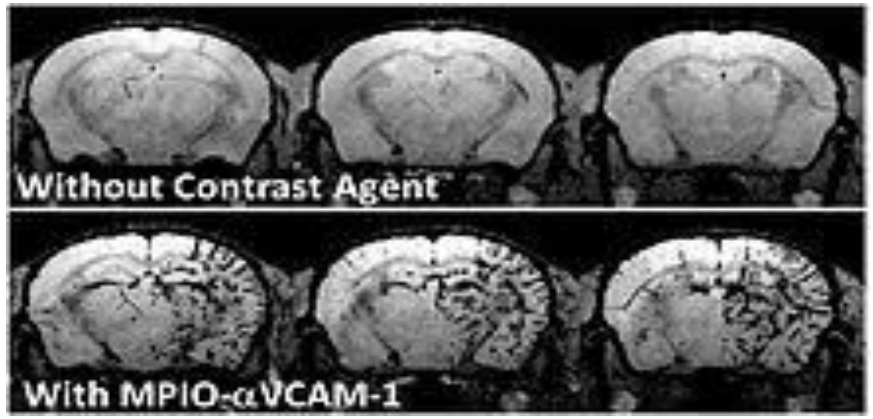
Imaging modalities

There are many different modalities that can be used for noninvasive molecular imaging. Each have their different strengths and weaknesses and some are more adept at imaging multiple targets than others.

1. **Magnetic resonance imaging**
2. **Optical imaging**
3. **Near Infrared imaging**
4. **Single photon emission computed tomography**
5. **Positron emission tomography**
6. **Probes and the imaging of molecular interactions**

Magnetic resonance imaging

- MRI has the advantages of having very high spatial resolution and is very adept at morphological imaging and functional imaging.
- MRI does have several disadvantages though. First, MRI has a sensitivity of around 10−3 mol/L to 10−5 mol/L which, compared to other types of imaging, can be very limiting.
- Improvements to increase MR sensitivity include increasing magnetic field strength, and hyperpolarization via optical pumping, dynamic nuclear polarization or para hydrogen induced polarization.



Molecular MRI of a mouse brain presenting acute inflammation in the right hemisphere. Whereas unenhanced MRI failed to reveal any difference between right en left hemispheres, injection of a contrast-agent targeted to inflamed vessels allows to reveal inflammation specifically in the right hemisphere.

Optical imaging

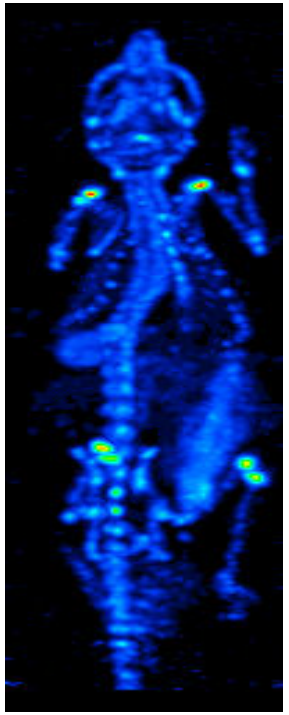
- There are a number of approaches used for optical imaging.
- The various methods depend upon fluorescence, bioluminescence, absorption or reflectance as the source of contrast.
- Optical imaging's most valuable attribute is that it and ultrasound do not have strong safety concerns like the other medical imaging modalities

Near Infrared imaging

- Fluorescent probes and labels are an important tool for optical imaging.
- Some researchers have applied NIR imaging in Rat model of acute myocardial infarction (AMI), using a peptide probe that can binds to apoptotic and necrotic cells.
- A number of near-infrared (NIR) fluorophores have been employed for in vivo imaging.

Single photon emission computed tomography

- The development of computed tomography in the 1970s allowed mapping of the distribution of the radioisotopes in the organ or tissue, and led to the technique now called single photon emission computed tomography (SPECT).



PECT image (bone tracer) of a mouse MIP

Positron emission tomography

- Positron emission tomography (PET) is a nuclear medicine imaging technique which produces a three-dimensional image or picture of functional processes in the body.

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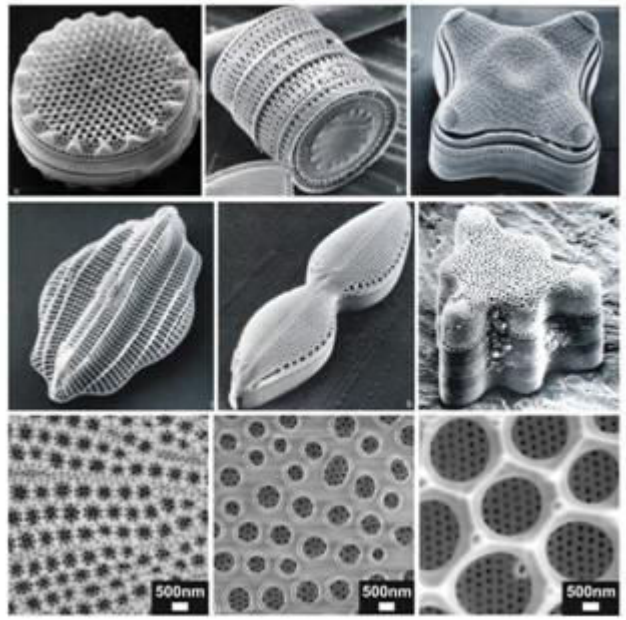
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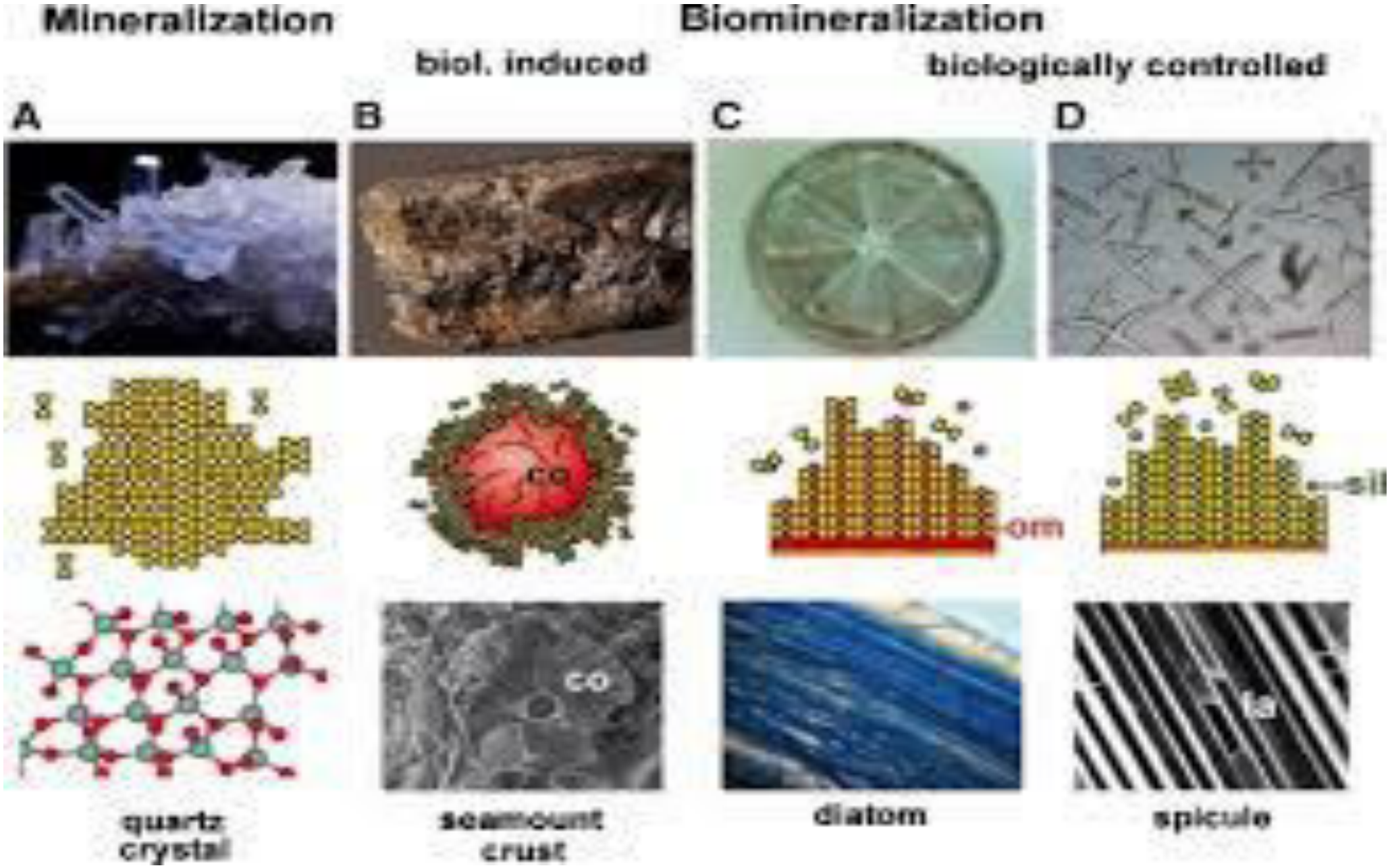
Bio-mineralization

Bio-mineralization

- Bio-mineralization is the process by which living organisms produce minerals, often to harden or stiffen existing tissues. Such tissues are called mineralized tissues.
- It is an extremely widespread phenomenon; all six taxonomic kingdoms contain members that are able to form minerals, and over 60 different minerals have been identified in organisms.
- Examples include silicates in algae and diatoms, carbonates in invertebrates, and calcium phosphates and carbonates in vertebrates.



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- Organisms have been producing mineralized skeletons for the past 550 million years.

- Other examples include copper, iron and gold deposits involving bacteria.

- Biologically-formed minerals often have special uses such as magnetic sensors in magneto tactic bacteria (Fe_3O_4), gravity sensing devices ($CaCO_3$, $CaSO_4$, $BaSO_4$) and iron storage and mobilization ($Fe_2O_3 \cdot H_2O$ in the protein ferritin).

- In terms of taxonomic distribution, the most common bio minerals are the phosphate and carbonate salts of calcium that are used in conjunction with organic polymers such as collagen and chitin to give structural support to bones and shells.

- The structures of these bio composite materials are highly controlled from the nanometer to the macroscopic level, resulting in complex architectures that provide multifunctional properties.

- Because this range of control over mineral growth is desirable for materials engineering applications, there is significant interest in understanding and elucidating the mechanisms of biologically controlled bio mineralization

- Among the most spectacular examples of bio mineralization are the intricately structured cell walls of diatoms, a large group of single-celled eukaryotic algae that are present in almost all water habitats.

- Diatom cell walls are made of amorphous, hydrated SiO_2 (silica) and exhibit highly regular porous patterns which are hierarchically arranged from the nano- to micrometer scale.

- The silica structures are produced by poly-condensation of $Si(OH)_4$ (silicic acid) molecules which occurs in a specific intracellular compartment, termed the silica deposition vesicle (SDV).

Evolution

- The first evidence of bio mineralization dates to some 750 million years ago, and sponge-grade organisms may have formed calcite skeletons 630 million years ago.
- But in most lineages, bio mineralization first occurred in the Cambrian or Ordovician periods.
- Organisms used whichever form of calcium carbonate was more stable in the water column at the point in time when they became bio mineralized, and stuck with that form for the remainder of their biological history.
- The stability is dependent on the Ca/Mg ratio of seawater, which is thought to be controlled primarily by the rate of sea floor spreading, although atmospheric CO2 levels may also play a role.

Thank you

Biophysical properties of the cytoskeleton

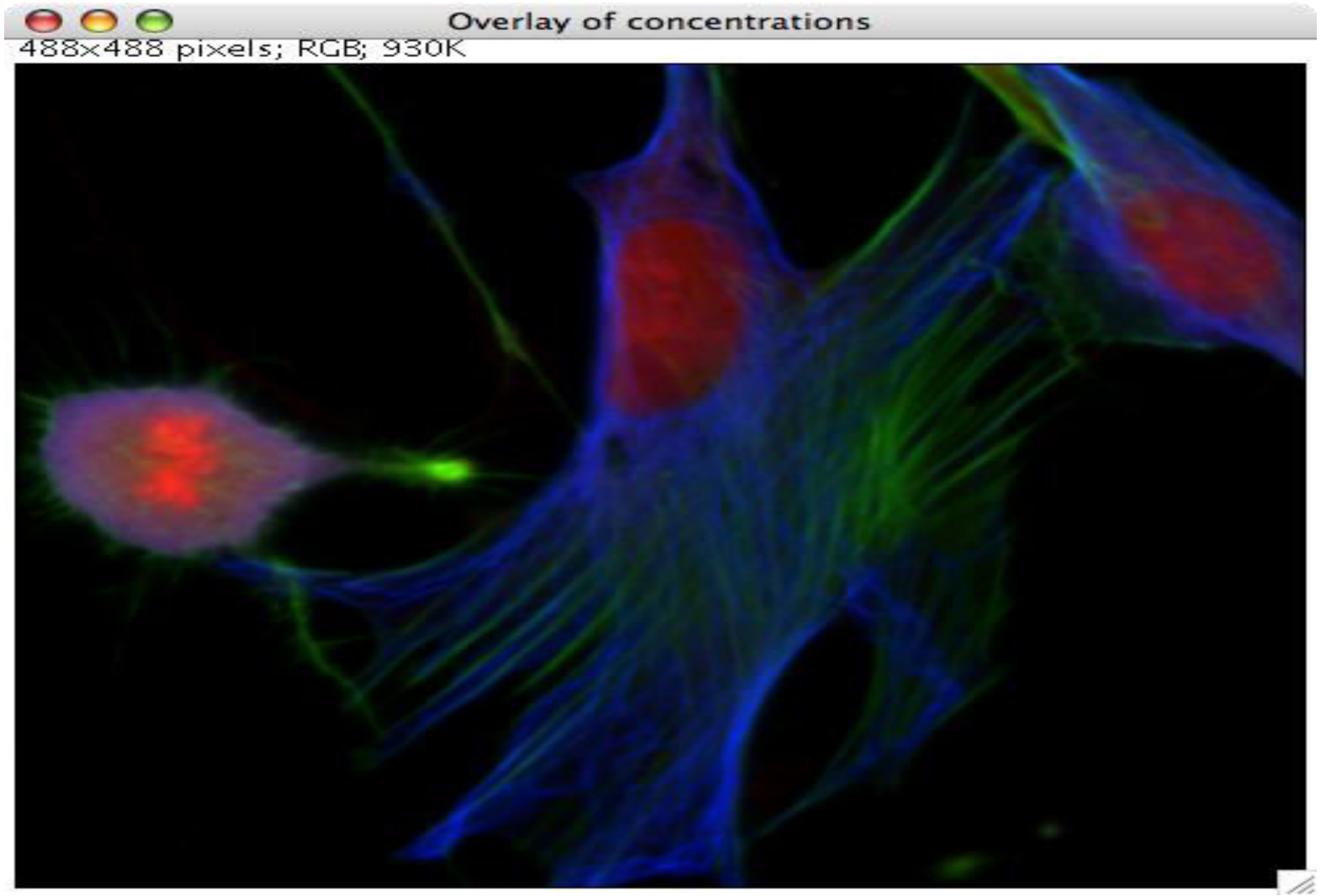
- The cytoskeleton is an umbrella term for structural components of cellular organization. The most prominent member of the cytoskeleton are:

- Microtubuli: stiff 25nm "rods"
- Intermediate filaments: coiled-coils
- Actin filaments: more flexible 8nm fibers, often branched

Cytoskeleton

Learn Biology

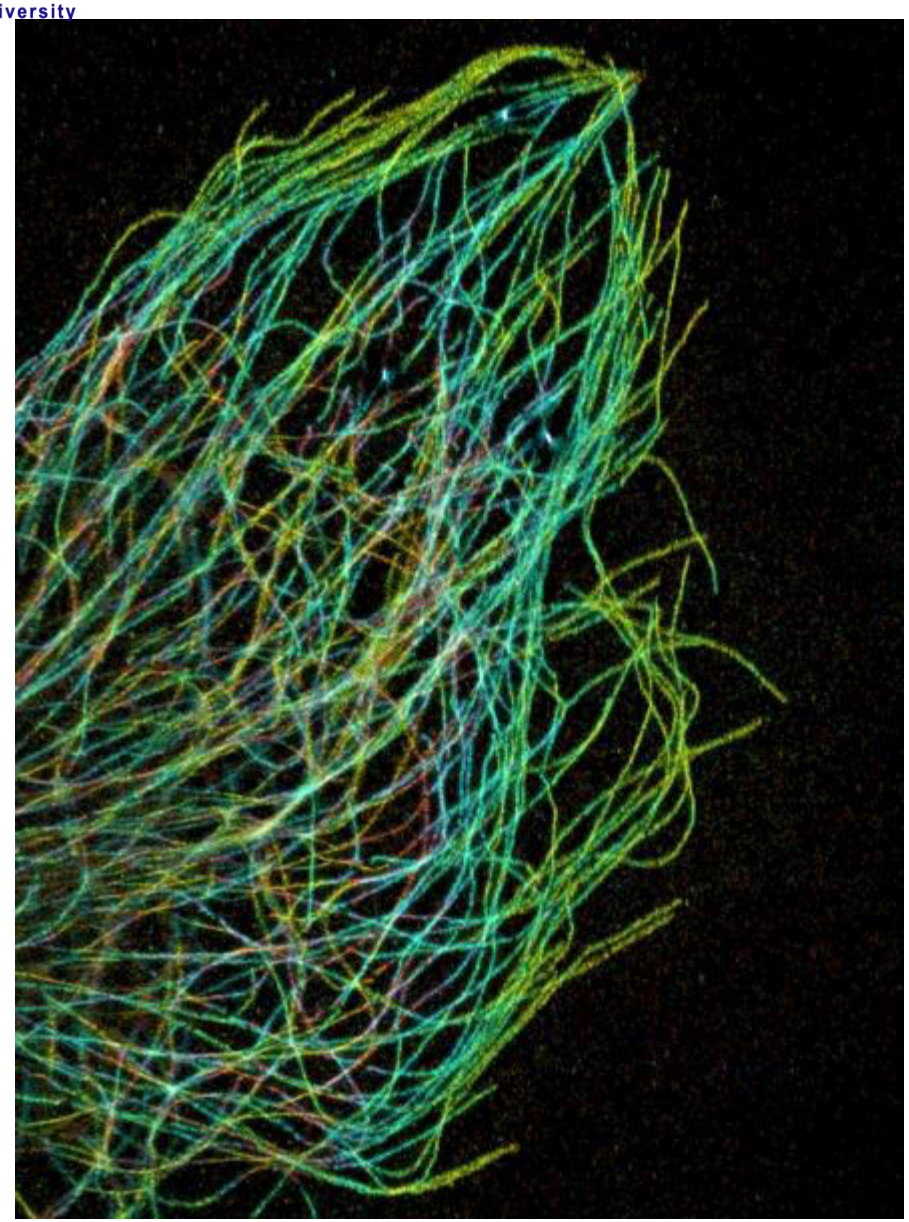
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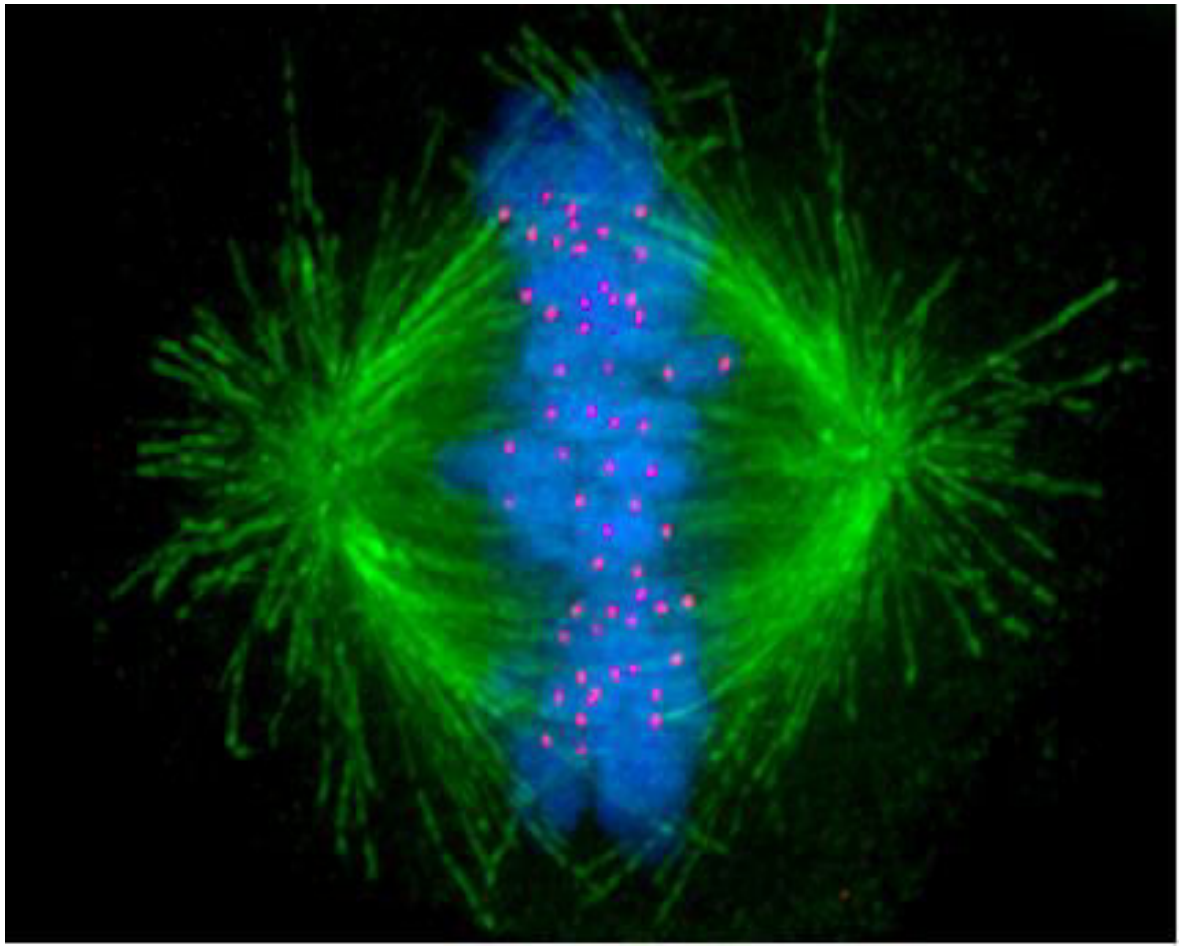


In this false-color image, microtubules are show in blue and actin in green.

Microtubuli

- Microtubuli are made from hetero-dimers of alpha and beta tubulin.
- They assemble into a walled tube with a circumference of 13 hetero-dimers and a diameter of about 25nm. They are polar. One end is known as plus end, the other is the minus end.
- Microtubuli arrange the separation of sister chromatids during cell division.
- For that purpose, several microtubuli are assembled into bundles that start at the spindle and connect to the kinetochores on the replicated sister chromatids.
- Microtubuli also serve as "cargo lanes" kinesin and dynein molecular motors. Kinesin motors move towards the plus end, while dynein motors run the other way.

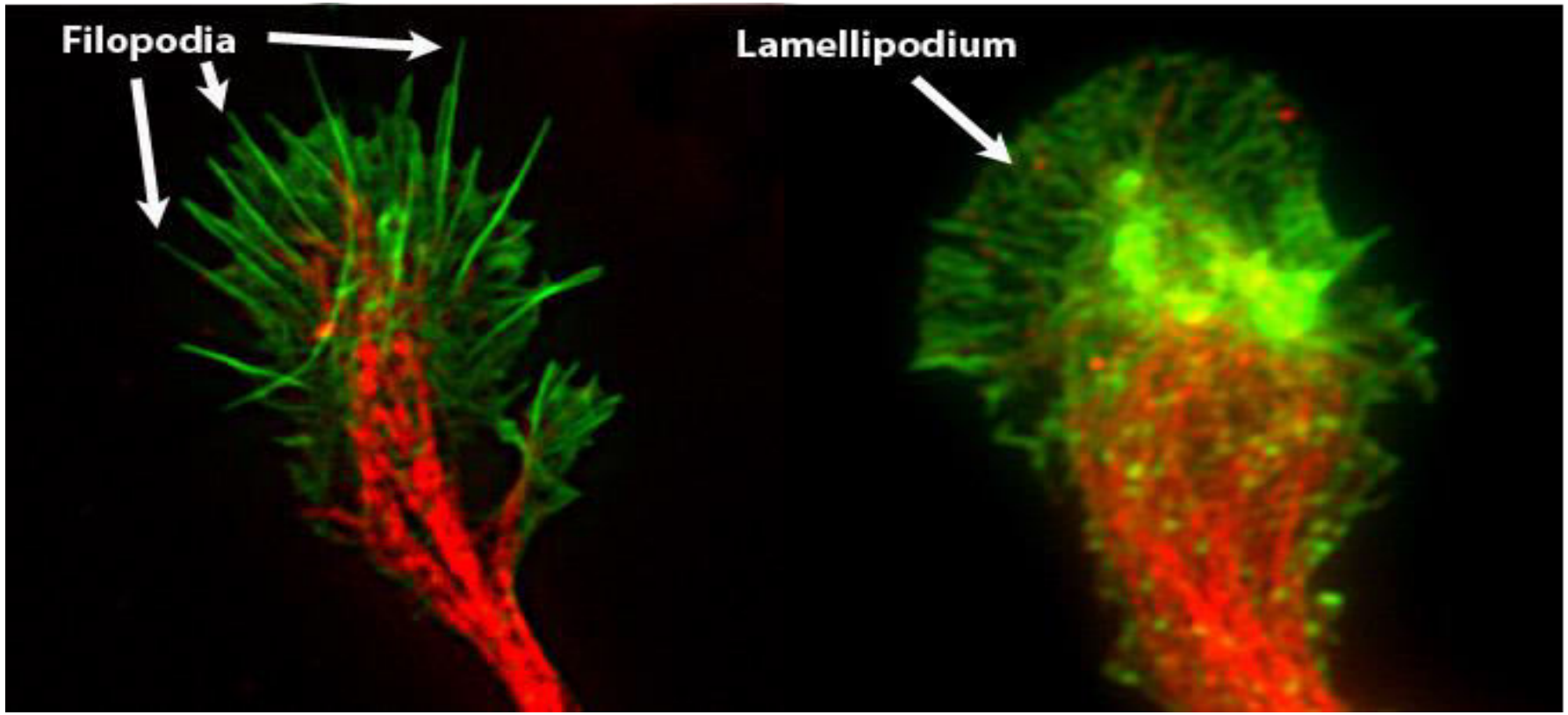




Actin

- Actin filaments are composed to G-actin monomers and two protofilaments are wrapped around each other.
- Myosin motors operate on and between actin fibers.
- There are many different types of myosins that carry cargo along fibers of slide actin fibers along each other.
- Actin filaments often form branches and interconnected networks through actin binding proteins that crosslink various fibers.
- The actin cytoskeleton is central to cell motility and muscle contractility. The myosin/actin network orchestrates many morphological changes during development.

Growth cones are pushed forward by the myosin/actin network, resulting in structures called filopodia and lamellipodia



Intermediate filaments

The term intermediate filament refers to a diverse group of coiled-coil proteins that are involved in variety of structural components of the cell, e.g. the nuclear lamina.

They are called intermediate because they have a ~10nm diameter, hence thicker than actin and thinner than microtubuli.

The precise function and assembly of intermediate filaments is much less understood than actin or microtubuli.

None of the proteins associated with these filaments seem to have motor activity. They are extremely stable and extensible.

They are often described as supporting membranes, for example the nuclear membrane or the cell membrane, and maintain membrane integrity by distributing tensile forces.

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Bioacoustics

Bioacoustics

- Bioacoustics is a cross-disciplinary science that combines biology and acoustics. Usually it refers to the investigation of sound production, dispersion and reception in animals (including humans).
- Acoustics is the branch of physics that deals with the study of all mechanical waves in gases, liquids, and solids including topics such as vibration, sound, ultrasound and infrasound.
- A scientist who works in the field of acoustics is an acoustician while someone working in the field of acoustics technology may be called an acoustical engineer.
- The application of acoustics is present in almost all aspects of modern society with the most obvious being the audio and noise control industries.

- This involves neurophysiological and anatomical basis of sound production and detection, and relation of acoustic signals to the medium they disperse through.

- The findings provide clues about the evolution of acoustic mechanisms, and from that, the evolution of animals that employ them.

- Bioacoustic techniques have recently been proposed as a non-destructive method for estimating biodiversity of an area

- In underwater acoustics and fisheries acoustics the term is also used to mean the effect of plants and animals on sound propagated underwater, usually in reference to the use of sonar technology for biomass estimation.

- The study of substrate-borne vibrations used by animals is considered by some a distinct field called biotremology.

- Biotremology is the study of production, dispersion and reception of mechanical vibrations by animals, and their effect on behaviour

History

- Bioacoustics as a scientific discipline was established by the **Ivan Regen** who began systematically to study insect sounds. In 1925 he used a special stridulatory device to play in a duet with an insect.

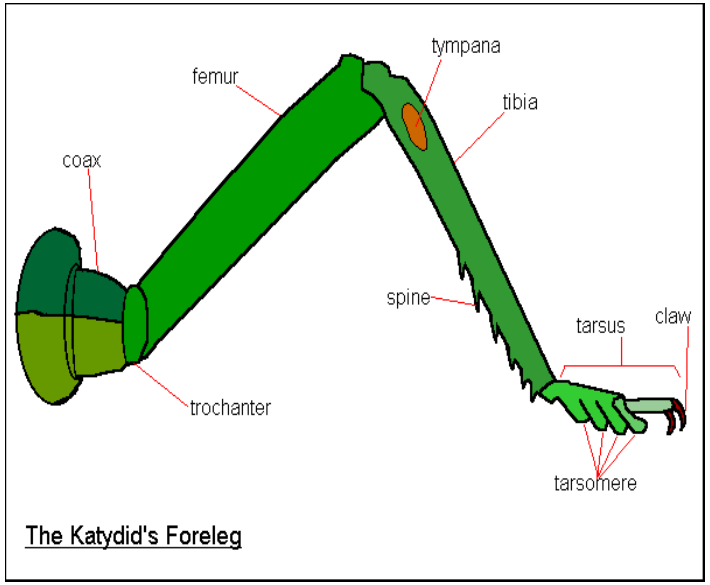
- Stridulation is the act of producing sound by rubbing together certain body parts. This behavior is mostly associated with insects, but other animals are known to do this as well, such as a number of species of fish, snakes and spiders.

- Later, he put a male cricket behind a microphone and female crickets in front of a loudspeaker. The females were not moving towards the male but towards the loudspeaker.

- Regen's most important contribution to the field apart from realization that insects also detect airborne sounds was the discovery of tympanal organ's function

Tympanal organ

- A tympanal organ is a hearing organ in insects, consisting of a membrane (tympanum) stretched across a frame backed by an air sac and associated sensory neurons.
- Sounds vibrate the membrane, and the vibrations are sensed.
- Tympanal organs occur in just about any part of the insect: the thorax, the base of the wing, the abdomen, the legs, etc., depending on the group of insects.



- Relatively crude electro-mechanical devices available at the time (such as phonographs) allowed only for crude appraisal of signal properties.

- More accurate measurements were made possible in the second half of the 20th century by advances in electronics and utilization of devices such as oscilloscopes and digital recorders

Phonograph

- The phonograph is a device for the mechanical recording and reproduction of sound. In its later forms, it is also called a gramophone or, a record player.



BIOACOUSTICS
AND THE VISUALIZATION OF INSECT SONG

A Master's Research Project submitted in conformity with the requirements for the degree of Master of Science in Biomedical Communications (MScBMC).

Offered through the Institute of Medical Science. Faculty of Medicine. University of Toronto in collaboration with Biomedical Communications. Department of Biology. University of Toronto Mississauga.

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Thank you

Methods in bio-acoustical research.

Methods in bioacoustical research

- Listening is still one of the main methods used in bioacoustical research.

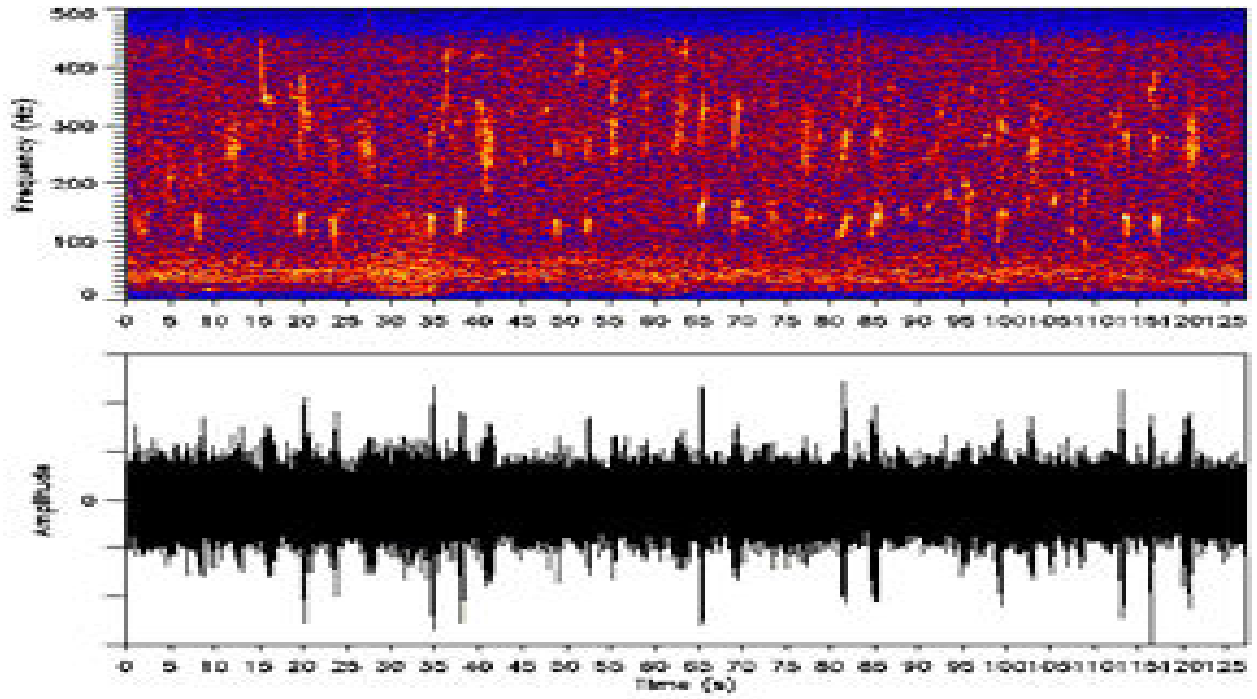
- Little is known about neurophysiological processes that play a role in production, detection and interpretation of sounds in animals, so animal behavior and the signals themselves are used for gaining insight into these processes.

Acoustic signals

- An experienced observer can use animal sounds to recognize a "singing" animal species, its location and condition in nature. Investigation of animal sounds also includes signal recording with electronic recording equipment.
- Due to the wide range of signal properties and media they propagate through, specialized equipment may be required instead of the usual microphone, such as a hydrophone (for underwater sounds), detectors of ultrasound (very high-frequency sounds) or infrasound (very low-frequency sounds), or a laser vibrometer (substrate-borne vibrational signals).
- Computers are used for storing and analysis of recorded sounds. Specialized sound-editing software is used for describing and sorting signals according to their intensity, frequency, duration and other parameters.

- Animal sound collections, managed by museums of natural history and other institutions, are an important tool for systematic investigation of signals.

- Many effective automated methods involving signal processing, data mining and machine learning techniques have been developed to detect and classify the bioacoustic signals



- Spectrogram (above) and oscillogram (below) of the humpback whale's calls



Hydrophone

Sound production, detection, and use in animals

- Scientists in the field of bioacoustics are interested in anatomy and neurophysiology of organs involved in sound production and detection, including their shape, muscle action, and activity of neuronal networks involved.

- Of special interest is coding of signals with action potentials in the latter.

- But since the methods used for neurophysiological research are still fairly complex and understanding of relevant processes is incomplete, more trivial methods are also used.

- Especially useful is observation of behavioral responses to acoustic signals.

- One such response is phonotaxis – directional movement towards the signal source. By observing response to well defined signals in a controlled environment, we can gain insight into signal function, sensitivity of the hearing apparatus, noise filtering capability, etc.

Biomass estimation

- Biomass estimation is a method of detecting and quantifying fish and other marine organisms using sonar technology.
- As the sound pulse travels through water it encounters objects that are of different density than the surrounding medium, such as fish, that reflect sound back toward the sound source.
- These echoes provide information on fish size, location, and abundance. The basic components of the scientific echo sounder hardware function is to transmit the sound, receive, filter and amplify, record, and analyze the echoes.
- While there are many manufacturers of commercially available "fish-finders," quantitative analysis requires that measurements be made with calibrated echo sounder equipment, having high signal-to-noise ratios.

Thank you

Animals and Plants Sounds

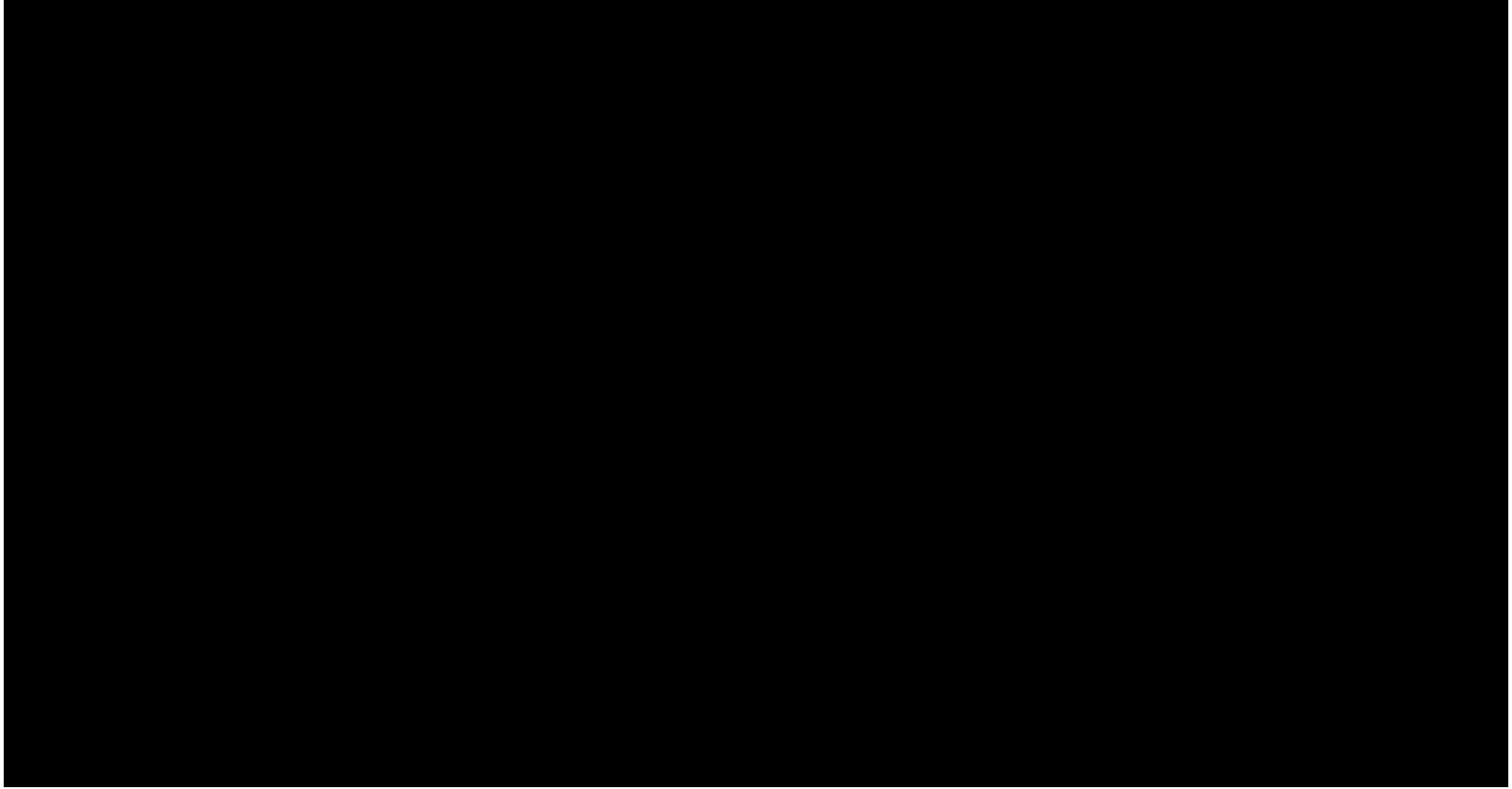
Animal sounds

- Sounds used by animals that fall within the scope of bioacoustics include a wide range of frequencies and media, and are often not "sound" in the narrow sense of the word (i.e. compression waves that propagate through air and are detectable by the human ear).

- Katydid crickets, for example, communicate by sounds with frequencies higher than 100 kHz, far into the ultrasound range.

- Lower, but still in ultrasound, are sounds used by bats for echolocation.

- On the other side of the frequency spectrum are low frequency-vibrations, often not detected by hearing organs, but with other, less specialized sense organs



- The examples include ground vibrations produced by elephants whose principal frequency component is around 15 Hz, and low- to medium-frequency substrate-borne vibrations used by most insect orders.

- Many animal sounds, however, do fall within the frequency range detectable by a human ear, between 20 and 20,000 Hz. Mechanisms for sound production and detection are just as diverse as the signals themselves.



Plant bioacoustics

- Measured sound emissions by plants as well as differential germination rates, growth rates and behavioral modifications in response to sound are well documented.

- Plants detect neighbors by means other than well-established communicative signals including volatile chemicals, light detection, direct contact and root signaling.

- Because sound waves travel efficiently through soil and can be produced with minimal energy expenditure, plants may use sound as a means for interpreting their environment and surroundings.

Buzz pollination

- Buzz pollination, or sonication, serves as an example of a behavioral response to specific frequencies of vibrations in plants.

- Some 2000 plants species, including Dodecatheon and Heliamphora have evolved buzz pollination in which they release pollen from anthers only when vibrated at a certain frequency created exclusively by bee flight muscles.

- The vibrations cause pollen granules to gain kinetic energy and escape from pores in the anthers



- Plants emit audio acoustic emissions between 10–240 Hz as well as ultrasonic acoustic emissions within 20–300 kHz.

- Evidence for plant mechano-sensory abilities are shown when roots are subjected to unidirectional 220 Hz sound and subsequently grow in the direction of the vibration source.

- Using electrograph vibrational detection, structured sound wave emissions were detected along the elongation zone of root tips of corn plants in the form of loud and frequent clicks.

- When plants are isolated from contact, chemical, and light signal exchange with neighboring plants they are still able to sense their neighbors and detect relatives through alternative mechanisms, among which sound vibrations could play an important role

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Prefixes

Prefixes

- Prefixes are useful for expressing units of physical quantities that are either very big or very small.

- A metric prefix is a unit prefix that precedes a basic unit of measure to indicate a multiple or fraction of the unit.

- Each prefix has a unique symbol that is prepended to the unit symbol.

- The prefix kilo-, for example, may be added to gram to indicate multiplication by one thousand: one kilogram is equal to one thousand grams.

- The prefix milli-, likewise, may be added to metre to indicate division by one thousand; one millimetre is equal to one thousandth of a metre.

- The SI prefixes are standardized for use in the International System of Units (SI) by the International Bureau of Weights and Measures (BIPM) in resolutions dating from 1960 to 1991.

- Since 2009, they have formed part of the International System of Quantities.

Prefixes	Value	Standard form	Symbol
Tera	1 000 000 000 000	10^{12}	T
Giga	1 000 000 000	10^{9}	G
Mega	1 000 000	10^{6}	M
Kilo	1 000	10^{3}	k
deci	0.1	10^{-1}	d
centi	0.01	10^{-2}	c
milli	0.001	10^{-3}	m
micro	0.000 001	10^{-6}	μ
nano	0.000 000 001	10^{-9}	n
pico	0.000 000 000 001	10^{-12}	p

- Each prefix name has a symbol that is used in combination with the symbols for units of measure.
- For example, the symbol for kilo- is 'k', and is used to produce 'km', 'kg', and 'kW', which are the SI symbols for kilometre, kilogram, and kilowatt, respectively.
- Where Greek letters are unavailable, the symbol for micro 'µ' is commonly replaced by 'u'.
- Prefixes corresponding to an integer power of one thousand are generally preferred. Hence 100 m is preferred over 1 hm (hectometre) or 10 dam (decametres).
- The prefixes hecto, deca, deci, and centi are commonly used for everyday purposes, and the centimetre (cm) is especially common.
- However, some modern building codes require that the millimetre be used in preference to the centimetre, because "use of centimetres leads to extensive usage of decimal points and confusion"

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Conversion of prefixes

Example 1:

The frequency of the radio wave is 350M Hz. What is the frequency of the radio wave in Hz?

Answer:

Mega (M) = 1,000,000 or 10^6

Therefore,

350MHz = 350 x 10^6Hz

Example 2:

The thickness of a film is 25nm. What is the thickness in unit meter?

Answer:

nano (n) = 0.000000001 or 10^-9

Therefore

25nm = 25 x 10^-9m

Example 3:

0.255 s is equal to how many ms?

Answer:

mili (m) = 0.001 or 10^-3

To write a normal number with prefixes, we divide the number with the value of the prefixes

0.0255 s = 0.0255 ÷ 10^-3 = 25.5 ms

Example 4:

Convert 265,500,000 W into GW.

Answer:

Gega (G) = 1,000,000,000 or 10^9

Therefore

265,500,000 W = 265,500,000 ÷ 10^9 = 0.2655GW

Thank you

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Dimensions and units

Dimensions and units

- Every physical quantity has abstract dimensions that tell us what kind of thing it is.
- Every kind of dimension can be measured using a variety of different units.
- The choice of units is arbitrary.

- Every physical quantity has its own identity.
- Such an identity is nothing but its quality with which it can be distinguished from all the other quantities.
- Such a unique quality possessed by a quantity is called dimension.
- Symbolically the dimension is expressed in the characteristic notation which is square bracket [ ].

For example, the dimension of length is expressed as [L] while the dimension of mass is expressed as [M] etc.

- Similar to the fundamental units, each derived unit also has a unique dimension associated with it.

- **For example,**

the volume can be mathematically expressed as

$$V = l \times b \times h$$

where l is the length, b is breadth and h is height.

All l, b and h are basically lengths measured in meters. The dimension of each l, b and h is [L] i.e. length. Hence equation of volume can be written in dimensional form as,

$$V = [L][L][L] = [L3].$$

- Thus [L3] is the dimension of the volume which indicates that calculations of volume involves the product of three lengths.

- Thus it can be said that the complete algebraic formula or relationship to obtain the derived unit from the fundamental units, is nothing but the dimension of the derived unit.

- The equality sign used to indicate the dimension indicates dimensional nation of the quality and is not related to actual numerical value of the quantity.

- Thus V = [L3] is the equality in terms of dimensions and should not be mixed up with actual numerical values.

Dimensional Formula of Physical Quantities & Physical Constants

- To express the dimensions of physical quantities in mechanics, the units of length, mass and time are denoted by bracketed capital letters [M],[L] and [T] respectively.

- If the dimensions of a physical quantity are a in mass, b in length and c in time, then dimensions of that physical, quantity are expressed in the following manner:

[Ma Lb Tc].

This is the "dimensional formula" of that quantity.

- Hence, dimensional formula of a physical quantity may be defined as the expression that indicates which of the fundamental units of mass; length and time enter into the derived unit of that quantity and with what powers. In other words, the dimensional formula tells us.

- 1) How and which of the fundamental units have been used to determine the quantity.

- 2) The nature of the dependence.

The System International, or SI units.

- In this system, lengths are measured in meters, masses in kilograms, time in seconds, and electric charge in coulombs.

- To clarify the distinction between dimensions and units, here are some examples:

1. Length has dimensions of L, by definition. In SI units we measure it in meters, abbreviated in this book as m.
2. Mass has dimensions of M, by definition. In SI units we measure it in kilograms, abbreviated as kg.
3. Time has dimensions of T, by definition. In SI units we measure it in seconds, abbreviated as s.
4. Velocity has dimensions of LT^{-1}. In SI units we measure it in $m\ s^{-1}$ (pronounced “meters per second”).
5. Acceleration has dimensions of LT^{-2}. In SI units we measure it in $m\ s^{-2}$.
6. Force has dimensions of MLT^{-2}. In SI units we measure it in $kg\ m\ s^{-2}$, which we also call newtons and abbreviate as N.

7. Energy has dimensions of ML2T−2. In SI units we measure it in kg m2 s−2, which we also call joules and abbreviate as J.

8. Electrical charge has dimensions of Q, by definition.

In SI units we measure it in units of coulombs, symbol C.

The flow rate of charge, or electric current, then must have dimensions of QT−1.

In SI units we measure it in units of coulombs per second, or coul s−1, also called amperes, abbreviated as A.

No.	Base Quantity	Unit (SI)	Symbol
1.	Length	metre	m
2.	Mass	kilogram	kg
3.	Time	second	s
4.	Electric current	ampere	A
5.	Temperature	kelvin	K
6.	Luminous intensity	candela	cd
7.	Amount of substance	mole	mol

No.	Base Quantity	Symbol of Dimension
1.	Length	[L]
2.	Mass	[M]
3.	Time	[T]
4.	Electric current	[I]
5.	Temperature	[θ]
6.	Luminous intensity	[J]
7.	Amount of substance	[N]

Q 1.7. Show that the expression vf=vi+at is dimensionally correct,where vi is the velocity at t=0, a is acceleration and vf is the velocity at time t.

Solution:

Prove that

$$v_f = v_i + at$$

We have

$$[LT^{-1}] = [LT^{-1}] + [LT^{-2}][T]$$

$$[LT^{-1}] = [LT^{-1}] + [LT^{-1}]$$

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Dimensionally both sides are equal, hence given equation $v_f = v_i + at$ is dimensionally correct.

Q 1.9. Show that the famous "Einstein equation" E=mc^2 is dimensionally consistent.

Solution:

$$E = mc^2$$

Dimensions of $E = [ML^2T^{-2}]$

$$[ML^2T^{-2}] = [M][LT^{-1}]^2$$

$$= [M][L^2T^{-2}]$$

$$= [ML^2T^{-2}]$$

Hence $E = mc^2$ is dimensionally consistent

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Use of Dimensional Analysis

- The dimensional equations have the following uses:

- 1) Convention of one system of units into another.
- 2) Checking the correctness of any equation (i.e, the accuracy of various formulae).
- 3) To derive the relationship between different physical quantities involved in a phenomenon.
- 4) a) To find the dimensions and units of a given physical quantity in a given system of units.

b)To find the dimensions and units of physical constants or variables in a given equation.

Thank you

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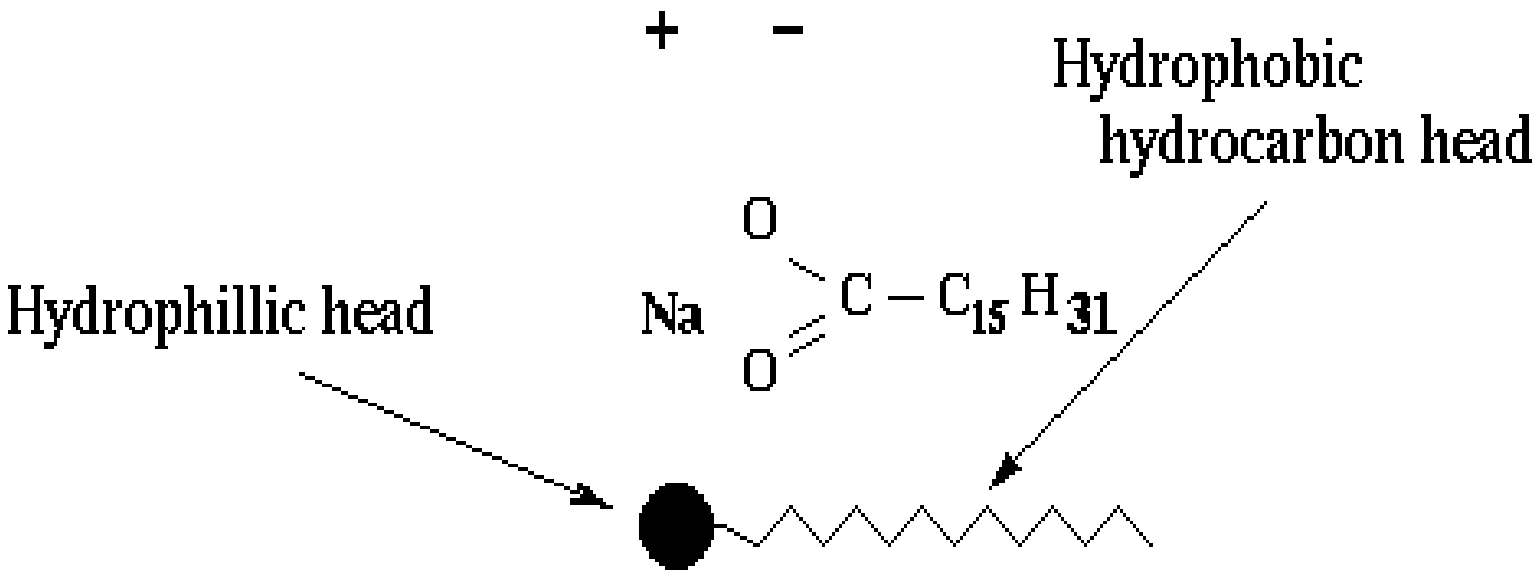
Amphiphile

Amphiphile

- An amphiphile is a chemical compound possessing both hydrophilic (water-loving, polar) and lipophilic (fat-loving) properties. Such a compound is called amphiphilic or amphipathic.

- This forms the basis for a number of areas of research in chemistry and biochemistry, notably that of lipid polymorphism.

- Common amphiphilic substances are soaps, detergents and lipoproteins.



A typical amphiphillic molecule.



Structure and properties

- The lipophilic group is typically a large hydrocarbon moiety, such as a long chain of the form CH3(CH2)n, with n > 4. The hydrophilic group falls into one of the following categories:

1. Charged groups

- **Anionic**. Examples, with the lipophilic part of the molecule represented by an R, are:
- carboxylates: RCO2−;
- sulfates: RSO4−;
- sulfonates: RSO3−.
- phosphates: The charged functionality in phospholipids.
- **Cationic.** Examples: ammoniums: RNH3+.

2. Polar, uncharged groups.

Examples are alcohols with large R groups, such as diacyl glycerol (DAG), and oligoethyleneglycols with long alkyl chains.

- Often, amphiphilic species have several lipophilic parts, several hydrophilic parts, or several of both. Proteins and some block copolymers are such examples.

- Amphiphilic compounds have lipophilic (typically hydrocarbon) structures and hydrophilic polar functional groups (either ionic or uncharged).

- As a result of having both lipophilic and hydrophilic portions, some amphiphilic compounds may dissolve in water and to some extent in non-polar organic solvents.

- When placed in an immiscible biphasic system consisting of aqueous and organic solvents, the amphiphilic compound will partition the two phases.

- The extent of the hydrophobic and hydrophilic portions determines the extent of partitioning.

Biological role

- Phospholipids, a class of amphiphilic molecules, are the main components of biological membranes.

- The amphiphilic nature of these molecules defines the way in which they form membranes.

- They arrange themselves into bilayers, by positioning their polar groups towards the surrounding aqueous medium, and their lipophilic chains towards the inside of the bilayer, defining a non-polar region between two polar ones.

Biological role

- Although phospholipids are principal constituents of biological membranes, there are other constituents, such as cholesterol and glycolipids, which are also included in these structures and give them different physical and biological properties.
- Many other amphiphilic compounds, such as pepducins, strongly interact with biological membranes by insertion of the hydrophobic part into the lipid membrane, while exposing the hydrophilic part to the aqueous medium, altering their physical behavior and sometimes disrupting them.

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Micelle

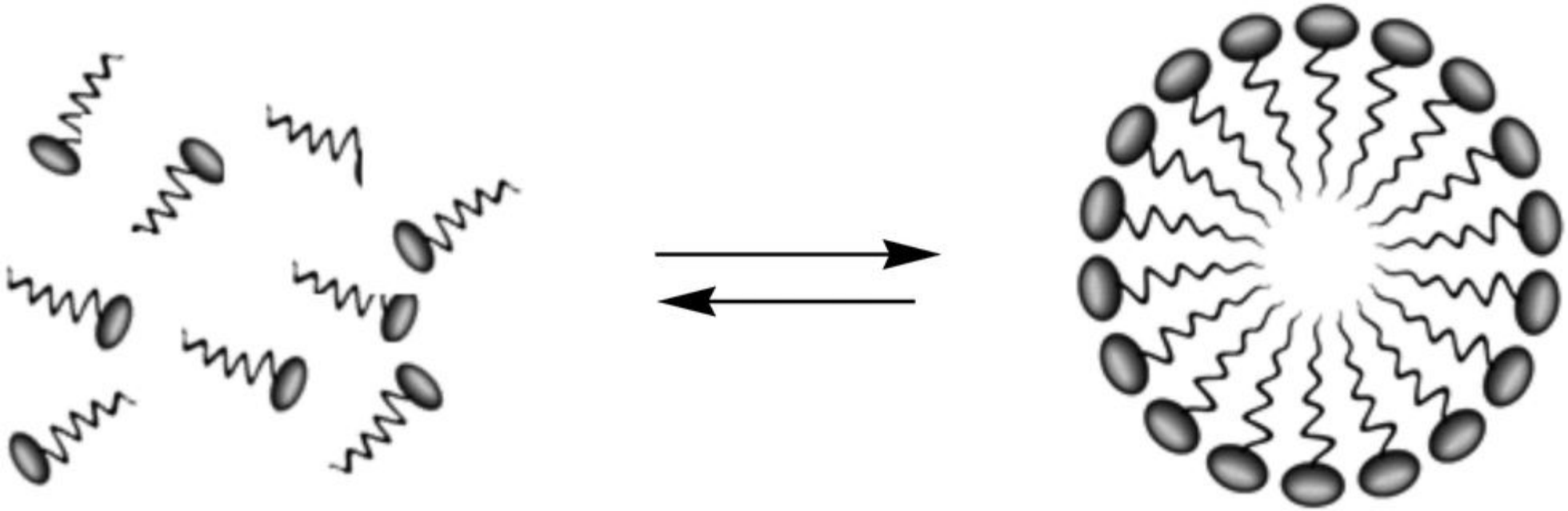
Micelle

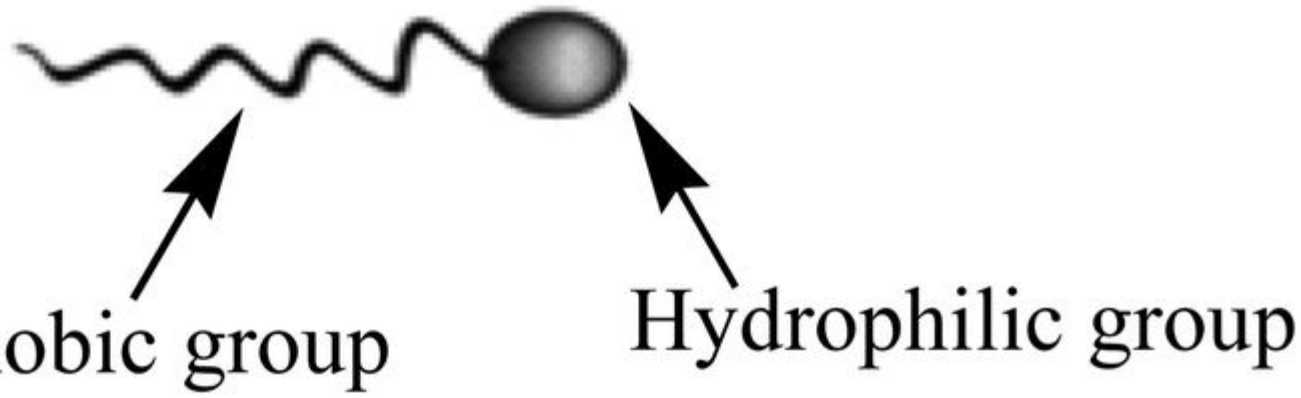
- A micelle is an aggregate (or supramolecular assembly) of surfactant molecules dispersed in a liquid colloid.

- A typical micelle in aqueous solution forms an aggregate with the hydrophilic "head" regions in contact with surrounding solvent, sequestering the hydrophobic single-tail regions in the micelle center. This phase is caused by the packing behavior of single-tail lipids in a bilayer.

- The difficulty filling all the volume of the interior of a bilayer, while accommodating the area per head group forced on the molecule by the hydration of the lipid head group, leads to the formation of the micelle. This type of micelle is known as a normal-phase micelle (oil-in-water micelle).

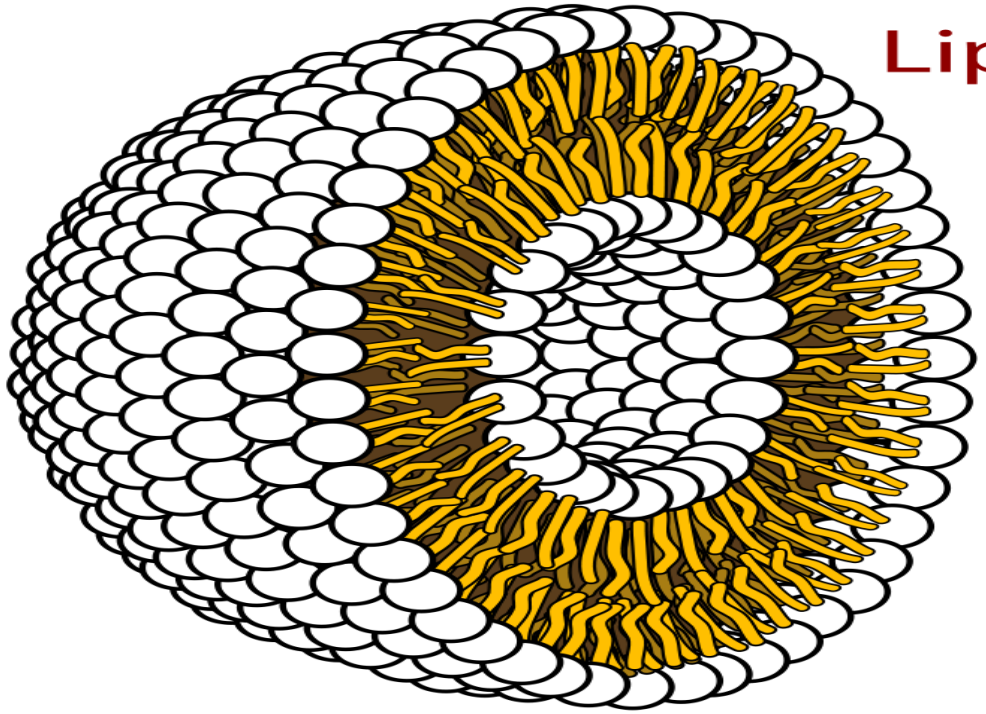
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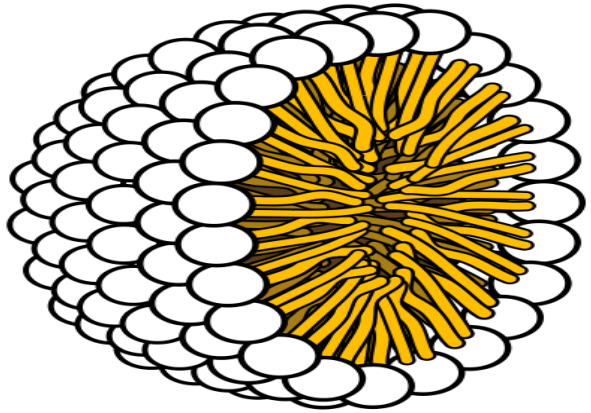


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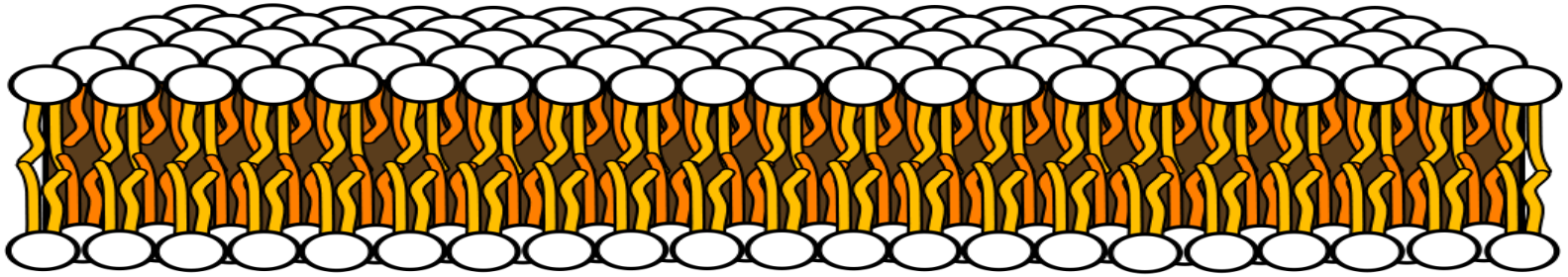
Liposome



Micelle



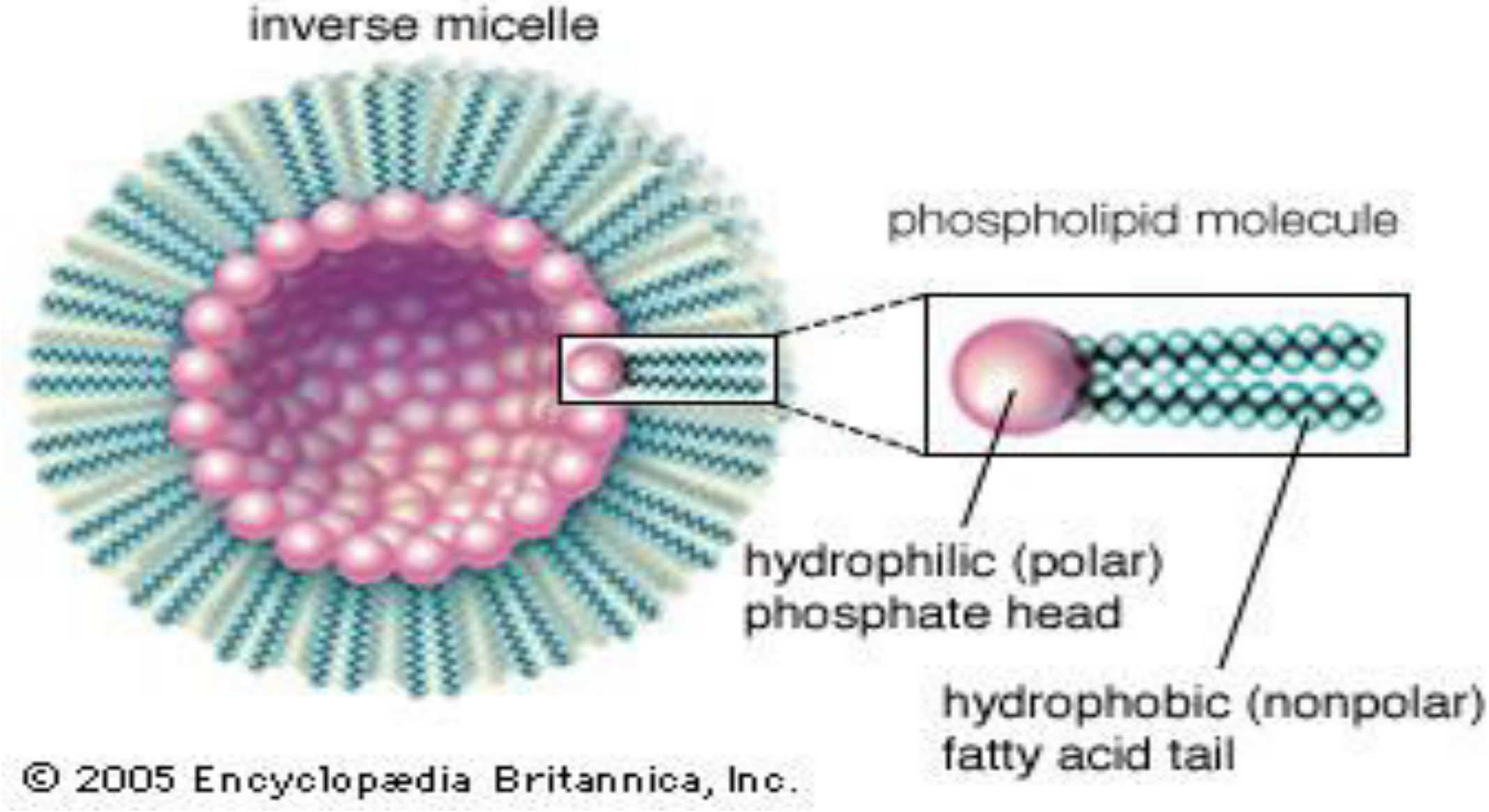
Bilayer sheet



Inverse micelles

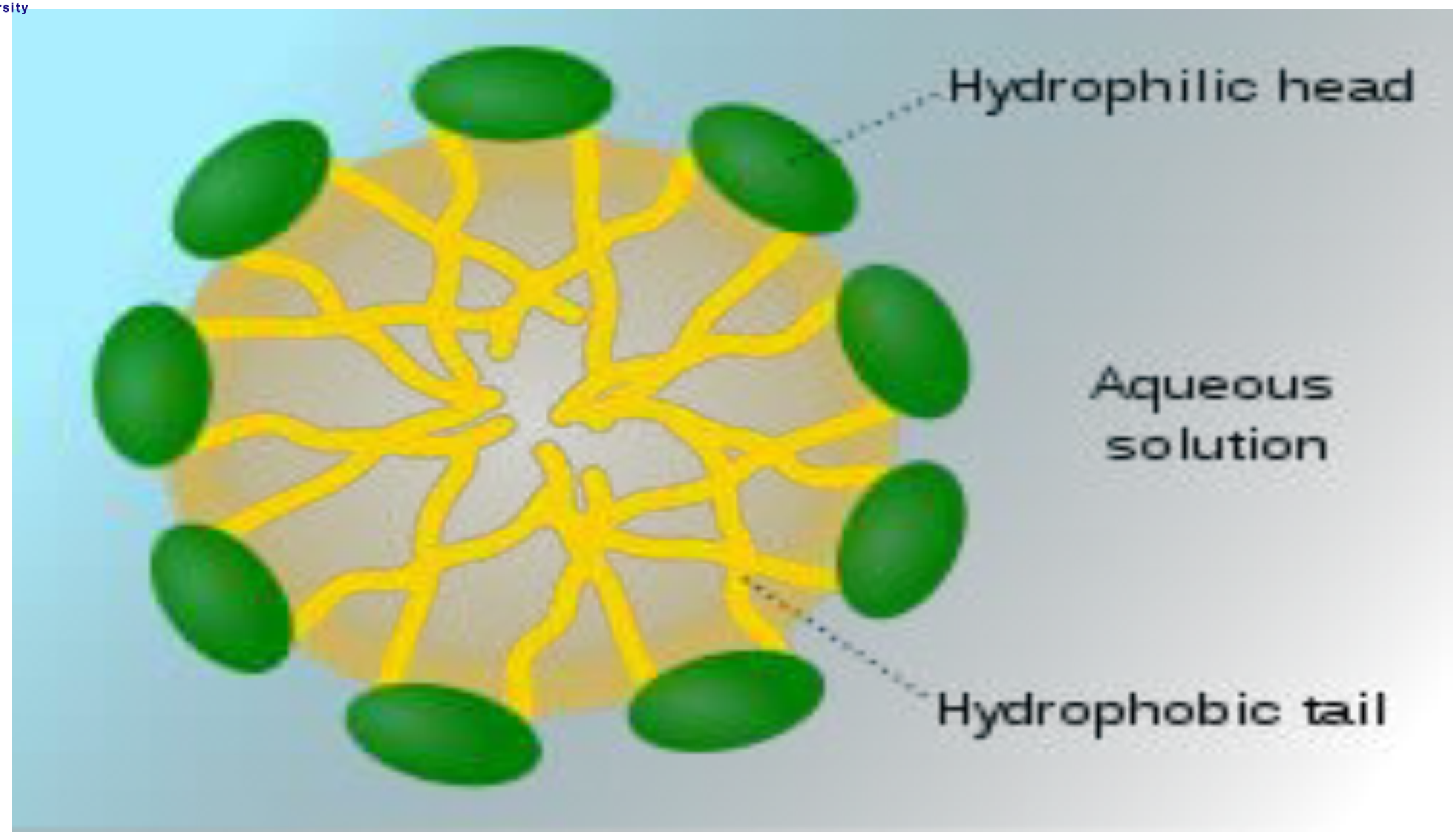
- Inverse micelles have the head groups at the centre with the tails extending out (water-in-oil micelle). Micelles are approximately spherical in shape.
- Other phases, including shapes such as ellipsoids, cylinders, and bilayers, are also possible.
- The shape and size of a micelle are a function of the molecular geometry of its surfactant molecules and solution conditions such as surfactant concentration, temperature, pH, and ionic strength.
- The process of forming micelles is known as micellization and forms part of the phase behavior of many lipids according to their polymorphism.

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Solvation

- Individual surfactant molecules that are in the system but are not part of a micelle are called "monomers".
- Micelles represent a molecular assembly, in which the individual components are thermodynamically in equilibrium with monomers of the same species in the surrounding medium.
- In water, the hydrophilic "heads" of surfactant molecules are always in contact with the solvent, regardless of whether the surfactants exist as monomers or as part of a micelle.
- However, the lipophilic "tails" of surfactant molecules have less contact with water when they are part of a micelle—this being the basis for the energetic drive for micelle formation


Hydrophilic head
Aqueous solution
Hydrophobic tail

- In a micelle, the hydrophobic tails of several surfactant molecules assemble into an oil-like core, the most stable form of which having no contact with water.

- By contrast, surfactant monomers are surrounded by water molecules that create a "cage" or solvation shell connected by hydrogen bonds. This water cage is similar to a clathrate and has an ice-like crystal structure and can be characterized according to the hydrophobic effect.

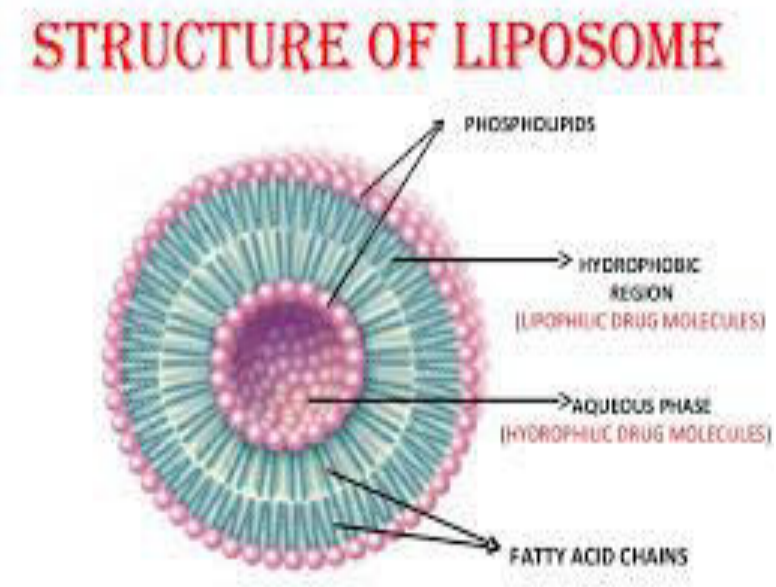
- The extent of lipid solubility is determined by the unfavorable entropy contribution due to the ordering of the water structure according to the hydrophobic effect.

Thank you

Liposome

Liposome

- Liposome is a spherical vesicle having at least one lipid bilayer.
- The liposome can be used as a vehicle for administration of nutrients and pharmaceutical drugs.
- Liposomes can be prepared by disrupting biological membranes (such as by sonication).



- Liposomes are most often composed of phospholipids, especially phosphatidyl choline, but may also include other lipids, such as egg phosphatidyl ethanolamine, so long as they are compatible with lipid bilayer structure.

- A liposome design may employ surface ligands for attaching to unhealthy tissue.

- The major types of liposomes are the multi-lamellar vesicle (MLV, with several lamellar phase lipid bilayers), the small uni-lamellar liposome vesicle (SUV, with one lipid bilayer), the large uni-lamellar vesicle (LUV), and the cochleate vesicle.

- A less desirable form are multi-vesicular liposomes in which one vesicle contains one or more smaller vesicles.

- Liposomes should not be confused with lysosomes, or with micelles and reverse micelles composed of monolayers.

- A liposome has an aqueous solution core surrounded by a hydrophobic membrane, in the form of a lipid bilayer; hydrophilic solutes dissolved in the core cannot readily pass through the bilayer.

- Hydrophobic chemicals associate with the bilayer.

- A liposome can be hence loaded with hydrophobic and/or hydrophilic molecules.

- To deliver the molecules to a site of action, the lipid bilayer can fuse with other bilayers such as the cell membrane, thus delivering the liposome contents; this is a complex and non-spontaneous event.

- By preparing liposomes in a solution of DNA or drugs (which would normally be unable to diffuse through the membrane) they can be (indiscriminately) delivered past the lipid bilayer, but are then typically distributed non-homogeneously.

- Liposomes are used as models for artificial cells.
- Liposomes can also be designed to deliver drugs in other ways.
- Liposomes that contain low (or high) pH can be constructed such that dissolved aqueous drugs will be charged in solution (i.e., the pH is outside the drug's pI range).
- As the pH naturally neutralizes within the liposome (protons can pass through some membranes), the drug will also be neutralized, allowing it to freely pass through a membrane.
- These liposomes work to deliver drug by diffusion rather than by direct cell fusion.

- The use of liposomes for transformation or transfection of DNA into a host cell is known as lipofection.

- In addition to gene and drug delivery applications, liposomes can be used as carriers for the delivery of dyes to textiles, pesticides to plants, enzymes and nutritional supplements to foods, and cosmetics to the skin.
- Liposomes are also used as outer shells of some microbubble contrast agents used in contrast-enhanced ultrasound.

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Enzymes

Enzymes

- Enzymes are very efficient catalysts for biochemical reactions. They speed up reactions by providing an alternative reaction pathway of lower activation energy.

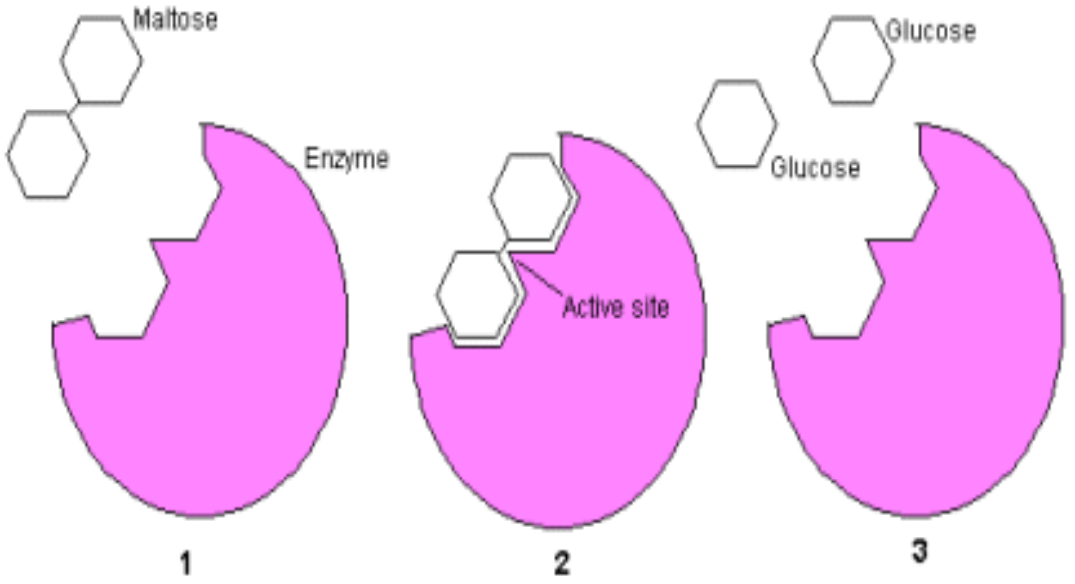
- Enzymes are made from **amino acids**, and they are proteins. When an enzyme is formed, it is made by stringing together between 100 and 1,000 amino acids in a very specific and unique order.

- The chain of amino acids then folds into a unique shape. That shape allows the enzyme to carry out specific chemical reactions -- an enzyme acts as a very efficient catalyst for a specific chemical reaction. The enzyme speeds that reaction up tremendously.

For example, the sugar maltose is made from two glucose molecules bonded together.

The enzyme **maltase is** shaped in such a way that it can break the bond and free the two glucose pieces.

The only thing maltase can do is break maltose molecules, but it can do that very rapidly and efficiently.



Maltose is made of two glucose molecules bonded together (1). The maltase enzyme is a protein that is perfectly shaped to accept a maltose molecule and break the bond (2). The two glucose molecules are released (3). A single maltase enzyme can break in excess of 1,000 maltose bonds per second, and will only accept maltose molecules.

- Inside a bacterium there are about 1,000 types of enzymes (lactase being one of them).

- All of the enzymes float freely in the cytoplasm waiting for the chemical they recognize to float by.

- There are hundreds or millions of copies of each different type of enzyme, depending on how important a reaction is to a cell and how often the reaction is needed.

- These enzymes do everything from breaking glucose down for energy to building cell walls, constructing new enzymes and allowing the cell to reproduce. Enzymes do all of the work inside cells.

Many enzymes consist of a protein and a non-protein (called the **cofactor**). The proteins in enzymes are usually globular.

Cofactors may be:

- organic groups that are permanently bound to the enzyme (**prosthetic groups**)
- cations - positively charged metal ions (**activators**), which temporarily bind to the active site of the enzyme, giving an intense positive charge to the enzyme's protein
- organic molecules, usually vitamins or made from vitamins (**coenzymes**), which are not permanently bound to the enzyme molecule, but combine with the enzyme-substrate complex temporarily.

How enzymes work

- For two molecules to react they must collide with one another. They must collide in the right direction (orientation) and with sufficient energy. Sufficient energy means that between them they have enough energy to overcome the energy barrier to reaction. This is called the activation energy.

- Enzymes have an active site. This is part of the molecule that has just the right shape and functional groups to bind to one of the reacting molecules. The reacting molecule that binds to the enzyme is called the substrate.

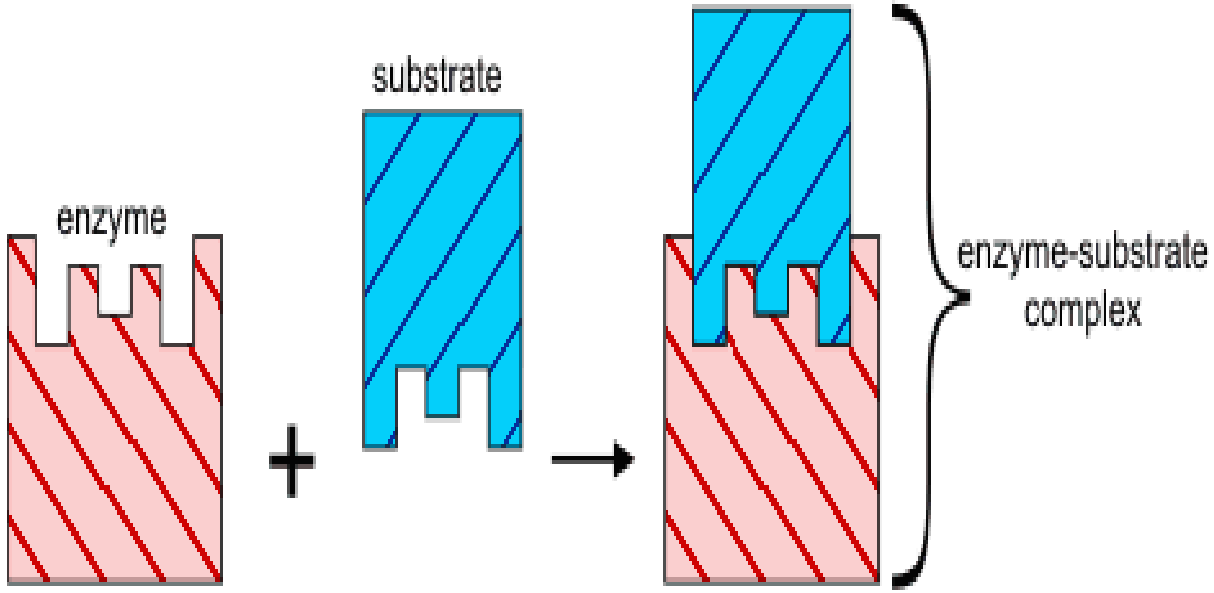
- An enzyme-catalysed reaction takes a different 'route'. The enzyme and substrate form a reaction intermediate. Its formation has a lower activation energy than the reaction between reactants without a catalyst.

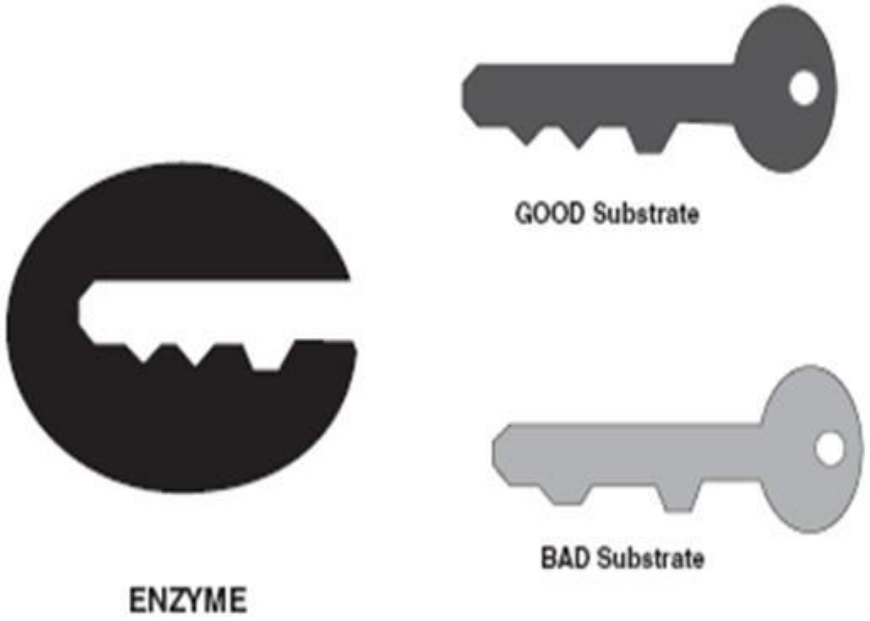
Lock and key hypothesis

- The substrate simply fits into the active site to form a reaction intermediate. Enzyme and Substrate fit together to make enzyme-substrate complex

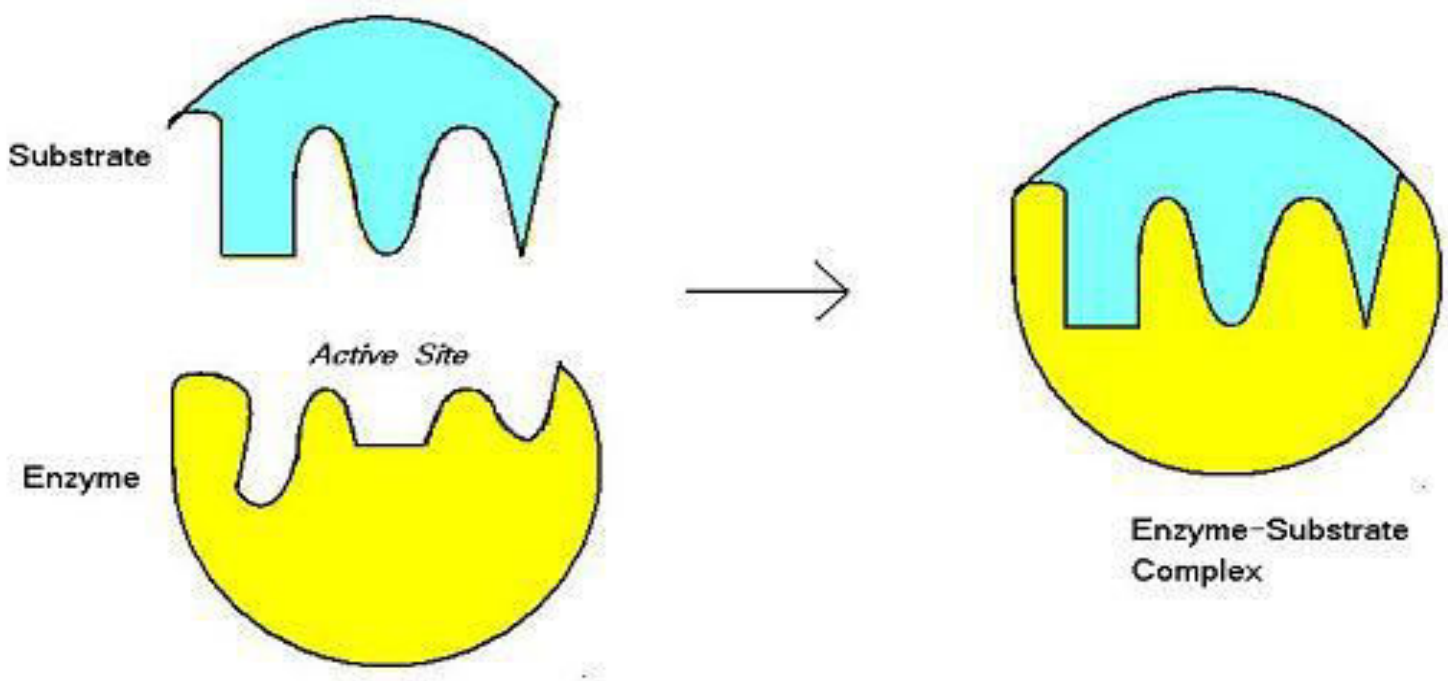
Induced fit hypothesis

- In this model the enzyme molecule changes shape as the substrate molecules gets close. The change in shape is 'induced' by the approaching substrate molecule.





Lock and key hypothesis



Induced-fit Model. - The enzyme active site forms a complementary shape to the substrate after binding.

- Enzymes are widely used commercially, for example in the detergent, food and brewing industries.
- Protease enzymes are used in 'biological' washing powders to speed up the breakdown of proteins in stains like blood and egg.
- Pectinase is used to produce and clarify fruit juices.

Problems using enzymes commercially include:

- they are water soluble which makes them hard to recover
- some products can inhibit the enzyme activity (feedback inhibition)

Immobilized enzymes

Enzymes can be immobilized by fixing them to a solid surface. This has a number of commercial advantages:

- the enzyme is easily removed
- the enzyme can be packed into columns and used over a long period
- speedy separation of products reduces feedback inhibition
- thermal stability is increased allowing higher temperatures to be used
- higher operating temperatures increase rate of reaction

There are four principal methods of immobilization currently in use:

- covalent bonding to a solid support
- adsorption onto an insoluble substance
- entrapment within a gel
- encapsulation behind a selectively permeable membrane

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Enzymes II

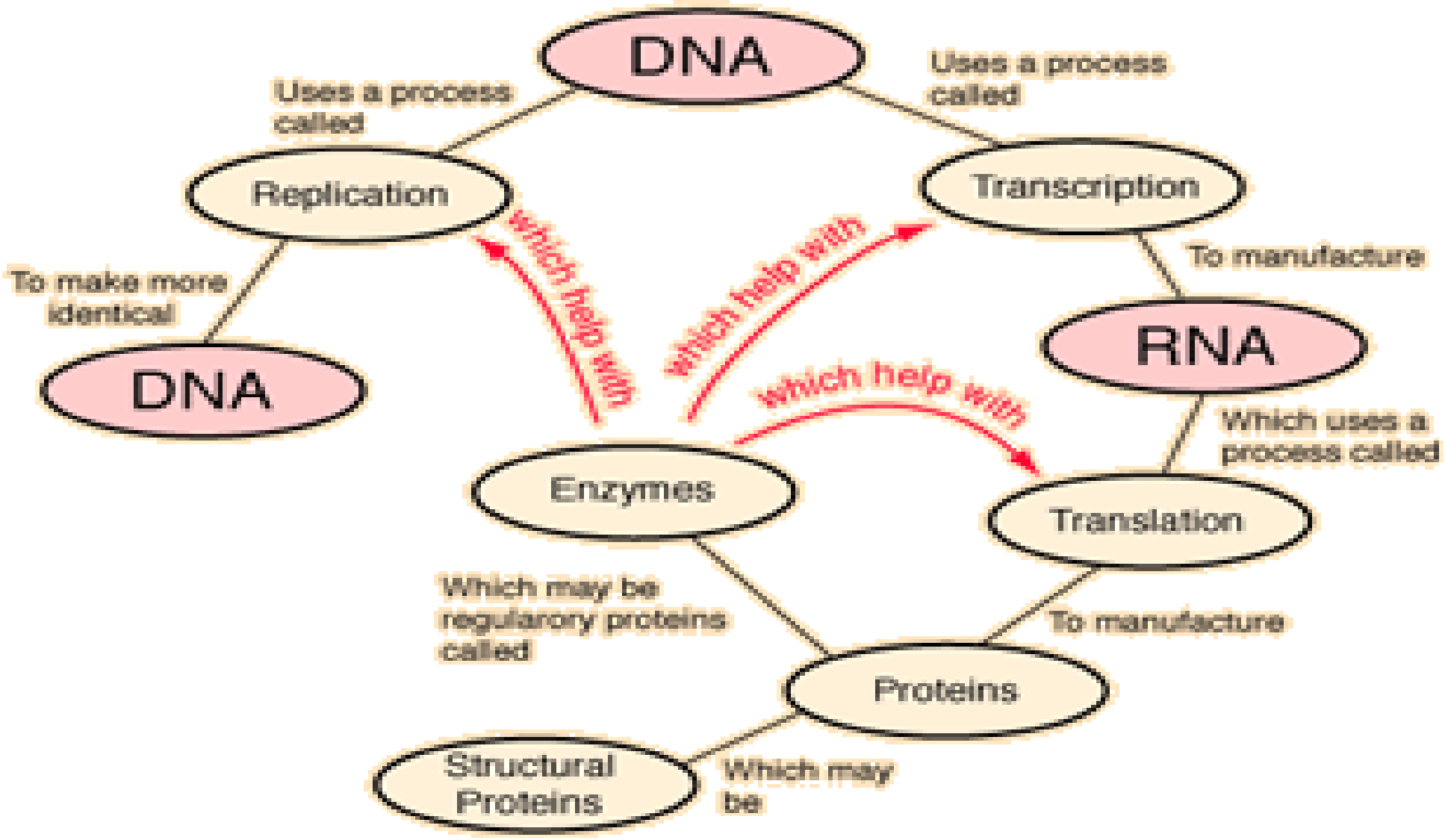
- Enzymes are regulatory proteins that assist in many of the reactions and processes in living cells.

- The DNA of a cell contains the master plan for the creation of the enzymes that are needed for life processes, and the enzymes assist in the replication of DNA and in fact help with making other copies of themselves and other proteins.

- One perspective on enzymes is that they facilitate many chemical reactions that otherwise would be too slow to sustain living systems.

- Examples have been found where an enzyme increased the rate of a biochemical reaction by more than a billion-fold!

- They may have binding sites which hold two or more other molecules to facilitate their binding to one another and also a binding site for ATP to provide the energy necessary for accomplishing the bonding.


DNA
Uses a process called
Uses a process called
Replication
Transcription
To make more identical
DNA
which help with
which help with
which help with
To manufacture
RNA
Which uses a process called
Enzymes
Translation
Which may be regularory proteins called
To manufacture
Proteins
Structural Proteins
Which may be

Inside a cell, hundreds of highly specialized enzymes carry out extremely specific tasks that the cell needs to live its life. Some of the more amazing enzymes found inside cells include:

- **Energy enzymes**
- A set of 10 enzymes allows a cell to perform **glycolysis**. Another eight enzymes control the **citric-acid cycle** (also known as the Krebs cycle). These two processes together allow a cell to turn glucose and oxygen into adenosine triphosphate, or ATP.

- In an oxygen-consuming cell like E. coli or a human cell, one glucose molecule forms 36 ATP molecules (in something like a yeast cell, which lives its life without oxygen, only glycolysis occurs and it produces only two ATP molecules per glucose molecule). ATP is a fuel molecule that is able to power enzymes by performing "uphill" chemical reactions.

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Restriction enzymes

- Many bacteria are able to produce restriction enzymes, which recognize very specific patterns in DNA chains and break the DNA at those patterns.

- When a virus injects its DNA into a bacterium, the restriction enzyme recognizes the viral DNA and cuts it, effectively destroying the virus before it can reproduce.

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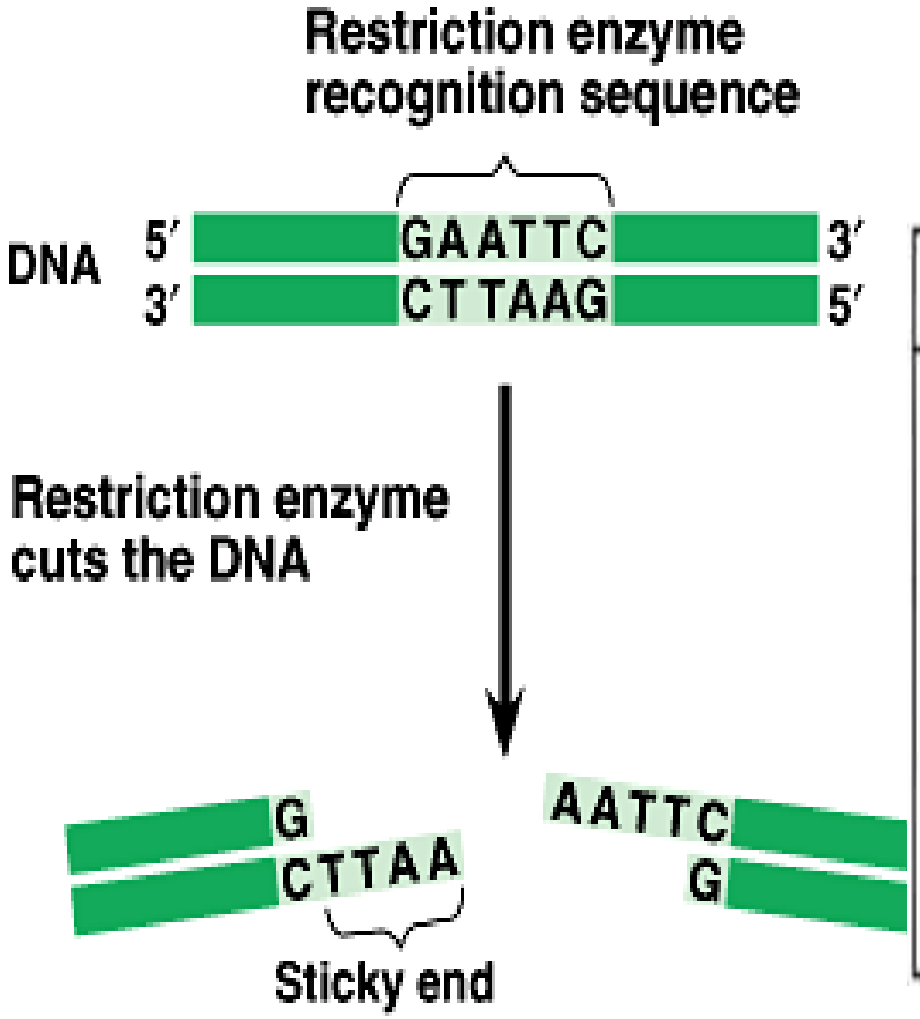
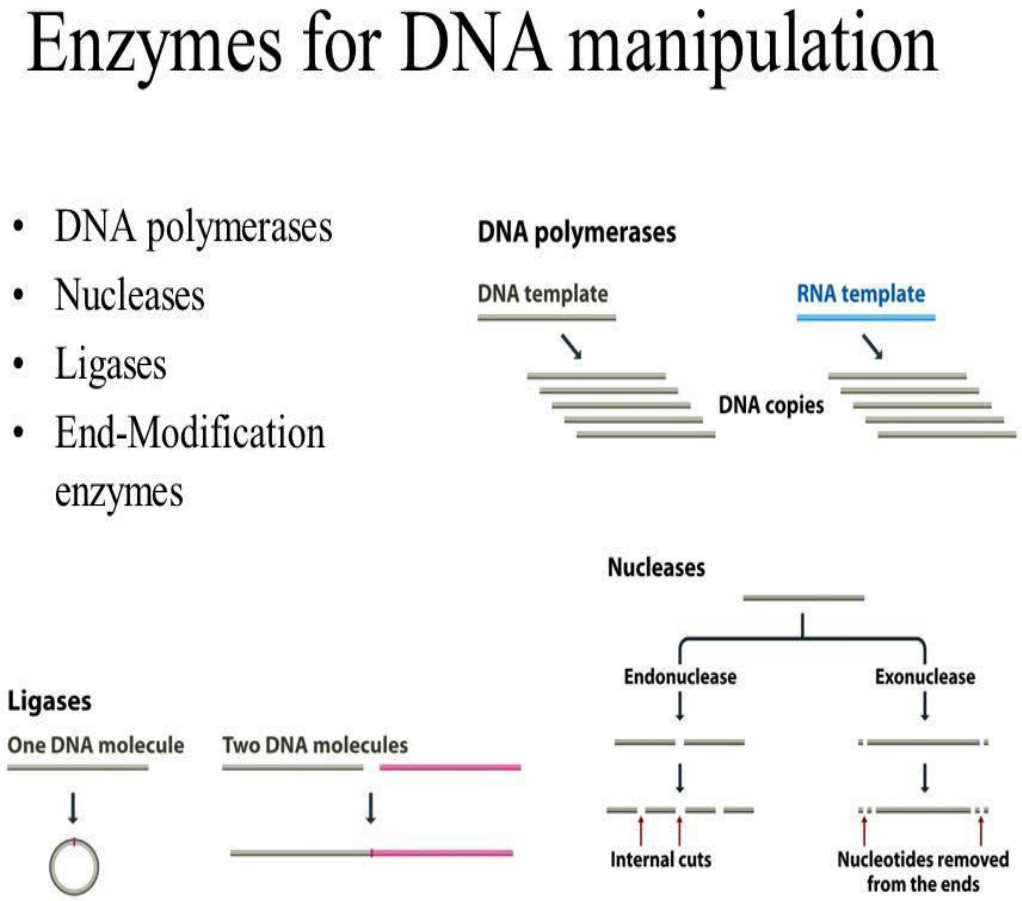


Table 13.1 Types of restriction endonucleases and their target sequences

	Name of restriction systems	Target sequence	Type of DNA end
1.	Bam HI (*Bacillus amyloliquifaciens*)	G↓GATCC	5′ sticky end
2.	EcoRI (*Escherichia coli*)	G↓AATTC	5′ sticky end
3.	Hae III (*Haemophilus aegyticus*)	GG↓GC	Blunt end
4.	Sma I (*Serratia marcescence*)	CCC↓GGG	Blunt end
5.	Pst I (*Providentia stuartii*)	CTGCA↓G	3′ sticky end
6.	Hind III (*Haemophilus influenzae*)	A↓AGCTT	5′ sticky end
7.	Alu I (*Arthrobacter luteus*)	A↓GCT	5′ sticky end

DNA-manipulation enzymes

- There are specialized enzymes that move along DNA strands and repair them.
- There are other enzymes that can untwist DNA strands to reproduce them (DNA polymerase).
- Still others can find small patterns on DNA and attach to them, blocking access to that section of DNA (DNA-binding proteins).



Enzyme-production enzymes

- All of these enzymes have to come from somewhere, so there are enzymes that produce the cell's enzymes.
- Ribonucleic acid (RNA), in three different forms (messenger RNA, transfer RNA and ribosomal RNA), is a big part of the process.

What Do Enzymes Do for You?

SECTOR	APPLICATION AREA	BENEFITS
• Detergents	Household washing and cleaning agents	Wash your clothes in cold water; make your teeth cleaner
• Textiles	Denim washing, silk polishing, leather goods softening	Stonewash your jeans; make cotton look and feel like silk; make your leather soft
• Food processing	Baking, brewing, fruit juice processing	Clarify your juice and beer; make bread better; turn corn starch into sugar syrup
• Pulp and paper	Starch conversion, pitch control, bleach-boosting, deinking, stickies control, slime control	Reduce production costs and improve quality

Enzyme	Applications
α-Amylase	Food processing
Avidin	In diagnostic kits
Cellulase	Production of alcohol from cellulose
β-glucanase	In brewing industry
β-glucuronidase	In diagnostic kits
Lignin peroxidase	In paper manufacture
Phytase	Improved phosphate utilization
Trypsin	Pharmaceutical
Xylanase	Biomass processing, paper and textile industries

Thank you

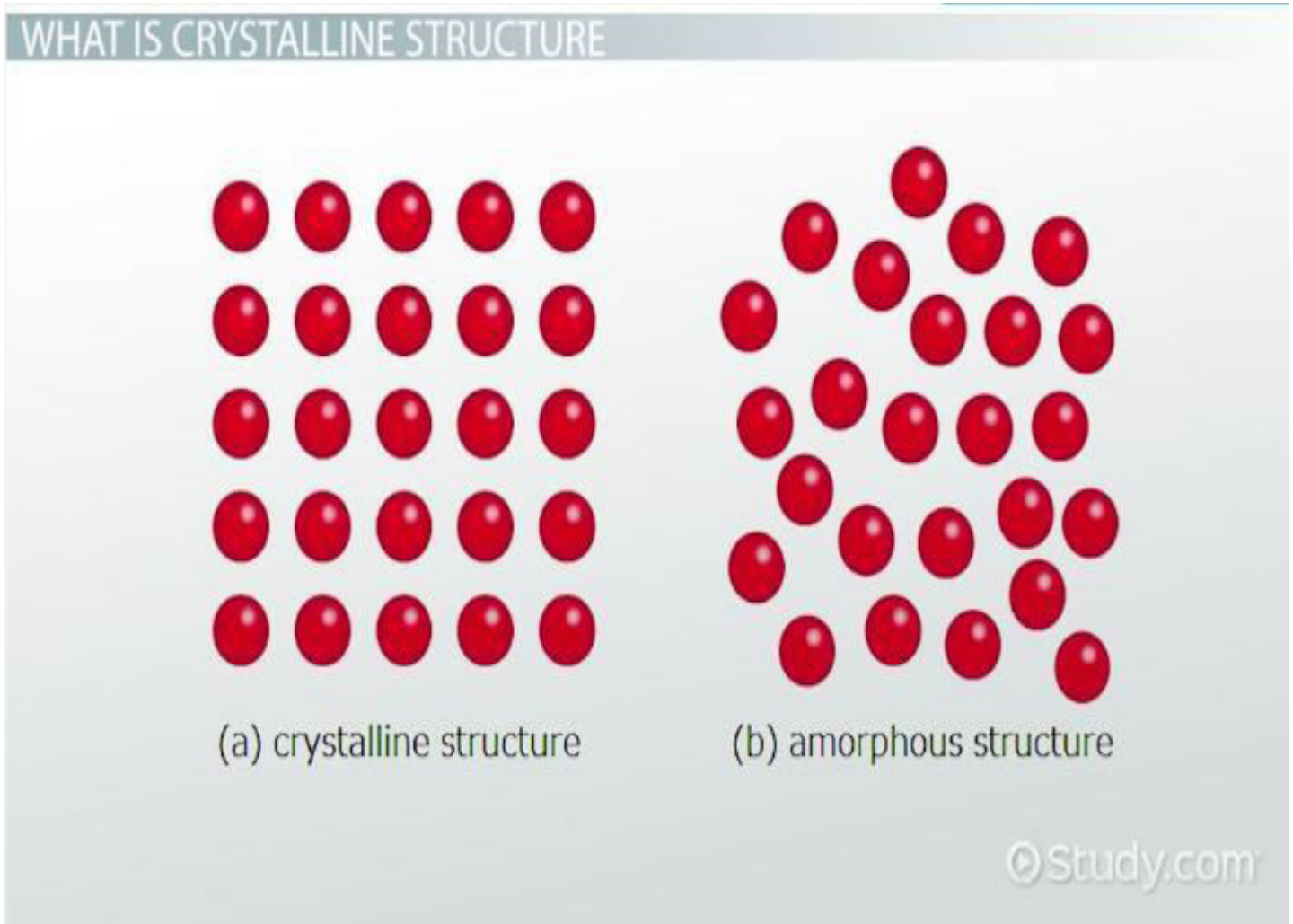
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Crystal engineering

Crystalline Structure

- A crystalline structure is any structure of ions, molecules, or atoms that are held together in an ordered, three-dimensional arrangement.
- Crystalline structure is one of two types of structural ordering of atoms, the other being the amorphous structure.
- The key difference in the crystalline and amorphous structure is the ordering of the structure.
- Crystalline structure can be thought of as the highest level of order that can exist in a material, while an amorphous structure is irregular and lacks the repeating pattern of a crystal lattice.

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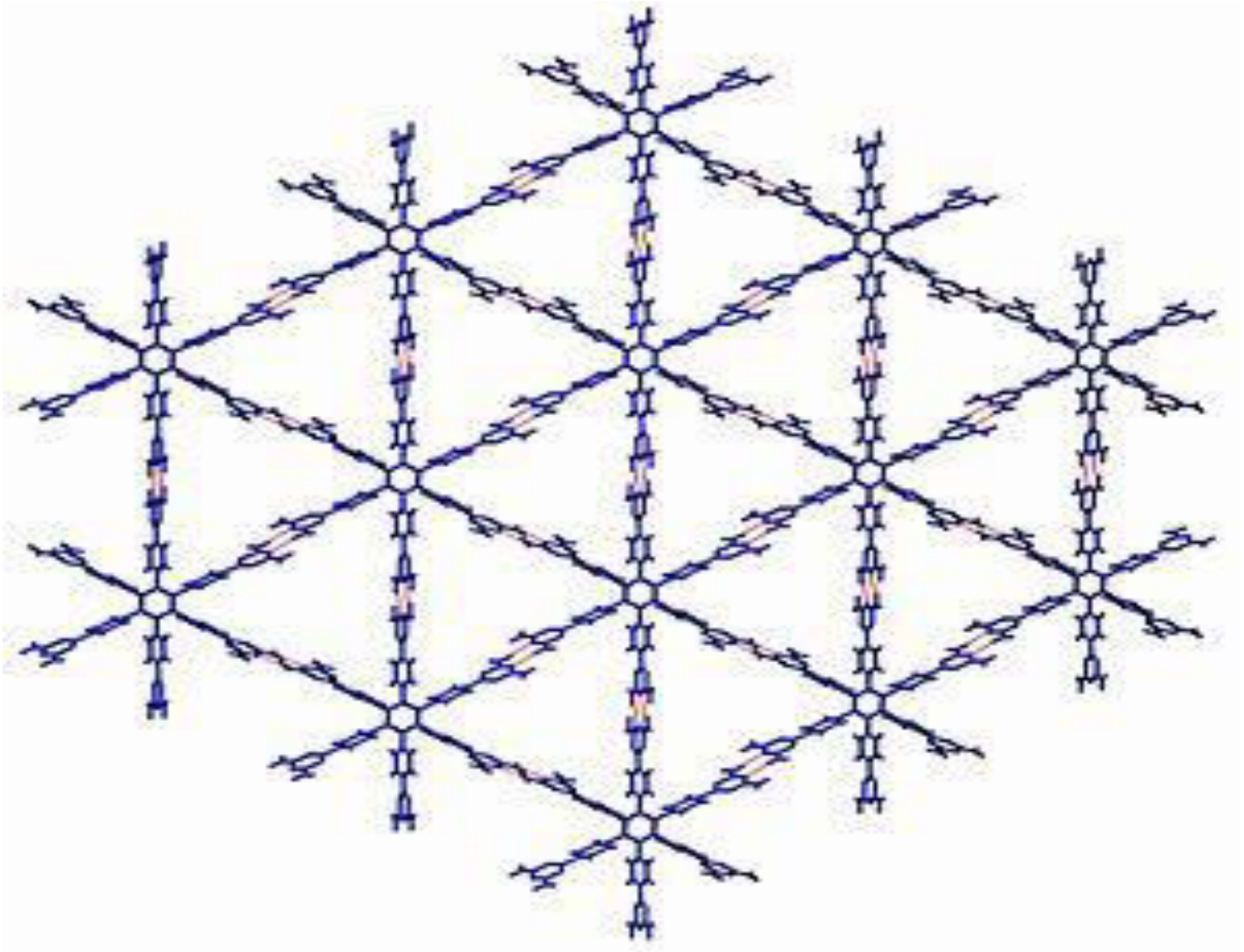
Crystal engineering

Crystal engineering is the design and synthesis of molecular solid state structures with desired properties, based on an understanding and use of intermolecular interactions.

The two main strategies currently in use for crystal engineering are based on:

- 1. hydrogen bonding
- 2. coordination bonding.

An example of crystal engineering using hydrogen bonding



History

- The term ‘crystal engineering’ was first used in 1971 by **Gerhard Schmidt** in connection with photo dimerization reactions in crystalline cinnamic acids.

- Since this initial use, the meaning of the term has broadened considerably to include many aspects of solid state supramolecular chemistry.

- Crystal engineering relies on -bonding to achieve the organization of molecules and ions in the solid state.
- Much of the initial work on purely organic systems focused on the use of hydrogen bonds, though with the recent extension to inorganic systems, the coordination bond has also emerged as a powerful tool.
- In addition to this, especially through studies during the last decade, the use of halogen bonds has proved beneficial in providing additional control in crystal design.
- Other intermolecular forces have all been exploited in crystal engineering studies, and ionic interactions can also be important.
- However, the two most common strategies in crystal engineering still employ only hydrogen bonds and coordination bonds.

- Molecular self-assembly is at the heart of crystal engineering, and it typically involves an interaction between complementary hydrogen bonding faces or a metal and a ligand.

- The supra-molecular synthon approach has been successfully applied in the synthesis of one dimensional tapes, two dimensional sheets and three dimensional structures.

- The CSD (Cambridge Structural Database) today contains atomic positional parameters for nearly 800 000 crystal structures, and this forms the basis for heuristic or synthon based or "experimental" crystal engineering.

- A major development in the field of crystal engineering in the last decade is related to the development of design strategies for bi-component and higher multi-component crystals (also known as co-crystals).

- The design of co-crystals is a difficult task as it involves recognition between different molecules which might be completely different in shape and size.

- Therefore, the more the number of components in a crystal, the more challenging it is to synthesize.

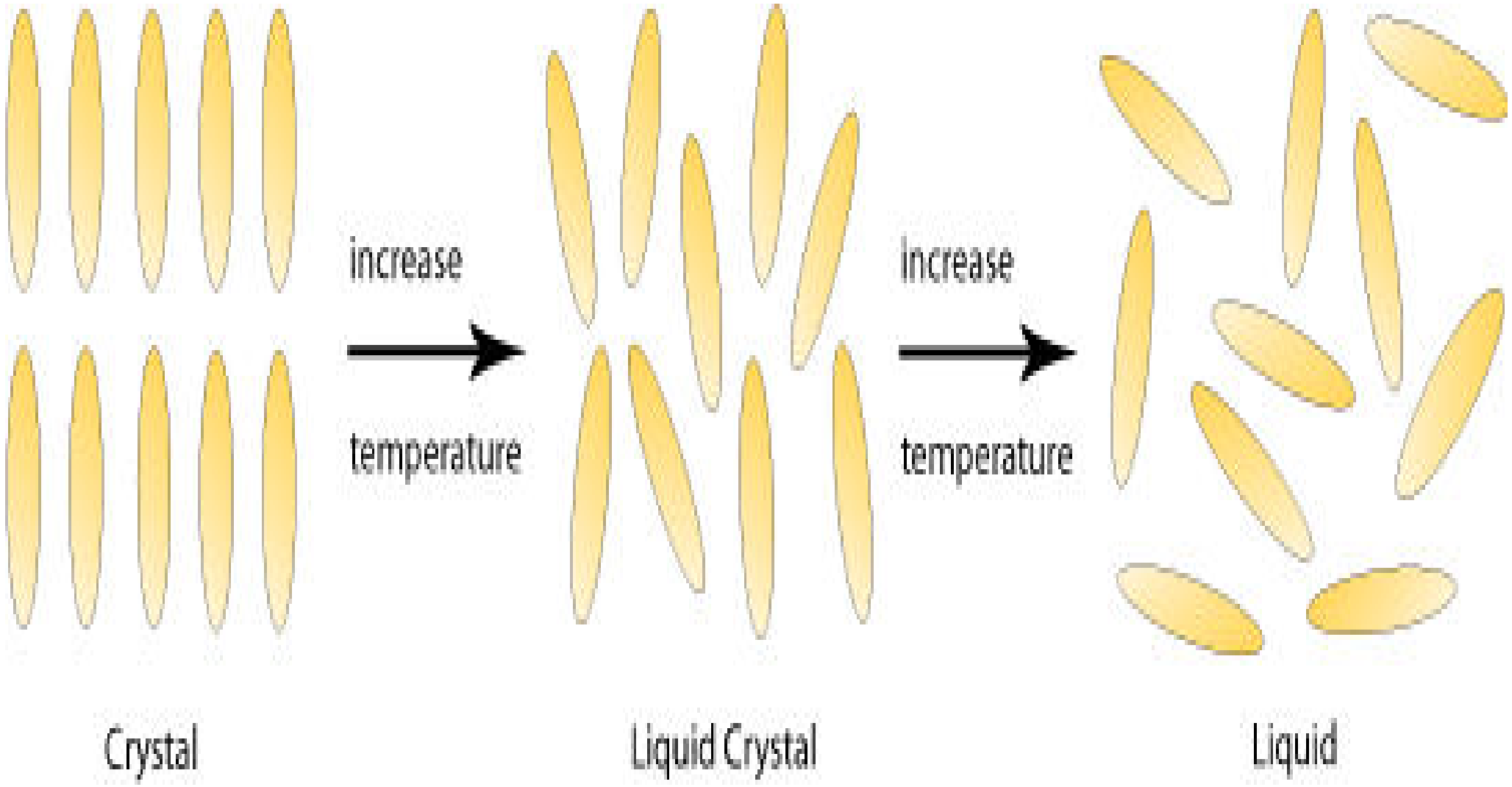
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Liquid crystals

Liquid crystal

- Liquid crystal, substance that blends the structures and properties of the normally disparate liquid and crystalline solid states.
- Liquids can flow, for example, while solids cannot, and crystalline solids possess special symmetry properties that liquids lack.
- Ordinary solids melt into ordinary liquids as the temperature increases—e.g., ice melts into liquid water. Some solids actually melt twice or more as temperature rises.
- Between the crystalline solid at low temperatures and the ordinary liquid state at high temperatures lies an intermediate state, the liquid crystal


increase
temperature
increase
temperature
Crystal
Liquid Crystal
Liquid

- Liquid crystals share with liquids the ability to flow but also display symmetries inherited from crystalline solids.

- The resulting combination of liquid and solid properties allows important applications of liquid crystals in the displays of such devices as wristwatches, calculators, portable computers, and flat-screen televisions.

- A true liquid is isotropic, meaning that its properties are uniform in all directions— the result of its molecules being in constant random motion.

- Crystalline solids, in contrast, are anisotropic; optical- and other properties such as thermal and electrical conductivity vary with direction.

- A liquid crystal phase has many of the physical attributes of a liquid, but its molecular units are sufficiently ordered to give rise to some anisotropy, most notably in their optical properties.

- As with so many scientific discoveries, it all started with an unexpected observation. In 1888, an **Austrian plant physiologist**, working in Prague, attempted to measure the melting point of a cholesterol derivative that he had extracted from a plant.

- To his surprise, he found that this substance appeared to have two melting points.

- At 145° C the crystalline solid first melted into a cloudy liquid, and then at 178° the cloudiness suddenly disappeared, leaving the clear, transparent liquid that one ordinarily expects after melting.

Properties of Liquid Crystals

- Liquid crystal phases are generally cloudy in appearance, which means that they scatter light in much the same way as colloids such as milk.

- This light scattering is a consequence of fluctuating regions of non-uniformity as small groups of molecules form and disperse.

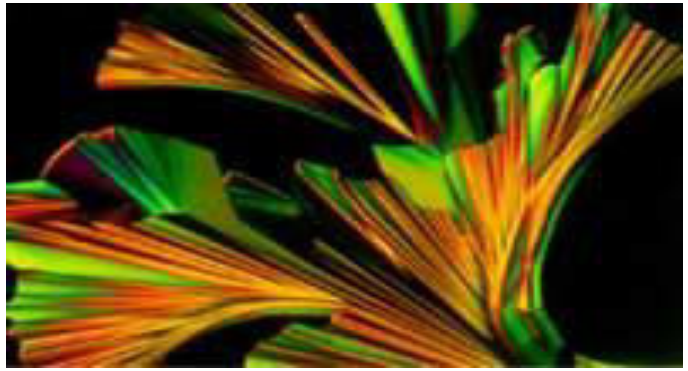


- The anisotropy of liquid crystals causes them to exhibit birefringence. That is, light that enters the crystal is broken up into two oppositely-polarized rays that travel at different velocities.

- Observation of a bi-refringent material between crossed polarizing filters reveals striking patterns and color effects.

Birefringent Materials.

Crystalline materials may have different indices of refraction associated with different crystallographic directions



- Liquid crystals, like all other kinds of matter, are subject to thermal expansion.

- As the temperature rises, the average spacing between the aligned molecules of a nematic phase increases, thus causing the e-ray to be increasingly retarded with respect to the o-ray.

- If a suitable liquid crystal mixture is painted onto the surface of a patient's body, it can often reveal the sites of infection or tumors, which cause increases or reductions in local blood flow giving rise to temperature anomalies.

- Inexpensive thermometers can be made by printing a succession of suitably formulated LC mixtures on a paper or plastic strip which is held in contact with the surface whose temperature is to be measured.

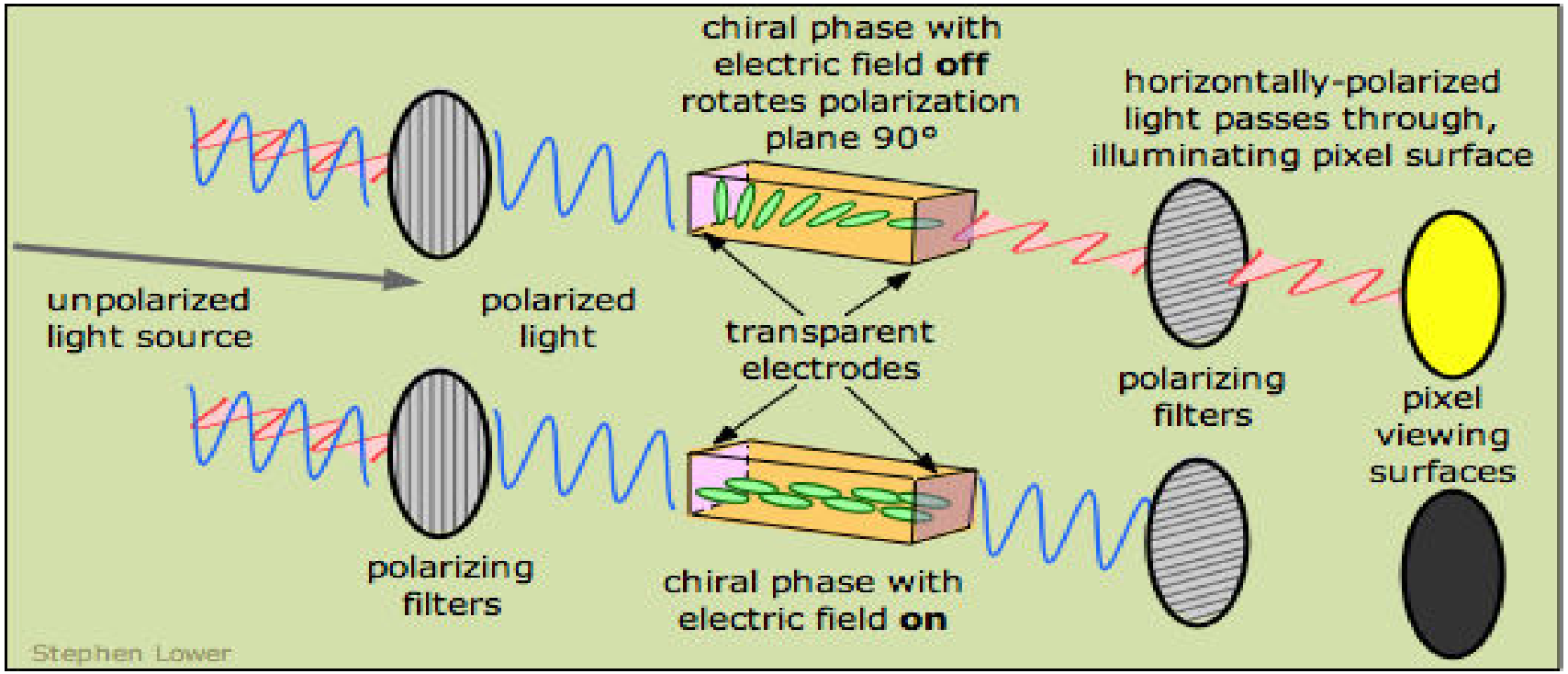
- The first working liquid crystal display ("LCD") was demonstrated at RCA in 1968.

- In the following year, James Fergason of Kent State University (OH) discovered the twisted nematic field effect which allowed a much higher-quality display and led to the first commercial LCD wristwatch in 1979.

- LCDs were first used in calculators in the late 1970s, but they are now widely encountered in computer- and television displays.

- The thickness of the chiral phase is such that polarized light passing through it is rotated by 90°, which corresponds to the orientation of the right-hand polarizing filter.

- So in this state, the light passes through and illuminates the pixel surface.

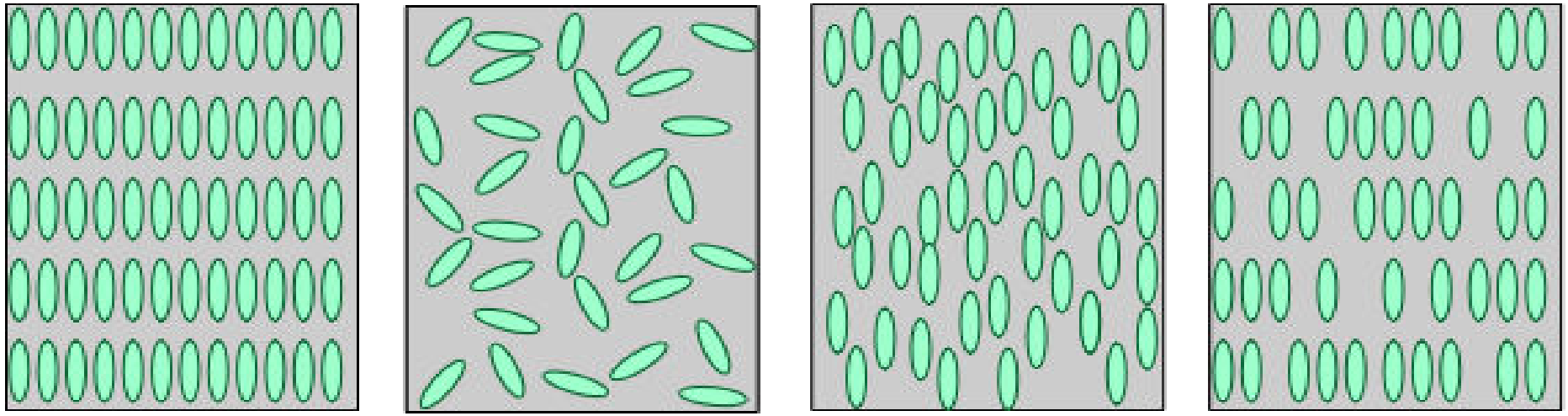


When an electric field is imposed on the liquid crystal phase, the component molecules line up in the field and the chirality is lost. Light passing through the cell does not undergo rotation of its polarization plane, and is therefore stopped by the right polarizing filter, turning the display off.

Thank you

Nematic and Smectic Phases

- There are many classes and sub-classes of liquid crystals, but for the purposes we will divide them into the two kinds depicted on the right side of this Figure below which also compares them with the two conventional condensed phases of matter:



Solid phase: orientation and periodicity

Liquid phase: no orientation or periodicity

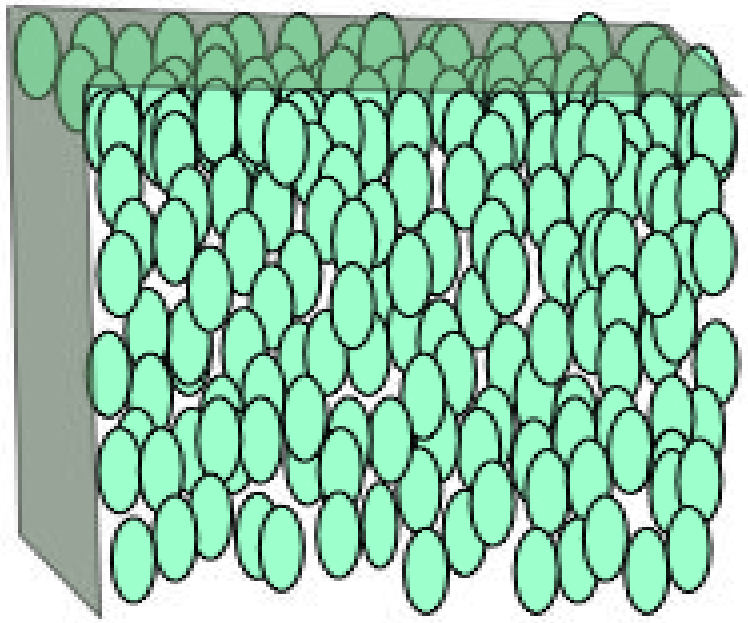
Nematic phase: orientation, no periodicity

Smectic phase: orientation with some periodicity

nematic phase

- In a *nematic phase* (the term means "thread-like") the molecules are aligned in the same direction but are free to drift around randomly, very much as in an ordinary liquid.

- Owing to their polarity, the alignment of the rod-like molecules can be controlled by applying an electric field; this is the physical basis for liquid crystal displays and certain other electrooptic devices.

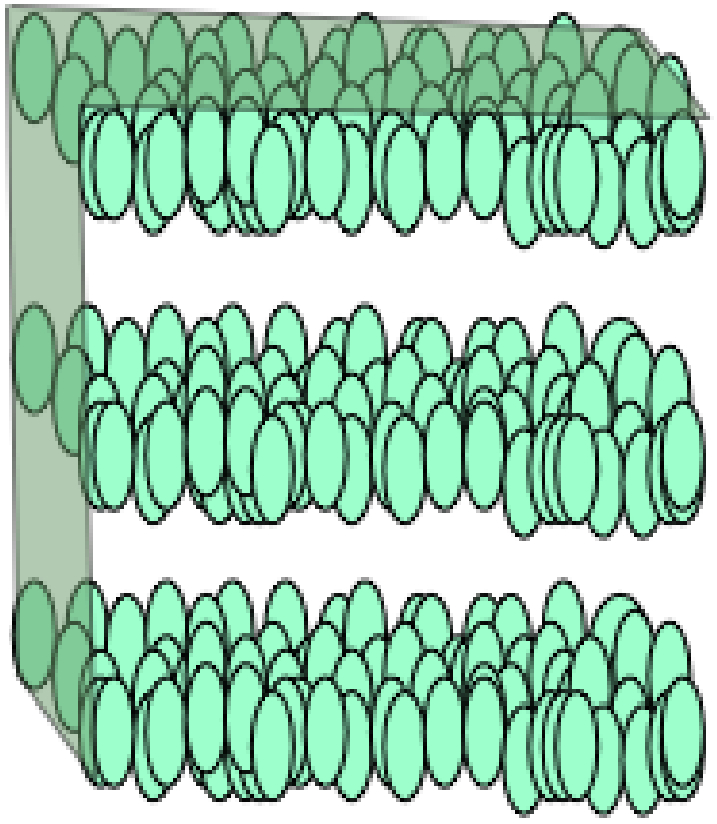


Active Nematic Liquid Crystal



smectic ("soap-like") phases

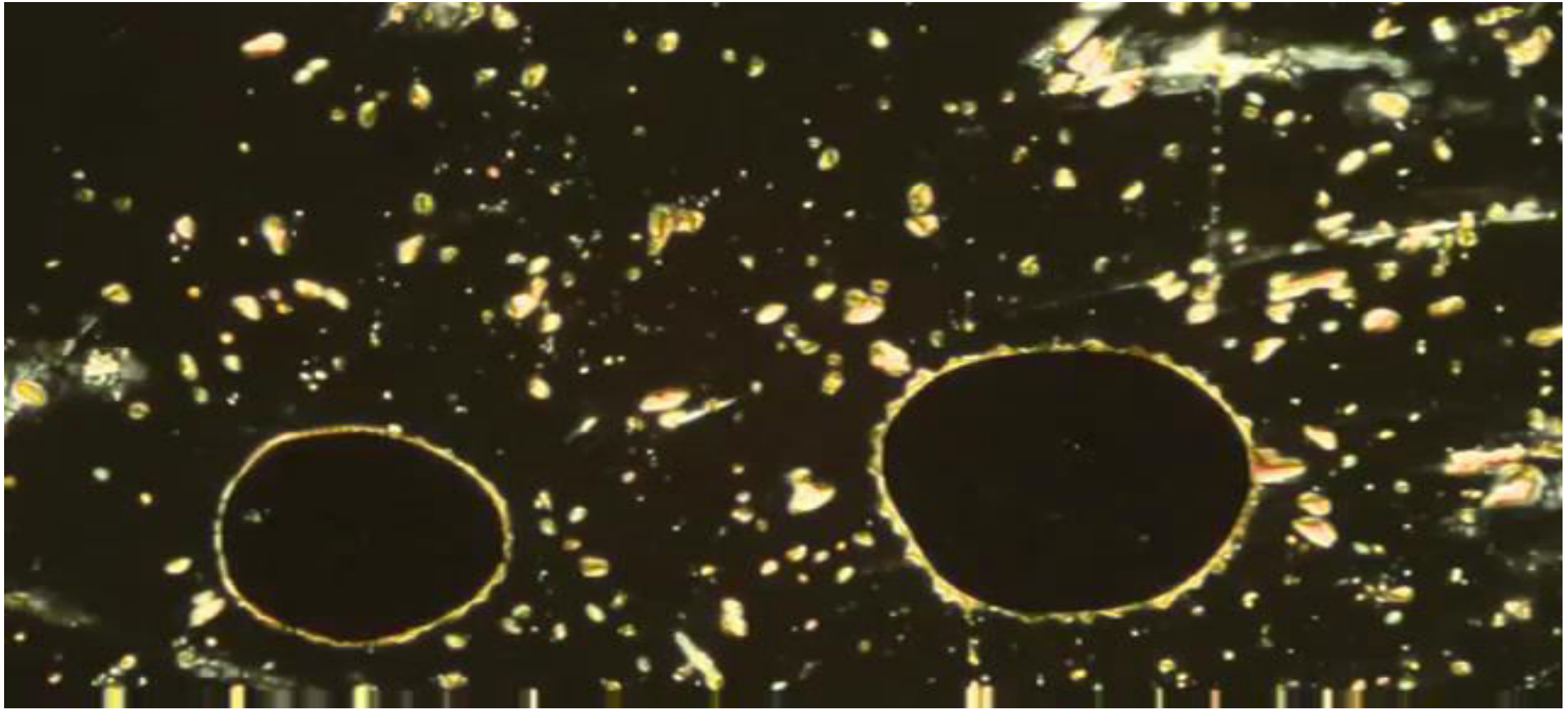
- In *smectic* ("soap-like") phases the molecules are arranged in layers, with the long molecular axes approximately perpendicular to the laminar planes.
- The only long-range order extends along this axis, with the result that individual layers can slip over each other (hence the "soap-like" nature) in a manner similar to that observed in graphite.
- Within a layer there is a certain amount of short-range order. There are a large number of sub-categories of smectic phases which we will not go into here.



smectic

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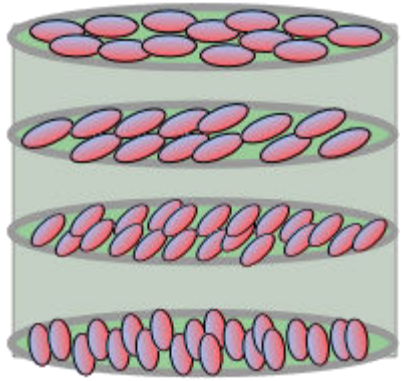
Smectic growth



Chiral phases

- Special cases of nematic and smectic phases are sometimes formed by molecules that display chirality — that is, they can exist in either left- or right-handed forms that cannot be superposed on each other.

- In the resulting *chiral phase*, successive molecules positioned along the long axis are rotated around this axis, giving rise to a periodicity that repeats itself at distances corresponding to a complete rotation.



chiral
(cholesteric)

- These twisted phases are able to rotate the plane of polarized light that passes along the axis. If the molecules are polar, this twisting can be turned off by imposing an external electric field at either end of the long axis.

- Besides the very important application of this property (known as *ferro-electricity*) to liquid crystal displays, these materials can be used to make electro-optic shutters which can be switched open and closed in microseconds.

Phase transformation from an isotropic liquid to a chiral nematic liquid crystal



Thank you

Liquid crystal compounds

Liquid crystal compounds

- Liquid-crystal-forming compounds are widespread and quite diverse.
- Soap can form a type of smectic known as a lamellar phase, also called neat soap.
- In this case it is important to recognize that soap molecules have a dual chemical nature.
- One end of the molecule (the hydrocarbon tail) is attracted to oil, while the other end (the polar head) attaches itself to water.

- When soap is placed in water, the hydrocarbon tails cluster together, while the polar heads adjoin the water.

- Small numbers of soap molecules form spherical or rodlike micelles, which float freely in the water, while concentrated solutions create bilayers, which stack along some direction just like smectic layers.

- Indeed, the name smectic is derived from the Greek word for soap. The slippery feeling caused by soap reflects the ease with which the layers slide across one another.

- Many biological materials form liquid crystals.
- Myelin, a fatty material extracted from nerve cells, was the first intensively studied liquid crystal.
- The tobacco mosaic virus, with its rod like shape, forms a nematic phase.
- In cholesterol the nematic phase is modified to a cholesteric phase characterized by continuous rotation of the direction of molecular alignment.

- An intrinsic twist of the cholesterol molecule, rather like the twist of the threads of a screw, causes this rotation.

- Since the molecular orientation rotates steadily, there is a characteristic distance after which the orientation repeats itself.

- This distance is frequently comparable to the wavelength of visible light, so brilliant color effects result from the diffraction of light by these materials.

Biological liquid crystals

- Lyotropic liquid-crystalline phases are abundant in living systems, the study of which is referred to as lipid polymorphism.
- Accordingly, lyotropic liquid crystals attract particular attention in the field of biomimetic chemistry.
- In particular, biological membranes and cell membranes are a form of liquid crystal.
- Their constituent molecules (e.g. phospholipids) are perpendicular to the membrane surface, yet the membrane is flexible.

- Many other biological structures exhibit liquid-crystal behavior.

- For instance, the concentrated protein solution that is extruded by a spider to generate silk is, in fact, a liquid crystal phase.

- The precise ordering of molecules in silk is critical to its renowned strength.

- DNA and many polypeptides can also form LC phases and this too forms an important part of current academic research.

Mineral liquid crystals

- Examples of liquid crystals can also be found in the mineral world, most of them being lyotropics. The first discovered was Vanadium(V) oxide, by Zocher in 1925.
- Since then, few others have been discovered and studied in detail.
- The existence of a true nematic phase in the case of the smectite clays family was raised by Langmuir in 1938, but remained open for a very long time and was only solved recently.
- With the rapid development of Nano sciences, and the synthesis of many new anisotropic nanoparticles, the number of such mineral liquid crystals is quickly increasing

Thank you

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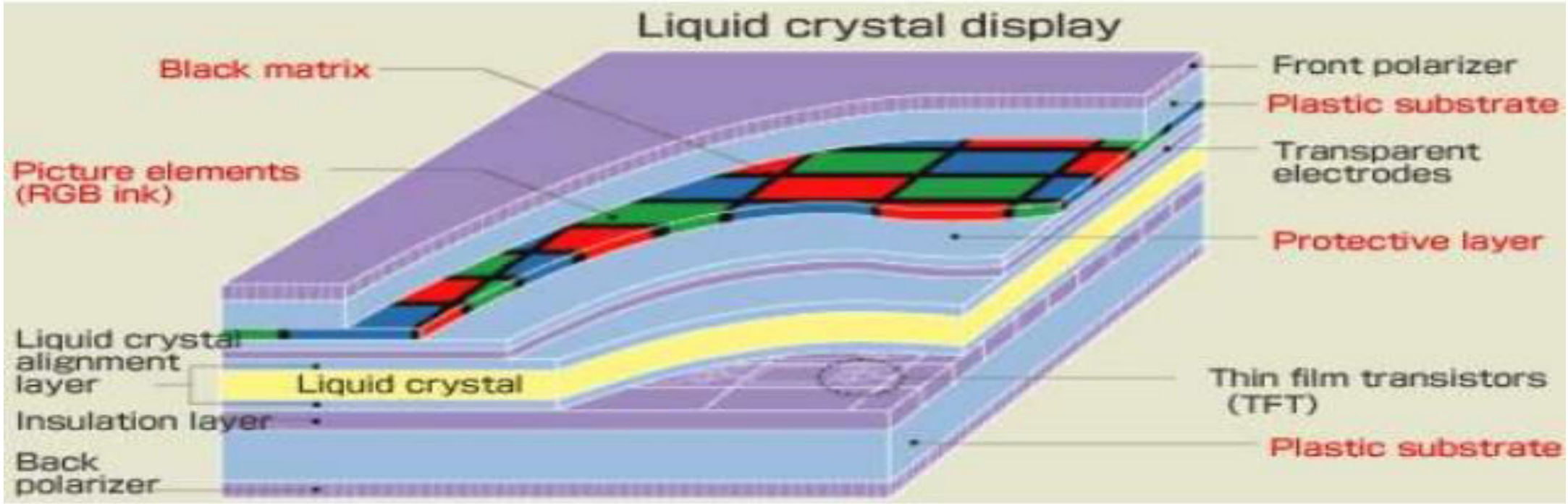
Applications of liquid crystals

Applications of liquid crystals

- Liquid crystals find wide use in liquid crystal displays, which rely on the optical properties of certain liquid crystalline substances in the presence or absence of an electric field.

- Liquid crystal tunable filters are used as electro-optical devices, e.g., in hyper-spectral imaging.

Liquid Crystal Displays



Liquid crystal display (LCD) is a flat electronic display panel used as a visual display aid. It uses liquid crystals to create visual effects on screen. Liquid crystal is a state of matter between solids and liquids. LCDs do not generate light on their own but rely on sunlight or room light to generate images with help of liquid crystal.

Advantages Of LCD

- **Brightness**: Produces very bright images due to high peak intensity. Very suitable for environments that are brightly lit.
- **Emissions**: Produce considerably lower electric, magnetic and electromagnetic fields than CRTs.
- **Geometric Distortion :** Geometric distortion at the native resolutions. Minor distortion can occur for other resolutions.
- **Power Consumption:**Energy efficient. Consume less than 1/3 the power of a comparable CRT. Consume less electricity than a CRT and produce little heat.
- **Physical Aspects:**Take up about 40% less desk space. LCDs are thin and compact.
- **Screen Shape :**Completely flat screen.
- **Sharpness:** At the native resolution, the image is perfectly sharp. Adjustments are required at all other resolutions which can result in measurable degradation to the image.

- Thermotropic chiral LCs whose pitch varies strongly with temperature can be used as crude liquid crystal thermometers, since the color of the material will change as the pitch is changed.

- Liquid crystal color transitions are used on many aquarium and pool thermometers as well as on thermometers for infants or baths.

- Other liquid crystal materials change color when stretched or stressed. Thus, liquid crystal sheets are often used in industry to look for hot spots, map heat flow, measure stress distribution patterns, and so on.

- Liquid crystal in fluid form is used to detect electrically generated hot spots for failure analysis in the semiconductor industry.

Liquid Crystal Thermometers





Liquid-crystal thermometers use liquid crystals that change color in response to temperature changes. Mixtures of liquid crystals are enclosed in separate partitions. Numbers on the partitions indicate temperatures according to the amount of heat present. Liquid-crystal thermometers include some indoor digital thermometers and fever thermometers, which are placed on the forehead.

Battery testing strips (used by DuraCell)



- Charge indicator for batteries. The principles for this application is similar to that used by liquid crystalline thermometers. Electrical current passing through the media generates heat. The change of relative orientation of the layers produce a color change.

- Polymer dispersed liquid crystal (PDLC) sheets and rolls are available as adhesive backed Smart film which can be applied to windows and electrically switched between transparent and opaque to provide privacy.

- Many common fluids, such as soapy water, are in fact liquid crystals. Soap forms a variety of LC phases depending on its concentration in water.

- Bowlic columns could be used for fast switches

Slugs and Snails (nature)



- Nature beat us to the punch in making use of liquid crystals. The mucus secreted by slugs behaves like liquid crystals. Its rod-shaped molecules align in varying degrees to control the viscosity of the mucus to adapt to different ground conditions.
- Liquid crystals can also be found in biological cell-membranes

Thank you

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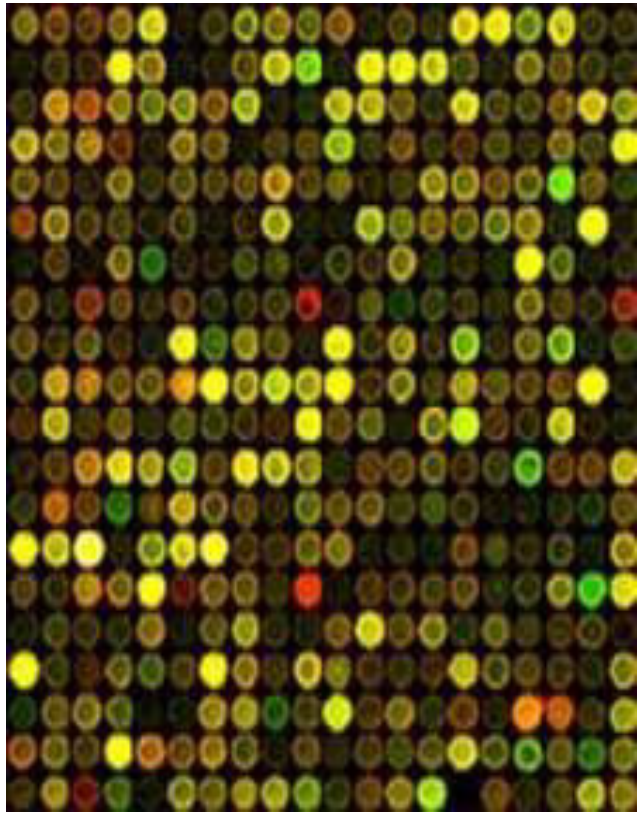
Microarray

Microarray

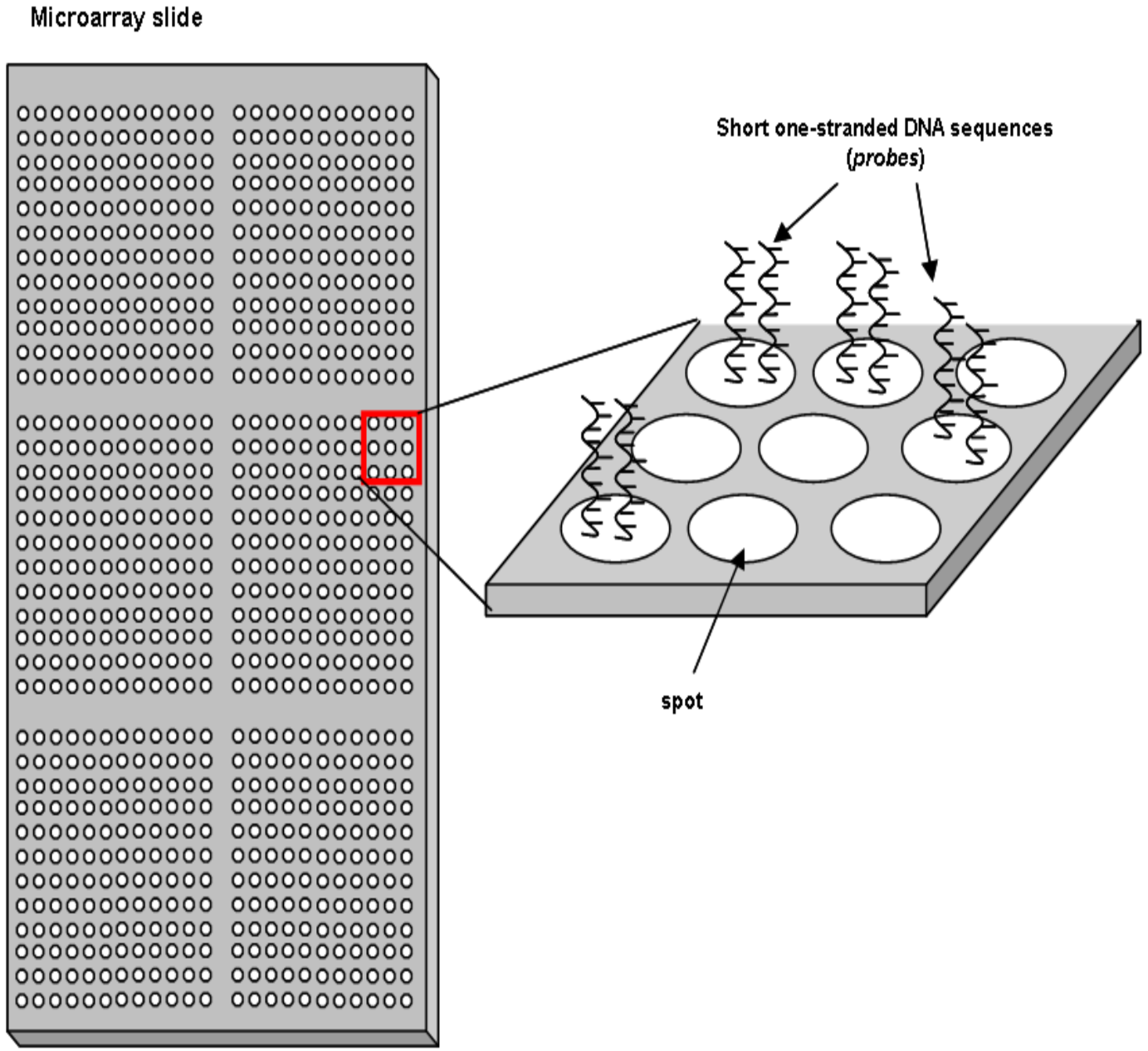
- A microarray is a multiplex lab-on-a-chip.

- It is a 2D array on a solid substrate (usually a glass slide or silicon thin-film cell) that assays large amounts of biological material using high-throughput screening miniaturized, multiplexed and parallel processing and detection methods.

- The concept and methodology of microarrays was first introduced and illustrated in antibody microarrays (also referred to as antibody matrix) by **Tse Wen Chang** in 1983.



digital photo of a microarray



Types of microarrays

- DNA microarrays, such as cDNA microarrays, oligonucleotide microarrays, BAC microarrays and SNP microarrays
- Protein microarrays
- Peptide microarrays, for detailed analyses or optimization of protein–protein interactions
- Tissue microarrays
- Cellular microarrays (also called transfection microarrays)
- Chemical compound microarrays
- Antibody microarrays
- Glycan arrays (carbohydrate arrays)
- Phenotype microarrays
- Reverse Phase Protein Microarrays, microarrays of lysates or serum
- Interferometric reflectance imaging sensor (IRIS)

DNA microarray

- A DNA microarray (also commonly known as DNA chip or biochip) is a collection of microscopic DNA spots attached to a solid surface.

- Scientists use DNA microarrays to measure the expression levels of large numbers of genes simultaneously or to genotype multiple regions of a genome

Protein microarray

- A protein microarray (or protein chip) is a high-throughput method used to track the interactions and activities of proteins, and to determine their function, and determining function on a large scale.

- Its main advantage lies in the fact that large numbers of proteins can be tracked in parallel.

Peptide microarray

- A peptide microarray (also commonly known as peptide chip or peptide epitope microarray) is a collection of peptides displayed on a solid surface, usually a glass or plastic chip.

- Peptide chips are used by scientists in biology, medicine and pharmacology to study binding properties and functionality and kinetics of protein-protein interactions in general.

Tissue microarray

- Tissue microarrays (also TMAs) consist of paraffin blocks in which up to 1000 separate tissue cores are assembled in array fashion to allow multiplex histological analysis.

Cellular microarray

- A cellular microarray (or cell microarray) is a laboratory tool that allows for the multiplex interrogation of living cells on the surface of a solid support.

- The support, sometimes called a "chip", is spotted with varying materials, such as antibodies, proteins, or lipids, which can interact with the cells, leading to their capture on specific spots

Chemical compound microarray

- A chemical compound microarray is a collection of organic chemical compounds spotted on a solid surface, such as glass and plastic.

- This microarray format is very similar to DNA microarray, protein microarray and antibody microarray.

- In chemical genetics research, they are routinely used for searching proteins that bind with specific chemical compounds, and in general drug discovery research, they provide a multiplex way to search potential drugs for therapeutic targets.

Antibody microarray

- An antibody microarray (also known as antibody array) is a specific form of protein microarray. In this technology, a collection of capture antibodies are spotted and fixed on a solid surface such as glass, plastic, membrane, or silicon chip, and the interaction between the antibody and its target antigen is detected.

- Antibody microarrays are often used for detecting protein expression from various bio fluids including serum, plasma and cell or tissue lysates.

- Antibody arrays may be used for both basic research and medical and diagnostic applications

Glycan array

- Glycan arrays, contain carbohydrate compounds that can be screened with lectins, antibodies or cell receptors to define carbohydrate specificity and identify ligands.

- Glycan array screening works in much the same way as other microarray that is used for instance to study gene expression DNA microarrays or protein interaction Protein microarrays.

Phenotype microarray

- The phenotype microarray approach is a technology for high-throughput phenotyping of cells.
- A phenotype microarray system enables one to monitor simultaneously the phenotypic reaction of cells to environmental challenges or exogenous compounds in a high-throughput manner.
- The phenotypic reactions are recorded as either end-point measurements or respiration kinetics similar to growth curves.

Thank you

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DNA microarray

- A DNA microarray (also commonly known as DNA chip or biochip) is a collection of microscopic DNA spots attached to a solid surface.
- Scientists use DNA microarrays to measure the expression levels of large numbers of genes simultaneously or to genotype multiple regions of a genome.
- Each DNA spot contains picomoles (10−12 moles) of a specific DNA sequence, known as probes (or reporters or oligos).
- These can be a short section of a gene or other DNA element that are used to hybridize a cDNA or cRNA (also called anti-sense RNA) sample (called target) under high-stringency conditions.
- Probe-target hybridization is usually detected and quantified by detection of fluorophore-, silver-, or chemiluminescence-labeled targets to determine relative abundance of nucleic acid sequences in the target.

Principle

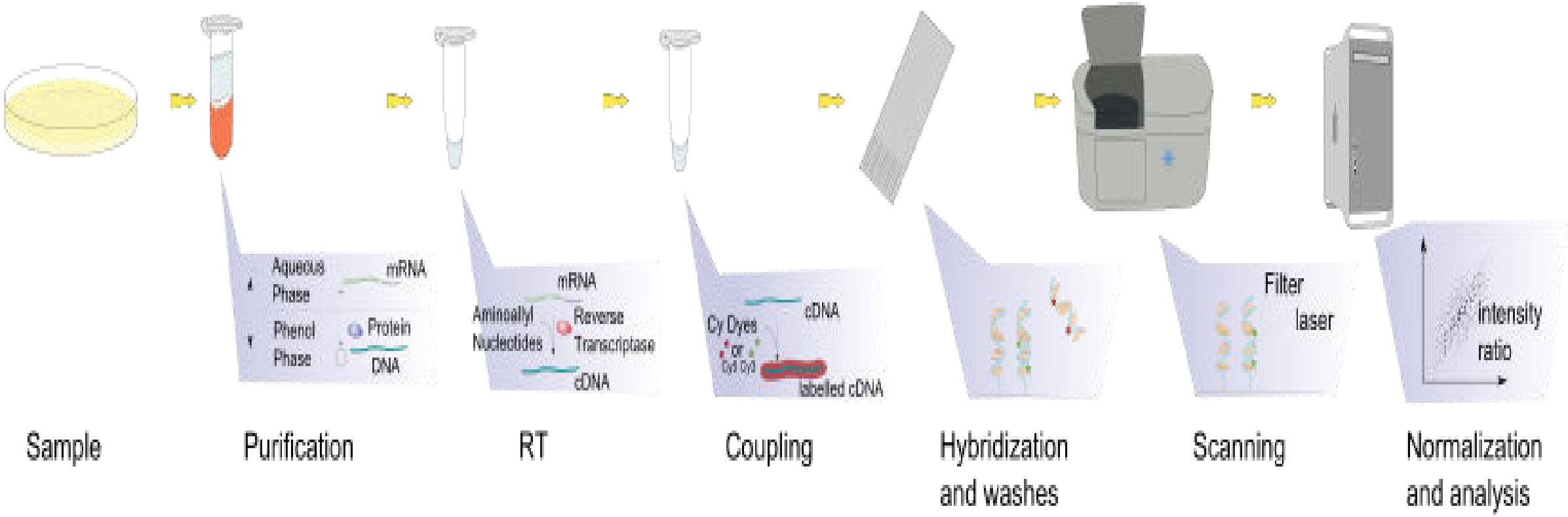
- The core principle behind microarrays is hybridization between two DNA strands, the property of complementary nucleic acid sequences to specifically pair with each other by forming hydrogen bonds between complementary nucleotide base pairs.

- A high number of complementary base pairs in a nucleotide sequence means tighter non-covalent bonding between the two strands.

- After washing off non-specific bonding sequences, only strongly paired strands will remain hybridized.

- Fluorescently labeled target sequences that bind to a probe sequence generate a signal that depends on the hybridization conditions (such as temperature), and washing after hybridization.
-
- Total strength of the signal, from a spot (feature), depends upon the amount of target sample binding to the probes present on that spot.

- Microarrays use relative quantitation in which the intensity of a feature is compared to the intensity of the same feature under a different condition, and the identity of the feature is known by its position.



The steps required in a microarray experiment

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Basic protocol

Step 1: Isolate and purify mRNA from samples of interest

Since we are interested in comparing gene expression, one sample usually serves as control, and another sample would be the experiment (healthy vs. disease, etc)

Step 2: Reverse transcribe and label the mRNA.

- In order to detect the transcripts by hybridization, they need to be labeled, and because starting material maybe limited, an amplification step is also used.

- Labeling usually involves performing a reverse transcription (RT) reaction to produce a complementary DNA strand (cDNA) and incorporating a florescent dye that has been linked to a DNA nucleotide, producing a fluorescent cDNA strand.

- Disease and healthy samples can be labeled with different dyes and cohybridized onto the same microarray in the following step.

- Some protocols do not label the cDNA but use a second step of amplification, where the cDNA from RT step serves as a template to produce a labeled cRNA strand.

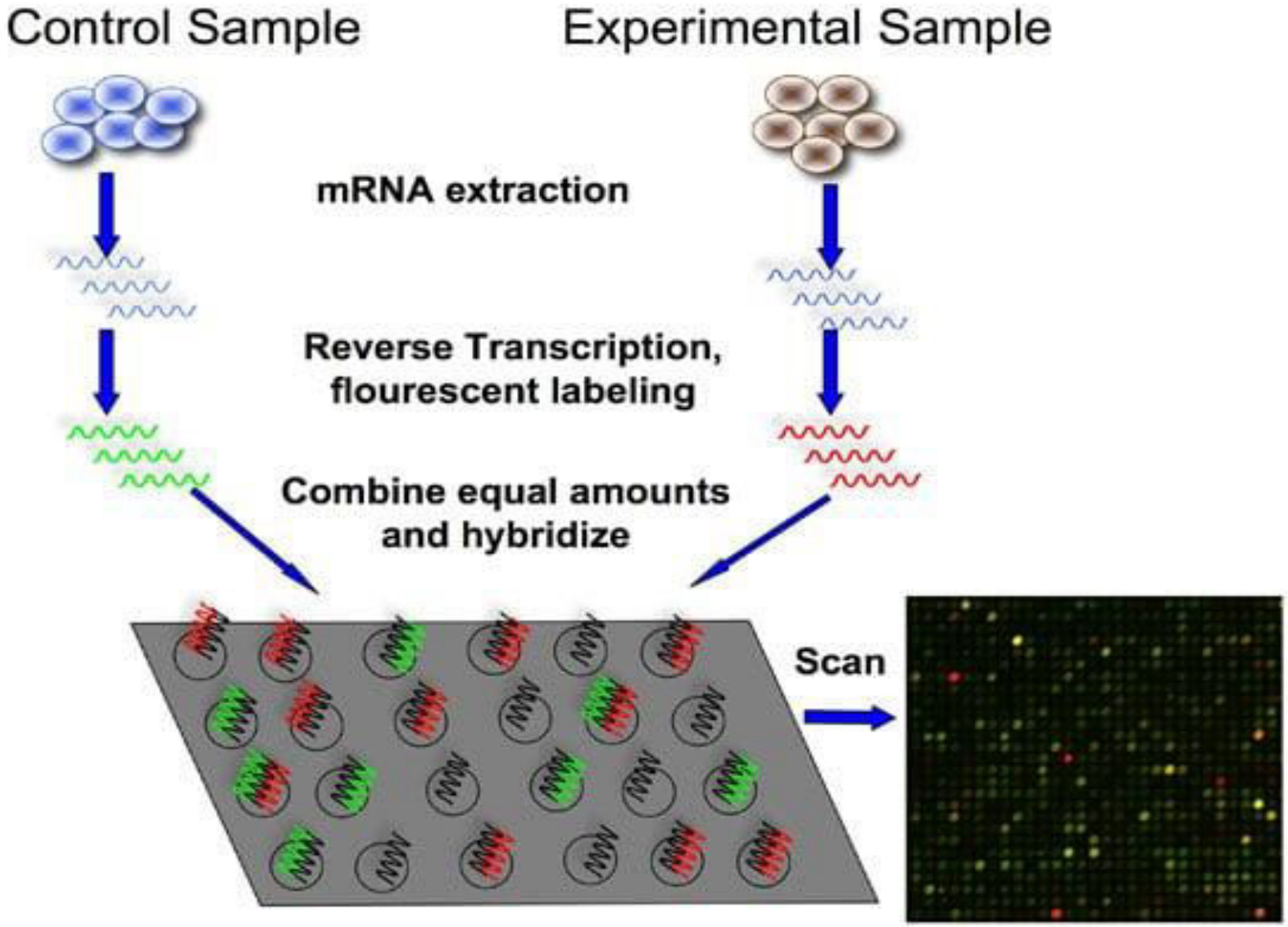
Step 3: Hybridize the labeled target to the microarray.

- This step involves placing labeled cDNAs onto a DNA microarray where it will hybridize to their synthetic complementary DNA probes attached on the microarray.

- A series of washes are used to remove non-bound sequences.

Step 4: Scan the microarray and quantitate the signal.

- The fluorescent tags on bound cDNA are excited by a laser and the fluorescently labeled target sequences that bind to a probe generate a signal.
- The total strength of the signal depends upon the amount of target sample binding to the probes present on that spot.
- Thus, the amount of target sequence bound to each probe correlates to the expression level of various genes expressed in the sample.
- The signals are detected, quantified, and used to create a digital image of the array.


Control Sample
Experimental Sample
mRNA extraction
Reverse Transcription,
flourescent labeling
Combine equal amounts
and hybridize
Scan

Types of DNA microarrays

DNA microarrays are of four types:

- **cDNA microarrays:** uses complementary DNA strands formed by transcription of mRNA
- **Oligo DNA microarrays:** uses chemically synthesized oligo DNA as probes
- **BAC microarrays:** uses template amplified by polymerase chain reaction as the probe
- **SNP microarrays:** used to detect polymorphisms within a population

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DNA Microarrays

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DNA Microarray

Applications

- To study transcriptomes and proteomes
- To diagnose pathogenic as well as genetic diseases in man
- To identify microbes in the environment with the help of species-specific probes
- To genotype genomes through single nucleotide polymorphism (SNP) analysis
- To detect gene expression of mRNAs of a particular cell at different times
- To measure changes in the level of gene expression
- To observe DNA mutations
- To study genomic gains and losses

Limitations of DNA microarrays

- The results take a lot of time to analyze as the amount of data collected from each array will be huge
- The results may be too complex to interpret and are not always quantitative
- The results are not always reproducible
- The technology is too expensive
- The arrays provide an indirect measure of relative concentration
- Especially for complex mammalian genomes, it is often difficult to design arrays in which multiple related DNA/RNA sequences do not bind to the same probe on the array
- A DNA array can only detect sequences that the array was designed to detect

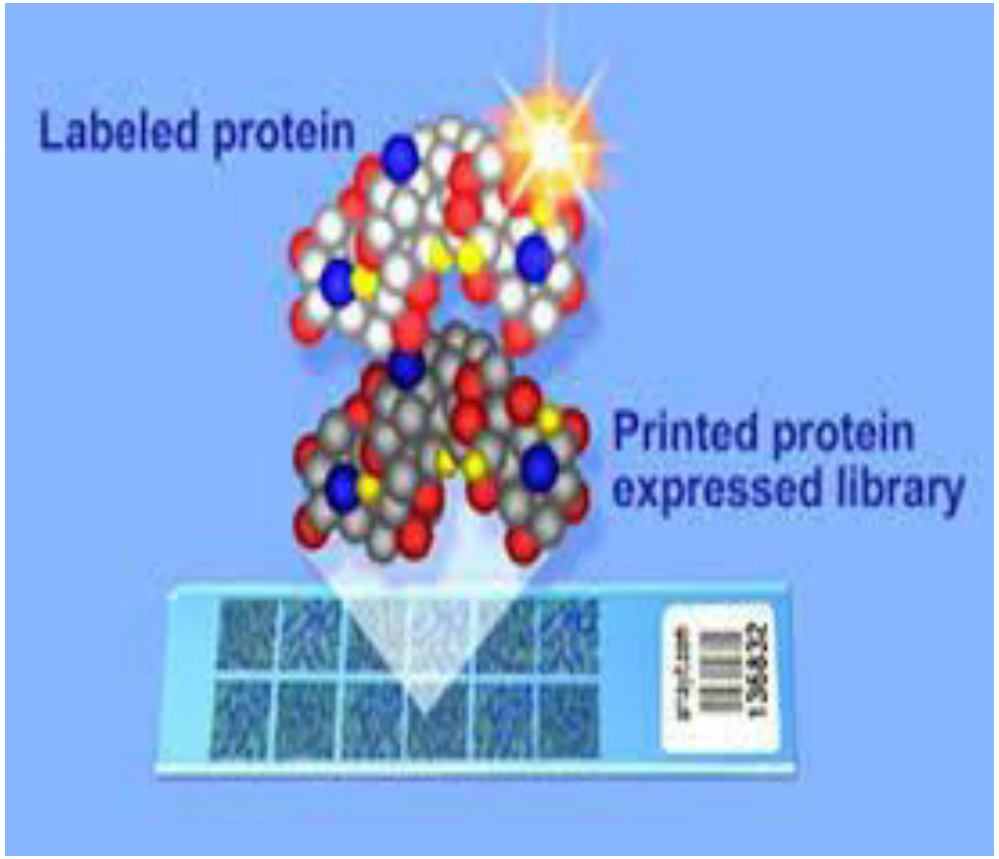
Thank you

Protein microarray

Protein microarray

- A protein microarray (or protein chip) is a high-throughput method used to track the interactions and activities of proteins, and to determine their function, and determining function on a large scale.

- Its main advantage lies in the fact that large numbers of proteins can be tracked in parallel.



Principle

- The chip consists of a support surface such as a glass slide, nitrocellulose membrane, bead, or microtiter plate, to which an array of capture proteins is bound.

- Probe molecules, typically labeled with a fluorescent dye, are added to the array. Any reaction between the probe and the immobilized protein emits a fluorescent signal that is read by a laser scanner.

- Protein microarrays are rapid, automated, economical, and highly sensitive, consuming small quantities of samples and reagents.

- Protein microarrays were developed due to the limitations of using DNA microarrays for determining gene expression levels in proteomics.

- The quantity of mRNA in the cell often doesn't reflect the expression levels of the proteins they correspond to. Since it is usually the protein, rather than the mRNA, that has the functional role in cell response, a novel approach was needed.

- Additionally post-translational modifications, which are often critical for determining protein function, are not visible on DNA microarrays.

- Protein microarrays replace traditional proteomics techniques such as 2D gel electrophoresis or chromatography, which were time consuming, labor-intensive and ill-suited for the analysis of low abundant proteins.

Steps

1. The proteins are arrayed onto a solid surface such as microscope slides, membranes, beads or microtitre plates. The function of this surface is to provide a support onto which proteins can be immobilized.

2. The chosen solid surface is then covered with a coating that must serve the simultaneous functions of immobilizing the protein, preventing its denaturation.

3. Immobilizing agents include layers of aluminum or gold, hydrophilic polymers, and polyacrylamide gels, or treatment with amines, aldehyde or epoxy. Thin-film technologies like physical vapor deposition (PVD) and chemical vapor deposition (CVD) are employed to apply the coating to the support surface.

4. An aqueous environment is essential at all stages of array manufacture and operation to prevent protein denaturation. Therefore, sample buffers contain a high percent of glycerol.

5. In the most common type of protein array, robots place large numbers of proteins or their ligands onto a coated solid support in a pre-defined pattern. This is known as robotic contact printing or robotic spotting. Another fabrication method is ink-jetting, a drop-on-demand, non-contact method of dispersing the protein polymers onto the solid surface in the desired pattern

6. The capture molecules arrayed on the solid surface may be antibodies, antigens, aptamers (nucleic acid-based ligands), affibodies (small molecules engineered to mimic monoclonal antibodies), or full length proteins.

Sources of such proteins include cell-based expression systems for recombinant proteins, purification from natural sources, production in vitro by cell-free translation systems, and synthetic methods for peptides

7. Protein array detection methods must give a high signal and a low background. The most common and widely used method for detection is fluorescence labeling which is highly sensitive, safe and compatible with readily available microarray laser scanners. Other labels can be used, such as affinity, photochemical or radioisotope tags

Applications

There are five major areas where protein arrays are being applied:

1. diagnostics,
2. proteomics,
3. protein functional analysis,
4. antibody characterization, and
5. treatment development

- Diagnostics involves the detection of antigens and antibodies in blood samples; the profiling of sera to discover new disease biomarkers; the monitoring of disease states and responses to therapy in personalized medicine; the monitoring of environment and food.
- Proteomics pertains to protein expression profiling i.e. which proteins are expressed in the lysate of a particular cell.
- Protein functional analysis is the identification of protein–protein interactions (e.g. identification of members of a protein complex), protein–phospholipid interactions, small molecule targets, enzymatic substrates (particularly the substrates of kinases) and receptor ligands.
- Antibody characterization is characterizing cross-reactivity, specificity and mapping epitopes.
- Treatment development involves the development of antigen-specific therapies for autoimmunity, cancer and allergies; the identification of small molecule targets that could potentially be used as new drugs.

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Peptide microarray

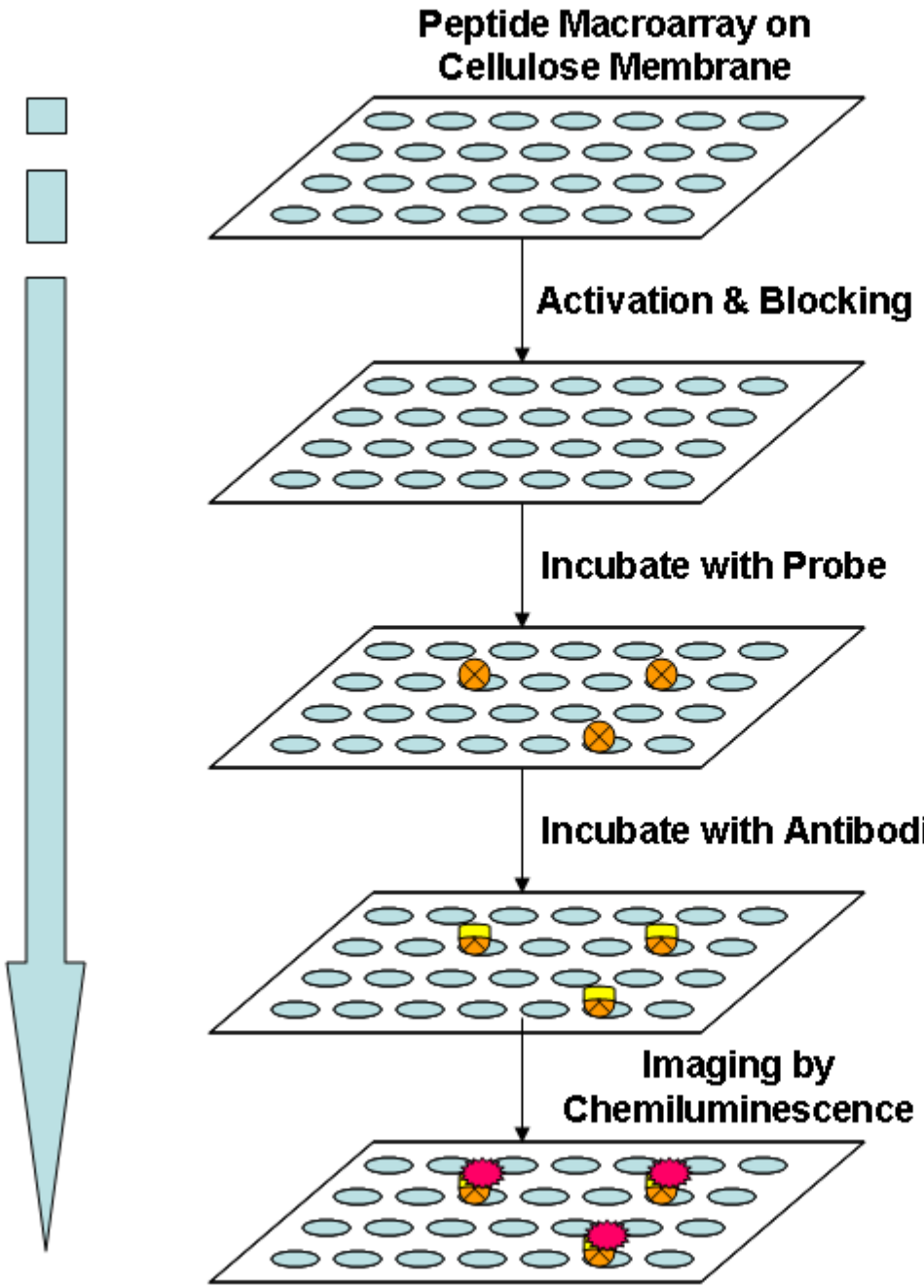
Peptide microarray

- A peptide microarray (also commonly known as peptide chip or peptide epitope microarray) is a collection of peptides displayed on a solid surface, usually a glass or plastic chip.
- Peptide chips are used by scientists in biology, medicine and pharmacology to study binding properties and functionality and kinetics of protein-protein interactions in general.

- In basic research, peptide microarrays are often used to profile an enzyme (like kinase, phosphatase, protease, acetyltransferase, histone deacetylase etc.), to map an antibody epitope or to find key residues for protein binding.
- Practical applications are seromarker discovery, profiling of changing humoral immune responses of individual patients during disease progression, monitoring of therapeutic interventions, patient stratification and development of diagnostic tools and vaccines.

Principle/ Procedure

- The assay principle of peptide microarrays is similar to an ELISA protocol.
- The peptides (up to tens of thousands in several copies) are linked to the surface of a glass chip typically the size and shape of a microscope slide.
- This peptide chip can directly be incubated with a variety of different biological samples like purified enzymes or antibodies, patient or animal sera, cell lysates and then be detected through a label-dependent fashion,
- for example, by a primary antibody that targets the bound protein or modified substrates.
- After several washing steps a secondary antibody with the needed specificity is applied.
- Usually, the secondary antibody is tagged by a fluorescence label that can be detected by a fluorescence scanner.
- Other label-dependent detection methods includes chemiluminescence, colorimetric or autoradiography.



Intuitive

Straightforward

Western Blot Like Procedure

No Need of Microarray Scanner

Only Digital Image Station

No Centralized Data Analysis

Full Control of Experiment & Data Analysis

Peptide microarrays show several advantages over protein microarrays:

- Ease and cost of synthesis
- Extended shelf stability
- Detection of binding events on epitope level, enabling study of i.e. epitope spreading
- Flexible design for peptide sequence (i.e. posttranslational modifications, sequence diversity, non-natural amino acids ...) and immobilization chemistries
- Higher batch-to-batch reproducibility

Applications of peptide microarrays

Immunology

- Mapping of immuno-dominant regions in antigens or whole proteomes
- Sero-marker discovery
- Monitoring of clinical trials
- Profiling of antibody signatures and epitope mapping
- Finding neutralizing antibodies

Enzyme profiling

- Identification of substrates for orphan enzymes
- Optimization of known enzyme substrates
- Elucidation of signal transduction pathways
- Detection of contaminating enzyme activities
- Consensus sequence and key residues determination
- Identifying sites for protein-protein interactions within a complex

Thank you

Tissue microarray and Cellular microarray

Tissue microarray

- Tissue microarrays (also TMAs) consist of paraffin blocks in which up to 1000 separate tissue cores are assembled in array fashion to allow multiplex histological analysis.



A Tissue MicroArray Block

Procedure

- In the tissue microarray technique, a hollow needle is used to remove tissue cores as small as 0.6 mm in diameter from regions of interest in paraffin-embedded tissues such as clinical biopsies or tumor samples.
- These tissue cores are then inserted in a recipient paraffin block in a precisely spaced, array pattern.
- Sections from this block are cut using a microtome, mounted on a microscope slide and then analyzed by any method of standard histological analysis.
- Each microarray block can be cut into 100 – 500 sections, which can be subjected to independent tests.
- Tests commonly employed in tissue microarray include immunohistochemistry, and fluorescent in situ hybridization.
- Tissue microarrays are particularly useful in analysis of cancer samples.

- The use of tissue microarrays in combination with immunohistochemistry has been a preferred method to study and validate cancer biomarkers in various defined cancer patient cohorts.

- The possibility to assemble a large number of representative cancer samples from a defined patient cohort that also has a corresponding clinical database, provides a powerful resource to study how different protein expression patterns correlate with different clinical parameters.

- Since patient samples are assembled into the same block, sections can be stained with the same protocol to avoid experimental variability and technical artefacts.

- Clinical cancer patient cohorts and corresponding tissue microarray sets have been used to study diagnostic, prognostic and treatment predictive cancer biomarkers in most forms of cancer, including lung, breast, colorectal and renal cell cancer

Cellular microarray

- A cellular microarray (or cell microarray) is a laboratory tool that allows for the multiplex interrogation of living cells on the surface of a solid support.
- The support, sometimes called a "chip", is spotted with varying materials, such as antibodies, proteins, or lipids, which can interact with the cells, leading to their capture on specific spots.
- Combinations of different materials can be spotted in a given area, allowing not only cellular capture, when a specific interaction exists, but also the triggering of a cellular response, change in phenotype, or detection of a response from the cell, such as a specific secreted factor.

Chemical compound microarray

- A chemical compound microarray is a collection of organic chemical compounds spotted on a solid surface, such as glass and plastic. This microarray format is very similar to DNA microarray, protein microarray and antibody microarray.

- In chemical genetics research, they are routinely used for searching proteins that bind with specific chemical compounds, and in general drug discovery research, they provide a multiplex way to search potential drugs for therapeutic targets.

Polymer Microarrays

- Polymer microarrays have been developed to allow screening for new polymeric materials to direct different tissue lineages.

- Research has also been directed towards studying the surface chemistry of these arrays to determine which surface chemistries control cell adhesion, although concerns have been raised as to the influence of the substrate on measurements and the questionable statistical interpretation of results.

Thank you

Main research goals in biophysics

Biophysics study makes exciting advancements for the future of DNA sequencing

- New technology that optimizes DNA sequencing using nanophysics and electric currents has been developed by scientists.

- Their work offers a method for loading DNA into sequencing wells with orders of magnitude higher efficiencies.

Research areas

Major areas of research include:

- Biophysical approaches to cell biology
- Complex biological systems
- Computational and theoretical biophysics
- Membrane biophysics
- Protein engineering and synthetic biology
- Proteomics and genomics
- Structural biology

Biophysical approaches to cell biology

- Atom-level analyses using physics-based methods and fundamental principles of physical chemistry.
- Protein folding in vivo, protein sorting, and macromolecule dynamics
- Molecular basis of electrical signaling and synaptic plasticity.
- Cellular energy metabolism.
- Engineering and physical principles underlying cellular morphogenesis
- Mechanisms of chromatin remodeling and trans membrane signaling.
- Regulation of transcription and translation.

Brain Imaging and Bioelectronics

- We can infer much about how a neural system functions by quantifying its output (behavior) in relation to its input (stimuli).
- But to completely understand a neural system, one must also have methods to measure its neural activity. Most such methods are invasive.
- However, invasive methods

(1) alter the system that it sets out to measure

(2) often involve anesthesia that alters neural activity

(3) may inflict suffering on animals, and

(4) usually can't be used to study humans.

Non-invasive functional neuro-imaging, although lagging behind invasive methods in spatial and temporal resolution, are mostly devoid of these drawbacks, and therefore are extremely valuable in the study of neural systems.

Systems Neuroscience

- Systems neuroscience addresses the function of neural circuits in intact organisms.
- The ultimate goal of this area of research is to understand the function of the nervous system at a variety of levels, from single cells to entire networks that mediate complex behaviors such as vision, audition and motor responses.
- Because systems neuroscience addresses such a broad range of issues, it integrates experimental, analytic and theoretical techniques from a wide range of disciplines.
- In particular many important analytic and theoretical approaches were developed first in physics and engineering and have been modified for application to neurobiology.
- At the finest scale of analysis, systems researchers investigate how single cells function as computational units. These studies assess the roles of channel distribution, electrical cable properties and dendritic architecture in neural computation.

Molecular Microscopy and Optical Probes

- Another large area of active research in the Biophysics Group is molecular microscopy.
- At the heart of the initiative is the idea that the next generation of microscopy will depend on novel optical probes made of proteins and nucleic acids, rather than organic chemicals.
- The ability to visualize the targeting within cells of native macromolecules has already become key to understanding their biological function.
- The next step is to understand how they assemble into the complexes that carry out cellular functions, and to monitor their activities in real-time in vivo with microscopy so that the spatial location of the activity can be understood.

- Fluorescent indicator dyes have revolutionized our understanding of cellular signaling by providing continuous measurements of physiological events in single cells and cell populations with high temporal and spatial resolution.

- Until now, these organic dyes have been limited in scope because they must be synthesized chemically and introduced as hydrolysable esters or by microinjection, meaning that they either fill all cells in a tissue entirely, or that only one or a small number of cells are labeled.

- In absence of targeting of the organic dyes to specific cell types, optical signals of interest are often drowned out either by indicator background from inactive cells, or by signals in cells that are not the focus of interest.

- Moreover, local signals due to protein-protein interactions are not measurable in the bulk cytoplasm, which organic dyes fill.

Virtual University

Thank you

Cell Signaling and Cellular Physiology

- Cell signaling and cell physiology are areas of research that are gaining prominence due to rapid advances in biophysical methods and the integration of these methods with molecular biology, biochemistry and genetics.

- These advances have allowed studies of molecular processes in the context of the integrated behavior of cells and their organelles, and of tissues and organs, and whole organisms.

- In addition, the application of theoretical mathematical models to this research is providing unique insights into how the ensemble behaviors of ion channels, pumps and cellular signaling systems determine cell behavior, and how groups of cells cooperatively determine the behavior of complex tissues such as the brain.

Computational Biology and Genomics

- Computational methods are used to discover previously undetected molecular and cellular biological activities, to design computational screens for genetic control elements, and to provide a detailed physical analysis of the biochemical and genetic networks that govern cellular development with the goal of engineering cell and tissue systems.

- Computational approaches are also being used to determine the rules of protein folding, in protein design (with an emphasis on enzymes), to understand the dynamics of signaling networks responsible for cell differentiation and motility, to model the energetics of force generation and structural rearrangement in motor proteins, ion channels and pumps, to model population/disease dynamics, and to model information processing by neural networks in vision, audition and learning, and to design systems for computer vision.

Biophysics is a thriving discipline, yet biophysics also has an identity problem—due to the wide range of research topics that properly fall under the general title of biophysics—and biophysicists often find themselves challenged when asked to describe what the term actually represents.

Thank you